

Research continuation and complete source proofs

Split-Zero Cohomology and Arithmetic Weight Control

12 September 2026

This reader integrates the Split-Zero carrier, finite arithmetic operator comparisons, the historical packet ghost, chain retractions, arithmetic moments, Gaussian resolvents, and the current orthogonal boundary calculation. It retains the original arithmetic functions, measures, scalar actions, support fibres, and complete jets. Part I gives the local continuation with proofs. Part II preserves the complete selected source notes, their exact revisions, and their chronology.

Contents

I	Integrated mathematical continuation	4
1	Exact Split-Zero carriers and the local spectral bridge	4
1.1	Scalar support and independent channel amplitudes	4
1.2	Tesseractine sectors and a supported critical fixed locus	6
1.3	Exact sector, holonomy and positive-moment maps	7
1.4	The theta-packet form and its scalar comparison	9
1.5	Projective zeta at all zeros and at its pole	9
1.6	Matrix packets and their full relation module	10
2	The historical packet ghost and its full arithmetic jet quotient	11
3	Historical Hurwitz reconstruction and the complete Laurent jet bridge	15
3.1	Source receipts and exact locators	15
3.2	Every compact Hurwitz input is determined by one simple-zero germ	16
3.3	The complete zero cluster removes the simplicity restriction	19
3.4	The historical period map preserves every finite spectral jet	20
3.5	Exact conjugate-reciprocal involution on the jets and conormal	22
3.6	Period collisions: the exact unlabelled image and its missing fibres	23
3.7	Two prime periods recover the complete finite spectral algebra	24
3.8	Split-Zero support is preserved by the exact maps	26
3.9	All-order Hurwitz motion transported to the period and tesseractine coordinates	27
3.10	Mathematical scope	28
4	Reconstruction with changes of support index	28
4.1	The internal quotient and the actual comparison kernel	30
5	The original theta complex and its arithmetic packet	30
5.1	Source spaces and fixed arithmetic factors	31
5.2	The exact operator range, including its inverse	32
5.3	The original extension supplies the boundary coefficient	33
5.4	Actual representative corrections and rank-two control	34
6	Certified moments from the original arithmetic theta source	35
6.1	Source polynomials and the exact reflection map	35
6.2	An absolutely convergent double series for every entry	36
6.3	A complete finite truncation certificate	37
6.4	Exact adjoint, parity, and finite-boundary checks	38
6.5	Reproducible certified finite arithmetic data	39
7	Exact source-metric optimization and aggregate arithmetic control	40
7.1	The fixed arithmetic maps and their boundary	40
7.2	A source-density calculation with its topology retained	41

7.3	Every positive packet metric has an exact finite source lift	41
7.4	Sharp control with all nilpotent jets present	43
7.5	The exact tensor cost of unrestricted metrics	45
7.6	A stronger bound on the actual orthogonal sequence	45
7.7	Original supported maps and the precise result delivered	46
8	The actual theta minimum as a confluent arithmetic kernel	47
8.1	Original functions, maps, coordinates, and measures	47
8.2	Exact constrained minimum over the admitted source	48
8.3	Monic arithmetic polynomials and their exact recurrence	49
8.4	Exact relative weight control from the same kernel	50
8.5	Exact change when one more theta direction is admitted	51
8.6	The original measure supplies a full tail recurrence	52
8.7	Metric dependence and the precise mathematical gain	53
9	The full jet kernel, finite resolvents, and the actual analytic tail	54
9.1	A finite compression in the original Mellin coordinate	54
9.2	Exact quadrature and the full confluent kernel	55
9.3	The infinite multiplication operator and its genuine tail	57
10	Quantitative convergence of the arithmetic Gaussian resolvents	58
10.1	The Gaussian rule inherits the original exponential moments	59
10.2	Exact moments give a quantitative analytic-strip estimate	60
10.3	Every confluent derivative and the original finite algebra	61
11	The exact terminal step and its retained boundary energy	63
11.1	A step written entirely in the original polynomial packet	63
11.2	The complete boundary-energy increment	64
11.3	The exact rational terminal step and its polynomial continuation	65
11.4	The extra jets required to recover the metric from the tail	66
11.5	Actual quartet packets close the energy in two volume ratios	68
11.6	What the recurrence proves about boundary energy	69
11.7	An exact positive-moment counterexample to consecutive energy monotonicity . .	69
12	The exact metric departure and the cost of complete jets	74
12.1	A quantitative constraint on every retained nilpotent chain	75
13	Coherent packets, joint tensor layers, and their exact spectral cost	76
13.1	The arithmetic unit and the coherent one-variable maps	76
13.2	The original reflection and the exact diagonal coefficient	77
13.3	The joint total-degree source and its relation layer	78
13.4	Nested packets on the joint source: the additional projection	79
13.5	Positive layer rank and the complete tensor spectral departure	80
14	Density of the original joint boundaries and determinant dissipation	82

14.1 Polynomial density with the original weighted measures	82
14.2 Strong limits of the actual source representatives	83
14.3 The accumulated determinant cost of the same boundary sequence	85
15 Which completion retains the arithmetic jets?	86
15.1 Logarithmic coordinates and the original measure	86
15.2 Exponential weights and exact jet norms	87
15.3 The energy term carried by the weighted topology	88
15.4 The tessarine coordinate in the actual weight form	88
16 The current arithmetic route and its next calculation	89
16.1 A single chain of exact comparison maps	89
16.2 Topology, duality and the two sides of a weight estimate	89
16.3 Provenance and mathematical status	91
II Complete source proof notes and chronology	94
A PR 8: Finite operator comparison	95
A.1 Exact finite operator quotients of the tau-base theta cohomology	95
B PR 9: Chain descent	107
B.1 Chain-level arithmetic descent over the tau base	107
C PR 10: Global retraction	117
C.1 A global retraction of the original theta complex over the tau base	117
D PR 11: Homotopy control	127
D.1 Homotopy choices and source-preserving arithmetic control	127
E PR 13: Orthogonal boundary control	133
E.1 Orthogonal theta boundaries and rank-two arithmetic control	133
F Derived Split-Zero mathematics	139
F.1 SplitZero: reconstruction, internal homology, and balanced finite jets	139
G PR 14: Coherent arithmetic interpolation and joint tensor boundary control	145
G.1 Coherent arithmetic interpolation and joint tensor boundary control	145

Part I

Integrated mathematical continuation

1 Exact Split-Zero carriers and the local spectral bridge

Throughout this section, semiring homomorphisms preserve additive zero and multiplicative unit. Let R be a nonzero commutative ring and $\mathbb{B} = (\{0, 1\}, \vee, \wedge)$. Define

$$G(R) = R \sqcup \{\tau\}, \quad e = 0_R \in R \subset G(R).$$

Addition and multiplication of supported elements are those of R , including when their result is zero. The additional element satisfies

$$\tau + r = r + \tau = r, \quad \tau + \tau = \tau, \quad \tau r = r\tau = \tau, \quad \tau^2 = \tau.$$

Thus the semiring zero is τ , and $e \neq \tau$.

1.1 Scalar support and independent channel amplitudes

Proposition 1.1 (Pair carrier and scalar reflection). *There is a semiring isomorphism*

$$\eta_R : G(R) \longrightarrow \{(0, 0)\} \cup (R \times \{1\}) \subset R \times \mathbb{B}, \quad \tau \mapsto (0, 0), \quad r \mapsto (r, 1).$$

The coordinate maps

$$\begin{aligned} p_R : G(R) &\rightarrow R, & p_R(\tau) &= 0, & p_R(r) &= r, \\ \chi_R : G(R) &\rightarrow \mathbb{B}, & \chi_R(\tau) &= 0, & \chi_R(r) &= 1 \end{aligned}$$

are semiring homomorphisms. The map p_R is universal among semiring homomorphisms from $G(R)$ to a ring.

Proof. In the product semiring, two supported pairs have sum $(r + s, 1)$ and product $(rs, 1)$. These identities retain support when the amplitude vanishes. The pair $(0, 0)$ is an additive identity and an absorbing multiplicative zero. This proves closure and all operation identities, and the displayed map is bijective. Its coordinate projections give p_R, χ_R . If $f : G(R) \rightarrow A$ has ring target, then $e + e = e$ gives $f(e) + f(e) = f(e)$, hence $f(e) = 0_A$. Its restriction to the supported copy of R is therefore a unital ring homomorphism $h : R \rightarrow A$. The equality $f = hp_R$ holds on R and on τ , and determines h . Conversely every such h gives hp_R . $\square$

The larger carrier $R \times \mathbb{B}$ contains $(1, 0)$; its exact relation to $G(R)$ is the inclusion above. The pair presentation requires $R \times \{1\}$.

For a nontrivial bounded distributive lattice L , define

$$\begin{aligned} G_L(R) &= \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\} \subset R \times L, \\ (r, \lambda) + (s, \mu) &= (r + s, \lambda \vee \mu), & (r, \lambda)(s, \mu) &= (rs, \lambda \wedge \mu). \end{aligned}$$

Write $z_\lambda = (0, \lambda)$, $\tau_L = z_{0_L}$, and $e_L = z_{1_L}$.

Proposition 1.2 (Lattice reflection with its original coefficient ring). *The displayed carrier is a commutative semiring. The projections*

$$p_L : G_L(R) \rightarrow R, \quad \chi_L : G_L(R) \rightarrow L$$

are respectively the universal ring reflection and the support homomorphism. In particular the amplitude codomain of $G_{\mathbb{B}^2}(\mathbb{C})$ is $\mathbb{C}$.

Proof. A summand with nonzero amplitude has top support, so its sum has top support. A factor with lower support has zero amplitude, so a product of lower support has zero amplitude. These facts prove closure. All laws restrict from the product semiring; distributivity uses that of L . Both projections preserve operations, zero and unit. If $f : G_L(R) \rightarrow A$ has ring target, $e_L + z_\lambda = e_L$ gives $f(z_\lambda) = 0$ by cancellation in A . Restriction to the supported copy of R is a ring homomorphism h and $f = hp_L$. Conversely each h gives this factorization. Surjectivity of p_L gives uniqueness. $\square$

Let $j : \mathbb{C} \rightarrow G(\mathbb{C})$ include the supported copy. It preserves addition, multiplication and unit; it takes ordinary zero to e , whereas the semiring zero is τ . Put

$$M = G(\mathbb{C})^2, \quad p^{(2)} = (p_{\mathbb{C}}, p_{\mathbb{C}}) : M \rightarrow \mathbb{C}^2, \quad \chi^{(2)} = (\chi_{\mathbb{C}}, \chi_{\mathbb{C}}) : M \rightarrow \mathbb{B}^2.$$

For $\epsilon \in \mathbb{B}$ let $u_0 = \tau, u_1 = e$.

Proposition 1.3 (Exact scalar-lattice and two-amplitude embeddings). *There are injective semiring homomorphisms*

$$b : G_{\mathbb{B}^2}(\mathbb{C}) \hookrightarrow M, \quad B : G_{\mathbb{B}^2}(\mathbb{C}^2) \hookrightarrow M$$

defined by

$$\begin{aligned} b(c, (1, 1)) &= (j(c), j(c)), \\ b(0, (\lambda_1, \lambda_2)) &= (u_{\lambda_1}, u_{\lambda_2}), \\ B((a, b), (1, 1)) &= (j(a), j(b)), \\ B((0, 0), (\lambda_1, \lambda_2)) &= (u_{\lambda_1}, u_{\lambda_2}). \end{aligned}$$

Their amplitude and support identities are

$$p^{(2)}b = \Delta_{\mathbb{C}}p_{\mathbb{B}^2}, \quad \chi^{(2)}b = \chi_{\mathbb{B}^2}, \quad p^{(2)}B = p_{\mathbb{B}^2}, \quad \chi^{(2)}B = \chi_{\mathbb{B}^2},$$

where $\Delta_{\mathbb{C}}(c) = (c, c)$.

Proof. The pair presentation identifies M with

$$\{((a, b), (\lambda_1, \lambda_2)) \in \mathbb{C}^2 \times \mathbb{B}^2 : \lambda_1 = 0 \Rightarrow a = 0, \quad \lambda_2 = 0 \Rightarrow b = 0\},$$

with componentwise operations. The map b restricts the product semiring map $(c, \lambda) \mapsto ((c, c), \lambda)$. Lower support in its source forces $c = 0$, so it lands in the displayed subset. It retains the amplitude and both support bits and is injective. The map B restricts the identity coordinate map on $\mathbb{C}^2 \times \mathbb{B}^2$; lower support in its source forces both amplitudes to vanish, so it too lands in M and is injective. The prescriptions agree at the top-support zero. Both maps preserve zero and unit. Projection proves all four identities. $\square$

The element $(j(1), \tau) \in M$ is outside both images. There is no function $r : M \rightarrow G_{\mathbb{B}^2}(\mathbb{C}^2)$ satisfying both $p_{\mathbb{B}^2}r = p^{(2)}$ and $\chi_{\mathbb{B}^2}r = \chi^{(2)}$: it would send that element to $((1, 0), (1, 0))$, outside its target carrier. This identifies the precise obstruction to a reverse map preserving both projections.

The synchronized carrier has the exact map

$$G(\mathbb{C}^2) \cong G(\mathbb{C}) \times_{\mathbb{B}} G(\mathbb{C}) \hookrightarrow M.$$

It takes τ to (τ, τ) and a supported (a, b) to $(j(a), j(b))$. Equality of the two support characters leaves exactly those cases, proving bijectivity onto the fibre product. Operations agree componentwise. Its image has masks $(0, 0), (1, 1)$, while the other masks in M carry independent mixed amplitudes.

The map $p^{(2)}$ is the universal ring reflection of M . Indeed, (e, e) is additively idempotent, so every ring-target map kills it. Adding (e, e) to $x \in M$ gives the fully supported element of amplitude $p^{(2)}x$. The map is consequently determined by a ring homomorphism from the supported copy of $\mathbb{C}^2$, proving the universal property. A semiring isomorphism $G_{\mathbb{B}^2}(\mathbb{C}) \cong M$ would induce a ring isomorphism $\mathbb{C} \cong \mathbb{C}^2$. This is impossible because $(1, 0)$ is a nontrivial idempotent of $\mathbb{C}^2$, and $\mathbb{C}$ has only 0, 1. The positive relation between the original carriers is b , with diagonal amplitude projection.

1.2 Tesserine sectors and a supported critical fixed locus

Reserve τ for split absence and use κ for conjugation. Let

$$\sigma(s) = 1 - s, \quad \kappa(s) = \bar{s}, \quad K_4 = \{1, \sigma, \kappa, \sigma\kappa\}.$$

The real tesserine algebra and its complexification are

$$A_{\mathbb{R}} = \mathbb{R}[K_4] = \mathbb{R}[\sigma, \kappa]/(\sigma^2 - 1, \kappa^2 - 1), \quad A_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} A_{\mathbb{R}}.$$

The polynomial algebra in this presentation is commutative.

Proposition 1.4 (Character algebra and quartet coordinates). *For $\epsilon, \delta \in \{1, -1\}$ let*

$$E_{\epsilon\delta} = \frac{1}{4}(1 + \epsilon\sigma)(1 + \delta\kappa).$$

Character evaluation gives an algebra isomorphism

$$\mathcal{F} : A_{\mathbb{C}} \rightarrow \mathbb{C}^4, \quad x \mapsto (\text{ev}_{\epsilon,\delta}(x))_{\epsilon,\delta},$$

with inverse $(d_{\epsilon\delta}) \mapsto \sum d_{\epsilon\delta} E_{\epsilon\delta}$. For the exact quartet element

$$Q(s) = s \cdot 1 + (1 - s)\sigma + \bar{s}\kappa + (1 - \bar{s})\sigma\kappa, \quad s = \beta + i\gamma,$$

the four character values are

$$c_{++} = 2, \quad c_{+-} = 0, \quad c_{-+} = 2(2\beta - 1), \quad c_{--} = 4i\gamma.$$

Projection to the last two specified characters is a unital algebra map $\pi : A_{\mathbb{C}} \rightarrow \mathbb{C}^2$. The composite πQ is a real affine bijection

$$\mathbb{C} \rightarrow \mathbb{R} \times i\mathbb{R}, \quad s \mapsto (a, b) = (c_{-+}(s), c_{--}(s)), \quad s = \frac{1}{2} + \frac{1}{4}(a + b).$$

Proof. Evaluation of $E_{\epsilon\delta}$ at (ϵ', δ') is $\frac{1}{4}(1 + \epsilon'\sigma)(1 + \delta'\kappa)$. It is one when both signs agree and zero otherwise. This proves the two inverse formulas and all orthogonal idempotent identities, retaining the factors of four. Character evaluation preserves the algebra operations. Directly,

$$\begin{aligned} c_{++} &= s + (1 - s) + \bar{s} + (1 - \bar{s}) = 2, \\ c_{+-} &= s + (1 - s) - \bar{s} - (1 - \bar{s}) = 0, \\ c_{-+} &= s - (1 - s) + \bar{s} - (1 - \bar{s}) = 2(s + \bar{s}) - 2 = 4\beta - 2, \\ c_{--} &= s - (1 - s) - \bar{s} + (1 - \bar{s}) = 2(s - \bar{s}) = 4i\gamma. \end{aligned}$$

The proposed inverse recovers $\beta + i\gamma$. Conversely every real a and purely imaginary b is recovered by substituting that inverse into the two character expressions. $\square$

The quartet image has real affine dimension two. Its real linear span has dimension three, spanned by E_{++}, E_{-+}, iE_{--} , retaining the constant character coordinate. The real tesserine algebra has dimension four. The maps $Q, \mathcal{F}, \pi$ and the displayed affine inverse specify the exact relationship of these objects.

Proposition 1.5 (Supported sector lift and fixed locus). *Define*

$$\Psi : \mathbb{C} \rightarrow M, \quad \Psi(s) = (j(c_{-+}(s)), j(c_{--}(s))).$$

It has $\chi^{(2)}\Psi(s) = (1, 1)$ for all s . The $G(\mathbb{C})$ -linear automorphisms

$$T_{\sigma}(x, y) = ((-1)x, (-1)y), \quad T_{\kappa}(x, y) = (x, (-1)y)$$

of the free semimodule M give

$$\Psi(\sigma s) = T_{\sigma}\Psi(s), \quad \Psi(\kappa s) = T_{\kappa}\Psi(s).$$

Here $-1 = j(-1)$ is a supported unit. Their product satisfies

$$\text{Fix}(T_{\sigma}T_{\kappa}) = \{\tau, e\} \times G(\mathbb{C}), \quad \Psi^{-1} \text{Fix}(T_{\sigma}T_{\kappa}) = \{s : \Re s = \frac{1}{2}\}.$$

Proof. The sector formulas give $(a, b)(1 - s) = (-a, -b)$ and $(a, b)(\bar{s}) = (a, -b)$. Multiplication by $j(-1)$ negates supported amplitudes and fixes τ . It preserves addition, commutes with scalar multiplication and squares to identity. This proves the automorphism and equivariance statements. Its fixed points are τ and supported $j(c)$ with $c = -c$. Since $2 \neq 0$ in $\mathbb{C}$, the latter is precisely e . The product action is $((-1)x, y)$, proving its fixed locus. On the image of Ψ the first coordinate is supported and is fixed exactly when $4\beta - 2 = 0$. $\square$

The scalar-lattice channel atoms have images $b(z_{10}) = (e, \tau)$ and $b(z_{01}) = (\tau, e)$. The separate support swap has fixed masks $(0, 0), (1, 1)$. Since $\chi^{(2)}\Psi$ is constantly $(1, 1)$, it cannot detect the critical line by that swap. The support action induced by $T_\sigma T_\kappa$ is identity; its amplitude action is $(-a, b)$. The preceding proposition supplies the exact fixed-locus test with the first-channel supported zero retained.

1.3 Exact sector, holonomy and positive-moment maps

Fix a prime p , $\ell_p = \log p > 0$, and $u \in S^1$. For an arithmetic packet, take $u = \psi(\text{Frob}_p)$ with $L/\mathbb{Q}$ finite abelian, p unramified, and $\psi : \text{Gal}(L/\mathbb{Q}) \rightarrow S^1$ a character. All powers use $p^w = \exp(w\ell_p)$ with the positive real logarithm.

Proposition 1.6 (Sector-to-holonomy map with constants retained). *The analytic function*

$$H_{p,u} : \mathbb{C}^2 \rightarrow \mathbb{C}^\times, \quad H_{p,u}(a, b) = u \exp\left(\frac{\ell_p}{4}(a + b)\right)$$

gives a function $H_{p,u}p^{(2)}$ on M . On the quartet image,

$$H_{p,u}(c_{-+}(s), c_{--}(s)) = up^{s-1/2} =: z_{p,u}(s),$$

$$\frac{\log |z_{p,u}(s)|}{\ell_p} = \frac{c_{-+}(s)}{4} = \Re s - \frac{1}{2}, \quad |z_{p,u}(s)|^2 - 1 = \exp\left(\frac{\ell_p}{2}c_{-+}(s)\right) - 1.$$

Proof. The exact inverse for the sector coordinates gives $s - \frac{1}{2} = (a + b)/4$. Substitution proves the exponential identity. On that image a is real and b is purely imaginary. Since $|u| = 1$, taking absolute values yields $\exp(\ell_p a/4)$. Taking its real logarithm and, separately, its square proves the two remaining identities. $\square$

The radial exponent is $c_{-+}/4$; the exponent $c_{-+}/2$ belongs to the squared modulus. The map $H_{p,u}$ is used with its analytic-function type, without any assertion that it preserves semiring addition.

Proposition 1.7 (Rank-one packet). *For $z \in \mathbb{C}$ define*

$$v_2(z) = \begin{pmatrix} 1 \\ z \end{pmatrix}, \quad T_2(z) = v_2(z)v_2(z)^* = \begin{pmatrix} 1 & \bar{z} \\ z & |z|^2 \end{pmatrix}, \quad P_z(w) = w - \bar{z}.$$

Then $T_2(z)$ is Hermitian positive semidefinite of rank one and

$$\ker T_2(z) = \mathbb{C}(-\bar{z}, 1)^T.$$

It is Toeplitz exactly when $|z| = 1$, equivalently when the unique root of P_z lies on S^1 . For $z = z_{p,u}(s)$ this is equivalent to $\Re s = \frac{1}{2}$.

Proof. For $x \in \mathbb{C}^2$, $x^*T_2(z)x = |v_2(z)^*x|^2 \geq 0$. The vector $v_2(z)$ is nonzero and hence its outer product has rank one. The adjoint kernel equation is $x_0 + \bar{z}x_1 = 0$, giving the displayed space. A two-by-two matrix is Toeplitz exactly when its diagonal entries agree, here $1 = |z|^2$. The root $\bar{z}$ has modulus $|z|$. The final equivalence follows from the preceding proposition and $\ell_p > 0$. $\square$

Theorem 1.8 (Positive finite packets and their exact diagonal defect). *Let $r \geq 1$, $\alpha_j > 0$, and $z_j \in \mathbb{C}$. For $n \geq 1$ put*

$$v_n(z) = (1, z, \dots, z^{n-1})^\top, \quad M_n = \sum_{j=1}^r \alpha_j v_n(z_j) v_n(z_j)^*.$$

The matrix M_n is Hermitian positive semidefinite. For $n \geq 3$, it is Toeplitz exactly when every $|z_j| = 1$. With matrix indices starting at zero,

$$\mathcal{D}(M_n) := (M_n)_{22} - 2(M_n)_{11} + (M_n)_{00} = \sum_{j=1}^r \alpha_j (|z_j|^2 - 1)^2.$$

For these packets and $n \geq 3$, vanishing of $\mathcal{D}$ already characterizes Toeplitzness. If the z_j are pairwise distinct and $r \leq n$, the rank is r . If $r < n$, coefficient vectors identify the kernel with

$$\left(\prod_{j=1}^r (w - \bar{z}_j) \right) \mathbb{C}[w]_{\leq n-1-r} \subset \mathbb{C}[w]_{\leq n-1}.$$

In particular $r = n - 1$ gives corank one and exactly the conjugated roots $\bar{z}_1, \dots, \bar{z}_{n-1}$.

Proof. The identity

$$x^* M_n x = \sum_j \alpha_j |v_n(z_j)^* x|^2$$

proves positivity and shows that its kernel is the simultaneous kernel of all $v_n(z_j)^*$. The first three diagonal entries are $\sum \alpha_j$, $\sum \alpha_j |z_j|^2$, and $\sum \alpha_j |z_j|^4$. Their displayed linear combination gives the defect without rescaling any weight. Toeplitzness equates these entries, forcing zero defect. Every weight is positive and every square nonnegative, hence all $|z_j| = 1$. Conversely that condition gives $(M_n)_{k\ell} = \sum_j \alpha_j z_j^{k-\ell}$, a Toeplitz matrix. The same argument proves the zero-defect characterization.

Let V have columns $v_n(z_j)$. Its first r rows have Vandermonde determinant $\prod_{i < j} (z_j - z_i) \neq 0$, so V has rank r . The established kernel identity is $\ker M_n = \ker V^*$, proving the rank. For $P_x(w) = \sum_{k=0}^{n-1} x_k w^k$, the kernel equations say $P_x(\bar{z}_j) = 0$. Distinct linear factors divide P_x exactly when these equations hold: division by one factor preserves the other root equations since that factor is nonzero at those roots. This proves the product and degree bound. At $r = n - 1$, the remaining factor is a scalar; for a nonzero kernel vector it is nonzero, so there are no further roots. $\square$

The size bound is sharp for general positive packets. At $n = 2$, the data $(\alpha_1, \alpha_2) = (4, 1)$ and $(z_1, z_2) = (1/2, 2)$ give

$$M_2 = 4 \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/4 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}.$$

This is positive definite and Toeplitz, while neither atom is on S^1 . At size three the diagonal is $(5, 5, 65/4)$ and the defect is $45/4$. One atom retains the two-by-two criterion. Zero-weight atoms are invisible; repeated atoms combine their weights before applying the distinct-column rank count.

The exact sector defect is therefore

$$\mathcal{D}(M_n) = \sum_{j=1}^r \alpha_j \left[\exp \left(\frac{\log p_j}{2} c_{-+}(s_j) \right) - 1 \right]^2.$$

The function chain

$$M \xrightarrow{p^{(2)}} \mathbb{C}^2 \xrightarrow{H_{p,u}} \mathbb{C}^\times \xrightarrow{z \mapsto v_n(z) v_n(z)^*} \text{Herm}_{\geq 0}(n)$$

has explicit maps at every step. Positive finite summation supplies the preceding theorem.

1.4 The theta-packet form and its scalar comparison

Proposition 1.9 (Eigenpair form quotient and sector channel). *For every complex d -by- d matrix A , Hermitian matrix G , and nonzero eigenvector v with $Av = \rho v$, let $W = A^*G + GA - G$. Then*

$$v^*Wv = (2\Re\rho - 1)v^*Gv.$$

*On the domain of these data where $v^*Gv \neq 0$, the quotient*

$$\omega(A, G, v, \rho) = \frac{v^*Wv}{v^*Gv} = 2\Re\rho - 1 = \frac{c_{-+}(\rho)}{2}$$

is real. Its associated holonomy satisfies

$$|z_{p,u}(\rho)|^2 = \exp(\ell_p\omega), \quad z_{p,u}(\rho) = u \exp\left(\frac{\ell_p}{4}(c_{-+}(\rho) + c_{--}(\rho))\right).$$

Proof. Since $(Av)^* = \bar{\rho}v^*$,

$$v^*(A^*G + GA - G)v = (Av)^*Gv + v^*G(Av) - v^*Gv = (\bar{\rho} + \rho - 1)v^*Gv.$$

This proves the unquotiented identity even when the right factor is zero. Division on the stated domain gives ω , and $c_{-+}(\rho) = 4\Re\rho - 2$ gives the factor $1/2$. The holonomy identities follow from the exact sector exponential above. $\square$

This is the typed map from an eigenpair of the form packet to its Split-Zero sector and local moment matrix. The moment matrix is positive for every parameter, so that positivity cannot supply a further zero-defect conclusion. For $\rho = 1, u = 1$,

$$z = \sqrt{p}, \quad T_2(z) = \begin{pmatrix} 1 & \sqrt{p} \\ \sqrt{p} & p \end{pmatrix} \geq 0,$$

and its diagonal entries differ. On the form side $A = (1), G = (1), v = (1)$ give $W = (1)$ and $\omega = 1$; the exact maps take these data to the same off-critical packet. Any arithmetic identity forcing the actual family of W to vanish on its eigenvectors must therefore be established from that family's construction. These maps retain that requirement rather than replacing it by Gram positivity.

1.5 Projective zeta at all zeros and at its pole

A pair in $G(\mathbb{C})^2$ is unimodular exactly when at least one amplitude is nonzero. Such a coordinate is a supported unit. If both amplitudes are zero, every linear combination has zero amplitude and cannot be the unit. Let $\mathbb{P}_{\text{um}}^1(G(\mathbb{C}))$ denote unimodular pairs modulo common unit multiplication.

Proposition 1.10 (Projective amplitude section and reciprocal). *The map*

$$p_* : \mathbb{P}_{\text{um}}^1(G(\mathbb{C})) \rightarrow \mathbb{P}^1(\mathbb{C}), \quad [x : y] \mapsto [p(x) : p(y)]$$

has an injective section

$$\iota : \mathbb{P}^1(\mathbb{C}) \rightarrow \mathbb{P}_{\text{um}}^1(G(\mathbb{C})), \quad [a : b] \mapsto [j(a) : j(b)],$$

whose image is the full-support component. With

$$\mathcal{R} : G(\mathbb{C}) \rightarrow \mathbb{P}_{\text{um}}^1(G(\mathbb{C})), \quad x \mapsto [1 : x], \quad r : \mathbb{C} \rightarrow \mathbb{P}^1(\mathbb{C}), \quad c \mapsto [1 : c],$$

one has $p_\mathcal{R} = rp_{\mathbb{C}}$ and $\mathcal{R}j = \iota r$.*

Proof. Unimodularity ensures the projected pair is nonzero. A common supported unit projects to a nonzero complex scalar, so p_* is well defined. A nonzero ordinary pair becomes a unimodular supported pair; changing its representative lifts to common supported unit multiplication. Thus ι is well defined, and $p_*\iota$ is identity. Every full-support pair is the supported lift of its amplitude pair. The pair $(1, x)$ is always unimodular, and projection of its coordinates proves both remaining identities. $\square$

In particular $\mathcal{R}(e) = [1 : e] = \infty_e$ and $\mathcal{R}(\tau) = [1 : \tau] = \infty_\tau$. Both project to $[1 : 0]$, while their Boolean projective supports are $[1 : 1]$ and $[1 : 0]$. Similarly $[e : 1]$ and $[\tau : 1]$ project to the ordinary zero while retaining different supports.

Proposition 1.11 (Two exact charts for the supported zeta map). *For the meromorphic zeta function with its simple pole of residue one at 1, the full-support lift is*

$$Z_{\text{sp}} : \mathbb{C} \rightarrow \mathbb{P}_{\text{um}}^1(G(\mathbb{C})), \quad Z_{\text{sp}}(s) = \begin{cases} [j(\zeta(s)) : 1], & s \neq 1, \\ [1 : e], & s = 1. \end{cases}$$

It equals $\iota\hat{\zeta}$. At every zeta zero ρ its value is $[e : 1]$. Near the pole it has the chart

$$Z_{\text{sp}}(s) = [1 : j(h(s))], \quad h(s) = 1/\zeta(s), \quad h(1) = 0, \quad h'(1) = 1.$$

The reciprocal is a finite-amplitude coordinate away from the zeta zeros, including its extension at 1. At a zero of order m , it has a pole of order m .

Proof. Put $t = s - 1$ and write $\zeta(1+t) = t^{-1} + g(t)$ with g holomorphic near zero. Then $h(1+t) = t/(1+tg(t))$. Its denominator has value one at zero, so this is holomorphic there with value zero and derivative one. On the overlap where $\zeta \neq 0$, common multiplication by $j(\zeta(s))$ gives $[1 : j(h(s))] = [j(\zeta(s)) : 1]$. This proves the transition and pole value. The first chart directly gives $[e : 1]$ at each zero. At a zero of order m , write $\zeta(\rho+t) = t^m u(t)$ with $u(0) \neq 0$. Its reciprocal is $t^{-m} u(t)^{-1}$, proving its pole order; the first chart retains the original order m . $\square$

The swap $J_*[x : y] = [y : x]$ is induced by an invertible permutation matrix and commutes with ι, p_* . Hence J_*Z_{sp} is the global reciprocal projective map, sending zeta zeros to ∞_e and the pole to $[e : 1]$. Analytic statements use the complex structure transported to the full-support component by ι ; no topology on the extra support components is needed.

1.6 Matrix packets and their full relation module

Theorem 1.12 (Block packet with explicit hypotheses). *Let V be a complex Hilbert space of finite dimension $d > 0$ with fixed positive-definite Hermitian inner product. Let $g \in \text{GL}(V)$ and $m \geq 1$. Use that metric and its orthogonal direct sums for all adjoints. Define*

$$W_m(g) : V \rightarrow V^{m+1}, \quad v \mapsto (v, gv, \dots, g^m v), \quad \mathbb{T}_m(g) = W_m(g)W_m(g)^*,$$

$$D_m(g) : V^m \rightarrow V^{m+1}, \quad \begin{cases} (D_m(g)y)_0 = -g^* y_1, \\ (D_m(g)y)_k = y_k - g^* y_{k+1} \quad (1 \leq k \leq m-1), \\ (D_m(g)y)_m = y_m. \end{cases}$$

The matrix $\mathbb{T}_m(g)$ is positive semidefinite and

$$\ker \mathbb{T}_m(g) = \text{im } D_m(g), \quad \text{rank } \mathbb{T}_m(g) = d, \quad \text{corank } \mathbb{T}_m(g) = md.$$

It is block Toeplitz exactly when g is unitary for the fixed metric.

Proof. The Gram factorization gives positivity and $\ker \mathbb{T}_m = \ker W_m^*$. The first component of $W_m v$ is v , so W_m is injective and the kernel dimension is $(m+1)d - d = md$. The adjoint is

$$W_m^*(v_0, \dots, v_m) = \sum_{k=0}^m (g^*)^k v_k.$$

Substitution of $D_m(y)$ gives

$$-g^* y_1 + \sum_{k=1}^{m-1} ((g^*)^k y_k - (g^*)^{k+1} y_{k+1}) + (g^*)^m y_m = 0.$$

Thus its image lies in the kernel. If $D_m(y) = 0$, the last component gives $y_m = 0$, and backward substitution gives each preceding $y_k = 0$. Its image dimension is md , proving equality with the kernel. The block of indices j, k is $g^j (g^*)^k$. For unitary g this is g^{j-k} , hence block Toeplitz. Conversely block Toeplitzness equates blocks $(0, 0), (1, 1)$, giving $I = gg^*$. Invertibility yields $g^* = g^{-1}$ and unitarity. $\square$

At $m = 0$ the only block is I , so the bound $m \geq 1$ is essential. At $m = 1$, the full kernel is $\{(-g^* y, y) : y \in V\}$; its vector polynomials are $(wI - g^*)y$. Taking determinant gives the exact scalar map

$$wI - g^* \mapsto \det(wI - g^*) = \overline{\det(\bar{w}I - g)}.$$

The equality follows by conjugating the determinant expansion and transposing. It retains conjugate eigenvalues and multiplicities. For the standard metric and $J_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ with $t > 0$, the polynomial is $(w - 1)^2$ but

$$J_t J_t^* = \begin{pmatrix} 1+t^2 & t \\ t & 1 \end{pmatrix} \neq I.$$

Its block packet consequently fails Toeplitzness, although the determinant roots lie on the unit circle. The kernel remains $\{(-J_t^* y, y) : y \in \mathbb{C}^2\}$, retaining the matrix information that determinant forgets.

2 The historical packet ghost and its full arithmetic jet quotient

The source of the packet construction in this section is the response dated 24 June 2026, 11:13:04 AM in the SSD export *Globalization Semiring Analysis*, whose exact path, hash and response-line locators are recorded in `logbook/historical_split_transcripts_20260912.md`. The source proves the permutation-packet calculation. We give that calculation completely, then construct its comparison with the full local algebras of the arithmetic packet. In particular, we retain the nilpotent jet operators and distinguish the source's complex-linear permutation from arithmetic conjugation by explicit maps.

Let Z be a finite set of complex numbers stable under $c(\rho) = \bar{\rho}$ and $\sigma(\rho) = 1 - \rho$. Assign a positive integer m_ρ constant on each orbit. Write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad E_h = \mathbb{C}[s]/(h), \quad A[f] = [sf].$$

We use the original coordinate s , without replacing its multiplicative operator by its semisimple part. Put

$$L_Z = \bigoplus_{\rho \in Z} \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}), \quad e_{\rho,k} = t_\rho^k \quad (0 \leq k < m_\rho)$$

in the indicated direct summand. The coefficient inner product makes all $e_{\rho,k}$ orthonormal; it is an explicitly chosen reference inner product, not the theta-source metric.

Proposition 2.1 (Original-coordinate identification with all jets). *The map*

$$T : E_h \longrightarrow L_Z, \quad T[f]_\rho = \sum_{k=0}^{m_\rho-1} \frac{f^{(k)}(\rho)}{k!} t_\rho^k$$

is a complex-algebra isomorphism. In these coordinates

$$TAT^{-1} = A_{ss} + N, \quad A_{ss}e_{\rho,k} = \rho e_{\rho,k}, \quad Ne_{\rho,k} = \begin{cases} e_{\rho,k+1} & k+1 < m_\rho, \\ 0 & k+1 = m_\rho. \end{cases}$$

Proof. Taylor multiplication modulo $t_\rho^{m_\rho}$ proves multiplicativity. Its kernel consists exactly of polynomials divisible by every $(s - \rho)^{m_\rho}$ and hence by their product h , since the factors are pairwise coprime. For completeness an inverse is explicit. Set $h_\rho = h/(s - \rho)^{m_\rho}$, and let a_ρ be the Taylor polynomial of $1/h_\rho(s)$ at ρ through degree $m_\rho - 1$. Then $E_\rho = [a_\rho h_\rho]$ has residue 1 at the ρ factor and 0 at every other factor. Thus

$$T^{-1}e_{\rho,k} = E_\rho[(s - \rho)^k]$$

gives the inverse. Multiplication by $s = \rho + t_\rho$ gives the displayed operator, including its last nilpotent column. $\square$

In every subsequent operator formula on L_Z , the symbol A denotes the transported operator $TAT^{-1} : L_Z \rightarrow L_Z$ of Proposition 2.1. The polynomial-coordinate quotient at the end is equipped with the original multiplication-by- s operator.

Define complex-linear permutations C_0, S_0 and a complex-linear parity operator H on this complete basis by

$$C_0e_{\rho,k} = e_{\bar{\rho},k}, \quad S_0e_{\rho,k} = e_{1-\rho,k}, \quad He_{\rho,k} = (-1)^k e_{\rho,k}.$$

Also let K be the conjugate-linear map $K(\sum a_{\rho,k}e_{\rho,k}) = \sum \bar{a}_{\rho,k}e_{\rho,k}$. All four maps commute, square to the identity, and preserve products in L_Z , with K conjugate-linear on scalars.

Proposition 2.2 (Typed arithmetic involutions). *Under T , ordinary conjugation of polynomial coefficients is the conjugate-linear map $\mathcal{C} = C_0K$, and pullback by $s \mapsto 1 - s$ is the complex-linear map $\mathcal{S} = S_0H$. In particular*

$$C_0 = \mathcal{C}K, \quad S_0 = \mathcal{S}H, \quad \mathcal{C}A = A\mathcal{C}, \quad \mathcal{S}AS = I - A.$$

Here the equality involving $\mathcal{C}$ is an equality of conjugate-linear maps. Moreover $C_0N = NC_0$, $S_0N = NS_0$, and $HN = -NH$.

Proof. If $f^\#(s) = \overline{f(\bar{s})}$, then $(f^\#)^{(k)}(\rho)/k! = \overline{f^{(k)}(\bar{\rho})/k!}$, giving $\mathcal{C} = C_0K$. The k th derivative of $f(1 - s)$ at ρ is $(-1)^k f^{(k)}(1 - \rho)$, giving $\mathcal{S} = S_0H$. The ideal (h) is preserved by both operations: conjugation fixes the monic polynomial h , and $h(1 - s) = (-1)^{\deg h} h(s)$. Conjugating coefficients fixes the indeterminate s , whereas substituting $1 - s$ replaces multiplication by s by multiplication by $1 - s$. This proves the two identities for A . The assertions about N follow on each basis vector, including the vector of largest jet degree, which both sides send to zero. $\square$

Theorem 2.3 (Historical ghost with complete multiplicities). *Set*

$$R = C_0S_0, \quad \Gamma = C_0 - S_0, \quad P_- = \frac{I - R}{2}, \quad P_+ = \frac{I + R}{2}.$$

These satisfy

$$\Gamma^* = \Gamma, \quad \Gamma^2 = 4P_-, \quad \Gamma e_{\rho,k} = 0 \iff \Re \rho = \frac{1}{2}.$$

On a nonreal off-axis quartet of common multiplicity m , Γ has eigenvalues -2 and 2 each with multiplicity m , and 0 with multiplicity $2m$. Furthermore

$$\text{rank } P_- = 2m, \quad \text{Tr}(\Gamma^*\Gamma) = 8m.$$

On a critical-line orbit $\Gamma = P_- = 0$ at every jet degree.

Proof. The permutations C_0, S_0 are commuting selfadjoint involutions in the declared reference inner product. Consequently $(C_0 - S_0)^2 = 2I - 2C_0S_0$. The equality $C_0e_{\rho,k} = S_0e_{\rho,k}$ says exactly that the two labelled basis vectors have the same root, namely $\bar{\rho} = 1 - \rho$, which is equivalent to $2\Re\rho = 1$.

On an off-axis nonreal quartet order the roots as $(\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho})$. At every fixed jet degree the four joint (C_0, S_0) eigenspaces have characters $(1, 1), (1, -1), (-1, 1), (-1, -1)$. Explicit joint eigenvectors are

$$e_{\rho,k} + \epsilon e_{\bar{\rho},k} + \delta e_{1-\rho,k} + \epsilon \delta e_{1-\bar{\rho},k}, \quad \epsilon, \delta \in \{1, -1\}.$$

They form a basis: their pairwise inner products are zero and each has squared norm 4. The eigenvalue of Γ is $\epsilon - \delta$. Thus its squared eigenvalues are 0, 4, 4, 0, giving the rank and trace with every one of the m jet degrees included. On a critical orbit the two permutations agree on every basis vector. $\square$

For real off-axis orbits the same proof gives rank m and squared trace $4m$, since the orbit has two elements. Actual nontrivial zeta zeros in $0 < \Re s < 1$ are nonreal: for real $0 < s < 1$ the alternating series $\eta(s) = \sum_{n \geq 1} (-1)^{n-1} n^{-s}$ is strictly positive by grouping successive pairs, and $\zeta(s) = \eta(s)/(1 - 2^{1-s}) < 0$. For completeness the paired terms satisfy

$$(2n-1)^{-s} - (2n)^{-s} = s \int_{2n-1}^{2n} x^{-s-1} dx.$$

They converge uniformly on each compact subset of $\Re s > 0$, because on such a subset $|s| \leq M$, $\Re s \geq \delta > 0$, and the absolute value is at most $M(2n-1)^{-1-\delta}$. Thus η is analytic there. The product identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$ follows first in $\Re s > 1$ by splitting even terms and then by analytic continuation. Division is used here only on the real interval $0 < s < 1$, where the denominator is strictly negative. Hence the quartet formula applies to off-axis arithmetic packets.

Proposition 2.4 (The exact group-algebra quotient). *For $W = \langle c, \sigma \mid c^2 = \sigma^2 = 1, c\sigma = \sigma c \rangle$ the homomorphism*

$$q : \mathbb{C}[W] \longrightarrow \mathbb{C}[u]/(u^2 - 1), \quad c \longmapsto u, \quad \sigma \longmapsto u$$

is surjective, with

$$I_q := \ker q = (c - \sigma)\mathbb{C}[W] = \mathbb{C}(c - \sigma) \oplus \mathbb{C}(1 - c\sigma).$$

Under the representation $c \mapsto C_0, \sigma \mapsto S_0, I_q L_Z = P_- L_Z$. Thus $L_Z/P_- L_Z$ is the largest quotient of the W -representation on which $C_0 = S_0$.

Proof. In the basis $(1, c, \sigma, c\sigma)$, q sends $a + bc + d\sigma + ec\sigma$ to $(a + e) + (b + d)u$. Its kernel is therefore spanned by $c - \sigma$ and $1 - c\sigma$. Multiplication by c sends $c - \sigma$ to $1 - c\sigma$, and multiplication by any generator stays in their span. This proves the ideal formula. Since $P_- = \Gamma^2/4$ and $\Gamma = \Gamma P_-$, their images agree and equal $I_q L_Z$. A quotient kills I_q precisely when it kills this image; this proves its universal property without any invariance assumption about the arithmetic generator. $\square$

The generator acts on the complete local rings. The next calculation constructs exactly the extra kernel needed to descend that action.

Theorem 2.5 (Arithmetic saturation of the historical ghost quotient). *Put*

$$Be_{\rho,k} = (\Re\rho - \tfrac{1}{2})e_{\rho,k}, \quad Z_{\text{crit}} = \{\rho \in Z : \Re\rho = \tfrac{1}{2}\}, \quad L_{\text{off}} = \bigoplus_{\rho \notin Z_{\text{crit}}} \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}).$$

The smallest A -invariant subspace containing $I_q L_Z$ is exactly

$$I_q L_Z + A(I_q L_Z) = L_{\text{off}}.$$

On each off-axis orbit this sum is direct. The largest quotient on which both the original generator A descends and $C_0 = S_0$ is therefore the exact algebra and W -module quotient

$$L_Z \longrightarrow L_{Z_{\text{crit}}}, \quad (f_\rho)_\rho \longmapsto (f_\rho)_{\rho \in Z_{\text{crit}}}.$$

Its kernel is L_{off} , including every local multiplicity. In polynomial coordinates this is the quotient

$$\mathbb{C}[s]/(h) \longrightarrow \mathbb{C}[s]/(h_{\text{crit}}), \quad [f]_h \longmapsto [f]_{h_{\text{crit}}}, \quad h_{\text{crit}} = \prod_{\rho \in Z_{\text{crit}}} (s - \rho)^{m_\rho},$$

whose kernel is $(h_{\text{crit}})/(h)$; the target is zero when $h_{\text{crit}} = 1$.

Proof. The permutation R sends ρ to $1 - \bar{\rho}$, preserving its imaginary part and reversing $\Re \rho - 1/2$. It preserves the jet index and commutes with N . Accordingly $RB = -BR$ and

$$A = \frac{1}{2}I + B + iY + N, \quad Ye_{\rho,k} = \Im \rho e_{\rho,k}, \quad RY = YR, \quad RN = NR.$$

This is an exact identity for A , with N retained. If $v \in P_-L_Z$, then $(\frac{1}{2}I + iY + N)v \in P_-L_Z$ whereas $Bv \in P_+L_Z$. Therefore

$$P_+Av = Bv \quad (v \in P_-L_Z).$$

On each off-axis orbit all diagonal entries of B are nonzero, so $B : P_-L_Z \rightarrow P_+L_Z$ is an isomorphism on that orbit, its inverse being the restriction of the original diagonal operator B^{-1} . For $w = w_- + w_+$ on that orbit, take $v = B^{-1}w_+$; then $P_+Av = w_+$ and $w - Av \in P_-L_Z$. Thus every w lies in $P_-L_Z + A(P_-L_Z)$. If $Av \in P_-L_Z$ with $v \in P_-L_Z$, then $Bv = P_+Av = 0$ and $v = 0$, proving directness.

On a critical orbit $P_- = 0$. Both A and all permutations preserve the direct sum of off-axis factors, so the subspace just calculated is invariant and is precisely the smallest A -invariant subspace containing P_-L_Z . If a linear map F intertwines A with an operator on its target and satisfies $FC_0 = FS_0$, its kernel contains I_qL_Z and is A -invariant. It consequently contains L_{off} . Conversely the displayed coordinate projection intertwines A , preserves products and has $C_0 = S_0$ on its target. This proves the universal quotient assertion, also for targets carrying all the stated structures. The polynomial quotient and its kernel follow from $h_{\text{crit}} \mid h$ and Proposition 2.1. $\square$

Proposition 2.6 (Exact displacement and nilpotent contribution). *In the declared reference coefficient metric, let $W_{\text{ref}} = A^* + A - I$. Then*

$$W_{\text{ref}} = 2B + N + N^*, \quad \frac{W_{\text{ref}} - RW_{\text{ref}}R}{2} = 2B, \quad \frac{W_{\text{ref}} + RW_{\text{ref}}R}{2} = N + N^*.$$

For the original tessarine coordinate $c_{-+}(\rho) = 4(\Re \rho - 1/2)$, the first displayed component equals multiplication by $c_{-+}/2$.

Proof. The adjoint of A_{ss} is diagonal with entries $\bar{\rho}$, so $A_{\text{ss}}^* + A_{\text{ss}} - I = 2B$. Because R is a self-adjoint permutation commuting with N , it also commutes with N^* . Conjugation by R reverses B . Adding and subtracting the two resulting identities proves both formulas, including the entire nilpotent term. The tessarine identification uses $c_{-+} = 2(2\Re \rho - 1) = 4B$ with its original factor 4. $\square$

The precise map from this reference calculation to a theta-source metric retains the full metric matrix. Let G_E be that positive Hermitian matrix in the original polynomial basis of E_h ; then in the declared coefficient basis of L_Z it is

$$G_L = (T^{-1})^* G_E T^{-1}.$$

Writing the original polynomial-coordinate generator as A_E and the transported one as $A_L = TA_ET^{-1}$ for this formula, one obtains

$$\begin{aligned} W_E &= A_E^* G_E + G_E A_E - G_E, \\ W_L &= A_L^* G_L + G_L A_L - G_L = (T^{-1})^* W_E T^{-1}, \\ W_L &= G_L W_{\text{ref}} + [A_L^*, G_L], \quad [A_L^*, G_L] = A_L^* G_L - G_L A_L^*. \end{aligned}$$

The first two identities follow by inserting $A_L = TA_ET^{-1}$ and cancelling adjacent factors $T^{-1}T$ and $T^*(T^{-1})^*$. The last identity follows by expanding $G_L(A_L^* + A_L - I) + (A_L^* G_L - G_L A_L^*)$. Thus both nilpotent and metric-commutator contributions are present in the exact arithmetic weight; no zero-location estimate is supplied by discarding either term.

Finally form the split-zero semiring $G(\text{End}_{\mathbb{C}} L_Z)$, using the associative endomorphism ring. The original ghost has supported lift $\Gamma^\bullet$; when the comparison cancels it is $e = 0^\bullet$, whereas the absent comparison is τ . Conjugation by T extends to a semiring isomorphism

$$G(\text{End}_{\mathbb{C}} E_h) \longrightarrow G(\text{End}_{\mathbb{C}} L_Z), \quad X^\bullet \longmapsto (TXT^{-1})^\bullet, \quad \tau \longmapsto \tau.$$

It preserves addition, composition, the unit, e , and τ directly. Thus the complete jet comparison preserves supported cancellation. Theorem 2.5 states exactly which arithmetic factors are removed if that cancellation is imposed as a generator-compatible quotient. It does not assert that those factors are absent from the original arithmetic cohomology.

The quotient itself induces an endomorphism map on an exact specified domain. Define

$$\mathcal{D} = \{X \in \text{End}_{\mathbb{C}} L_Z : X(L_{\text{off}}) \subseteq L_{\text{off}}\}.$$

It is a unital subalgebra, and the induced quotient action is a surjective unital homomorphism

$$d : \mathcal{D} \longrightarrow \text{End}_{\mathbb{C}} L_{Z_{\text{crit}}}, \quad \ker d = \text{Hom}_{\mathbb{C}}(L_Z, L_{\text{off}}).$$

Indeed invariance of L_{off} is exactly what makes the quotient action well defined; the kernel condition says exactly that every value of X lies in L_{off} . Any endomorphism of the critical direct summand lifts by making it zero on the off-axis direct summand, proving surjectivity. Thus

$$G(d) : G(\mathcal{D}) \longrightarrow G(\text{End}_{\mathbb{C}} L_{Z_{\text{crit}}})$$

is the explicit split-zero descent map. Both A and Γ belong to $\mathcal{D}$; it sends $A^\bullet$ to the supported critical generator, $\Gamma^\bullet$ to the supported zero e , and τ to τ . This statement includes the zero target algebra when the critical summand is absent.

3 Historical Hurwitz reconstruction and the complete Laurent jet bridge

This section reconstructs a compact probability measure from the complete local Hurwitz zero cluster at any nontrivial zeta zero. It also proves the complete finite-jet relation with the marked Laurent period, including the involution coefficients and the exact collision quotient. Two distinct prime periods recover the entire finite spectral algebra with every multiplicity retained.

3.1 Source receipts and exact locators

1. The original *Split-Zero CUE Hurwitz phase space* source has SHA256 95CCA3275F2012E32F9E506EC0211273D5E7DA1FD72AD90F7167AA37094E94CC. Its exact excerpts accompany this edition in `sources/historical/hurwitz_laurent/`: `compact_hurwitz_and_residues.tex`

reproduces source lines 722–856; `zero_motion_and_bernstein.tex` reproduces lines 1110–1257; and `all_moments_claim.tex` reproduces lines 1907–1912. The statements below retain the original coordinate s , displacement t , support coordinate y , support bound L , and moments $m_n(\mu)$.

2. The *Mathematical Functor Construction* transcript has SHA256 8F94D3A8172F8585837A4E4F6060B5F8DAC38CF627C9689335F13792B7710D85. Its complete mathematical proof block, source lines 14142–15894, is retained as `sources/historical/hurwitz_laurent/marked_period_laurent_conormal_proofs.md`. Theorems 3, 7, 11 and 13 give the marked Laurent augmentation and period maps; Theorems 15–18 give the packet and Laurent involution; Theorems 20–25 give the marked conormal chart. Within that source, lines 15143–15227 prove $J_u^* = J_{-u}$, lines 15397–15502 give the $d \log |a_\infty|$ chart, and lines 15504–15631 give the augmentation-conormal diagram.
3. The corresponding standalone *Terzo-CC-CVS-JT deformation intertwiner* witness has SHA256 BAF7DB6506710CA9AA5B7705C7D5687B4E66536C08A9FE77549ABBC041BC9044. Its Laurent deformation appears at source lines 490–542 and its conormal calculation at lines 638–760. The complete proofs needed here are reproduced below and in the accompanying transcript excerpt.

The historical source gives the forward zero-motion recursion and the marked conormal map. The proofs below give the inverse moment recursion, the complete multiple-zero version, the full nilpotent Laurent isomorphism, and its collision quotient. Source line numbers and hashes identify the particular source editions; these comparisons make no claim of global mathematical priority.

3.2 Every compact Hurwitz input is determined by one simple-zero germ

Keep the original family

$$Z_{\mu,t}(s) = \int_{[0,L]} \zeta(s, 1+ty) d\mu(y), \quad m_n(\mu) = \int_{[0,L]} y^n d\mu(y), \quad m_0(\mu) = 1.$$

Here μ is a Borel probability measure, L is any finite support bound, and t is the original displacement; no rescaling of either variable is made. For $|t|L < 1$, put

$$b_n(s) = \frac{(-1)^n}{n!} (s)_n \zeta(s+n), \quad Z_{\mu,t}(s) = \sum_{n \geq 0} m_n(\mu) b_n(s) t^n. \quad (\text{H1})$$

The expansion is the source's compact Taylor formula. To verify its analytic scope directly, for $\text{Re } s > 1$ expand each $(k+ty)^{-s}$ in its binomial Taylor series. On compact s -sets and $|t|L \leq r < 1$, the ratios $|ty|/k \leq r$ give normally convergent double sums. Integration is termwise valid. The identity extends meromorphically in s ; near a nontrivial zero ρ all displayed coefficients are holomorphic. Joint holomorphicity and normal convergence in the strip can be checked directly: for a compact set K in $0 < \text{Re } s < 1$, the product

$$\left| \frac{(s)_n}{n!} \right| = \prod_{j=1}^n \left| 1 + \frac{s-1}{j} \right| \leq \prod_{j=1}^n \left(1 + \frac{C_K}{j} \right) \leq e^{C_K} (n+1)^{C_K}$$

is polynomially bounded uniformly in $s \in K$. The factor $\zeta(s+n)$ is uniformly bounded for $n \geq 1$ by the absolutely convergent Dirichlet series at $1 + \min_{s \in K} \text{Re } s$. Since $m_n \leq L^n$, the terms of (H1) are bounded by a constant times $(n+1)^{C_K} (L|t|)^n$. Thus the series is normally convergent when $L|t| < 1$, including on a surrounding compact annulus for any contour used below. Cauchy estimates give the same local derivative control. For $n \geq 1$, $\text{Re}(\rho+n) > 1$ and none of $\rho, \rho+1, \dots, \rho+n-1$ is zero, so

$$b_n(\rho) \neq 0. \quad (\text{H2})$$

The zeta factor in (H2) is nonzero because on $\operatorname{Re} z > 1$ the logarithmic Euler-product series

$$\sum_{p \text{ prime}} \sum_{k \geq 1} \frac{p^{-kz}}{k}$$

converges absolutely, and its exponential is $\zeta(z)$ there. This conclusion uses the actual zero's strip $0 < \operatorname{Re} \rho < 1$; it does not assume RH.

Fix a simple nontrivial zero ρ , so $\zeta(\rho) = 0$ and $\zeta'(\rho) \neq 0$. The analytic implicit function theorem produces the unique local holomorphic germ

$$\rho_\mu(t) = \rho + w_\mu(t), \quad w_\mu(t) = \sum_{j \geq 1} c_j(\mu) t^j, \quad Z_{\mu,t}(\rho_\mu(t)) = 0. \quad (\text{H3})$$

This is the existing source theorem. Its simplicity hypothesis is retained explicitly, rather than silently extending a scalar branch through a multiple zero.

The inverse recurrence. For each integer $N \geq 1$, the input moment is recovered by the exact finite formula

$$m_N(\mu) = -\frac{1}{b_N(\rho)} \sum_{n=0}^{N-1} m_n(\mu) [t^{N-n}] b_n \left(\rho + \sum_{j=1}^N c_j(\mu) t^j \right). \quad (\text{H4})$$

Thus $(c_1, \dots, c_N)$ determines $(m_1, \dots, m_N)$ and conversely. In particular, two compact probability inputs have the same zero-trajectory jet through order N exactly when their first N moments agree.

Proof. Substitute (H3) in (H1). Its coefficient of t^N is

$$0 = \sum_{n=0}^N m_n(\mu) [t^{N-n}] b_n(\rho + w_\mu(t)).$$

The $n = N$ summand is exactly $m_N(\mu) b_N(\rho)$. It contains no positive-order coefficient of w . Every other summand uses only $m_0, \dots, m_{N-1}$ and $c_1, \dots, c_N$. Division is legitimate by (H2), proving (H4). In the same coefficient equation the only occurrence of c_N is $\zeta'(\rho) c_N$, from the $n = 0$ summand, because every other Taylor term has either a positive power of t already or at least two positive-order factors of w . This proves the forward triangular recurrence and both directions of the finite-jet assertion by induction. $\square$

The first two inverse formulas, with every sign retained, are

$$m_1 = \frac{\zeta'(\rho) c_1}{\rho \zeta(\rho + 1)}, \quad (\text{H5})$$

$$m_2 = -\frac{2}{\rho(\rho + 1)\zeta(\rho + 2)} \left[\zeta'(\rho) c_2 + \frac{\zeta''(\rho)}{2} c_1^2 - m_1(\zeta(\rho + 1) + \rho \zeta'(\rho + 1)) c_1 \right]. \quad (\text{H6})$$

These follow by $b_1(s) = -s\zeta(s + 1)$ and $b_2(s) = s(s + 1)\zeta(s + 2)/2$, not by changing the Hurwitz convention.

Exact measure reconstruction on the original interval. Assume $L > 0$. For every integer $M \geq 1$ and $0 \leq j \leq M$, set

$$\lambda_{M,j} = \binom{M}{j} \sum_{k=0}^{M-j} (-1)^k \binom{M-j}{k} \frac{m_{j+k}(\mu)}{L^{j+k}}, \quad \mu_M = \sum_{j=0}^M \lambda_{M,j} \delta_{Lj/M}. \quad (\text{H7})$$

Then every $\lambda_{M,j} \geq 0$, their sum is one, and μ_M converges weakly to μ . Thus (H4) and (H7) give an explicit reconstruction of the entire input from the zero germ, with the original support coordinate and bound present in every formula.

Proof. Expanding $(1 - y/L)^{M-j}$ proves

$$\lambda_{M,j} = \int_{[0,L]} \binom{M}{j} (y/L)^j (1 - y/L)^{M-j} d\mu(y).$$

This proves positivity, and the binomial theorem proves the sum is one. At fixed original y , let J have binomial law with M trials and success probability y/L . Its original-coordinate variable LJ/M has mean y and variance

$$\mathbb{E}[(LJ/M - y)^2] = \frac{y(L - y)}{M} \leq \frac{L^2}{4M}. \quad (\text{H8})$$

For continuous f on $[0, L]$ and any $\delta > 0$, split the expectation according to $|LJ/M - y| \leq \delta$. Uniform continuity and Chebyshev's inequality give

$$|\mathbb{E}f(LJ/M) - f(y)| \leq \omega_f(\delta) + 2\|f\|_\infty \frac{L^2}{4M\delta^2},$$

uniformly in y . First take $M \rightarrow \infty$, then $\delta \rightarrow 0$, and integrate against μ . This is weak convergence. For Lipschitz f , Cauchy-Schwarz gives the explicit bound $\text{Lip}(f)L/(2\sqrt{M})$. If $L = 0$, the input is exactly δ_0 , all positive moments vanish, and the original zero branch is constant; this exceptional fibre is retained separately. $\square$

Injectivity without a prespecified common support bound. If two compactly supported probability measures have the same zero germ at the chosen simple ρ , (H4) gives equal moments. Choose any common finite support bound for the two measures. Their measures (H7) are identical at every M , and their weak limits are therefore identical. Hence the map

$$\left\{ \begin{array}{c} \text{compact probability measures} \\ \text{on } [0, \infty) \end{array} \right\} \longrightarrow \left\{ \begin{array}{c} \text{holomorphic zero germs} \\ \text{at the fixed simple } \rho \end{array} \right\},$$

$$\mu \longmapsto \rho_\mu$$

is injective onto its actual image. This statement asserts no surjectivity onto arbitrary holomorphic germs.

The exact support endpoint is also recovered:

$$\sup \text{supp } \mu = \lim_{n \rightarrow \infty} m_n(\mu)^{1/n}. \quad (\text{H9})$$

Indeed, if the endpoint is L_μ , then $m_n \leq L_\mu^n$. For any $0 < a < L_\mu$, positive mass lies in $(a, L_\mu]$, so $m_n \geq a^n \mu((a, L_\mu])$. Taking n th roots and then $a \uparrow L_\mu$ proves the claim. The zero endpoint gives δ_0 and both sides zero.

The completed residue comparison has its own exact factorization. Define

$$\Lambda_{\mu,t}(s) = \pi^{-s/2} \Gamma(s/2) Z_{\mu,t}(s).$$

Then, for the same displacement domain $|t|L < 1$,

$$\text{Res}_{s=1} \Lambda_{\mu,t} = 1, \quad \text{Res}_{s=0} \Lambda_{\mu,t} = -1 - 2tm_1(\mu), \quad (\text{HRes1})$$

and consequently

$$\mathfrak{B}(\mu, t) := \text{Res}_{s=0} \Lambda_{\mu,t} + \text{Res}_{s=1} \Lambda_{\mu,t} = -2tm_1(\mu). \quad (\text{HRes2})$$

Proof. For $\operatorname{Re} \mathfrak{a} > 0$ and $\operatorname{Re} s > 1$, termwise integration of the absolutely convergent geometric series gives

$$\Gamma(s)\zeta(s, \mathfrak{a}) = \int_0^\infty x^{s-1} \frac{e^{-\mathfrak{a}x}}{1 - e^{-x}} dx.$$

At zero the integrand's second factor is $x^{-1} + (1/2 - \mathfrak{a}) + O(x)$, uniformly on compact subsets of the indicated parameter half-plane. Subtract these two terms on $(0, 1)$. The remaining integrals define a function $H(s, \mathfrak{a})$ holomorphic for $\operatorname{Re} s > -1$, and

$$\Gamma(s)\zeta(s, \mathfrak{a}) = \frac{1}{s-1} + \frac{1/2 - \mathfrak{a}}{s} + H(s, \mathfrak{a}).$$

The tail integral is holomorphic there because of exponential decay. Since $\Gamma(1) = 1$ and $\Gamma(s) = s^{-1} + O(1)$, the formula proves $\operatorname{Res}_{s=1} \zeta(s, \mathfrak{a}) = 1$ and $\zeta(0, \mathfrak{a}) = 1/2 - \mathfrak{a}$. For $\mathfrak{a} = 1 + ty$, the parameters stay in a compact subset of that half-plane as y ranges over the original support. Uniformity therefore permits integration of both coefficients against the probability measure. It gives residue one at $s = 1$ and $Z_{\mu,t}(0) = -1/2 - tm_1(\mu)$. Finally $\pi^{-1/2}\Gamma(1/2) = 1$, $\Gamma(s/2) = 2/s + O(1)$, and $\pi^{-s/2} = 1 + O(s)$ give (HRes1) and their sum (HRes2). $\square$

Thus the residue map factors through the explicitly recovered first moment:

$$\rho_\mu \mapsto \frac{\zeta'(\rho)c_1(\mu)}{\rho\zeta(\rho+1)} = m_1(\mu) \mapsto -2tm_1(\mu).$$

The full germ retains every higher moment through (H4), with measure reconstruction given by (H7).

3.3 The complete zero cluster removes the simplicity restriction

The preceding source theorem used a simple scalar branch. The full cluster gives an inverse for **every** nontrivial zero, with its original multiplicity retained. Fix a nontrivial zero ρ of exact order $r \geq 1$. Choose a closed disc about ρ entirely inside $0 < \operatorname{Re} s < 1$, whose boundary Γ contains no zero and whose interior contains no zero of ζ other than ρ . Thus the entire disc also excludes the pole at 1. Shrink the complex t -disc so $Z_{\mu,t}$ is nonzero on Γ and has exactly r zeros inside, with multiplicity. The normal-convergence estimate following (H1) supplies joint holomorphicity and uniform convergence to ζ on Γ ; Rouché's theorem then proves the exact zero count.

For $1 \leq j \leq r$ define the holomorphic power-sum germs

$$q_j(t) = \frac{1}{2\pi i} \oint_\Gamma (s - \rho)^j \frac{\partial_s Z_{\mu,t}(s)}{Z_{\mu,t}(s)} ds.$$

They are exactly the sums of the j th powers of the displacements of the r zeros, counted with multiplicity, by the residue theorem. Set $e_0(t) = 1$ and recursively

$$je_j(t) = \sum_{i=1}^j (-1)^{i-1} e_{j-i}(t) q_i(t).$$

Newton's identities show that the polynomial

$$P_\mu(s, t) = \sum_{j=0}^r (-1)^j e_j(t) (s - \rho)^{r-j} = (s - \rho)^r + \sum_{N \geq 1} P_N(s) t^N \quad (\text{H10})$$

has precisely the full zero cluster as its divisor, and every $P_N(s)$ has degree at most $r - 1$ in the original displacement $s - \rho$. The original point ρ and monic leading coefficient one are both retained. The polynomial is independent of any enumeration or Puiseux branch choices.

There is a jointly holomorphic unit $U_\mu(s, t)$ on a smaller disc such that

$$Z_{\mu,t}(s) = U_\mu(s, t)P_\mu(s, t), \quad U_\mu(s, t) = \sum_{N \geq 0} U_N(s)t^N, \quad U_0(s) = \frac{\zeta(s)}{(s - \rho)^r}. \quad (\text{H11})$$

For completeness, joint holomorphicity does not need to be assumed from the root parametrization. Choose an inner circle still enclosing all zeros for small t ; on that circle the quotient Z/P has nonzero denominator. Its Cauchy integral against $1/(\zeta - s)$ defines a jointly holomorphic function of (s, t) inside, since the integrand is jointly holomorphic on the fixed contour. For each fixed t , the zeros of Z and P agree with their exact multiplicities, so the quotient has removable singularities at the roots and the Cauchy integral equals Z/P . The quotient has no zero, because the two divisors already coincide. This proves (H11). The contour variable ζ here denotes a complex integration variable, not the Riemann zeta function.

Now the input moments have the exact recursive recovery

$$m_N(\mu) = \frac{1}{b_N(\rho)} \sum_{k=1}^N U_{N-k}(\rho)P_k(\rho), \quad (\text{H12})$$

followed by the exact analytic division

$$U_N(s) = \frac{m_N(\mu)b_N(s) - \sum_{k=1}^N U_{N-k}(s)P_k(s)}{(s - \rho)^r}. \quad (\text{H13})$$

In (H13) the numerator vanishes to order at least r ; its holomorphic extension at ρ has value equal to its r th derivative there divided by $r!$. Thus the formula retains every derivative needed by a multiple zero, rather than dividing a nonvanishing germ by a zero.

Proof. Comparing the coefficient of t^N in (H1) and (H11) gives

$$m_N(\mu)b_N(s) = U_N(s)(s - \rho)^r + \sum_{k=1}^N U_{N-k}(s)P_k(s).$$

Evaluation at $s = \rho$ and the nonvanishing (H2) prove (H12). Rearranging the same analytic equality proves (H13), including the divisibility and the value at the centre. Starting from the completely known U_0 , the pair (H12)–(H13) is triangular and recovers every moment from the polynomial-cluster germ.

Two measures with the same polynomial-cluster jet through order N therefore have the same moments through N , by induction. Conversely, equal moments through N give $Z_{\mu,t} - Z_{\nu,t} = O(t^{N+1})$ as holomorphic functions on the fixed circle. Since both denominators remain nonzero there, their logarithmic derivatives, contour power sums, Newton coefficients and hence polynomials (H10) agree through order N . This proves the converse without choosing scalar zero branches. Finally (H7) reconstructs the measure, so the cluster germ at any nontrivial zero determines the entire compact probability input. $\square$

For $r = 1$, $P_N(s) = -c_N$, so (H12)–(H13) reduce to the inverse branch recurrence. For $r > 1$, the contour construction keeps all r zeros and their multiplicities even when individual zeros have fractional-power trajectories. This is a fully proved extension of the historical source's simple-zero statement.

3.4 The historical period map preserves every finite spectral jet

The transcript's marked Frobenius character is

$$z_{p,\omega}(s) = \omega p^{s-1/2}, \quad p > 1, \quad \omega \in S^1, \quad a = \log p > 0. \quad (\text{J1})$$

For the original prime-circle construction, p is a rational prime and $\omega = \psi(\text{Frob}_p)$; the proof below only uses the displayed positive a and unit-modulus ω . It retains the mark and phase.

Fix $\rho \in \mathbb{C}$ and an integer multiplicity $m \geq 1$, and write

$$z = \omega p^{\rho-1/2}, \quad E_{\rho,m} = \mathbb{C}[X]/(X - \rho)^m, \quad L_{z,m} = \mathbb{C}[U, U^{-1}]/(U - z)^m.$$

The original spectral coordinate remains X , and the original Laurent period generator remains U .

Exact jet isomorphism. There is a complex-algebra isomorphism

$$\Phi_{\rho,m} : L_{z,m} \xrightarrow{\sim} E_{\rho,m}, \quad [U] \mapsto z \sum_{k=0}^{m-1} \frac{a^k (X - \rho)^k}{k!}. \quad (\text{J2})$$

Its inverse is

$$[X] \mapsto \rho + \frac{1}{a} \sum_{k=1}^{m-1} \frac{(-1)^{k+1}}{k} \left(\frac{U - z}{z} \right)^k. \quad (\text{J3})$$

For every $0 \leq k < m$, (J2) induces on the k th associated conormal grade the exact multiplier

$$[(U - z)^k] \mapsto (za)^k [(X - \rho)^k]. \quad (\text{J4})$$

Proof. The target of (J2) is a unit: it is z times a polynomial with constant term one and nilpotent remaining part. Its difference from z lies in $(X - \rho)$, so its m th power vanishes. The Laurent universal property and the quotient therefore define (J2). In $\mathbb{C}[t]/t^m$, truncated $\exp(t)$ and $\log(1 + t)$ are inverse operations. This identity can be proved without convergence: formal differentiation shows $(\log(\exp t))' = 1$ and its constant term is zero; likewise $\exp(\log(1 + t))/(1 + t)$ has derivative zero and constant term one. Reducing the resulting formal identities modulo t^m gives (J3) and both composites. Finally

$$\Phi(U - z) = za(X - \rho) + (X - \rho)^2 \left(z \sum_{j=2}^{m-1} \frac{a^j (X - \rho)^{j-2}}{j!} \right),$$

so its k th power modulo $(X - \rho)^{k+1}$ is exactly $(za)^k (X - \rho)^k$. For $m = 1$, the sums in (J3) are empty and both rings are the specified scalar fibres, as required. $\square$

For a finite primary spectral block $A = \rho I + N$ with $N^m = 0$, this means that its actual period operator is exactly

$$\omega \exp(a(A - \tfrac{1}{2}I)) = z \sum_{k=0}^{m-1} \frac{a^k N^k}{k!}. \quad (\text{J5})$$

There is no deletion of nilpotent multiplicity in this map. Indeed, $\exp(aN) - I = N(aI + NB(N))$; its second factor is an invertible polynomial in N , so its powers have exactly the same kernels as the corresponding powers of N . This proves preservation of all Jordan block lengths locally.

3.5 Exact conjugate-reciprocal involution on the jets and conormal

Define the original spectral involution

$$\rho^\dagger = 1 - \bar{\rho}, \quad z^\dagger = \omega p^{\rho^\dagger - 1/2} = 1/\bar{z}. \quad (\text{J6})$$

On polynomial classes define the conjugate-linear algebra isomorphism

$$\mathcal{S}_E : E_{\rho,m} \longrightarrow E_{\rho^\dagger,m}, \quad [f(X)] \longmapsto [\bar{f}(1 - X)]. \quad (\text{J7})$$

On Laurent classes define

$$\mathcal{S}_L : L_{z,m} \longrightarrow L_{z^\dagger,m}, \quad \left[\sum c_n U^n \right] \longmapsto \left[\sum \bar{c}_n U^{-n} \right]. \quad (\text{J8})$$

Both have square the identity after returning to the original labelled block, and the full finite-jet diagram commutes:

$$\boxed{\mathcal{S}_E \Phi_{\rho,m} = \Phi_{\rho^\dagger,m} \mathcal{S}_L.} \quad (\text{J9})$$

Proof. The polynomial generator transforms by

$$X - \rho \longmapsto 1 - X - \bar{\rho} = -(X - \rho^\dagger).$$

The Laurent generator transforms by the exact identity

$$U - z \longmapsto U^{-1} - \bar{z} = -\bar{z} U^{-1} (U - z^\dagger). \quad (\text{J10})$$

The factor $-\bar{z}U^{-1}$ is a unit, so the respective m th-power ideals map to one another and both maps are defined. Applying each twice fixes constants and the original generator. To check (J9), it suffices to check the Laurent generator and constants. Its left image is

$$\bar{z} \sum_{k=0}^{m-1} \frac{(-a)^k (X - \rho^\dagger)^k}{k!},$$

which is the inverse of $z^\dagger \exp(a(X - \rho^\dagger))$ modulo $(X - \rho^\dagger)^m$ and hence the right image. $\square$

For clarity, the entire nonlinear mixing of Laurent Taylor coefficients under (J8) is explicit. With $t' = U - z^\dagger$, for $1 \leq k < m$ one has

$$\boxed{\mathcal{S}_L((U - z)^k) = \sum_{\ell=k}^{m-1} (-1)^\ell \binom{\ell-1}{\ell-k} \bar{z}^{k+\ell} (t')^\ell.} \quad (\text{J11})$$

Proof. By (J10), the expression is

$$(-\bar{z})^k U^{-k} (t')^k = (-1)^k \bar{z}^{2k} (t')^k (1 + \bar{z}t')^{-k}.$$

The formal binomial expansion $(1 + v)^{-k} = \sum_{j \geq 0} (-1)^j \binom{k+j-1}{j} v^j$, truncated when the total degree reaches m , gives (J11). The constant basis vector maps to one separately. $\square$

In particular, the conormal maps are

$$[X - \rho] \longmapsto -[X - \rho^\dagger], \quad [U - z] \longmapsto -\bar{z}^2 [U - z^\dagger], \quad (\text{J12})$$

and their compatibility with (J4) is an exact equality of coefficients:

$$-\bar{z}^2 (z^\dagger a) = -\bar{z}a = \bar{z}\bar{a}(-1).$$

The higher grade k has factors $(-1)^k$ and $(-1)^k \bar{z}^{2k}$ respectively, with the corresponding $(za)^k$ comparison. This supplies the all-order anti-involution/conormal morphism requested by the historical construction; it does not erase the higher terms in (J11).

On the real analytic radial chart $s = \beta + i\gamma$, the same exact map gives

$$\nu = \log |z| = (\beta - \tfrac{1}{2}) \log p, \quad d\nu = (\log p) d\beta, \quad c_{-+} = 4(\beta - \tfrac{1}{2}), \quad \nu = \frac{\log p}{4} c_{-+}. \quad (\text{J13})$$

The source's marked transverse chart $j_S(\nu) = [((e^\nu, 1, \dots, 1), 0)]$ pulls its displayed form $\alpha = d \log |a_\infty|$ back to $d\nu$, since $\log(e^\nu) = \nu$. Thus the composite character→radial chart has pullback $\alpha = (\log p) d\beta$, with the original prime and tessarine factor retained. The local identity requires no conjecture about zeros. A global identification with any separately defined Weil quadratic form is not asserted by this local chart calculation; the exact algebraic and chart maps proved here are the morphisms actually available from this source block.

3.6 Period collisions: the exact unlabelled image and its missing fibres

Let Z be a finite set of distinct spectral points ρ , with original multiplicities $m_\rho \geq 1$, and

$$h(X) = \prod_{\rho \in Z} (X - \rho)^{m_\rho}, \quad E_h = \mathbb{C}[X]/h(X) \cong \prod_{\rho \in Z} E_{\rho, m_\rho}.$$

The last map is the CRT map $f \mapsto (f \bmod (X - \rho)^{m_\rho})$; pairwise coprimality gives its inverse by the Bezout identity. No multiplicity is discarded.

Set $z_\rho = \omega p^{\rho-1/2}$. The equality of two period values is exactly

$$z_\rho = z_\sigma \iff \rho - \sigma \in \frac{2\pi i}{\log p} \mathbb{Z}. \quad (\text{J14})$$

The proof is the exact kernel $2\pi i \mathbb{Z}$ of the complex exponential. In particular, a fixed prime's character cannot automatically be treated as globally injective on a zero packet. No noncollision claim for actual zeta zeros is inserted here.

For each distinct period value z , put $M_z = \max_{\rho: z_\rho=z} m_\rho$ and

$$P(U) = \prod_z (U - z)^{M_z}, \quad L_P = \mathbb{C}[U, U^{-1}]/P(U).$$

There is an injective algebra map

$$\Phi_Z : L_P \hookrightarrow E_h \quad (\text{J15})$$

whose ρ th component is Laurent reduction modulo $(U - z_\rho)^{m_\rho}$ followed by (J2). Its image consists exactly of the tuples which, after (J3) on each labelled primary block, have the same Laurent Taylor coefficients at a common period value through every order visible in both blocks. More explicitly, if $a_\rho \in E_{\rho, m_\rho}$, write

$$\Phi_{\rho, m_\rho}^{-1}(a_\rho) = \sum_{k=0}^{m_\rho-1} b_{\rho, k} (U - z_\rho)^k.$$

Then (a_ρ) lies in the image exactly when

$$b_{\rho,k} = b_{\sigma,k} \quad \text{whenever } z_\rho = z_\sigma \text{ and } 0 \leq k < \min(m_\rho, m_\sigma). \quad (\text{J16})$$

The exact dimension and cokernel dimension as complex vector spaces are

$$\dim \operatorname{im} \Phi_Z = \sum_z M_z, \quad \dim \operatorname{coker} \Phi_Z = \sum_\rho m_\rho - \sum_z M_z. \quad (\text{J17})$$

Proof. Use CRT also on P . For a fixed common value z , a Laurent polynomial vanishes on every corresponding spectral block exactly when it is divisible by $(U - z)^{m_\rho}$ for every such ρ , which is equivalent to divisibility by $(U - z)^{M_z}$. Therefore the kernel of the original Laurent evaluation map is exactly (P) , giving (J15). Reduction of one degree- $< M_z$ polynomial to each smaller truncation gives (J16). Conversely choose any one block of maximal multiplicity M_z ; its Laurent coefficients define the common polynomial, and (J16) forces all other components to be its truncations. These constructions prove the image equality, while the dimensions follow from the explicit polynomial bases. $\square$

Keeping the labels restores a full exact isomorphism:

$$\prod_{\rho \in Z} L_{z_\rho, m_\rho} \xrightarrow{\Pi \Phi_{\rho, m_\rho}} E_h. \quad (\text{J18})$$

Thus the historical period map and the current finite arithmetic jet object have a proved relation at both the labelled and unlabelled levels. The collision compatibility space is exactly the image (J16). An explicit quotient records every lost independent coefficient: choose one maximal block ρ_z over each z , with $m_{\rho_z} = M_z$, and define the complex-linear map

$$\Delta_Z : E_h \longrightarrow \bigoplus_z \bigoplus_{\substack{\rho: z_\rho = z \\ \rho \neq \rho_z}} \mathbb{C}^{m_\rho}, \quad (a_\rho) \longmapsto (b_{\rho,k} - b_{\rho_z,k})_{\substack{z, \rho \neq \rho_z \\ 0 \leq k < m_\rho}}. \quad (\text{J19})$$

It is surjective: set all maximal-block coefficients to zero and prescribe each remaining block's coefficients to be the desired output. Its kernel is exactly (J16). Consequently

$$0 \longrightarrow L_P \xrightarrow{\Phi_Z} E_h \xrightarrow{\Delta_Z} \bigoplus_z \bigoplus_{\substack{\rho: z_\rho = z \\ \rho \neq \rho_z}} \mathbb{C}^{m_\rho} \longrightarrow 0 \quad (\text{J20})$$

is an exact sequence of complex vector spaces. The first map is an algebra homomorphism; the last map is explicitly linear and is not incorrectly promoted to a ring homomorphism. Its target has the cokernel dimension in (J17).

If Z is stable under $\rho \mapsto 1 - \bar{\rho}$ with the same multiplicities, the maps (J7) and (J8), together with permutation of the labelled factors, give conjugate-linear involutions on both sides of (J18). Formula (J9) proves that (J18) intertwines them. The unlabelled construction (J15) is also stable: $z \mapsto 1/\bar{z}$ permutes the common period values and preserves M_z ; (J10) carries the defining ideal of one value to that of the paired value.

3.7 Two prime periods recover the complete finite spectral algebra

Retain a finite nonempty set Z of distinct spectral points with the original orders $m_\rho \geq 1$, and $E_h = \mathbb{C}[X]/h(X)$, where $h(X) = \prod_{\rho \in Z} (X - \rho)^{m_\rho}$. Choose distinct rational primes p, q and fixed marks $\omega_p, \omega_q \in S^1$. Put

$$a = \log p, \quad b = \log q, \quad u_\rho = \omega_p e^{a(\rho-1/2)}, \quad v_\rho = \omega_q e^{b(\rho-1/2)}.$$

Define the units $\mathfrak{A}_p, \mathfrak{A}_q \in E_h$ by their complete original spectral CRT components:

$$\begin{aligned} (\mathfrak{A}_p)_\rho &= u_\rho \sum_{k=0}^{m_\rho-1} \frac{a^k (X - \rho)^k}{k!}, \\ (\mathfrak{A}_q)_\rho &= v_\rho \sum_{k=0}^{m_\rho-1} \frac{b^k (X - \rho)^k}{k!}. \end{aligned} \tag{P1}$$

The period-algebra map therefore has the exact type

$$\mathcal{P}_{p,q} : \mathbb{C}[U^{\pm 1}, V^{\pm 1}] \longrightarrow E_h, \quad U \longmapsto \mathfrak{A}_p, \quad V \longmapsto \mathfrak{A}_q. \tag{P2}$$

This map uses the actual periods of the original generator; the notation $\mathcal{P}_{p,q}$ leaves the theta map Θ unchanged.

Theorem 3.1 (Complete two-prime Laurent presentation). *The map (P2) is surjective, with every local order retained. More precisely, in $\mathcal{L} = \mathbb{C}[U^{\pm 1}, V^{\pm 1}]$, put*

$$\begin{aligned} B_\rho(U) &= v_\rho \sum_{j=0}^{m_\rho-1} \binom{b/a}{j} \left(\frac{U - u_\rho}{u_\rho} \right)^j, \\ I_\rho &= ((U - u_\rho)^{m_\rho}, V - B_\rho(U)). \end{aligned} \tag{P3}$$

Here $\binom{b/a}{0} = 1$, and for $j \geq 1$ the coefficient is $(b/a)(b/a - 1) \cdots (b/a - j + 1)/j!$. The ideals I_ρ are pairwise comaximal, and

$$\begin{aligned} \ker \mathcal{P}_{p,q} &= \bigcap_{\rho \in Z} I_\rho = \prod_{\rho \in Z} I_\rho, \\ \mathcal{L} / \bigcap_{\rho \in Z} I_\rho &\xrightarrow{\sim} E_h. \end{aligned} \tag{P4}$$

The map on each primary quotient is the original finite exponential map (J2), with the second generator determined by (P3).

Proof. First the two marked periods separate distinct complex points. If their values at ρ, σ agree, there exist integers k, l with

$$a(\rho - \sigma) = 2\pi i k, \quad b(\rho - \sigma) = 2\pi i l.$$

Eliminating $\rho - \sigma$ gives $k \log q = l \log p$, hence $q^k = p^l$. Prime valuations in the positive rationals give $k = l = 0$, including negative integer exponents. Thus $\rho = \sigma$. Each fixed phase cancels only within its own prime character; neither mark was changed.

For a fixed ρ , set $y = (U - u_\rho)/u_\rho$. The quotient $\mathcal{L}/I_\rho$ is exactly $\mathbb{C}[y]/y^{m_\rho}$: its generators satisfy

$$U = u_\rho(1 + y), \quad V = v_\rho \sum_{j=0}^{m_\rho-1} \binom{b/a}{j} y^j.$$

Both expressions are units, since their constant terms are nonzero and their remaining parts are nilpotent. The Laurent inverses therefore impose no extra relation. These expressions give a map from the Laurent quotient to $\mathbb{C}[y]/y^{m_\rho}$, whose inverse sends y to $(U - u_\rho)/u_\rho$; their composites fix the displayed generators.

The map to E_{ρ, m_ρ} is

$$y \longmapsto \sum_{k=1}^{m_\rho-1} \frac{a^k (X - \rho)^k}{k!}, \quad X \longmapsto \rho + \frac{1}{a} \sum_{j=1}^{m_\rho-1} \frac{(-1)^{j+1}}{j} y^j \tag{P5}$$

for it and its inverse, respectively. The finite logarithm and exponential identities proved in (J2)–(J3) verify both composites. The identity $(1+y)^{b/a} = \exp((b/a) \log(1+y))$, reduced modulo y^{m_ρ} , verifies the prescribed V -image. For completeness its binomial coefficients are fixed by the formal differential equation $(1+y)F' = (b/a)F$, $F(0) = 1$: $(j+1)[y^{j+1}]F = (b/a-j)[y^j]F$. This recurrence gives exactly the coefficients in (P3). Thus I_ρ is precisely the kernel of the ρ -component of (P2), including order one.

For two different labels, either $u_\rho \neq u_\sigma$, or their v -values differ by the separation already proved. In the first case the powers of the two distinct linear U -factors have a polynomial Bezout identity, proving $I_\rho + I_\sigma = \mathcal{L}$. In the second case, $(V - v_\rho)^{m_\rho} \in I_\rho$: modulo I_ρ , $V - v_\rho = B_\rho(U) - v_\rho$ is a multiple of $U - u_\rho$. The corresponding powers of the two distinct linear V -factors again give a Bezout identity. Hence all primary kernels are pairwise comaximal.

The Chinese-remainder map now gives

$$\mathcal{L} / \bigcap_{\rho} I_{\rho} \xrightarrow{\sim} \prod_{\rho \in Z} \mathcal{L} / I_{\rho} \xrightarrow{\sim} \prod_{\rho \in Z} E_{\rho, m_{\rho}} \xleftarrow{\sim} E_h. \quad (\text{P6})$$

For clarity, the first map is onto because Bezout identities give classes ϵ_ρ equal to one in factor ρ and zero in every other factor; summing ϵ_ρ times prescribed representatives recovers any tuple. Its kernel is the intersection by definition. Pairwise comaximality also gives intersection equal to product, first for two ideals by multiplying a Bezout identity by an element of the intersection, and then by induction. These arguments prove every assertion in (P4).

The original coordinate itself has the explicit inverse image

$$[X] \mapsto \sum_{\rho \in Z} \epsilon_\rho \left(\rho + \frac{1}{a} \sum_{j=1}^{m_\rho-1} \frac{(-1)^{j+1}}{j} \left(\frac{U - u_\rho}{u_\rho} \right)^j \right). \quad (\text{P7})$$

Each term is exactly (P5) in its own primary factor; the other factors vanish through ϵ_ρ . Thus the inverse retains the original generator and all of its nilpotent jets. $\square$

This presentation is an exact statement for each finite packet. The single-prime collision quotient (J20) and the two-prime map (P6) specify their respective information. The factorwise split lifts below retain their individual support labels.

3.8 Split-Zero support is preserved by the exact maps

For any ring R , keep the original $G(R) = \{\tau\} \sqcup R^\bullet$, global zero τ , and supported zero $e_R = 0_R^\bullet$. Every ring map $f : R \rightarrow S$ gives

$$G(f)(\tau) = \tau, \quad G(f)(r^\bullet) = f(r)^\bullet. \quad (\text{S1})$$

This preserves addition and multiplication: if one input is τ , use its defining identity/absorber laws, and if both are supported, use the ring homomorphism identities with the supported tag retained. In particular $G(f)(e_R) = e_S \neq \tau$, even if f has a nontrivial kernel.

Apply (S1) separately to each map (J2), and then take the product of these lifted semiring isomorphisms. The resulting map is

$$\prod_{\rho \in Z} G(L_{z_\rho, m_\rho}) \xrightarrow{\sim} \prod_{\rho \in Z} G(E_{\rho, m_\rho}). \quad (\text{S2})$$

Every individual supported-zero coordinate is retained. The support map on both sides is the full vector of presence indicators in $\{0, 1\}^Z$; under (S2) it is unchanged.

For a finite nonempty index set $\mathcal{I}$, the comparison with applying G once to a product ring is the map

$$\begin{aligned} \iota_R : G \left(\prod_{i \in \mathcal{I}} R_i \right) &\longrightarrow \prod_{i \in \mathcal{I}} G(R_i), \\ \tau &\longmapsto (\tau)_i, \quad (r_i)_i^\bullet \longmapsto (r_i^\bullet)_i. \end{aligned} \quad (\text{S4})$$

It is an injective unital semiring homomorphism. For two supported inputs, the operations agree coordinate by coordinate by the ring laws. When a τ occurs, its additive-identity and multiplicative-absorber laws also agree coordinate by coordinate. The two sorts of image are disjoint because $\mathcal{I}$ is nonempty; equality of supported images recovers every r_i . Its support image is exactly the empty and full support vectors. For ring maps $f_i : R_i \rightarrow S_i$, direct evaluation proves

$$\left(\prod_i G(f_i) \right) \iota_R = \iota_S G \left(\prod_i f_i \right). \quad (\text{S5})$$

This is the exact natural comparison with the factorwise map (S2). If $\mathcal{I}$ is empty, the same comparison has the two-element semiring $G(0)$ as source and the one-element empty product as target, so it is not injective; its unique map identifies e and τ . The product map (S2) itself remains the identity between empty products in that case.

For the unlabelled map, a source factor $G(L_{z,M_z})$ maps to all spectral factors over z , using (J2) after the quotient $L_{z,M_z} \twoheadrightarrow L_{z,m_\rho}$. This is injective because one target block has multiplicity M_z . Its support image consists exactly of the vectors which are constant on every collision class $q^{-1}(z)$, where $q : Z \rightarrow \{z_\rho\}$ is $\rho \mapsto z_\rho$. A source τ gives τ in every such factor, while a supported element gives a supported element in every factor, including factors in which its amplitude becomes zero. The support embedding is exactly

$$q^{-1} : \mathcal{P}(\{z_\rho\}) \hookrightarrow \mathcal{P}(Z), \quad (\text{S3})$$

which preserves empty set, unions and intersections. The direct-image map $q_* : \mathcal{P}(Z) \rightarrow \mathcal{P}(\{z_\rho\})$ preserves unions and has $q_* q^{-1} = \text{id}$, but need not preserve intersections. For two distinct labels ρ, σ in the same collision class, $\{\rho\} \cap \{\sigma\} = \emptyset$ while $q_* \{\rho\} \cap q_* \{\sigma\} = \{z\}$. Thus (S3), rather than an unsupported lattice quotient isomorphism, is the exact support relation; the amplitude compatibility is (J16).

For a same-centre truncation $m' \leq m$, the quotient

$$L_{z,m} \twoheadrightarrow L_{z,m'}, \quad E_{\rho,m} \twoheadrightarrow E_{\rho,m'}$$

commutes with (J2) because the finite power series simply reduce to the corresponding lower nilpotence order. Its lifted map sends the supported nonzero class $(U - z)^{m'}$ to e when $m' < m$, and never to τ . This supplies a concrete example of the original distinction between a vanished jet amplitude and absent support, inside the current arithmetic finite-jet packet.

3.9 All-order Hurwitz motion transported to the period and tesserine coordinates

For the actual zero branch (H3), the exact period germ is

$$z_\mu(t) = \omega p^{\rho_\mu(t)-1/2} = z_\rho \exp \left((\log p) \sum_{n \geq 1} c_n(\mu) t^n \right). \quad (\text{C1})$$

Its N th coefficient is

$$[t^N]z_\mu(t) = z_\rho \sum_{k=1}^N \frac{(\log p)^k}{k!} \sum_{\substack{n_1+\dots+n_k=N \\ n_j \geq 1}} \prod_{j=1}^k c_{n_j}(\mu) \quad (N \geq 1). \quad (\text{C2})$$

The empty coefficient is z_ρ . Conversely, using the unique logarithm germ vanishing at $z_\mu/z_\rho = 1$,

$$c_N(\mu) = \frac{1}{\log p} [t^N] \log(z_\mu(t)/z_\rho). \quad (\text{C3})$$

Equations (C2)–(C3) are inverse because the same formal exponential/logarithm calculation used in (J2)–(J3) applies at every finite order. Combining (C3) with (H4) proves that the marked local period germ also determines the entire compact Hurwitz input.

For real t in the germ's domain,

$$\begin{aligned} \log |z_\mu(t)| &= (\log p) \left(\operatorname{Re} \rho - \frac{1}{2} + \sum_{n \geq 1} \operatorname{Re} c_n(\mu) t^n \right), \\ c_{-+}(\rho_\mu(t)) &= 4 \left(\operatorname{Re} \rho - \frac{1}{2} + \sum_{n \geq 1} \operatorname{Re} c_n(\mu) t^n \right). \end{aligned} \quad (\text{C4})$$

All coefficients, including tangential ones, are retained in (C1)–(C3). The real projection in (C4) is explicitly the radial/tesserine observable; its loss of imaginary coefficients is a specified map. For $t \neq 0$, the Hurwitz family generally lacks the original functional equation, so $1 - \rho_\mu(t)$ is not asserted to be another zero of that family. The tesserine coordinate formula is the proved coordinate map evaluated at the actual branch, with this spectral distinction preserved.

3.10 Mathematical scope

The maps (J2)–(J20) identify the full finite arithmetic algebra with its labelled Laurent jets and calculate the exact quotient of unlabelled single-prime observations. Equations (H12)–(H13) recover all moments from the actual zero cluster, and (H7) reconstructs the entire measure. These statements retain the original multiplicities, support coordinates and marked periods.

The constructions give exact algebraic and analytic comparison maps. They do not supply the uniform estimate on the arithmetic moment matrices required by the weight-control calculation elsewhere in this reader.

The accompanying source records retain the historical locators and independent proof reviews. The proofs in this section specify their own hypotheses and maps in full.

4 Reconstruction with changes of support index

The web formalization constructs reverse reconstruction, its fibrewise quotients, and an inverse on each actual semimodule. This section gives the explicit categorical comparison, including changes of support index, used to connect those objects to the arithmetic chart. No assumption of injective transition maps or zero bottom fibre is made.

Fix a commutative ring R and its split semiring $G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}$. A diagram object consists of a join-semilattice L with bottom and R -modules V_l , with linear maps $\rho_{lk} : V_l \rightarrow V_k$ for $l \leq k$, satisfying identity and composition. A morphism from (L, V) to (L', W) is a pair (α, f) , where $\alpha : L \rightarrow L'$ preserves bottom and joins with R -linear fibre maps

$$f_l : V_l \rightarrow W_{\alpha(l)}, \quad f_k \rho_{lk} = \rho'_{\alpha(l), \alpha(k)} f_l. \quad (\text{D1})$$

Composition is $(\beta, g)(\alpha, f) = (\beta\alpha, (g_{\alpha(l)}f_l)_l)$. These are the literal carriers and maps of the diagram category.

Theorem 4.1 (Full support-index reconstruction). *The diagram category just defined is equivalent to the category of $G(R)$ -semimodules and their $G(R)$ -linear maps. The forward functor has carrier*

$$\mathcal{T}(L, V) = \coprod_{l \in L} V_l$$

and operations

$$(l, x) + (k, y) = (l \vee k, \rho_{l, l \vee k}x + \rho_{k, l \vee k}y), \quad \tau(l, x) = (\perp, 0), \quad r^\bullet(l, x) = (l, rx). \quad (\text{D2})$$

Its global zero is $(\perp, 0)$. On morphisms it is $\mathcal{T}(\alpha, f)(l, x) = (\alpha(l), f_l x)$.

The inverse sends a semimodule M to

$$\begin{aligned} L_M &= eM, \quad l \vee k = l + k, \quad \perp = 0_M, \\ V_l &= \{m \in M : em = l\}, \quad \rho_{lk}(m) = m + k \quad (l + k = k). \end{aligned} \quad (\text{D3})$$

The natural comparison maps are

$$\mathcal{T}(L_M, V) \longrightarrow M, \quad (l, m) \mapsto m, \quad m \mapsto (em, m), \quad (\text{D4})$$

and the label-and-fibre isomorphism

$$L \longrightarrow e\mathcal{T}(L, V), \quad l \mapsto (l, 0), \quad x \in V_l \longmapsto (l, x). \quad (\text{D5})$$

Proof. For (D2), threefold addition transports both parenthesizations to $l \vee k \vee j$. Linearity and composition of the ρ 's make both amplitudes $\rho_l x + \rho_k y + \rho_j z$ in that same fibre. The bottom zero and commutativity follow likewise. For supported scalars the module laws follow from those of each V_l . Their addition law at a vector in fibre l is the ordinary R -module addition in V_l ; in particular $0_R^\bullet$ gives $(l, 0)$. The τ cases use the global bottom zero. Distributivity over vectors follows from linearity of each transition. These checks prove every semimodule axiom. Equation (D1) proves that $\mathcal{T}(\alpha, f)$ respects addition; its scalar and zero laws are immediate. Identity and composition hold on each pair.

For the reverse direction, $e^2 = e$, $e + e = e$, and $1 + e = 1$ in $G(R)$. Thus every $l \in eM$ is additively idempotent; eM is closed under addition, which supplies a semilattice with the specified bottom. For $m \in V_l$,

$$m + l = (1 + e)m = m, \quad m + (-1_R)^\bullet m = em = l.$$

The inherited addition in V_l , with zero l and inverse $(-1_R)^\bullet m$, is therefore an abelian group. For any $r \in R$, $e(r^\bullet m) = em = l$, since $er^\bullet = e$. Its scalar action satisfies every R -module law inherited from the semimodule, and $0_R m = em = l$ is its local zero. This proves that V_l is the specified module, without cancellation assumptions on all of M .

If $l + k = k$, then $e(m + k) = l + k = k$, so ρ_{lk} has the correct codomain. Its zero law is $l + k = k$; its additivity uses $k + k = k$. Also $r^\bullet k = k$, proving R -linearity. Identity follows from $m + l = m$, and composition from $k + j = j$. A split-linear map $F : M \rightarrow N$ takes eM to eN , with $\alpha(l) = F(l)$. It preserves joins and bottom, restricts to R -linear maps on each fibre, and $F(m + k) = F(m) + F(k)$ proves (D1). This defines the inverse functor.

Both arrows in (D4) are inverse as functions. Addition on the reconstructed carrier has amplitude

$$(m + (l \vee k)) + (n + (l \vee k)) = m + n + (l \vee k) = m + n,$$

because $e(m + n) = l \vee k$. Thus (D4) is additive; its scalar identities follow from (D2), including τ . It is a split-linear isomorphism. In the reconstructed total, $e(l, x) = (l, 0)$, so (D5) is an isomorphism of semilattices and of the original fibres. Its transition map is exactly $(l, x) + (k, 0) = (k, \rho_{lk}x)$, recovering the given ρ . For a morphism (α, f) , (D5) takes $(l, 0)$ to $(\alpha(l), 0)$; for F , (D4) sends a representative to $F(m)$. These equalities prove naturality of both inverse comparisons, including support-index changes. Hence they are a categorical equivalence. $\square$

4.1 The internal quotient and the actual comparison kernel

Let $B_l \subseteq V_l$ be transition-stable submodules. The quotient diagram has fibres V_l/B_l with their induced transitions. The map

$$q : \mathcal{T}(L, V) \rightarrow \mathcal{T}(L, V/B), \quad (l, x) \mapsto (l, [x]) \quad (\text{D6})$$

is onto and split-linear. It is the coequalizer of the two maps $\mathcal{T}(L, B) \rightrightarrows \mathcal{T}(L, V)$ given by inclusion and $(l, b) \mapsto (l, 0)$.

To prove the universal property against any split-linear target N , let f equalize these maps. If $x - y \in B_l$,

$$f(l, x) = f(l, y) + f(l, x - y) = f(l, y) + f(l, 0) = f(l, y).$$

The last equality is the image of $(l, y) + (l, 0) = (l, y)$; no subtraction or cancellation in N was used. Thus f descends as a function through (D6). Lifting representatives proves additivity, scalar laws and uniqueness. Different labels remain distinct under (D6). In particular a boundary is carried to its original support-labelled zero.

For a cochain map $f : C \rightarrow D$ of R -module complexes at a fixed degree, put $Z_C = \ker d_C$, $B_C = \text{im } d_C^{\text{prev}}$ inside Z_C , and $K_{\text{rep}} = \{z \in Z_C : f(z) \in B_D\}$. Then the map

$$K_{\text{rep}}/B_C \xrightarrow{\sim} \ker H(f), \quad [z] \mapsto [z] \quad (\text{D7})$$

is a linear isomorphism. It is well-defined since f maps boundaries to boundaries. The prequotient map $K_{\text{rep}} \rightarrow \ker H(f)$, $z \mapsto [z]$, has kernel exactly B_C ; every kernel class has a cycle representative, proving surjectivity. If $fa = bf$ for chain endomorphisms, $aK_{\text{rep}} \subseteq K_{\text{rep}}$ and $aB_C \subseteq B_C$; (D7) therefore intertwines the actual induced actions. If f is a morphism of diagrams of complexes, these constructions commute with the transitions and reconstruct by (D2). For the equivariant assertion across supports, a, b are diagram endomorphisms as well. The equation $d^2 = 0$ then becomes the map $(l, x) \mapsto (l, 0)$ from its original source degree to its original target degree. It is different from constant absence when $l \neq \perp$, and (D6)–(D7) are the exact comparison maps.

For completeness, the full semimodule category retains two precisely related kernel constructions. For $F = \mathcal{T}(\alpha, f) : M = \mathcal{T}(L, V) \rightarrow N = \mathcal{T}(L', W)$, put $L_0 = \alpha^{-1}(\perp')$. Then

$$\begin{aligned} \ker_{G(R)} F &= \mathcal{T}(L_0, (\ker f_l)_{l \in L_0}), \\ F^{-1}(eN) &= \mathcal{T}(L, (\ker f_l)_{l \in L}) = M \times_N eN. \end{aligned} \quad (\text{D8})$$

The inclusion of the first into the second is induced by $L_0 \hookrightarrow L$. Indeed L_0 contains bottom and is closed under joins. Naturality in (D1) takes each fibre kernel into the next one. The formula $F(l, x) = (\alpha(l), f_l x)$ equals the global zero exactly when $\alpha(l) = \perp'$ and $f_l x = 0$; it belongs to eN exactly when $f_l x = 0$. These pointwise descriptions prove both universal properties: any split-linear map satisfying the respective condition has its unique factorization through the displayed subsemimodule. Thus reconstruction of the all-label fibre kernels in (D7) gives the second object. If α is the identity, the categorical kernel is its bottom fibre, with the inclusion in (D8) retaining the exact comparison.

5 The original theta complex and its arithmetic packet

The September 12 source chain has advanced from a support dictionary to an actual characteristic-zero cochain construction. We retain its complete coefficient square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

In particular $p_{\mathbb{Z}}$ has infinite target. The absolute pointed base is $\{\tau, 1\}$, with τ absorbing for multiplication; it is not obtained by imposing $e = 1$. The map p_R sends both τ and e_R to 0_R , while the support map distinguishes them. All of the linear maps below lift on each given support fibre; their ordinary zero values lift to that fibre's supported zero.

For clarity, the fixed section called R_0 in the web notes is denoted R_{ref} here. The corrected sequence keeps its name R_m , including its genuinely different level-zero member. This is an explicit notation correspondence between the same maps.

5.1 Source spaces and fixed arithmetic factors

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_+) : p_{b,k}(F) = \sup_{x>0} x^b |D^k F(x)| < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0\}, \quad D = -x\partial_x.$$

Use the original Fourier, theta and Mellin transforms

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \quad (\text{A1})$$

The seminorms $p_{b,k}$ define the strong Schwartz topology. Logarithmic coordinates give completeness by uniform convergence of every derivative on compact intervals and the weighted bounds. The conditions defining V are closed in the even Schwartz space.

Proposition 5.1 (The fixed multiplier and injective theta map). *The map $\Theta : V \rightarrow \mathcal{B}$ is continuous and injective, $D\Theta = \Theta D$, and*

$$\Theta\widehat{\phi}(x) = x^{-1}\Theta\phi(1/x).$$

On $0 < \text{Re } s < 1$, define

$$H_\phi(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)} \int_0^\infty \phi(x) x^s \frac{dx}{x}.$$

Then

$$\mathcal{M}\Theta\phi = gH_\phi, \quad g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (\text{A2})$$

For the exact original source $\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$,

$$H_{\phi_*} = 1, \quad H_{P(D)\phi_*} = P, \quad \mathcal{M}\Theta(P(D)\phi_*) = gP. \quad (\text{A3})$$

Proof. Schwartz estimates give uniform convergence of each differentiated theta series on $x \geq \delta > 0$ and all bounds at infinity. Poisson summation and the two source moments give the displayed Fourier identity; applying the same bounds to $\widehat{\phi}$ gives all bounds at zero, with continuous dependence on Schwartz seminorms. Differentiating the series proves intertwining. For $x > 0$, the absolutely convergent divisor rearrangement gives

$$\frac{1}{2} \sum_{n \geq 1} \mu(n) \Theta\phi(nx) = \sum_{k \geq 1} \left(\sum_{n|k} \mu(n) \right) \phi(kx) = \phi(x).$$

Absolute convergence follows from $\sum_k d(k) |\phi(kx)| < \infty$. This proves injectivity.

In a right half-plane absolute integration gives $\mathcal{M}\Theta\phi = 2\zeta M_+\phi$. Both sides continue to the open strip, with the source's zero moments removing the relevant apparent singularities, which proves (A2). More explicitly $M_+\phi$ is holomorphic for $\text{Re } s > -2$, since evenness and $\phi(0) = 0$ give $\phi(x) =$

$O(x^2)$ at zero; the theta side is entire. The multiplier defining H_ϕ is holomorphic and nonzero on the open strip. The Gaussian integral gives

$$M_+\phi_*(s) = \pi^{-s/2}\{2\Gamma(s/2 + 2) - 3\Gamma(s/2 + 1)\} = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2).$$

The source moments follow from its value at zero and this formula at $s = 1$. Integration by parts proves $H_{D\phi} = sH_\phi$; all boundary terms vanish. These calculations prove (A3), with both factors of two retained. $\square$

The actual one-leg complex, in degrees zero and one, is $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$. Let $Q = \mathcal{B}/\Theta V$ and $q : \mathcal{B} \rightarrow Q$. The two-leg complex is

$$C_{+-} = [V \oplus V \xrightarrow{d} \mathcal{B}], \quad d(v, w) = \Theta(v - \widehat{w}). \quad (\text{A4})$$

Its source operator is $(D, 1 - D)$. Its degree-zero cycles are exactly $(\widehat{w}, w)$; its degree-one homology is Q . This follows from injectivity of theta and $\text{im } d = \Theta V$. Thus there is a concrete inclusion of C_+ into C_{+-} , with the extra degree-zero cycle retained through the map $w \mapsto (\widehat{w}, w)$.

5.2 The exact operator range, including its inverse

Proposition 5.2 (Source and target Euler inverses). *For $\rho \in \mathbb{C}$, on the indicated kernel define*

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy.$$

Then

$$(D - \rho)\mathcal{B} = \ker(\mathcal{M}(\cdot)(\rho)), \quad (D - \rho)S_\rho = 1, \quad \mathcal{M}S_\rho F(s) = \frac{\mathcal{M}F(s)}{s - \rho}. \quad (\text{A5})$$

The operator $D - \rho$ is injective. For $0 < \text{Re } \rho < 1$, the same formula, extended evenly, gives

$$(D - \rho)V = \ker(H(\cdot)(\rho)). \quad (\text{A6})$$

These inverses are continuous for the displayed source topologies.

Proof. Differentiation proves the right-inverse identity. For F with zero Mellin value the upper tail equals the negative lower tail. In $x = e^y$ coordinates, with $\sigma = \text{Re } \rho$, the upper bound $C_N e^{-Ny}$ becomes $C_N e^{-Ny}/(N - \sigma)$ for $N > \sigma$, and the lower bound $C'_N e^{Ny}$ becomes $C'_N e^{Ny}/(N + \sigma)$ for $N + \sigma > 0$. Differentiating uses $\partial_y S_\rho F(e^y) = -\rho S_\rho F(e^y) - F(e^y)$. These estimates prove every required seminorm bound and continuity. The homogeneous solution $cx^{-\rho}$ fails the target decay at one end unless $c = 0$, proving injectivity and uniqueness. Integration by parts proves the Mellin identity, with its removable value at ρ filled in.

For the source, $H_\phi(\rho) = 0$ implies $M_+\phi(\rho) = 0$, since its coefficient is nonzero in the open strip. Near zero the same inverse is

$$S_\rho \phi(x) = - \int_0^1 u^{\rho-1} \phi(xu) du.$$

Dominated differentiation gives smooth even extension and value zero. The upper tail gives Schwartz decay. Integration of $(D - \rho)S_\rho \phi = \phi$ on $\mathbb{R}$ gives $(1 - \rho) \int S_\rho \phi = \int \phi = 0$. Thus the second moment is preserved. The same estimates, with derivatives under the integral at zero, prove continuity on V . $\square$

For any polynomial $h(t) = \prod_i (t - \rho_i)^{m_i}$ with distinct centres in the open strip, iteration of (A5)–(A6) gives

$$h(D)\mathcal{B} = \ker(j_h \mathcal{M}), \quad h(D)V = \ker(j_h H), \quad E_h = \mathbb{C}[t]/(h). \quad (\text{A7})$$

Here j_h retains the full Taylor jets through the Chinese-remainder isomorphism. Order of inversion does not affect the answer, by uniqueness for the injective $h(D)$. The source jet map is onto by (A3). The target jet map is onto: the finitely many functions $y^k e^{\rho y} / k!$ are independent on any real interval; their Gram against a smooth nonnegative bump positive on an interval is invertible. Its inverse produces compactly supported smooth log functions with any prescribed vector of moments. Consequently the actual quotient square is

$$\begin{array}{ccc} V/h(D)V & \xrightarrow{\Theta} & \mathcal{B}/h(D)\mathcal{B} \\ j_h H \downarrow \wr & & \wr \downarrow j_h \mathcal{M} \\ E_h & \xrightarrow{\times j_h g} & E_h. \end{array} \quad (\text{A8})$$

Theorem 5.3 (Finite arithmetic kernel, quotient and packet section). *The actual theta sequence gives the exact sequence*

$$0 \longrightarrow Q[h(D)] \longrightarrow E_h \xrightarrow{\times j_h g} E_h \longrightarrow Q/h(D)Q \longrightarrow 0. \quad (\text{A9})$$

For a nonempty finite packet Z of actual nontrivial zeros with their full orders, put

$$h = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E = E_h, \quad v = g/h, \quad \varepsilon = j_h(h/g).$$

The element ε is a unit. Let $R_Z(u)$ be the unique polynomial of degree $< d$ representing εu , and set

$$R_{\text{ref}} u = S_h^{\mathcal{B}} \Theta(R_Z(u)(D)\phi_*), \quad J_Z F = j_h \mathcal{M} F, \quad \sigma = q R_{\text{ref}}.$$

Then

$$\mathcal{M} R_{\text{ref}} u = (g/h) R_Z(u), \quad J_Z R_{\text{ref}} = 1, \quad J_Z \Theta = 0, \quad Q = Q[h(D)] \oplus h(D)Q. \quad (\text{A10})$$

The projector is $\sigma \bar{J}_Z$; its image is $Q[h(D)]$, and its kernel is $h(D)Q$.

Proof. If $h(D)[F] = 0$, write the unique $h(D)F = \Theta\phi$ and send $[F]$ to $j_h H\phi$. Changing F by a theta relation changes ϕ by $h(D)V$, so this is well-defined. If its value is zero, (A7) gives $\phi = h(D)\psi$, and injectivity yields $F = \Theta\psi$. Its image lies in $\ker(j_h g)$ by (A8). Conversely a polynomial representative P of a vector in that kernel gives the class $[S_h \Theta(P(D)\phi_*)]$. It is defined by (A7), and changing P by hT changes the lift by $\Theta(T(D)\phi_*)$. These are inverse maps. The cokernel description in (A9) follows by quotienting the right side of (A8), proving exactness with its original arrows.

For the packet, $j_h g = 0$, and v is nonvanishing at every chosen zero because the complete order was retained. The Mellin formula proves $J_Z R_{\text{ref}} = j_h v \varepsilon = 1$. The kernel-to-quotient comparison of (A9) is multiplication by $j_h v$, an isomorphism. Its inverse specifies σ . Thus every class has a unique sum of a class in $Q[h(D)]$ and one in $h(D)Q$, proving (A10) and the projector assertions. $\square$

5.3 The original extension supplies the boundary coefficient

With $A = M_t$ on E , define $\ell(u) = [t^{d-1}]R_Z(u)$. Polynomial division, without changing R_Z or its unit, gives

$$D R_{\text{ref}} - R_{\text{ref}} A = f_0 \ell, \quad f_0 = \Theta\phi_*. \quad (\text{A11})$$

Indeed $t R_Z(u) - R_Z(tu)$ is divisible by the monic h , and its degree is at most d ; its quotient is precisely $\ell(u)$. Apply (A3) and Mellin injectivity, or multiply by the injective $h(D)$, to obtain (A11). For one simple zero this coefficient is $1/g'(\rho) = 1/(2\xi'(\rho))$.

The invariant subspace $\mathcal{B}_Z = \Theta V + R_{\text{ref}} E$ fits into the actual extension

$$0 \rightarrow V \xrightarrow{\Theta} \mathcal{B}_Z \xrightarrow{J_Z} E \rightarrow 0. \quad (\text{A12})$$

It splits complex-linearly through R_{ref} ; in these coordinates its operator is

$$\begin{pmatrix} D_V & \phi_* \ell \\ 0 & A \end{pmatrix}.$$

The extension class, using the resolution $\mathbb{C}[t] \xrightarrow{h} \mathbb{C}[t]$, is the class of $R_Z(1)(D)\phi_*$ in $V/h(D)V$. Under (A7) it is exactly $\varepsilon \neq 0$. This follows by applying $h(D)$ to the chosen lift $R_{\text{ref}}(1)$. It has no equivariant section into $\mathcal{B}$, since every equivariant image of E would be killed by $h(D)$, which is injective there. Its exact replacement is the equivariant roof

$$E[-1] \xleftarrow{\simeq} [V \rightarrow \mathcal{B}_Z] \longrightarrow [V \rightarrow \mathcal{B}]. \quad (\text{A13})$$

The left arrow is J_Z in degree one and zero in degree zero; exactness of (A12) proves it a quasi-isomorphism. Thus no relation or homotopy is suppressed by the ordinary direct-sum notation.

5.4 Actual representative corrections and rank-two control

Throughout this subsection $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$, with inner product conjugate-linear in the first slot. Coefficient spaces carry the Euclidean inner products in the retained bases; every adjoint below uses these specified metrics. Put $\phi_j = D^j \phi_*$, $f_j = D^j f_0$, and $\Phi_m(c) = \sum_{j=0}^m c_j \phi_j$. For $F_m = \Theta \Phi_m : \mathbb{C}^{m+1} \rightarrow \mathcal{H}$, define

$$\begin{aligned} H_m &= F_m^* F_m, & C_m &= F_m^* R_{\text{ref}}, & B_m &= -H_m^{-1} C_m, \\ \Pi_m &= F_m H_m^{-1} F_m^*, & R_m &= R_{\text{ref}} + F_m B_m = (1 - \Pi_m) R_{\text{ref}}. \end{aligned} \quad (\text{A14})$$

The columns f_j are independent: Mellin would turn a dependence into $gP = 0$. Thus $H_m > 0$. The correction is an original theta boundary, giving $qR_m = \sigma$, $J_Z R_m = 1$, and $F_m^* R_m = 0$. For every original coefficient matrix $B : E \rightarrow \mathbb{C}^{m+1}$,

$$\begin{aligned} (R_{\text{ref}} + F_m B)^* (R_{\text{ref}} + F_m B) &= G_m + (B - B_m)^* H_m (B - B_m), \\ G_m &= R_m^* R_m = R_{\text{ref}}^* R_{\text{ref}} - C_m^* H_m^{-1} C_m > 0. \end{aligned} \quad (\text{A15})$$

Expansion proves the identity; injectivity from $J_Z R_m = 1$ proves its strict positivity.

Let $I_m e_j = e_j$ and $S_m e_j = e_{j+1}$, both with codomain $\mathbb{C}^{m+2}$. In the retained source coordinates,

$$K_m = e_0 \ell + S_m B_m - I_m B_m A, \quad D R_m - R_m A = F_{m+1} K_m. \quad (\text{A16})$$

Integration by parts in dx , with the source boundary term $[x \bar{F} H]_0^\infty = 0$, gives

$$W_m = A^* G_m + G_m A - G_m = -(R_m^* F_{m+1} K_m + K_m^* F_{m+1}^* R_m).$$

All but the last source column pair to zero. With $z_m = R_m^* f_{m+1}$ and $b_m = e_m^* B_m$ this is exactly

$$W_m = -z_m b_m - b_m^* z_m^*. \quad (\text{A17})$$

In particular rank $W_m \leq 2$. Factoring $G_m^{-1} W_m$ through the two columns gives its nonzero spectrum as that of

$$-\begin{pmatrix} c_m & v_m \\ u_m & \bar{c}_m \end{pmatrix}, \quad u_m = z_m^* G_m^{-1} z_m, \quad v_m = b_m G_m^{-1} b_m^*, \quad c_m = b_m G_m^{-1} z_m.$$

Cauchy-Schwarz gives $u_m v_m \geq |c_m|^2$. Hence its exact two-sided allowance is

$$\epsilon_m = |\operatorname{Re} c_m| + \sqrt{u_m v_m - (\operatorname{Im} c_m)^2}, \quad -\epsilon_m G_m \leq W_m \leq \epsilon_m G_m. \quad (\text{A18})$$

This includes ranks zero and one. For a reflection-stable packet, $\operatorname{Tr}(G_m^{-1} W_m) = 2 \operatorname{Re} \operatorname{Tr} A - d = 0$. Every eigenvector retains the equality

$$\frac{u^* W_m u}{u^* G_m u} = 2 \operatorname{Re} \rho - 1. \quad (\text{A19})$$

The next sections compute the actual arithmetic moment problem behind G_m , sharpen its constraints, and determine the topology of the original jet observation. They use (A11)–(A19) rather than an assigned metric in place of the theta representative.

6 Certified moments from the original arithmetic theta source

The moment matrices in the arithmetic kernel formulas can be calculated directly from the specified theta source. This section gives an exact incomplete-gamma series, proves an explicit bound for the entire omitted double tail, and applies directed interval arithmetic to the retained terms. The construction retains the original factor $g = 2\xi$, the measure dx , the Euler operator $D = -x\partial_x$, and every polynomial coefficient. No information about the location of a zeta zero enters this calculation.

6.1 Source polynomials and the exact reflection map

Keep

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad f_j = D^j \Theta \phi_*, \quad \Theta \phi(x) = 2 \sum_{n \geq 1} \phi(nx). \quad (\text{AM.1})$$

Introduce polynomials with their original integer coefficients by

$$P_0(z) = 4z^2 - 6z, \quad P_{j+1}(z) = 2z(P_j(z) - P'_j(z)), \quad Q_j(z) = \sum_{k=0}^j (-1)^k \binom{j}{k} P_k(z). \quad (\text{AM.2})$$

Write p_{ja} and q_{ja} for the coefficients of z^a . Both polynomials have degree $d_j = j + 2$, with leading coefficients 2^{j+2} and $(-1)^j 2^{j+2}$, respectively, and both have zero constant coefficient. The formula defining Q_j is an explicit change of polynomial coordinates; neither family is discarded.

Define the linear involution on the actual Hilbert observation by

$$\mathcal{T} : L^2(\mathbb{R}_{>0}, dx) \longrightarrow L^2(\mathbb{R}_{>0}, dx), \quad (\mathcal{T}F)(x) = x^{-1}F(1/x). \quad (\text{AM.3})$$

It is unitary and self-adjoint, satisfies $\mathcal{T}^2 = 1$, and, on the strong Schwartz space $\mathcal{B}$,

$$D\mathcal{T} = \mathcal{T}(1 - D), \quad \mathcal{T}f_0 = f_0, \quad \mathcal{T}f_j = (1 - D)^j f_0. \quad (\text{AM.4})$$

Consequently, the complete source series and its reflected series are

$$f_j(x) = 2 \sum_{n \geq 1} P_j(\pi n^2 x^2) e^{-\pi n^2 x^2}, \quad \tilde{f}_j(x) := (1 - D)^j f_0(x) = 2 \sum_{n \geq 1} Q_j(\pi n^2 x^2) e^{-\pi n^2 x^2}, \quad (\text{AM.5})$$

and $f_j(1/x) = x \tilde{f}_j(x)$.

Proof. For $z = \pi x^2$, applying $-x\partial_x$ to $P(z)e^{-z}$ gives $2z(P - P')e^{-z}$. This proves the recurrence and (AM.5). The leading coefficients and zero constant coefficients follow by induction. Under the Fourier convention $\widehat{\phi}(\xi) = \int \phi(x) e^{-2\pi i x \xi} dx$, differentiating the Fourier transform of the Gaussian gives

$$\widehat{x^2 e^{-\pi x^2}}(\xi) = \left(\frac{1}{2\pi} - \xi^2 \right) e^{-\pi \xi^2}, \quad \widehat{x^4 e^{-\pi x^2}}(\xi) = \left(\xi^4 - \frac{3\xi^2}{\pi} + \frac{3}{4\pi^2} \right) e^{-\pi \xi^2}.$$

Multiplication by the exact coefficients $4\pi^2, -6\pi$ therefore gives $\widehat{\phi}_* = \phi_*$. Its value at zero is zero, and its integral is $\widehat{\phi}_*(0) = 0$. The theta-Fourier identity already proved for the source V then gives $\mathcal{T}f_0 = f_0$.

Substitution $u = 1/x$ gives $\int |x^{-1}F(1/x)|^2 dx = \int |F(u)|^2 du$, so $\mathcal{T}$ is unitary. It is an involution and hence self-adjoint. Direct differentiation proves $D\mathcal{T} = \mathcal{T}(1 - D)$, including the constant 1 arising from the original factor x^{-1} . Induction proves the other identities in (AM.4). Local uniform convergence of each differentiated Gaussian series follows on $x \geq \delta > 0$ by a polynomial times Gaussian majorant. The series gives decay of all Euler derivatives at infinity, and its exact reflected expression gives the corresponding decay at zero. Thus these operations apply to the original functions in $\mathcal{B}$. $\square$

6.2 An absolutely convergent double series for every entry

For integers $i, j \geq 0$, retain the raw moment entry

$$H_{ij} = \langle D^i f_0, D^j f_0 \rangle_{L^2(dx)} = \int_{\mathbb{R}} \overline{(1/2 + it)^i} (1/2 + it)^j |2\xi(1/2 + it)|^2 \frac{dt}{2\pi}. \quad (\text{AM.6})$$

The equality is the original Mellin Plancherel identity. In particular H_{ij} is real because the source functions are real. For $L \geq 0$ and $\lambda > 0$, set

$$I_L(\lambda) = \int_1^\infty x^{2L} e^{-\lambda x^2} dx = \frac{\Gamma(L + 1/2, \lambda)}{2\lambda^{L+1/2}}, \quad \Gamma(a, \lambda) = \int_\lambda^\infty u^{a-1} e^{-u} du. \quad (\text{AM.7})$$

Theorem 6.1 (Exact original arithmetic moment series). *With $\lambda_{nm} = \pi(n^2 + m^2)$, the following double series converges absolutely, including after the finite coefficient sums are replaced by their absolute values:*

$$H_{ij} = 2 \sum_{n,m \geq 1} \sum_{a=1}^{d_i} \sum_{b=1}^{d_j} (p_{ia} p_{jb} + q_{ia} q_{jb}) \pi^{a+b} n^{2a} m^{2b} \cdot \frac{\Gamma(a + b + 1/2, \lambda_{nm})}{\lambda_{nm}^{a+b+1/2}}. \quad (\text{AM.8})$$

Equivalently, it is the Gram integral on the positive half-domain

$$H_{ij} = \int_1^\infty (f_i(x) f_j(x) + \tilde{f}_i(x) \tilde{f}_j(x)) dx. \quad (\text{AM.9})$$

Thus the precise map to the positive Gram presentation is

$$\mathbb{C}[s] \longrightarrow L^2([1, \infty), dx)^{\oplus 2}, \quad P \longmapsto (P(D)f_0, P(1-D)f_0)|_{[1, \infty)}. \quad (\text{AM.10})$$

It preserves the original polynomial inner product (AM.6) and is injective. The signs in (AM.8) are retained; the positivity is the Gram positivity of this specified map.

Proof. Split the integral in (AM.6) at $x = 1$. On $(0, 1)$, set $x = 1/u$. The two factors u supplied by (AM.5) cancel the Jacobian u^{-2} exactly. This gives (AM.9). For absolute convergence, write

$$C_j = \sum_{a=1}^{d_j} |p_{ja}| \pi^a, \quad E_j = \sum_{a=1}^{d_j} |q_{ja}| \pi^a. \quad (\text{AM.11})$$

For $n, x \geq 1$, the polynomial bounds are

$$|P_j(\pi n^2 x^2)| \leq C_j n^{2d_j} x^{2d_j}, \quad |Q_j(\pi n^2 x^2)| \leq E_j n^{2d_j} x^{2d_j}.$$

Since $n^2 + m^2 \geq 2$ and $x^2 \geq 1$,

$$(n^2 + m^2)x^2 \geq \frac{1}{2}(n^2 + m^2) + x^2.$$

The absolute integrand series is therefore bounded by

$$4(C_i C_j + E_i E_j) x^{2(d_i + d_j)} e^{-\pi x^2} \left(\sum_{n \geq 1} n^{2d_i} e^{-\pi n^2/2} \right) \left(\sum_{m \geq 1} m^{2d_j} e^{-\pi m^2/2} \right),$$

which is integrable on $[1, \infty)$. This proves all required sum-integral interchanges. Substitution $u = \lambda x^2$ gives (AM.7). The two theta factors contribute 4, and (AM.7) contributes $1/2$, leaving exactly the factor 2 in (AM.8).

For complex polynomial coefficients, conjugate the coefficients of the first factor in the Gram integral. Equation (AM.9) then proves the inner-product assertion in (AM.10). If a nonzero polynomial P were in its kernel, its first component would vanish on $[1, \infty)$. Put $U(z) = \sum_j P_j(z)c_j$ when $P(s) = \sum_j c_j s^j$. The distinct degrees and nonzero leading coefficients of the P_j imply $U \neq 0$. Multiplying the vanishing theta series by $e^{\pi x^2}/2$ gives

$$U(\pi x^2) = - \sum_{n \geq 2} U(\pi n^2 x^2) e^{-\pi(n^2-1)x^2}.$$

The right side decays exponentially times a fixed polynomial as $x \rightarrow \infty$, by the same Gaussian bound. A nonzero polynomial on the left cannot have that decay. This contradiction proves injectivity. $\square$

6.3 A complete finite truncation certificate

Let $H_{ij}^{[N]}$ be (AM.8) with both theta indices restricted to $1 \leq n, m \leq N$, keeping all coefficients p_{ia}, q_{ia} . For the calculation of this entry choose an integer $N \geq \max(d_i, d_j)$. Define, for $k = i, j$,

$$\begin{aligned} r_{k,N} &= \left(\frac{N+2}{N+1} \right)^{2d_k} e^{-\pi(2N+3)}, & T_{k,N} &= \frac{(N+1)^{2d_k}}{1 - r_{k,N}}, \\ A_{k,N} &= \sum_{n=1}^N n^{2d_k} e^{-\pi(n^2-1)} + T_{k,N} e^{-\pi((N+1)^2-1)}. \end{aligned} \quad (\text{AM.12})$$

All these numbers are specified positive real constants. The choice of N ensures $r_{k,N} < 1$, since

$$\log r_{k,N} < \frac{2d_k}{N+1} - \pi(2N+3) < 2 - 5\pi < 0.$$

Theorem 6.2 (Rigorous complete double-tail bound). *For every $i, j \geq 0$ and $N \geq \max(i, j) + 2$,*

$$\boxed{|H_{ij} - H_{ij}^{[N]}| \leq \mathcal{E}_{ij,N} := 4(C_i C_j + E_i E_j)(T_{i,N} A_{j,N} + A_{i,N} T_{j,N}) \cdot I_{d_i+d_j}(\pi((N+1)^2 + 1))}. \quad (\text{AM.13})$$

In particular the bound tends to zero as $N \rightarrow \infty$, for each fixed pair i, j , and provides an explicit finite certificate for every raw arithmetic moment.

Proof. For $x \geq 1$, the ratio of successive terms of $n^{2d_k} e^{-\pi n^2 x^2}$ is at most

$$\left(\frac{n+1}{n} \right)^{2d_k} e^{-\pi(2n+1)}.$$

The logarithm of this expression has derivative $-2d_k/(n(n+1)) - 2\pi < 0$ for positive real n . For $n \geq N+1$ every ratio is consequently at most $r_{k,N}$. Summing the geometric majorant yields

$$\sum_{n > N} n^{2d_k} e^{-\pi n^2 x^2} \leq T_{k,N} e^{-\pi(N+1)^2 x^2}, \quad \sum_{n \geq 1} n^{2d_k} e^{-\pi n^2 x^2} \leq A_{k,N} e^{-\pi x^2}. \quad (\text{AM.14})$$

For the second inequality, divide by $e^{-\pi x^2}$, bound each retained factor $e^{-\pi(n^2-1)x^2}$ at $x = 1$, and apply the first inequality to the tail.

Write $F_{k,N}$ for the finite P_k -series and $\tilde{F}_{k,N}$ for the finite Q_k -series. Equations (AM.11) and (AM.14) give

$$\begin{aligned} |f_k(x)|, |F_{k,N}(x)| &\leq 2C_k x^{2d_k} A_{k,N} e^{-\pi x^2}, \\ |f_k(x) - F_{k,N}(x)| &\leq 2C_k x^{2d_k} T_{k,N} e^{-\pi(N+1)^2 x^2}. \end{aligned}$$

The reflected inequalities have exactly E_k in place of C_k . In each product of (AM.9), use the exact decomposition $ab - a_N b_N = (a - a_N)b + a_N(b - b_N)$. The resulting bound on the absolute difference of integrands is

$$4(C_i C_j + E_i E_j)(T_{i,N} A_{j,N} + A_{i,N} T_{j,N}) x^{2(d_i + d_j)} e^{-\pi((N+1)^2 + 1)x^2}.$$

Its integral is exactly the right side of (AM.13). Finally, the $A_{k,N}$ remain bounded for fixed k , the $T_{k,N}$ grow at most polynomially, and the integral decays exponentially in $(N + 1)^2$. For completeness, with $L = d_i + d_j$ fixed and $\lambda \geq 2L$, the inequality $\log x \leq (x^2 - 1)/2$ for $x \geq 1$ gives

$$I_L(\lambda) \leq e^{-\lambda} \int_1^\infty e^{-(\lambda-L)(x^2-1)} dx \leq \frac{e^{-\lambda}}{2(\lambda-L)}.$$

Here the last step uses $x^2 - 1 \geq 2(x - 1)$. This proves the asserted limit with an additional elementary upper bound when desired. $\square$

The finite matrix $(H_{ij}^{[N]})_{0 \leq i, j \leq m}$ is itself positive definite for every $N \geq 1$. Indeed it is the Gram matrix of $(F_{j,N}, \tilde{F}_{j,N})$ on the same half-domain; the injectivity proof in (AM.10) applies verbatim to the finite sum. This fact does not assign an ordering to its difference from the full matrix. The relation to the full matrix is the entrywise enclosure (AM.13), whose signs and error bounds are explicit.

6.4 Exact adjoint, parity, and finite-boundary checks

On $\mathcal{B}$ in $L^2(dx)$, integration by parts gives $\langle DF, H \rangle = \langle F, (1 - D)H \rangle$. Therefore the complete moments satisfy

$$H_{i+1,j} + H_{i,j+1} = H_{ij}, \quad H_{ij} = \sum_{k=0}^i (-1)^k \binom{i}{k} H_{0,j+k}, \quad H_{01} = \frac{1}{2} H_{00}. \quad (\text{AM.15})$$

All boundary terms vanish because the original functions have rapid decay at both ends. More explicitly,

$$(Df_i)f_j + f_i(Df_j) - f_i f_j = -\partial_x(x f_i f_j),$$

whose integral on $(0, \infty)$ is zero. Moving each of the i copies of D in the first argument to the second gives the middle identity; symmetry gives the last one.

The finite sums retain an exact boundary correction at $x = 1$:

$$H_{i+1,j}^{[N]} + H_{i,j+1}^{[N]} - H_{ij}^{[N]} = F_{i,N}(1)F_{j,N}(1) - \tilde{F}_{i,N}(1)\tilde{F}_{j,N}(1). \quad (\text{AM.16})$$

To prove this, use $DF_{j,N} = F_{j+1,N}$ for the first component and $D\tilde{F}_{j,N} = \tilde{F}_{j,N} - \tilde{F}_{j+1,N}$ for the second. Applying the preceding integration-by-parts calculation on $[1, \infty)$ gives the first displayed endpoint product for the first component and its negative for the second. The endpoints at infinity vanish term by term. In particular the exact finite identity $H_{01}^{[N]} = H_{00}^{[N]}/2$ follows even before the omitted tail is enclosed, since $F_{0,N} = \tilde{F}_{0,N}$.

One may also retain the exact triangular coordinate map

$$B_{ja} = \begin{cases} \binom{j}{a} (-1/2)^{j-a}, & a \leq j, \\ 0, & a > j, \end{cases} \quad C = BHB^T. \quad (\text{AM.17})$$

This is the Gram matrix of $u_j = (D - 1/2)^j f_0$, with inverse coordinate matrix $(B^{-1})_{ja} = \binom{j}{a} (1/2)^{j-a}$. No original moment is lost. Since $\mathcal{T}u_j = (-1)^j u_j$, unitarity proves $C_{ij} = 0$ for odd $i + j$. The operator $D - 1/2$ is skew-symmetric on this displayed strong Schwartz domain, so for $i + j = 2q$,

$$C_{ij} = (-1)^{i+q} \|u_q\|^2, \quad \|u_q\|^2 = \int_{\mathbb{R}} t^{2q} |2\xi(1/2 + it)|^2 \frac{dt}{2\pi} > 0. \quad (\text{AM.18})$$

For the first equality move all i powers to the second argument, and compare with moving the q powers in $\langle u_q, u_q \rangle$. Mellin Plancherel gives the second equality. Strict positivity follows from the nonzero original function and the polynomial injectivity already proved.

6.5 Reproducible certified finite arithmetic data

The companion program `scripts/check_arithmetic_moments.py` implements the exact integer recurrence (AM.2), evaluates every retained term of (AM.8) by Arb directed ball arithmetic, and enlarges its radius by an upward-rounded value of (AM.13). Thus the final balls enclose the infinite-source entries, including rounding error and both omitted theta tails. The default run uses $0 \leq i, j \leq 5, N = 7$, and 110 decimal working digits; its machine-readable output is `checks/arithmetic_moments.json`.

Let $\Delta_k = \det(H_{ij})_{0 \leq i, j < k}$ and $\Delta_0 = 1$. The squared norm h_k of the monic degree- k polynomial for the original measure μ is

$$h_k = \frac{\Delta_{k+1}}{\Delta_k}. \quad (\text{AM.19})$$

Indeed the coefficients below the leading term solve the Gram system making that monic polynomial orthogonal to degrees below k . Substitution of that unique solution into its squared norm gives the Schur complement of the leading k -by- k Gram block. The block determinant identity gives (AM.19). The interval calculation certifies strict positivity of all six leading principal determinants recorded in the companion certificate and all six corresponding h_k ; those are the actual arithmetic norms used by the orthogonal recurrence, through the finite degrees stated.

Here are explicit certified intervals from that run. Each row means the entire interval $[c - r, c + r]$, with the printed decimal centre c and radius r interpreted exactly. Rounding the stored Arb balls to the displayed decimals was performed outwards.

raw moment	c	r
H_{00}	1.279007247846485140480	$4.67 \cdot 10^{-22}$
H_{01}	0.6395036239232425702398	$3.33 \cdot 10^{-23}$
H_{02}	-12.73579749060893715027	$2.81 \cdot 10^{-21}$
H_{11}	13.37530111453217972051	$2.57 \cdot 10^{-21}$
H_{12}	6.687650557266089860256	$2.85 \cdot 10^{-22}$
H_{22}	389.8820543216399270934	$1.65 \cdot 10^{-20}$

Symmetry supplies the other entries of this original three-by-three moment block. The six original monic squared norms are enclosed by

k	c for h_k	r
0	1.279007247846485140480	$4.67 \cdot 10^{-22}$
1	13.05554930257055843539	$2.69 \cdot 10^{-21}$
2	250.0089768190499145175	$4.88 \cdot 10^{-20}$
3	6694.666685525627506773	$4.76 \cdot 10^{-19}$
4	224298.1953314520556463	$9.95 \cdot 10^{-18}$
5	9383436.471794727053469	$2.54 \cdot 10^{-16}$

The analytic double-tail bound is less than $2.961369 \cdot 10^{-63}$ for every entry of the six-by-six matrix, and is less than $1.150030 \cdot 10^{-83}$ for H_{00} . These bounds account for omitted source terms; the displayed intervals additionally include finite-term arithmetic and decimal conversion errors.

The checker also evaluates (AM.15) and the independent exact finite-boundary identity (AM.16), and detects removal of either the reflected half-domain, the theta factor 2, or the adjoint constant 1. Its checks execute identically under ordinary Python and Python's optimization flag. Finite interval certificates do not assert an infinite-degree weight estimate; (AM.8)–(AM.14) give the exact arithmetic input and its complete error control for calculating further degrees.

7 Exact source-metric optimization and aggregate arithmetic control

This section continues the actual theta complexes in the September 12 chain-descent, global-comparison, Deligne-control, and orthogonal-boundary notes. It proves an aggregate bound on the distinguished orthogonal representatives and computes exactly the optimization over all finite polynomial theta corrections. Every matrix below is recorded in the original polynomial coordinates of the packet. Auxiliary comparison maps and metrics are written explicitly.

7.1 The fixed arithmetic maps and their boundary

Retain

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^k F(x)| < \infty \ (b \in \mathbb{Z}, k \geq 0)\}, \quad D = -x\partial_x, \\ \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \quad q : \mathcal{B} \longrightarrow Q, \quad \mathcal{H} = L^2(\mathbb{R}_{>0}, dx). \end{aligned}$$

The Hilbert inner product is conjugate-linear in its first variable. The original transform conventions are

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \quad \widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx.$$

The original source vector and its iterates are

$$\phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \phi_j = D^j \phi_*, \quad f_j = \Theta \phi_j, \quad \mathcal{M}f_j(s) = s^j g(s), \quad g = 2\xi.$$

Fix a nonempty finite set Z of actual nontrivial zeros with their full multiplicities m_ρ , and retain

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h_Z, \quad E = \mathbb{C}[t]/(h_Z), \quad A = M_t.$$

The coordinate basis of E is $1, t, \dots, t^{d-1}$. Let j_h denote the full Taylor-jet/Chinese-remainder map, let $\varepsilon_Z = j_h(h_Z/g)$, and write $R_Z(u)$ for the unique polynomial of degree less than d representing $\varepsilon_Z u$. The finite source construction gives

$$R_{\text{ref}} u = S_{h_Z}^{\mathcal{B}} \Theta(R_Z(u)(D)\phi_*), \quad JF = j_h \mathcal{M}F, \quad \sigma_Z = q R_{\text{ref}},$$

$$J R_{\text{ref}} = 1_E, \quad J\Theta = 0, \quad D R_{\text{ref}} - R_{\text{ref}} A = f_0 \ell_Z, \quad \ell_Z(u) = [t^{d-1}] R_Z(u). \quad (\text{TC1})$$

Here $S_\rho^{\mathcal{B}} F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy$, on its zero-Mellin-moment domain, and its iterated inverse defines $S_{h_Z}^{\mathcal{B}}$. Thus all local units in g/h_Z and the factor two in $g = 2\xi$ remain present.

For $L \geq 0$ define the actual finite source maps

$$\Phi_L : \mathbb{C}^{L+1} \longrightarrow V, \quad c \longmapsto \sum_{j=0}^L c_j \phi_j, \quad F_L = \Theta \Phi_L : \mathbb{C}^{L+1} \longrightarrow \mathcal{B}.$$

Every allowed representative in the family considered here has the form

$$R = R_{\text{ref}} + F_L C, \quad C : E \longrightarrow \mathbb{C}^{L+1}. \quad (\text{TC2})$$

In particular $qR = \sigma_Z$ and $JR = 1_E$. Let $I_L, S_L : \mathbb{C}^{L+1} \rightarrow \mathbb{C}^{L+2}$ send the standard coordinate vector e_j respectively to e_j, e_{j+1} . Directly,

$$DR - RA = \Theta K_R, \quad K_R = \phi_* \ell_Z + \Phi_{L+1}(S_L C - I_L C A) : E \longrightarrow V. \quad (\text{TC3})$$

This formula keeps the entire source boundary. Its class in Q is zero through $q\Theta = 0$.

Put $G_R = R^*R$, a positive definite Hermitian matrix in the fixed packet coordinates. For $F, H \in \mathcal{B}$, integration by parts gives

$$\langle DF, H \rangle + \langle F, DH \rangle = \langle F, H \rangle,$$

because $[x\overline{F(x)}H(x)]_0^\infty = 0$. Substitution of (TC3) therefore proves the exact identity

$$W_R := A^*G_R + G_RA - G_R = -(R^*\Theta K_R + (\Theta K_R)^*R). \quad (\text{TC4})$$

The adjoints here are adjoints of maps from finite-dimensional coefficient spaces to $\mathcal{H}$; no Hilbert extension of Θ is presumed.

7.2 A source-density calculation with its topology retained

We give the density argument used below, so the realization theorem has its full analytic input available. The map

$$\mathcal{U} : \mathcal{H} \longrightarrow L^2(\mathbb{R}, dy), \quad (\mathcal{U}F)(y) = e^{y/2}F(e^y)$$

is unitary by $x = e^y$. Fourier transformation with kernel e^{ity} and measure $dt/(2\pi)$ sends $\mathcal{U}f_j$ to $(1/2 + it)^j g(1/2 + it)$.

The Gaussian theta series defining $f_0(e^z)$ converges holomorphically when $|\operatorname{Im} z| < \pi/4$. On every closed narrower strip, $\operatorname{Re}(e^{2z}) \geq c_a e^{2\operatorname{Re} z}$, with $c_a = \cos(2a) > 0$; its polynomial Gaussian summands consequently have superexponential decay as $\operatorname{Re} z \rightarrow +\infty$. The source ϕ_* is Fourier-fixed: the Gaussian is Fourier-fixed and the transform sends $D(D-1)$ to $(1-D)(-D) = D(D-1)$. Poisson summation and its two zero moments give $f_0(x) = x^{-1}f_0(1/x)$. Thus the holomorphic function $k(z) = e^{z/2}f_0(e^z)$ obeys $k(z) = k(-z)$, first on the real axis and then throughout the strip. The same decay holds at the negative end. Moving the Fourier contour to $\operatorname{Im} z = a$ sign t has vanishing vertical edges and proves, for every $0 < a < \pi/4$,

$$|g(1/2 + it)| \leq C_a e^{-a|t|}. \quad (\text{TC5})$$

Let $d\mu(t) = |g(1/2 + it)|^2 dt/(2\pi)$. This finite positive measure has a positive exponential moment. If $v \in L^2(\mu)$ is orthogonal to every polynomial, the finite complex measure $d\nu = \bar{v} d\mu$ has a smaller positive exponential moment, by Cauchy-Schwarz. Its Fourier transform is holomorphic on a strip and all derivatives at zero vanish, by the polynomial orthogonality. Its power series and analytic continuation make it identically zero. Uniqueness of the Fourier transform of finite measures gives $\nu = 0$, hence $v = 0$ in $L^2(\mu)$. Polynomials are therefore dense in $L^2(\mu)$. Multiplication by $g(1/2 + it)$ is a unitary map from $L^2(\mu)$ onto $L^2(dt/(2\pi))$: its norm equality is the definition of μ ; surjectivity follows by division off the measure-zero zero set of the nonzero entire function g . It follows that

$$\overline{\bigcup_{L \geq 0} F_L \mathbb{C}^{L+1}}^{\mathcal{H}} = \mathcal{H}. \quad (\text{TC6})$$

This is a statement in the displayed Hilbert topology. The identities $JR = 1$ below continue to hold in the original test-function space.

7.3 Every positive packet metric has an exact finite source lift

Define

$$H_m = F_m^* F_m, \quad C_m = F_m^* R_{\text{ref}}, \quad B_m = -H_m^{-1} C_m, \quad \Pi_m = F_m H_m^{-1} F_m^*, \quad R_m = R_{\text{ref}} + F_m B_m. \quad (\text{TC7})$$

The columns of F_m are linearly independent, because a relation has Mellin transform $g(s)P(s) = 0$ and hence $P = 0$. Thus $H_m > 0$, Π_m is the orthogonal projection onto the stated finite theta space, and

$$F_m^* R_m = 0, \quad G_m := R_m^* R_m = R_{\text{ref}}^* R_{\text{ref}} - C_m^* H_m^{-1} C_m > 0. \quad (\text{TC8})$$

Its strict positivity follows from $JR_m = 1$. Equation (TC6) proves $\Pi_m \rightarrow 1$ strongly. Since E is finite dimensional, $R_m \rightarrow 0$ in $\text{Hom}(E, \mathcal{H})$, and $G_m \rightarrow 0$ in matrix norm.

Exact fixed-level image. At every specified level m , the entire image of the source-Gram map is

$$\{(R_{\text{ref}} + F_m C)^*(R_{\text{ref}} + F_m C) : C : E \rightarrow \mathbb{C}^{m+1}\} = \{G_m + \Delta : \Delta \succeq 0, \text{rank } \Delta \leq m+1\}. \quad (\text{TC8a})$$

Indeed completing the square, using $F_m^* R_m = 0$, gives

$$(R_{\text{ref}} + F_m C)^*(R_{\text{ref}} + F_m C) = G_m + (C - B_m)^* H_m (C - B_m). \quad (\text{TC8b})$$

The extra term is positive semidefinite and has the stated rank bound. Conversely, a positive semidefinite Δ of rank at most $m+1$ has a factorization $\Delta = T^* T$ with $T : \mathbb{C}^d \rightarrow \mathbb{C}^{m+1}$: diagonalize the finite Hermitian matrix Δ , retain its nonnegative eigenvalues, and insert their square roots into $m+1$ rows. Taking $C = B_m + H_m^{-1/2} T$ proves the converse. The auxiliary factorization records a coefficient map; it changes none of the original packet data.

Exact realization theorem. For every specified positive definite Hermitian matrix G in the original polynomial coordinates of E , there exist a finite L and a coefficient map $C : E \rightarrow \mathbb{C}^{L+1}$ for which

$$R = R_{\text{ref}} + F_L C, \quad JR = 1, \quad qR = \sigma_Z, \quad R^* R = G. \quad (\text{TC9})$$

The construction is as follows. Choose $m \geq d-1$ with $G_m \prec G$, set $L = m$, and let $I_{d-1,m} : \mathbb{C}^d \rightarrow \mathbb{C}^{m+1}$ pad a vector with zeros. The principal matrix $I_{d-1,m}^* H_m I_{d-1,m}$ is exactly H_{d-1} . Put

$$\Delta = G - G_m > 0, \quad C = B_m + I_{d-1,m} H_{d-1}^{-1/2} \Delta^{1/2}. \quad (\text{TC10})$$

Both square roots are the uniquely specified positive Hermitian roots of finite positive matrices on the displayed coefficient spaces, using their fixed standard Hermitian forms and the fixed polynomial-coordinate Hermitian form on E . The resulting map remains expressed in the original packet coordinates. Indeed

$$R = R_m + F_m I_{d-1,m} H_{d-1}^{-1/2} \Delta^{1/2}, \quad R_m^* F_m = 0,$$

so its Gram is

$$R^* R = G_m + \Delta^{1/2} H_{d-1}^{-1/2} H_{d-1} H_{d-1}^{-1/2} \Delta^{1/2} = G.$$

All added columns are finite complex linear combinations of the original $D^j \phi_*$ followed by Θ . This proves (TC9) inside the declared source family, with a finite coefficient degree.

The extension class has the literal type $\text{Ext}_{\mathbb{C}[t]}^1(E, V) \simeq V/h_Z(D)V$; its arithmetic coordinate ε_Z has type E . The exact map between these types is

$$\iota_h : E \longrightarrow V/h_Z(D)V, \quad [p(t)] \longmapsto [p(D)\phi_*]. \quad (\text{TC10a})$$

It is well-defined and $\mathbb{C}[t]$ -linear because replacing p by $p + h_Z a$ adds $h_Z(D)a(D)\phi_*$. It is an isomorphism, with the explicit inverse

$$\kappa_h : V/h_Z(D)V \longrightarrow E, \quad [\psi] \longmapsto j_h \left(\frac{\mathcal{M}\psi}{\mathcal{M}\phi_*} \right). \quad (\text{TC10b})$$

The denominator is

$$\mathcal{M}\phi_*(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2),$$

which is holomorphic and nonzero near every $\rho \in Z$. Every $\psi \in V$ has $\psi(x) = O(x^2)$ at zero, so its Mellin transform is holomorphic for $\operatorname{Re} s > -2$. Mellin integration by parts gives $\mathcal{M}(p(D)\psi) = p(s)\mathcal{M}\psi$ near Z ; this proves well-definedness of κ_h and $\kappa_h\iota_h = 1_E$. For the other inverse law, let $\psi \in V$ have vanishing Mellin jets along h_Z . For the first factor $t - \rho$, its source inverse on $x > 0$ is

$$S_\rho\psi(x) = x^{-\rho} \int_x^\infty \psi(y)y^{\rho-1} dy = - \int_0^1 \psi(tx)t^{\rho-1} dt.$$

The second expression and $0 < \operatorname{Re} \rho < 1$ prove smooth even extension to zero, with value zero; differentiating in every order is justified by the integrable factor $t^{\operatorname{Re} \rho - 1}$. The first expression gives Schwartz decay at infinity. Integration of $(D - \rho)S_\rho\psi = \psi$ on the full real line gives $(1 - \rho) \int S_\rho\psi = \int \psi = 0$, so $S_\rho\psi \in V$. Its Mellin transform is $\mathcal{M}\psi(s)/(s - \rho)$, so the remaining zero jets permit iteration through every factor of h_Z . Consequently $\ker(j_h\mathcal{M}|_V) = h_Z(D)V$. Apply this to $\psi - p(D)\phi_*$, where $p = j_h(\mathcal{M}\psi/\mathcal{M}\phi_*)$, to obtain $\iota_h\kappa_h = 1$. Thus the class of the actual arithmetic extension is precisely $\iota_h(\varepsilon_Z)$, which is nonzero. For every source change (TC2), this class is unchanged. Explicitly,

$$h_Z(D)R(1) = \Theta(R_Z(1)(D)\phi_* + h_Z(D)\Phi_L C(1)),$$

and the second term is zero in $V/h_Z(D)V$. There is still no strictly D -equivariant section: an intertwiner $T : E \rightarrow \mathcal{B}$ has image killed by $h_Z(D)$, whereas $h_Z(D)$ is injective on $\mathcal{B}$. To see this injectivity, $(D - \rho)F = 0$ implies $F(x) = cx^{-\rho}$; the bounds defining $\mathcal{B}$ force $c = 0$, and one iterates over the factors of h_Z . The exact map relating any section to the original arithmetic action is (TC3), with its retained nonzero extension class.

7.4 Sharp control with all nilpotent jets present

For a positive definite G , set

$$W_A(G) = A^*G + GA - G, \quad \epsilon_A(G) = \sup_{u \neq 0} \frac{|u^*W_A(G)u|}{u^*Gu}, \quad \eta_Z = \max_{\rho \in Z} |2 \operatorname{Re} \rho - 1|. \quad (\text{TC11})$$

For every eigenvector $Au = \rho u$,

$$u^*W_A(G)u = (2 \operatorname{Re} \rho - 1)u^*Gu. \quad (\text{TC12})$$

Consequently $\epsilon_A(G) \geq \eta_Z$, for every positive G . We now construct metrics arbitrarily close to that bound without removing a single nilpotent jet.

Fix $\epsilon > \eta_Z$, and fix any $G_0 > 0$, both in the original coordinates. Define

$$\begin{aligned} F(t) &= e^{-t}e^{tA^*}G_0e^{tA}, \\ G_\epsilon &= \int_{-\infty}^\infty e^{-\epsilon|t|}F(t) dt, \\ W_A(G_\epsilon) &= \epsilon \int_{-\infty}^\infty \operatorname{sign}(t)e^{-\epsilon|t|}F(t) dt. \end{aligned} \quad (\text{TC13})$$

The last identity is a conclusion, proved below. The factor e^{-t} records the original weight-one character; the action is still $e^{tA} = U_Z(e^t)$.

Let $e_\rho \in E$ be the exact Chinese-remainder idempotent and write $P_\rho = M_{e_\rho}$ and $N_\rho = (A - \rho I)P_\rho$. On $P_\rho E$, $N_\rho^{m_\rho} = 0$. Directly,

$$e^{tA} = \sum_{\rho \in Z} e^{t\rho} \sum_{j=0}^{m_\rho-1} \frac{t^j}{j!} N_\rho^j P_\rho. \quad (\text{TC14})$$

Thus every matrix entry in the integral is a finite sum of polynomial-exponential functions. The real part of each exponent $\bar{\rho} + \sigma - 1$ has absolute value at most η_Z . This proves absolute convergence of (TC13), including its derivative, and proves the endpoint limits needed for integration by parts. In particular $G_\epsilon > 0$. Since $F'(t) = A^*F(t) + F(t)A - F(t)$, integration by parts on the negative and positive half-lines gives (TC13). The two values at $t = 0$ cancel exactly. Both half-line integrals are strictly positive matrices, so

$$-\epsilon G_\epsilon \prec W_A(G_\epsilon) \prec \epsilon G_\epsilon. \quad (\text{TC15})$$

For completeness the construction requires no unevaluated analytic integral once the packet data and G_0 are specified. Put

$$I_k(z, \epsilon) = k! \left((\epsilon - z)^{-k-1} + (-1)^k (\epsilon + z)^{-k-1} \right). \quad (\text{TC16})$$

The substitutions $t \mapsto -t$ on the negative half-line and $\int_0^\infty t^k e^{-at} dt = k! a^{-k-1}$ show that

$$G_\epsilon = \sum_{\rho, \sigma \in Z} \sum_{i=0}^{m_\rho-1} \sum_{j=0}^{m_\sigma-1} \frac{I_{i+j}(\bar{\rho} + \sigma - 1, \epsilon)}{i!j!} \cdot (N_\rho^i P_\rho)^* G_0 (N_\sigma^j P_\sigma). \quad (\text{TC17})$$

All local Jordan powers and every cross term remain in this formula. Combining (TC12), (TC15), and the actual realization (TC9) proves

$$\inf_{G>0} \epsilon_A(G) = \inf_{\substack{L \geq 0 \\ C: E \rightarrow \mathbb{C}^{L+1}}} \epsilon_A((R_{\text{ref}} + F_L C)^*(R_{\text{ref}} + F_L C)) = \eta_Z. \quad (\text{TC18})$$

The equality concerns all finite polynomial corrections. It does not assert that the distinguished orthogonal sequence (TC7) attains the same infimum.

Exact attainment criterion. The infimum in (TC18) is achieved precisely when every eigenvalue on either boundary $|2 \operatorname{Re} \rho - 1| = \eta_Z$ is semisimple. For $E = \mathbb{C}[t]/h_Z$, this means $m_\rho = 1$ for those zeros.

To prove necessity, a bound $\epsilon_A(G) \leq \eta_Z$ integrates to

$$\|e^{tA} u\|_G \leq e^{(1+\eta_Z)t/2} \|u\|_G \quad (t \geq 0),$$

and applying the other inequality backward gives

$$\|e^{-tA} u\|_G \leq e^{(-1+\eta_Z)t/2} \|u\|_G \quad (t \geq 0).$$

On a generalized eigenspace on the upper boundary the first estimate bounds $e^{t(A-\rho I)}$ uniformly for $t \geq 0$, up to a unit-modulus scalar. A nonzero nilpotent block has a vector whose exponential contains a nonzero highest power of t , hence is unbounded. The lower boundary is treated by the backward estimate. This proves necessity.

For sufficiency, use the unchanged CRT direct-sum map $u \mapsto (P_\rho u)_\rho$. Make its summands mutually orthogonal by taking the pullback sum of chosen positive metrics on them. On a semisimple boundary summand, $A = \rho I$, so the control form is exactly $(2 \operatorname{Re} \rho - 1)$ times that summand's metric. For each interior summand choose a number strictly between $|2 \operatorname{Re} \rho - 1|$ and η_Z , and use (TC13) on that summand. Its control is strictly smaller than η_Z . The sum therefore has the required bound. When $\eta_Z = 0$ there are only boundary summands, and the same construction gives zero control exactly. The source realization (TC9) then supplies an actual finite corrected representative for this metric. Its boundary (TC3) persists; zero metric control only states that its Hermitian cross-pairing in (TC4) is zero.

7.5 The exact tensor cost of unrestricted metrics

For the original ordered tensor space $E^{\otimes n}$, retain

$$A_n = \sum_{j=1}^n I^{\otimes(j-1)} \otimes A \otimes I^{\otimes(n-j)}, \quad W_{A_n, n}(G) = A_n^* G + G A_n - nG. \quad (\text{TC19})$$

Its eigenvalues, with algebraic multiplicity, are all ordered sums $\rho_1 + \dots + \rho_n$. This follows by tensoring complete invariant flags, or directly by the CRT decomposition and the nilpotent sum on each ordered product block. Hence

$$\max_{\lambda \in \text{Spec } A_n} |2 \operatorname{Re} \lambda - n| = n\eta_Z. \quad (\text{TC20})$$

The upper bound is the triangle inequality; equality uses the ordered repeated tuple $(\rho, \dots, \rho)$ for a zero attaining η_Z . Repeating (TC12)–(TC17) with the weight factor e^{-nt} proves the exact value

$$\inf_{G > 0} \sup_{v \neq 0} \frac{|v^* W_{A_n, n}(G) v|}{v^* G v} = n\eta_Z. \quad (\text{TC21})$$

The nilpotent sum on a product block is retained throughout this argument. Thus an unrestricted choice of a product-image metric, even a non-product metric, has a precisely calculated linear spectral cost. Establishing sublinear excess for the actual arithmetic image requires mathematical information about its spectrum or a further comparison which constrains that spectrum. Equations (TC18) and (TC21) are calculations of the optimization; they supply no missing estimate by choosing representatives.

7.6 A stronger bound on the actual orthogonal sequence

Return to the specific R_m in (TC7), and take Z stable under $\rho \mapsto 1 - \bar{\rho}$, including multiplicities. Write

$$b_m = e_m^* B_m : E \rightarrow \mathbb{C}, \quad z_m = R_m^* f_{m+1} : \mathbb{C} \rightarrow E.$$

The source boundary of R_m lies in the span of $f_0, \dots, f_{m+1}$. Orthogonality to the first $m+1$ vectors gives directly from (TC3)–(TC4)

$$W_m = A^* G_m + G_m A - G_m = -z_m b_m - b_m^* z_m^*, \quad \operatorname{rank} W_m \leq 2. \quad (\text{TC22})$$

Reflection stability gives

$$\operatorname{Tr}(G_m^{-1} W_m) = 2 \operatorname{Re} \operatorname{Tr} A - d = 0. \quad (\text{TC23})$$

Let $\epsilon_m = \epsilon_A(G_m)$. The G_m -self-adjoint map $G_m^{-1} W_m$ therefore has at most two nonzero eigenvalues, namely $\epsilon_m, -\epsilon_m$; if the rank is zero both are zero.

Aggregate displacement theorem. For every finite m in this original orthogonal source sequence,

$$\epsilon_m \geq \sum_{\rho \in Z} m_\rho \max(2 \operatorname{Re} \rho - 1, 0) = \sum_{\rho \in Z} m_\rho |\operatorname{Re} \rho - \tfrac{1}{2}|. \quad (\text{TC24})$$

Here the second equality uses the given reflection involution and its preservation of multiplicities.

For a full proof of the first inequality, take a complete A -invariant flag, keeping all generalized eigenvectors. Choose a G_m -orthonormal frame adapted to that flag and denote its explicit coordinate map by $T : \mathbb{C}^d \rightarrow E$. It satisfies $T^* G_m T = I$; it is only a map used in the proof, while the recorded A, G_m, W_m remain in the original coordinates. The matrix $T^{-1} A T$ is upper triangular with diagonal entries the zeros ρ , repeated m_ρ times. Therefore the Hermitian matrix $H = T^* W_m T$ has diagonal

entries $\delta_j = 2 \operatorname{Re} \rho_j - 1$. By (TC22)–(TC23) its positive part H_+ has trace ϵ_m . Let P be the orthogonal coordinate projection onto the indices with $\delta_j > 0$. As $0 \leq P \leq I$ and $H \leq H_+$,

$$\sum_{\delta_j > 0} \delta_j = \operatorname{Tr}(PH) \leq \operatorname{Tr}(PH_+) \leq \operatorname{Tr}(H_+) = \epsilon_m. \quad (\text{TC25})$$

Each trace inequality follows by inserting the positive square root of P or H_+ , so no sign is inferred from a product of noncommuting matrices. This proves (TC24).

More generally, whenever the positive rank of a control form is r_+ , the same argument gives $\sum_j \max(\delta_j, 0) \leq r_+ \epsilon_A(G)$. In the present source construction $r_+ \leq 1$, which is why the aggregate bound has no dimension loss.

The source's exact scalar certificate is retained:

$$u_m = z_m^* G_m^{-1} z_m, \quad v_m = b_m G_m^{-1} b_m^*, \quad c_m = b_m G_m^{-1} z_m, \quad \epsilon_m^2 = u_m v_m - (\operatorname{Im} c_m)^2. \quad (\text{TC26})$$

Indeed the two nonzero eigenvalues are those of $-\begin{pmatrix} c_m & v_m \\ u_m & c_m \end{pmatrix}$, and (TC23) gives $\operatorname{Re} c_m = 0$. If $a_{m+1} = (1 - \Pi_m) f_{m+1}$ and $\eta_{m+1} = \|a_{m+1}\|^2$, the exact rank-one Gram update gives

$$G_{m+1} = G_m - \eta_{m+1}^{-1} z_m z_m^*, \quad u_m = \eta_{m+1} \left(1 - \frac{\det G_{m+1}}{\det G_m} \right). \quad (\text{TC27})$$

The determinant identity is the rank-one determinant lemma, obtained by choosing a basis whose first vector spans $G_m^{-1} z_m$, or by expanding the determinant multilinearly. Thus a fully arithmetic version of (TC24) is

$$\eta_{m+1} \left(1 - \frac{\det G_{m+1}}{\det G_m} \right) v_m - (\operatorname{Im} c_m)^2 \geq \left(\sum_{\rho \in Z} m_\rho \left| \operatorname{Re} \rho - \frac{1}{2} \right| \right)^2. \quad (\text{TC28})$$

Every entry on the left is an actual theta Gram or cross-Gram entry with the original source representative. This identifies a stronger target for the arithmetic moment estimates than a bound for a single eigenvector.

7.7 Original supported maps and the precise result delivered

The passage from a finite source correction C to its representative, Gram, and boundary is the explicit chain of maps

$$C \longmapsto R_{\text{ref}} + \Theta \Phi_L C \longmapsto (qR, JR, R^*R, K_R) = (\sigma_Z, 1_E, G_R, K_R). \quad (\text{TC29})$$

For every active support label λ , the linear maps lift as $(\lambda, u) \mapsto (\lambda, Ru)$ and $(\lambda, u) \mapsto (\lambda, K_R u)$. The quotient of the boundary is $(\lambda, \Theta K_R u) \mapsto (\lambda, 0_Q)$; external absence remains the external zero τ . For the two-leg differential $d(\phi, \psi) = \Theta(\phi - \widehat{\psi})$, the two primitives are $(K_R u, 0)$ and $(0, -\widehat{K_R u})$. Their difference $(K_R u, \widehat{K_R u})$ is the retained degree-zero cycle. Thus all source changes in the optimization have actual typed cochain lifts with the same support synchronization as the input programme.

The new conclusions are the exact fixed-level Gram image (TC8a), the finite source realization (TC9), the sharp source-metric infimum with its attainment criterion (TC18), the exact tensor infimum (TC21), and the aggregate arithmetic bound (TC24)–(TC28). The metric realization leaves the extension unit, the residue/Jacobian pairing, the quotient classes and the generator unchanged. The original distinguished orthogonal metrics remain a smaller, arithmetically specified family. Their relative estimates, including all shrinking denominators in (TC26), are the next concrete arithmetic calculation.

8 The actual theta minimum as a confluent arithmetic kernel

This section continues the orthogonal-boundary construction of the 12 September 2026 tau workbench, through PR 13. Its new calculation identifies the permitted finite source corrections with a constrained polynomial minimum, evaluates that minimum by a complete jet kernel, and proves the exact two-vector boundary formula for that kernel. The arithmetic packet, Euler operator, local units, and multiplicities are retained throughout.

8.1 Original functions, maps, coordinates, and measures

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \quad D = -x\partial_x.$$

Let $\mathcal{B}$ be the space of smooth functions on $\mathbb{R}_{>0}$ such that $\sup_{x>0} x^b |D^k F(x)| < \infty$ for every integer b and every $k \geq 0$. The original maps and function are

$$\Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x},$$

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad f_0 = \Theta\phi_*, \quad g(s) = 2\xi(s), \quad \mathcal{M}f_0 = g.$$

Write $\phi_j = D^j \phi_*$ and $f_j = D^j f_0 = \Theta\phi_j$. The Hilbert observation is $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$; the first argument of every inner product is conjugate-linear. The actual Mellin-line map

$$\mathcal{U} : \mathcal{H} \longrightarrow L^2(\mathbb{R}, dt/(2\pi)), \quad \mathcal{U}F(t) = \mathcal{M}F(1/2 + it)$$

is unitary. Indeed $x = e^y$ gives the unitary map $F \mapsto e^{y/2} F(e^y)$ to $L^2(dy)$, and the displayed Mellin transform is its Fourier transform with the positive exponential e^{ity} . Plancherel has measure $dt/(2\pi)$ in this convention. Integration by parts on $\mathcal{B}$ gives $\mathcal{U}DF = (1/2 + it)\mathcal{U}F$.

Fix a nonempty finite packet Z of actual nontrivial zeros of g , including the full multiplicity m_ρ of each selected zero. Put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad E = \mathbb{C}[s]/(h), \quad A = M_s : E \longrightarrow E.$$

The basis of E remains $1, s, \dots, s^{d-1}$. Let j_h denote polynomial remainder together with its equivalent complete Taylor-jet description. Set

$$v(s) = \frac{g(s)}{h(s)}, \quad \varepsilon = (j_h v)^{-1} \in E^\times, \quad U_\varepsilon = M_\varepsilon.$$

The function v is entire and has nonzero value at each selected centre after the corresponding cancellation. The matrix U_ε retains all derivatives of g/h , including the factor 2 in $g = 2\xi$. It commutes with A . For $u \in E$, let $R_Z(u)$ be the unique polynomial of degree below d representing εu . Write $R_{\text{ref}} : E \rightarrow \mathcal{B}$ for the source note's fixed finite section R_0 . This alias records its identity and avoids colliding with the $m = 0$ corrected member of the sequence R_m . The fixed section is characterized by

$$\mathcal{M}R_{\text{ref}}u = vR_Z(u), \quad J_Z R_{\text{ref}} = I_E, \quad J_Z \Theta = 0. \quad (\text{TB.1})$$

These are the actual finite section and jets of the tau construction; the proofs below do not replace them with a finite model.

Introduce the two specified measures on the same Mellin line

$$d\mu(t) = |g(1/2 + it)|^2 \frac{dt}{2\pi}, \quad d\nu_Z(t) = |v(1/2 + it)|^2 \frac{dt}{2\pi}. \quad (\text{TB.2})$$

Both have every polynomial moment. For μ this follows from the Gaussian-series Mellin decay proved in PR 13; division by the fixed polynomial h preserves that decay outside a compact set, and the removable values of v are bounded on the compact set. The densities are positive except on discrete sets, because g and v are nonzero entire functions. Consequently the squared norm of every nonzero polynomial in either measure is strictly positive. No positive arithmetic residue or Weil pairing has been inserted in (TB.2).

8.2 Exact constrained minimum over the admitted source

For $m \geq 0$, the original permitted changes of representative are

$$R_B u = R_{\text{ref}} u + \sum_{j=0}^m f_j(Bu)_j, \quad B : E \longrightarrow \mathbb{C}^{m+1}. \quad (\text{TB.3})$$

Set $N = d + m$. Denote by $\mathcal{P}_N$ the polynomials of degree at most N , with their coefficient columns in the monomial basis, and define the surjection

$$C_N : \mathcal{P}_N \longrightarrow E, \quad C_N P = j_h P. \quad (\text{TB.4})$$

Let H_N^ν be its actual moment Gram matrix:

$$(H_N^\nu)_{ij} = \int_{\mathbb{R}} \overline{(1/2 + it)^i} (1/2 + it)^j d\nu_Z(t), \quad 0 \leq i, j \leq N. \quad (\text{TB.5})$$

It is positive definite by (TB.2). Define

$$K_N = C_N (H_N^\nu)^{-1} C_N^*. \quad (\text{TB.6})$$

This is positive definite: C_N is onto, so C_N^* is injective, and its nonzero vectors have positive quadratic form under $(H_N^\nu)^{-1}$.

Theorem 8.1 (The actual source minimum). *The orthogonally corrected representative R_m of PR 13 has the following explicit numerator and Gram matrix:*

$$p_{N,u} = (H_N^\nu)^{-1} C_N^* K_N^{-1} U_\varepsilon u, \quad \mathcal{M} R_m u = v p_{N,u}, \quad (\text{TB.7})$$

$$\boxed{G_m = R_m^* R_m = U_\varepsilon^* K_N^{-1} U_\varepsilon.} \quad (\text{TB.8})$$

The polynomial correction

$$B_m(u)(s) = \frac{p_{N,u}(s) - R_Z(u)(s)}{h(s)} \quad (\text{TB.9})$$

has degree at most m . Its coefficients give exactly the map $B_m : E \rightarrow \mathbb{C}^{m+1}$ in the original source family (TB.3). Thus (TB.7) specifies the original theta boundary itself.

Proof. Taking Mellin transforms in (TB.3) gives

$$\mathcal{M} R_B u = v(s) \left(R_Z(u)(s) + h(s) \sum_{j=0}^m (Bu)_j s^j \right).$$

The expressions in parentheses range over precisely

$$\{P \in \mathcal{P}_N : C_N P = U_\varepsilon u\}. \quad (\text{TB.10})$$

Indeed the displayed expressions have the required residue. Conversely the difference between a polynomial in (TB.10) and $R_Z(u)$ is divisible by h , and the degree bound makes its quotient a polynomial of degree at most m . Mellin Plancherel identifies its squared Hilbert norm with $P^* H_N^\nu P$; this equality includes the measure $dt/(2\pi)$.

The vector $p = p_{N,u}$ in (TB.7) obeys $C_N p = U_\varepsilon u$. For $z \in \ker C_N$,

$$z^* H_N^\nu p = z^* C_N^* K_N^{-1} U_\varepsilon u = 0.$$

Therefore every competitor $p + z$ has the exact squared norm

$$(p + z)^* H_N^\nu (p + z) = u^* U_\varepsilon^* K_N^{-1} U_\varepsilon u + z^* H_N^\nu z. \quad (\text{TB.11})$$

Positivity proves uniqueness of the minimum. The original orthogonal projection construction is the unique minimum over the same affine source family, so its representative is (TB.7). Polarization proves (TB.8), while (TB.10) proves (TB.9). Every equality has been proved before application of the quotient; after applying q and J_Z one still has $qR_m = \sigma_Z$ and $J_Z R_m = I_E$. $\square$

The formula also extends to $m = -1$, $N = d - 1$, with no allowed source correction: C_{d-1} is the identity in the retained polynomial coordinates, and (TB.7) is precisely $R_Z(u)$.

8.3 Monic arithmetic polynomials and their exact recurrence

For this subsection use ν_Z , with no symmetry assumption on the selected packet. Let q_n be its monic polynomial of degree n orthogonal to all smaller degrees, and let $\kappa_n = \|q_n\|_{\nu_Z}^2 > 0$. The polynomials are explicitly the result of subtracting the projection of s^n onto smaller monomials. This is a specified unitriangular change of auxiliary polynomial basis; no coordinate of E , no arithmetic operator, and no polynomial coefficient has been replaced by an orthonormal coordinate.

Lemma 8.2 (The retained vertical-line recurrence). *There are $b_n \in \mathbb{C}$, with $b_n + \bar{b}_n = 1$, such that*

$$sq_0 = q_1 + b_0 q_0, \quad sq_n = q_{n+1} + b_n q_n - \frac{\kappa_n}{\kappa_{n-1}} q_{n-1} \quad (n \geq 1). \quad (\text{TB.12})$$

Proof. On the original line $s = 1/2 + it$, one has $\bar{s} = 1 - s$. Thus multiplication by s has adjoint multiplication by $1 - s$ on polynomial arguments. The coefficient of q_{n+1} in sq_n is one because both are monic. If $j \leq n - 2$, then

$$\langle q_j, sq_n \rangle_{\nu_Z} = \langle (1 - s)q_j, q_n \rangle_{\nu_Z} = 0,$$

since the first polynomial has degree at most $n - 1$. For $j = n - 1$ the same identity is $-\kappa_n$; division by κ_{n-1} gives the displayed negative coefficient. Finally

$$(b_n + \bar{b}_n)\kappa_n = \langle q_n, sq_n \rangle + \langle sq_n, q_n \rangle = \kappa_n.$$

This proves all coefficients, including their signs. No assertion that $b_n = 1/2$ is needed for a general packet. $\square$

Define the retained complete residue columns

$$v_n = j_h q_n \in E. \quad (\text{TB.13})$$

Changing the auxiliary monomial basis in (TB.6) to the stated monic basis and retaining its diagonal Gram matrix gives

$$K_N = \sum_{n=0}^N \frac{v_n v_n^*}{\kappa_n}. \quad (\text{TB.14})$$

Equivalently, this is the complete jet of the polynomial kernel

$$\mathcal{K}_N(z, w) = \sum_{n=0}^N \frac{q_n(z) \overline{q_n(w)}}{\kappa_n}. \quad (\text{TB.15})$$

In Taylor-jet coordinates at each centre, v_n has entries $q_n^{(k)}(\rho)/k!$ for $0 \leq k < m_\rho$. The map from these entries to the retained basis of E is the inverse confluent Vandermonde/Chinese-remainder map. In particular the factorials and all repeated-root derivatives remain in the construction.

Theorem 8.3 (Confluent kernel boundary formula). *For every $N \geq d - 1$,*

$$\boxed{AK_N + K_N A^* - K_N = \frac{v_{N+1} v_N^* + v_N v_{N+1}^*}{\kappa_N}}. \quad (\text{TB.16})$$

Before taking any jets the corresponding polynomial identity is

$$(z + \bar{w} - 1) \mathcal{K}_N(z, w) = \frac{q_{N+1}(z) \overline{q_N(w)} + q_N(z) \overline{q_{N+1}(w)}}{\kappa_N}. \quad (\text{TB.17})$$

Proof. Apply j_h to (TB.12). The result replaces its left side by Av_n , because $j_h(sq_n) = A(j_h q_n)$. Substitute these relations into

$$AK_N + K_N A^* - K_N = \sum_{n=0}^N \frac{(Av_n) v_n^* + v_n (Av_n)^* - v_n v_n^*}{\kappa_n}.$$

The diagonal terms vanish by $b_n + \bar{b}_n = 1$. The negative contribution involving v_{n-1} has coefficient $-1/\kappa_{n-1}$, and cancels exactly the preceding positive contribution. The sole uncanceled term is the $n = N$ upper edge, giving (TB.16). The identical substitution before applying j_h proves (TB.17) for all complex z, w . This is a polynomial identity; no division by $z + \bar{w} - 1$ is performed at its zero set, and differentiating it proves the full confluent version. $\square$

8.4 Exact relative weight control from the same kernel

The original integration-by-parts form remains

$$W_m = A^* G_m + G_m A - G_m = -\{R_m^* (DR_m - R_m A) + (DR_m - R_m A)^* R_m\}. \quad (\text{TB.18})$$

The boundary term $[x \bar{F} H]_0^\infty$ vanishes for $F, H \in \mathcal{B}$. Put

$$L_N = AK_N + K_N A^* - K_N.$$

Commutation of A with U_ε in (TB.8) proves the exact comparison

$$\boxed{W_m = U_\varepsilon^* K_N^{-1} L_N K_N^{-1} U_\varepsilon}. \quad (\text{TB.19})$$

For clarity, multiplication gives

$$K_N^{-1} L_N K_N^{-1} = K_N^{-1} A + A^* K_N^{-1} - K_N^{-1},$$

which is exactly the middle factor in (TB.18). This proves the morphism between the inverse-kernel and original Hilbert metrics.

Define the actual real and complex numbers

$$\alpha_N = v_{N+1}^* K_N^{-1} v_{N+1}, \quad \beta_N = v_N^* K_N^{-1} v_N, \quad \gamma_N = v_N^* K_N^{-1} v_{N+1}. \quad (\text{TB.20})$$

Then the least symmetric allowance for the actual pair (G_m, W_m) is

$$\epsilon_m = \frac{|\operatorname{Re} \gamma_N| + \sqrt{\alpha_N \beta_N - (\operatorname{Im} \gamma_N)^2}}{\kappa_N}. \quad (\text{TB.21})$$

Here $-\epsilon_m G_m \preceq W_m \preceq \epsilon_m G_m$. To prove minimality, the invertible change $y = K_N^{-1} U_\varepsilon u$ turns its Rayleigh quotient into $y^* L_N y / (y^* K_N y)$. By (TB.16) the nonzero eigenvalues of this Hermitian comparison are the eigenvalues of the two-by-two matrix

$$\frac{1}{\kappa_N} \begin{pmatrix} \gamma_N & \beta_N \\ \alpha_N & \overline{\gamma_N} \end{pmatrix}.$$

Its trace and determinant give

$$\lambda_\pm = \frac{\operatorname{Re} \gamma_N \pm \sqrt{\alpha_N \beta_N - (\operatorname{Im} \gamma_N)^2}}{\kappa_N}.$$

Cauchy-Schwarz in K_N^{-1} gives $|\gamma_N|^2 \leq \alpha_N \beta_N$, so the radicand is nonnegative; its square root is at least $|\operatorname{Re} \gamma_N|$. The maximum absolute eigenvalue is therefore (TB.21), including rank-one and zero cases.

For a packet closed under $\rho \mapsto 1 - \bar{\rho}$, with the multiplicities retained, $2 \operatorname{Re} \operatorname{tr} A = d$. Taking the trace in (TB.16) after multiplying by K_N^{-1} gives

$$0 = \operatorname{tr}(K_N^{-1} L_N) = \frac{2 \operatorname{Re} \gamma_N}{\kappa_N}. \quad (\text{TB.22})$$

Thus its real part is zero without any coordinate choice. Every eigenvector $Au = \rho u$ still satisfies

$$\frac{u^* W_m u}{u^* G_m u} = 2 \operatorname{Re} \rho - 1. \quad (\text{TB.23})$$

In particular (TB.21) has the intrinsic lower bound $\epsilon_m \geq \max_{\rho \in Z} |2 \operatorname{Re} \rho - 1|$. All of its quantities are now specified by arithmetic moments, complete jets, and the original local unit.

8.5 Exact change when one more theta direction is admitted

The step $m \mapsto m + 1$ means $N \mapsto N + 1$. Set $w = v_{N+1}$, $k = \kappa_{N+1}$, and $\theta = w^* K_N^{-1} w \geq 0$. Formula (TB.14) gives

$$K_{N+1} = K_N + \frac{w w^*}{k}. \quad (\text{TB.24})$$

Multiplication directly verifies the inverse identity

$$K_{N+1}^{-1} = K_N^{-1} - \frac{K_N^{-1} w w^* K_N^{-1}}{k + \theta}.$$

Consequently

$$G_{m+1} = G_m - \frac{U_\varepsilon^* K_N^{-1} w w^* K_N^{-1} U_\varepsilon}{k + \theta}, \quad (\text{TB.25})$$

$$\frac{\det G_{m+1}}{\det G_m} = \frac{k}{k + \theta} > 0. \quad (\text{TB.26})$$

The inverse formula follows by multiplying by (TB.24); If $w = 0$, both updates are identities and the determinant ratio is one. For $w \neq 0$, the determinant formula follows by extending $K_N^{-1} w$ to a basis, in which the rank-one map $K_N^{-1} w w^* / k$ has one possible nonzero eigenvalue θ/k . Hence the determinant factor of K_{N+1} relative to K_N is $1 + \theta/k$.

These are the same Hilbert metrics at adjacent relation levels. Subtract the two polynomial numerators in (TB.7) and divide their difference by h , as in (TB.9). This gives a source polynomial P_u of degree at most $m + 1$. The actual representative difference is $R_{m+1}u - R_m u = \Theta(P_u(D)\phi_*)$, an element of ΘW_{m+1} , where $W_{m+1} = \text{span}(\phi_*, D\phi_*, \dots, D^{m+1}\phi_*)$. The quotient map $\mathcal{B}/\Theta W_m \rightarrow \mathcal{B}/\Theta W_{m+1}$ has kernel $\Theta W_{m+1}/\Theta W_m$; injectivity of Θ identifies this kernel with the original one-dimensional space W_{m+1}/W_m . The scalar e sends a class to the zero in its same support fibre, while τ sends it to external absence. None of (TB.24)–(TB.26) changes those maps.

8.6 The original measure supplies a full tail recurrence

There is a second exact description which uses the unchanged measure μ of (TB.2). Let Q_n denote its monic orthogonal polynomials and $h_n = \|Q_n\|_\mu^2$. The reality and functional equation of g make this measure invariant under $s \mapsto 1 - s$; uniqueness of the monic orthogonal polynomial then proves $Q_n(1 - s) = (-1)^n Q_n(s)$. The recurrence proof above therefore gives

$$sQ_n = Q_{n+1} + \frac{1}{2}Q_n - \frac{h_n}{h_{n-1}}Q_{n-1} \quad (n \geq 1). \quad (\text{TB.27})$$

For $n = 0$, omit the last term.

Define the rational function-valued map and coefficient rows

$$r(u)(s) = \frac{R_Z(u)(s)}{h(s)}, \quad d_n u = \frac{1}{h_n} \int_{\mathbb{R}} \overline{Q_n(1/2 + it)} r(u)(1/2 + it) d\mu(t). \quad (\text{TB.28})$$

At an actual line zero a rational pole in this expression is interpreted in $L^2(\mu)$; its product with g is the entire function $vR_Z(u)$. In particular its norm and every displayed pairing are finite. Polynomials are dense in $L^2(\mu)$: the positive exponential moment from PR 13 makes the Fourier transform of an orthogonal annihilator analytic in a strip, and all its derivatives vanish at zero, so Fourier uniqueness annihilates it. Therefore Parseval gives

$$\mathcal{U}R_m u = g \sum_{n>m} Q_n d_n u, \quad \boxed{G_m = \sum_{n>m} h_n d_n^* d_n}. \quad (\text{TB.29})$$

The series is a norm-convergent positive matrix series.

Let $\ell_Z(u) = [s^{d-1}]R_Z(u)$ be the original rank-one extension coefficient. Polynomial division gives the exact identity $sr - rA = \ell_Z$. Pair it with Q_n , and use the adjoint relation $s^* = 1 - s$ and (TB.27). With $d_{-1} = 0$, the result is

$$\boxed{\frac{h_{n+1}}{h_n} d_{n+1} = d_{n-1} + d_n \left(\frac{1}{2}I - A \right) - \delta_{n0} \ell_Z}. \quad (\text{TB.30})$$

This computation uses only finite pairings of the rational function and polynomials; it does not differentiate an unproved termwise infinite expansion.

The polynomial correction in (TB.3) has last row $b_m = -d_m$, since its polynomial is $-\sum_{j=0}^m Q_j d_j$ and Q_m is monic. Furthermore $R_m^* f_{m+1} = h_{m+1} d_{m+1}^*$. Substitution in the exact PR 13 boundary contraction gives

$$\boxed{W_m = h_{m+1}(d_{m+1}^* d_m + d_m^* d_{m+1})}. \quad (\text{TB.31})$$

Thus the moment-kernel description (TB.16)–(TB.21) and the full tail description (TB.29)–(TB.31) calculate the same operator and metric through explicitly proved maps.

Theorem 8.4 (The full source defect and its observation of the packet). *The entire original derivative boundary, including its orthogonal lower source term, is*

$$\boxed{DR_m - R_m A = -\frac{h_{m+1}}{h_m} Q_m(D) f_0 d_{m+1} - Q_{m+1}(D) f_0 d_m}. \quad (\text{TB.35})$$

An exact primitive in V is obtained by replacing f_0 with ϕ_* in this formula. The finite observation

$$\mathcal{O}_m : E \longrightarrow \mathbb{C}^{2d}, \quad u \longmapsto (d_m A^k u, d_{m+1} A^k u)_{k=0}^{d-1} \quad (\text{TB.36})$$

is injective. In particular, its two retained source rows, together with the unchanged arithmetic action, distinguish every vector and every nilpotent jet in the original packet.

Proof. Use the finite expression $r_m = r - \sum_{j=0}^m Q_j d_j$ and compute

$$s r_m - r_m A = \ell_Z - \sum_{j=0}^m (s Q_j d_j - Q_j d_j A).$$

Insert (TB.27) and group coefficients of Q_j . For $0 \leq j < m$ the coefficient is zero by (TB.30); the $j = 0$ equation includes precisely the term ℓ_Z . The coefficient at Q_m is $-h_{m+1} d_{m+1}/h_m$, and that at Q_{m+1} is $-d_m$. Multiplying by g and applying the inverse Mellin map proves (TB.35). Its displayed primitive belongs to V , since every $D^j \phi_*$ does, and its theta image is the entire right side. Thus the term $Q_m(D) f_0$ remains an actual supported boundary even though its pairing with R_m vanishes.

Suppose that $\mathcal{O}_m u = 0$. Since $h(A) = 0$, polynomial division by h proves $d_m A^k u = d_{m+1} A^k u = 0$ for every $k \geq 0$. Equation (TB.30) for $n = m + 1$, applied to each $A^k u$, gives $d_{m+2} A^k u = 0$ for every k . Induction with the same recurrence gives $d_n A^k u = 0$ for all $n > m$ and $k \geq 0$. In particular every term of the positive series in (TB.29) vanishes on u , so $u^* G_m u = 0$. The exact left inverse $J_Z R_m = I_E$ makes G_m positive definite, hence $u = 0$. This proves injectivity without assuming distinct roots or removing a Jordan direction. $\square$

8.7 Metric dependence and the precise mathematical gain

Positivity in (TB.8) is the positivity of the original $L^2(dx)$ norm on an injective representative map. For the same-packet residue and dual-metric comparisons (TB.32)–(TB.33), assume explicitly that $Z = Z^\dagger$ under $\rho \mapsto 1 - \bar{\rho}$, with every full order retained. The preceding Christoffel-kernel and source-boundary results allow arbitrary packets. Reflection on a general packet is the conjugate-linear map $E_Z \rightarrow E_{Z^\dagger}$ defined in Section 16. The original arithmetic residue matrix S , satisfying $S^* = -S$ and $A^* S + S A = S$, is still related to its dual metric by

$$G_m^D = S^* G_m^{-1} S, \quad W_m^D = -S^* G_m^{-1} W_m G_m^{-1} S. \quad (\text{TB.32})$$

The original arithmetic Weil form, with $J_g = M_{j_h g'}$, is

$$u^* S J_g v = \langle u, (G_m^{-1} S J_g) v \rangle_{G_m}. \quad (\text{TB.33})$$

Thus the map relating the positive source metric to the arithmetic form is explicitly $G_m^{-1} S J_g$. A positive matrix G_m does not supply its positivity.

For each fixed packet, the original polynomial theta span is dense in $\mathcal{H}$, as proved in PR 13. Hence $G_m \rightarrow 0$. Equations (TB.8) and (TB.26) then imply

$$\lambda_{\min}(K_{d+m}) \longrightarrow +\infty, \quad \prod_{m=m_0}^M \frac{\kappa_{d+m+1}}{\kappa_{d+m+1} + v_{d+m+1}^* K_{d+m}^{-1} v_{d+m+1}} \longrightarrow 0. \quad (\text{TB.34})$$

Indeed $K^{-1} = U_\varepsilon^{-*} G_m U_\varepsilon^{-1} \rightarrow 0$, and the product telescopes to the positive determinant ratio. Absolute decay of the Gram matrix thus has a specified arithmetic-kernel counterpart. The relative quantity remains the exact expression (TB.21), with the spectral lower bound (TB.23).

The new result is a source-constrained, multiplicity-preserving arithmetic kernel formula and its exact finite relative control. It makes the next calculation depend on the concrete ν_Z -moments and the two retained complete jet vectors, rather than on freely selected positive matrices. No estimate

that makes (TB.21) tend to zero for all actual packets has been proved here. A claim of spectral purity would require proving that estimate on these arithmetic quantities; (TB.23) identifies exactly what such a result would establish.

For standard background on orthogonal polynomial recurrence and kernel identities, see NIST DLMF, Section 18.2, <https://dlmf.nist.gov/18.2>. All identities used above, including the vertical-line signs, full jet maps, and metric comparisons, have been proved here for the retained objects.

9 The full jet kernel, finite resolvents, and the actual analytic tail

Retain the arithmetic measure, coordinates and maps of Section 8:

$$s = \frac{1}{2} + it, \quad d\nu_Z(t) = |g(s)/h(s)|^2 \frac{dt}{2\pi}, \quad E = \mathbb{C}[s]/(h), \quad A = M_s, \quad c = j_h 1.$$

In particular, every selected zero has its complete multiplicity in h , and c is the coordinate column of the unit in the retained monomial basis of E . Write q_k for the monic orthogonal polynomials of ν_Z , $\kappa_k = \|q_k\|_{\nu_Z}^2$, and $a_k = \kappa_k/\kappa_{k-1} > 0$ for $k \geq 1$. The exact recurrence is

$$sq_k = q_{k+1} + b_k q_k - a_k q_{k-1}, \quad b_k + \bar{b}_k = 1, \quad (\text{KR.1})$$

where the last term is omitted at $k = 0$.

9.1 A finite compression in the original Mellin coordinate

Fix $n \geq d$, so $N = n - 1$ is the degree bound in the preceding kernel construction. On the coefficient space with basis $q_0, \dots, q_{n-1}$, put

$$H_n = \text{diag}(\kappa_0, \dots, \kappa_{n-1}), \quad (J_n)_{k,k} = b_k, \quad (J_n)_{k+1,k} = 1, \quad (J_n)_{k,k+1} = -a_{k+1}. \quad (\text{KR.2})$$

All unspecified entries are zero, and indices start at zero. Multiplication of the displayed matrices proves

$$J_n^* H_n + H_n J_n = H_n. \quad (\text{KR.3})$$

The isometry from this coefficient space to its polynomial image is $u \mapsto \sum_k q_k u_k$. For a standard coefficient Hermitian form, the corresponding isometry is

$$\mathcal{I}_n u = \sum_{k=0}^{n-1} \frac{q_k}{\sqrt{\kappa_k}} u_k, \quad B_n = H_n^{1/2} J_n H_n^{-1/2}. \quad (\text{KR.4})$$

This explicitly stated auxiliary coordinate map leaves A , the packet coordinates, q_k , and κ_k unchanged. The matrix B_n has lower off-diagonal entry $\sqrt{a_{k+1}}$, upper off-diagonal entry $-\sqrt{a_{k+1}}$, and satisfies $B_n^* + B_n = I$.

Lemma 9.1 (Nodes and endpoint resolvents). *The characteristic polynomial of J_n , and hence of B_n , is $q_n(z)$. Its roots $\lambda_{n,1}, \dots, \lambda_{n,n}$ are distinct and lie on $\text{Re } z = 1/2$. There are positive numbers $\omega_{n,j}$ and $\sigma_{n,j}$ such that*

$$\begin{aligned} m_n(z) &= \kappa_0 e_0^* (zI - B_n)^{-1} e_0 = \sum_{j=1}^n \frac{\omega_{n,j}}{z - \lambda_{n,j}}, \\ \sum_j \omega_{n,j} &= \kappa_0, \\ F_n(z) &= e_{n-1}^* (zI - B_n)^{-1} e_{n-1} = \frac{q_{n-1}(z)}{q_n(z)} = \sum_{j=1}^n \frac{\sigma_{n,j}}{z - \lambda_{n,j}}, \\ \sum_j \sigma_{n,j} &= 1. \end{aligned} \quad (\text{KR.5})$$

Here m_n and F_n are respectively the initial and terminal observations of the same finite compression.

Proof. Expanding the tridiagonal determinant along its last row gives $D_{k+1}(z) = (z - b_k)D_k(z) + a_k D_{k-1}(z)$, with $D_0 = 1, D_1 = z - b_0$. This is (KR.1), proving $D_n = q_n$. The matrix $(B_n - \frac{1}{2}I)/i$ is Hermitian. If an eigenvector has first coordinate zero, its first row equation forces its second coordinate to be zero; recurrence through the rows forces every coordinate to vanish. Thus each eigenspace has dimension at most one. Hermitian diagonalization gives a complete eigenbasis, hence all eigenvalues are simple. The same argument beginning at the last row shows every eigenvector has nonzero last coordinate. If P_j are its rank-one orthogonal eigenprojections, set $\omega_{n,j} = \kappa_0 e_0^* P_j e_0 > 0$ and $\sigma_{n,j} = e_{n-1}^* P_j e_{n-1} > 0$. Their sums and the two spectral expansions follow from $\sum_j P_j = I$. The last diagonal cofactor of $zI - B_n$ is $q_{n-1}(z)$, proving the ratio in (KR.5). $\square$

Define $\delta(z) = |\operatorname{Re} z - 1/2|$. For $\delta(z) > 0$, (KR.3)–(KR.4) give $\|(zI - B_n)^{-1}\| \leq \delta(z)^{-1}$: for a vector u , the real part of $u^*(zI - B_n)u$ is $(\operatorname{Re} z - 1/2)\|u\|^2$, and Cauchy–Schwarz then gives the lower singular-value bound. In particular $|F_n(z)| \leq \delta(z)^{-1}$, and

$$\operatorname{Re} F_n(z) = (\operatorname{Re} z - \tfrac{1}{2}) \sum_j \frac{\sigma_{n,j}}{|z - \lambda_{n,j}|^2}. \quad (\text{KR.6})$$

9.2 Exact quadrature and the full confluent kernel

Theorem 9.2 (A polynomial formula valid at every packet centre). *Put*

$$L_{n,j}(z) = \frac{q_n(z)}{(z - \lambda_{n,j})q'_n(\lambda_{n,j})}. \quad (\text{KR.7})$$

This denotes the polynomial obtained by cancelling the displayed linear factor. For every polynomial P of degree at most $2n - 1$,

$$\int P(s) d\nu_Z = \sum_j \omega_{n,j} P(\lambda_{n,j}). \quad (\text{KR.8})$$

Consequently the complete arithmetic kernel is

$$K_{n-1} = \sum_{j=1}^n \frac{L_{n,j}(A) c c^* L_{n,j}(A)^*}{\omega_{n,j}}. \quad (\text{KR.9})$$

Formula (KR.9) includes all Taylor derivatives at all selected zeros, even when a selected critical zero equals a quadrature node.

Proof. For degree below n , the vector $P(J_n)e_0$ records exactly the coefficients of P in the monic orthogonal basis: in the successive multiplications no term of degree n occurs. Its initial coefficient is $\kappa_0^{-1} \int P d\nu_Z$. Thus the spectral expansion in (KR.5) proves (KR.8) in that degree range. For a general P in the asserted range, polynomial division gives $P = q_n U + R$, where $\deg U \leq n - 1$ and $\deg R < n$. On the line, the conjugate of $U(s)$ is again a polynomial in s , namely $\overline{U(1 - \bar{s})}$. Orthogonality therefore gives $\int q_n U d\nu_Z = 0$. The same multiple vanishes at all the nodes, proving (KR.8).

The polynomials $L_{n,j}$ form the interpolation basis of $\mathcal{P}_{n-1}$. Apply (KR.8) to $\overline{L_{n,j}(s)} L_{n,k}(s)$, interpreted as a polynomial on the line; its degree is at most $2n - 2$. Their Gram matrix is $\operatorname{diag}(\omega_{n,j})$. In this basis the residue map has column j $L_{n,j} = L_{n,j}(A)c$. The definition $K = CH^{-1}C^*$ now proves (KR.9). This is a basis change in the polynomial source only. Polynomial evaluation at A retains its nilpotent powers, so the formula gives the complete jets without any distinct-centre or simple-zero assumption. $\square$

To identify the quadrature endpoints exactly, one has

$$\sigma_{n,j} = \frac{\omega_{n,j} |q_{n-1}(\lambda_{n,j})|^2}{\kappa_{n-1}} = \frac{\kappa_{n-1}}{\omega_{n,j} |q'_n(\lambda_{n,j})|^2}. \quad (\text{KR.10})$$

Indeed an eigenvector of B_n at λ has coordinates proportional to $\overline{q_k(\lambda)}/\sqrt{\kappa_k}$: substitution into the recurrence proves this assertion, including the endpoint row. The squared ratio of its last and first coordinates proves the first equality. The preceding polynomial kernel formula gives $\mathcal{K}_{n-1}(\lambda_{n,j}, \lambda_{n,j}) = 1/\omega_{n,j}$. In the original Christoffel–Darboux polynomial identity, set $w = \lambda_{n,j}$, divide its common polynomial factor $z - \lambda_{n,j}$, then evaluate at that point. This gives

$$\frac{1}{\omega_{n,j}} = \frac{q'_n(\lambda_{n,j})\overline{q_{n-1}(\lambda_{n,j})}}{\kappa_{n-1}}.$$

Its squared modulus gives the second equality in (KR.10).

Theorem 9.3 (The exact resolvent metric for the packet). *Suppose, for this formula, that no selected centre equals one of the specified n quadrature nodes. Define maps on the original space E by*

$$Q_n = q_n(A), \quad X_{n,j} = (A - \lambda_{n,j}I)^{-1}c, \quad S_n = \sum_j \sigma_{n,j} X_{n,j} X_{n,j}^*, \quad r_n = F_n(A)c. \quad (\text{KR.11})$$

Then Q_n is invertible, $S_n > 0$, and

$$\boxed{K_{n-1} = \kappa_{n-1}^{-1} Q_n S_n Q_n^*, \quad AS_n + S_n A^* - S_n = cr_n^* + r_n c^*}. \quad (\text{KR.12})$$

In particular, the original relative-control quantity at $m = n - 1 - d$ is exactly

$$\epsilon_m = \sup_{y \neq 0} \frac{|y^*(cr_n^* + r_n c^*)y|}{y^* S_n y}. \quad (\text{KR.13})$$

The identity is valid whenever that index belongs to the original source sequence; $m = -1$ is its already specified fixed-section extension. Every off-line packet obeys the nonintersection condition for every n . At an intersecting critical centre, (KR.9) gives the exact polynomial continuation, with no deletion of that centre or its jets.

Proof. The finite algebra E has an inverse for a polynomial precisely when the polynomial is nonzero at every centre: the Euclidean algorithm gives an inverse modulo h , or, equivalently, the finite Taylor inverse at each centre gives the inverse under the complete Chinese-remainder map. This proves the asserted invertibilities. Substitution of (KR.7) and (KR.10) into (KR.9) proves the first identity in (KR.12). It proves $S_n > 0$, since the residue map is onto at $n \geq d$, hence $K_{n-1} > 0$. The identity $AX_{n,j} = \lambda_{n,j} X_{n,j} + c$, together with $\lambda_{n,j} + \bar{\lambda}_{n,j} = 1$, proves

$$AS_n + S_n A^* - S_n = c \left(\sum_j \sigma_{n,j} X_{n,j} \right)^* + \left(\sum_j \sigma_{n,j} X_{n,j} \right) c^*.$$

The partial fractions in (KR.5), evaluated in the finite algebra, identify the sum with $F_n(A)c$. Because Q_n commutes with A , both kernel and its weight form are congruent, with the same factor κ_{n-1}^{-1} , to S_n and its displayed weight form. The invertible change $y = Q_n^* x$ proves equality of their Rayleigh suprema. The previous exact map from K_{n-1} to G_m then proves (KR.13). All of these maps are stated in the original quotient coordinates; the Taylor expansion of each rational factor includes every power through $m_\rho - 1$. $\square$

The scalar form of (KR.13), with every inverse taken on the same retained packet coordinates, is

$$\alpha_n = r_n^* S_n^{-1} r_n, \quad \beta_n = c^* S_n^{-1} c, \quad \gamma_n = c^* S_n^{-1} r_n,$$

$$\boxed{\epsilon_m = |\operatorname{Re} \gamma_n| + \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}}. \quad (\text{KR.13a})$$

Indeed the nonzero eigenvalues of $S_n^{-1/2}(cr_n^* + r_nc^*)S_n^{-1/2}$ coincide, with multiplicities, with the nonzero eigenvalues of $\begin{pmatrix} \gamma_n & \beta_n \\ \alpha_n & \bar{\gamma}_n \end{pmatrix}$: factor the former through its two displayed vectors and interchange the two rectangular factors. Its characteristic polynomial gives $\text{Re } \gamma_n \pm \sqrt{\alpha_n\beta_n - (\text{Im } \gamma_n)^2}$. Cauchy-Schwarz gives $|\gamma_n|^2 \leq \alpha_n\beta_n$, so their greatest absolute value is (KR.13a), including zero or one-dimensional rank. The full congruence (KR.12) specifies exactly how the original q_n , κ_{n-1} and local unit enter the metric whose relative form is evaluated here.

9.3 The infinite multiplication operator and its genuine tail

Let $\mathcal{L} = L^2(\nu_Z)$, and let M be multiplication by s , with domain $\{f : sf \in \mathcal{L}\}$. It is a closed operator and $M^* = I - M$. These statements follow directly from the multiplication formula; its resolvent off the line is multiplication by $(z - s)^{-1}$, of norm at most $\delta(z)^{-1}$. The auxiliary vectors $\psi_k = q_k/\sqrt{\kappa_k}$ form a complete orthonormal basis. Completeness follows from the positive exponential moment of the actual arithmetic measure by the Fourier-annihilator proof already given for the theta measures.

Let P_n project onto $\psi_0, \dots, \psi_{n-1}$, and put $Q_n^{\text{tail}} = I - P_n$; this projection is distinct from the polynomial map $Q_n = q_n(A)$ in (KR.11). Define

$$M_n^{\text{tail}} = Q_n^{\text{tail}} M Q_n^{\text{tail}} \quad \text{on } Q_n^{\text{tail}} \text{Dom } M, \quad \zeta_n(z) = \langle \psi_n, (z - M_n^{\text{tail}})^{-1} \psi_n \rangle. \quad (\text{KR.14})$$

This tail resolvent exists at every off-line z , with norm at most $\delta(z)^{-1}$. Here is the domain justification. Every polynomial belongs to $\text{Dom } M$, so P_n preserves that domain and $Q_n^{\text{tail}} \text{Dom } M = \text{Dom } M \cap Q_n^{\text{tail}} \mathcal{L}$. The operator $(M - \frac{1}{2}I)/i$ is self-adjoint multiplication by t . Its off-diagonal blocks for $P_n \oplus Q_n^{\text{tail}}$ extend to a bounded, finite-rank self-adjoint operator, by (KR.1). Subtracting this operator preserves self-adjointness: for a self-adjoint T and bounded self-adjoint C , the definition of the adjoint gives $(T - C)^* = T^* - C$ on exactly $\text{Dom } T^*$. The resulting diagonal operator reduces the two projections. Its tail block is therefore self-adjoint on the displayed tail domain. Multiplication by i and addition of $1/2$ gives the claimed tail resolvent.

Theorem 9.4 (Exact finite-section, terminal, and tail comparison). *For $\delta(z) > 0$, let*

$$m(z) = \int \frac{d\nu_Z(t)}{z - s}, \quad H_n^{\text{obs}}(z) = \frac{1}{\kappa_n} \int \frac{|q_n(s)|^2}{z - s} d\nu_Z(t).$$

Then

$$H_n^{\text{obs}}(z) = \frac{\zeta_n(z)}{1 + a_n F_n(z) \zeta_n(z)}, \quad (\text{KR.15})$$

$$m(z) - m_n(z) = \frac{(-1)^n \kappa_n \zeta_n(z)}{q_n(z)^2 [1 + a_n F_n(z) \zeta_n(z)]} = \frac{(-1)^n}{q_n(z)^2} \int \frac{|q_n(s)|^2}{z - s} d\nu_Z(t). \quad (\text{KR.16})$$

Every denominator in these formulas is nonzero off the original Mellin line. The terminal finite observations additionally obey the exact Schur recurrence

$$F_1(z) = \frac{1}{z - b_0}, \quad F_{n+1}(z) = \frac{1}{z - b_n + a_n F_n(z)}. \quad (\text{KR.17})$$

Proof. Under $\mathcal{I}_n \oplus I$, the block operator M is

$$\begin{pmatrix} B_n & -\sqrt{a_n} e_{n-1} \langle \psi_n, \cdot \rangle \\ \sqrt{a_n} \psi_n e_{n-1}^* & M_n^{\text{tail}} \end{pmatrix}. \quad (\text{KR.18})$$

The signs follow directly from (KR.1); both coupling maps are bounded on the displayed summands. Eliminate the tail component in $z - M$. The finite Schur complement is $zI - B_n + a_n \zeta_n(z) e_{n-1} e_{n-1}^*$. For $\text{Re } z > 1/2$, the real part of its quadratic form is at least $\delta(z) \|u\|^2$, because $\text{Re } \zeta_n(z) > 0$; on the

other side all signs reverse. The sign of this scalar real part follows either from the self-adjoint tail resolvent or from the resolvent identity $\operatorname{Re}\langle v, (z - M_n^{\text{tail}})^{-1}v \rangle = (\operatorname{Re} z - 1/2)\|(z - M_n^{\text{tail}})^{-1}v\|^2$. Thus the Schur complement is invertible and $1 + a_n \zeta_n F_n \neq 0$. Direct multiplication verifies the rank-one inverse formula

$$R_n - \frac{a_n \zeta_n}{1 + a_n \zeta_n F_n} R_n e_{n-1} e_{n-1}^* R_n, \quad R_n = (zI - B_n)^{-1}. \quad (\text{KR.19})$$

The analogous bottom Schur calculation, paired against ψ_n , gives (KR.15), since $\langle \psi_n, (z - M)^{-1} \psi_n \rangle = H_n^{\text{obs}}(z)$.

The two endpoint cofactors of $zI - B_n$ give

$$e_0^* R_n e_{n-1} = \frac{(-1)^{n-1} \sqrt{\kappa_{n-1}/\kappa_0}}{q_n(z)}, \quad e_{n-1}^* R_n e_0 = \frac{\sqrt{\kappa_{n-1}/\kappa_0}}{q_n(z)}. \quad (\text{KR.20})$$

These cofactors are the products of the successive upper and lower entries, respectively; the formula also covers $n = 1$. Pair (KR.19) against $\sqrt{\kappa_0} e_0$, the coefficient vector of the unchanged function 1. Substitute (KR.20) and $a_n \kappa_{n-1} = \kappa_n$. This proves the first equality in (KR.16); (KR.15) proves the second. Finally (KR.17) is (KR.1) divided by q_{n+1} , equivalently the last-coordinate finite Schur complement. $\square$

There is also a direct polynomial check on every factor in (KR.16). The polynomial

$$p_n(z) = \int \frac{q_n(z) - q_n(s)}{z - s} d\nu_Z(t)$$

has degree below n . At a quadrature node its ratio residue is $p_n(\lambda_{n,j})/q'_n(\lambda_{n,j}) = \int L_{n,j} d\nu_Z = \omega_{n,j}$, so $m_n = p_n/q_n$. Orthogonality to the polynomial $(q_n(z) - q_n(s))/(z - s)$ gives

$$m - m_n = \frac{1}{q_n(z)^2} \int \frac{q_n(s)^2}{z - s} d\nu_Z.$$

The polynomial $(-1)^n \overline{q_n(1 - \bar{s})}$ is monic and has the same orthogonality conditions as q_n ; uniqueness makes it equal to q_n . On the line this says $q_n(s)^2 = (-1)^n |q_n(s)|^2$, giving (KR.16) again. In particular the explicit error inequality is

$$|m(z) - m_n(z)| \leq \frac{\kappa_n}{\delta(z) |q_n(z)|^2}. \quad (\text{KR.21})$$

The initial and terminal observations have now been related by the actual block maps (KR.18)–(KR.20), the tail (KR.14), and the exact remainder (KR.16). The relative form uses the terminal observation through (KR.11)–(KR.13). An estimate for the initial observation can be propagated through these equalities while retaining the factors q_n , κ_n , a_n , and the tail denominator. None of those arithmetic quantities is discarded in passing from one observation to the other.

10 Quantitative convergence of the arithmetic Gaussian resolvents

Retain the actual measure and every coordinate of Section 9:

$$s = \frac{1}{2} + it, \quad d\nu_Z(t) = \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{dt}{2\pi}, \quad E = \mathbb{C}[s]/(h), \quad A = M_s, \quad c = j_h 1.$$

The polynomial h contains the complete multiplicities of its selected actual zeros. The quotient g/h is therefore entire, including at the selected critical zeros. Use the initial Gaussian observation of

(KR.5) and (KR.8),

$$m_n(z) = \sum_{j=1}^n \frac{\omega_{n,j}}{z - \lambda_{n,j}}, \quad m(z) = \int \frac{d\nu_Z(t)}{z - s}, \quad \mathcal{E}_n(z) = m(z) - m_n(z), \quad \delta(z) = |\operatorname{Re} z - \tfrac{1}{2}|. \quad (\text{GC.1})$$

The Gaussian weights are positive and the distinct nodes lie on the original Mellin line, as proved in Section 9. Its exact quadrature (KR.8) and exact error (KR.16) are the inputs below. No asymptotic for q_n or κ_n is assumed.

For each fixed $0 < a < \pi/2$, the number

$$E_a = \max \left\{ \int e^{at} d\nu_Z(t), \int e^{-at} d\nu_Z(t) \right\} \quad (\text{GC.2})$$

is finite. Here is the connection to the proved arithmetic decay. The actual theta calculation in Section 15 gives $|g(1/2 + it)| \leq C_b e^{-b|t|}$ for every $0 < b < \pi/4$. Choose $a/2 < b < \pi/4$. Outside a sufficiently large compact interval, the monic polynomial h of degree d satisfies $|h(1/2 + it)| \geq c_h (1 + |t|)^d$. Thus the density multiplied by $e^{a|t|}$ is bounded there by a constant times $(1 + |t|)^{-2d} e^{-(2b-a)|t|}$. On the compact interval the entire quotient is bounded. Both parts are integrable, proving (GC.2) with the original factor $dt/(2\pi)$ retained.

10.1 The Gaussian rule inherits the original exponential moments

Lemma 10.1 (Exponential moment domination). *For every real v with $|v| \leq a$, write $\lambda_{n,j} = 1/2 + it_{n,j}$. Then*

$$\sum_j \omega_{n,j} e^{vt_{n,j}} \leq \int e^{vt} d\nu_Z(t) \leq E_a. \quad (\text{GC.3})$$

Proof. Let H be the real Hermite interpolation polynomial of degree at most $2n - 1$ whose values and first derivatives at the distinct real nodes $t_{n,j}$ equal those of $f(t) = e^{vt}$. Its existence and uniqueness follow from the invertibility of the interpolation map: a polynomial in its kernel has a double zero at each of the n nodes, hence is zero in degree below $2n$; domain and target both have dimension $2n$. For a real t outside the nodes the interpolation remainder is

$$f(t) - H(t) = \frac{f^{(2n)}(\xi)}{(2n)!} \prod_j (t - t_{n,j})^2 \geq 0 \quad (\text{GC.4})$$

for some real ξ between t and the nodes. To prove the remainder, subtract $C \prod_j (x - t_{n,j})^2$ from $f(x) - H(x)$, choosing C to produce an additional zero at $x = t$. The resulting function has the n double zeros and the additional zero. Repeated Rolle's theorem gives a zero of its $2n$ -th derivative; solving at that zero gives (GC.4). The last sign follows from $f^{(2n)}(x) = v^{2n} e^{vx} \geq 0$. At a node the remainder is zero.

Integrate the pointwise inequality. The exponential and every polynomial are integrable by (GC.2). Since H is a polynomial in $t = (s - 1/2)/i$ of degree at most $2n - 1$, (KR.8) yields

$$\int H(t) d\nu_Z(t) = \sum_j \omega_{n,j} H(t_{n,j}) = \sum_j \omega_{n,j} e^{vt_{n,j}}.$$

This proves the first inequality in (GC.3). Pointwise convexity of e^{vt} in $v \in [-a, a]$, followed by integration, proves the second. No bound on the largest quadrature node has been inserted. $\square$

10.2 Exact moments give a quantitative analytic-strip estimate

Define

$$\Phi(u) = \int e^{iut} d\nu_Z(t), \quad \Phi_n(u) = \sum_j \omega_{n,j} e^{iut_{n,j}}, \quad \Delta_n(u) = \Phi(u) - \Phi_n(u). \quad (\text{GC.5})$$

These functions are holomorphic on $|\operatorname{Im} u| < a$. Indeed, on each smaller closed strip, every differentiated integrand is dominated by a constant times $e^{at} + e^{-at}$, because $|t|^k e^{b|t|} \leq C_{k,b,a}(e^{at} + e^{-at})$ when $b < a$. Dominated convergence proves all the derivatives of Φ ; the finite sum is entire. Equation (GC.3), applied at $v = -\operatorname{Im} u$, gives

$$|\Delta_n(u)| \leq 2E_a \quad (|\operatorname{Im} u| < a), \quad \Delta_n^{(k)}(0) = 0 \quad (0 \leq k < 2n). \quad (\text{GC.6})$$

The derivative identities are exactly the moments in (KR.8).

Lemma 10.2 (The complete strip bound). *For $|\operatorname{Im} u| < a$,*

$$|\Delta_n(u)| \leq 2E_a \left| \tanh\left(\frac{\pi u}{4a}\right) \right|^{2n}. \quad (\text{GC.7})$$

Proof. The explicit holomorphic coordinate maps

$$w = \tanh\left(\frac{\pi u}{4a}\right), \quad u = \frac{2a}{\pi} \operatorname{Log} \frac{1+w}{1-w}$$

are inverse bijections between the strip and the unit disk. For $|w| < 1$, the fraction lies in the right half-plane, so its principal logarithm has imaginary part in $(-\pi/2, \pi/2)$, which proves the asserted strip range. Compose $\Delta_n/(2E_a)$ with this inverse. By (GC.6), the resulting disk function is bounded by one and has a zero of order at least $2n$ at zero. Divide by w^{2n} , filling in its removable value at zero. On each circle $|w| = r < 1$, its modulus is at most r^{-2n} ; the maximum principle gives this bound inside the circle. Letting r increase to one proves the required bound by $|w|^{2n}$, hence (GC.7). $\square$

Theorem 10.3 (Direct off-line convergence with an explicit rate). *For every $0 < a < \pi/2$, every $n \geq 1$, and every z off the original Mellin line,*

$$|m(z) - m_n(z)| \leq \frac{4aE_a}{\pi} \Gamma\left(\frac{2a\delta(z)}{\pi}\right) (4n)^{-2a\delta(z)/\pi}. \quad (\text{GC.8})$$

On a compact set K disjoint from that line the same bound holds uniformly after replacing $\delta(z)$ everywhere by $\delta_K = \min_{z \in K} \delta(z) > 0$.

Proof. Put $w = z - 1/2$. On $\operatorname{Re} w > 0$, the elementary Laplace identity and Fubini's theorem give

$$\mathcal{E}_n(z) = \int_0^\infty e^{-wu} \Delta_n(u) du. \quad (\text{GC.9})$$

For example the absolute integral for the original measure before subtraction is at most $\nu_Z(\mathbb{R})/\operatorname{Re} w$, so the interchange is justified. On $\operatorname{Re} w < 0$, the exact orientation is

$$\mathcal{E}_n(z) = - \int_0^\infty e^{wu} \Delta_n(-u) du. \quad (\text{GC.10})$$

These follow respectively from integrating $e^{-(w-it)u}$ and $-e^{(w-it)u}$. Set $b = \pi/(4a)$. Equation (GC.7) gives

$$|\mathcal{E}_n(z)| \leq 2E_a I_n(\delta(z)), \quad I_n(\delta) = \int_0^\infty e^{-\delta u} \tanh(bu)^{2n} du. \quad (\text{GC.11})$$

For $0 < x < 1$, direct integration gives

$$\log \frac{1-x}{1+x} = -2 \int_0^x \frac{dv}{1-v^2} \leq -2x.$$

Taking $x = e^{-2bu}$ yields $\tanh(bu)^{2n} \leq e^{-4ne^{-2bu}}$. With $\beta = \delta/(2b) = 2a\delta/\pi$, the same variable change gives

$$\begin{aligned} I_n(\delta) &\leq \frac{1}{2b} \int_0^1 x^{\beta-1} e^{-4nx} dx \\ &\leq \frac{1}{2b} (4n)^{-\beta} \Gamma(\beta), \quad \Gamma(\beta) = \int_0^\infty y^{\beta-1} e^{-y} dy. \end{aligned} \tag{GC.12}$$

Substitution in (GC.11) proves (GC.8). On the compact set one uses $e^{-\delta(z)u} \leq e^{-\delta_K u}$ in (GC.11) and then (GC.12) at δ_K ; no monotonicity of the gamma factor is assumed. This proves locally uniform convergence on both components of the off-line domain. $\square$

10.3 Every confluent derivative and the original finite algebra

Write $F^{[k]} = F^{(k)}/k!$. The exact error (KR.16), with its vertical-line sign, gives for all $k \geq 0$

$$\mathcal{E}_n^{[k]}(z) = (-1)^n \sum_{j=0}^k (q_n^{-2})^{[k-j]}(z) (-1)^j \int \frac{|q_n(s)|^2}{(z-s)^{j+1}} d\nu_Z(t). \tag{GC.13}$$

Every derivative under this integral is justified on a compact off-line neighborhood by its positive distance from the line and the finite measure $|q_n|^2 \nu_Z$. The product rule for Taylor coefficients has no binomial factors; those are precisely cancelled by the convention $F^{[k]} = F^{(k)}/k!$. For completeness its polynomial factor is explicitly

$$(q_n^{-2})^{[r]}(z) = \frac{(-1)^r}{q_n(z)^2} \sum_{r_1+\dots+r_n=r} \prod_{j=1}^n \frac{r_j+1}{(z-\lambda_{n,j})^{r_j}}. \tag{GC.14}$$

This follows by multiplying the Taylor series of the n original factors $(z - \lambda_{n,j})^{-2}$, retaining every cross term.

Differentiating (GC.9)–(GC.10) gives the independent estimate

$$|\mathcal{E}_n^{[k]}(z)| \leq \frac{2E_a}{k!} \int_0^\infty u^k e^{-\delta(z)u} \tanh(bu)^{2n} du. \tag{GC.15}$$

The integrable majorant $u^k e^{-\delta_K u}$ on a smaller off-line compact neighborhood justifies this differentiation. If $0 < \sigma < \delta$, the exponential series gives $u^k e^{-(\delta-\sigma)u} \leq k!/(\delta-\sigma)^k$. Equations (GC.11)–(GC.12) prove

$$|\mathcal{E}_n^{[k]}(z)| \leq \frac{4aE_a}{\pi(\delta-\sigma)^k} \Gamma\left(\frac{2a\sigma}{\pi}\right) (4n)^{-2a\sigma/\pi}, \quad 0 < \sigma < \delta \leq \delta(z). \tag{GC.16}$$

Taking $\delta = \delta_K$ makes this a simultaneous compact-uniform bound on every fixed finite list of derivatives.

Here is the complete algebra transport, including the part of the original packet outside this resolvent domain. Write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{d_\rho}, \quad H_\rho(s) = \frac{h(s)}{(s - \rho)^{d_\rho}}, \quad U_\rho(s) = \sum_{k=0}^{d_\rho-1} (H_\rho^{-1})^{[k]}(\rho) (s - \rho)^k,$$

and define the actual polynomial remainder

$$e_\rho = j_h(H_\rho U_\rho) \in E. \quad (\text{GC.17})$$

Its remainder is one modulo $(s - \rho)^{d_\rho}$ and zero modulo every other factor of h . The distinct factors are coprime, and a polynomial divisible by each is divisible by their product: successive use of the Bezout identity proves this last statement. It follows directly that

$$e_\rho e_\eta = 0 \ (\rho \neq \eta), \quad e_\rho^2 = e_\rho, \quad \sum_{\rho \in Z} e_\rho = 1. \quad (\text{GC.18})$$

Restriction of a polynomial to its truncated Taylor coefficients therefore gives the exact algebra isomorphism

$$E \longrightarrow \bigoplus_{\rho \in Z} \mathbb{C}[\epsilon_\rho]/(\epsilon_\rho^{d_\rho}), \quad j_h P \longmapsto \left(\sum_{k < d_\rho} P^{[k]}(\rho) \epsilon_\rho^k \right)_\rho. \quad (\text{GC.19})$$

Its inverse sends the coordinate ϵ_ρ^k to $e_\rho(s - \rho)^k$; composition is the identity by the just proved congruences. Multiplication is preserved because the Taylor coefficient of a product is the exact convolution of the two coefficient lists, truncated only by the displayed ideal.

Let $Z_{\text{off}} = \{\rho \in Z : \text{Re } \rho \neq 1/2\}$ and $Z_{\text{line}} = Z \setminus Z_{\text{off}}$. Define

$$e_{\text{off}} = \sum_{\rho \in Z_{\text{off}}} e_\rho, \quad e_{\text{line}} = 1 - e_{\text{off}}, \quad E_{\text{off}} = e_{\text{off}} E, \quad E_{\text{line}} = e_{\text{line}} E. \quad (\text{GC.20})$$

Then $E = E_{\text{off}} \oplus E_{\text{line}}$ as algebras with their respective idempotent units. The projection $u \mapsto e_{\text{off}} u$ has exactly kernel E_{line} and commutes with the original multiplication operator A . Under (GC.19), its restriction at every retained centre is $A_\rho = \rho I + N_\rho$, where multiplication by ϵ_ρ gives $N_\rho^{d_\rho} = 0$ and retains all lower powers. This constructs the precise map from the full packet to the off-line resolvent domain, together with its full kernel and the operator on that kernel. An empty off-line set gives the zero summand, with the full original packet in the kernel.

For a function f holomorphic near Z_{off} , define

$$f(A_{\text{off}}) = \sum_{\rho \in Z_{\text{off}}} \sum_{k=0}^{d_\rho-1} f^{[k]}(\rho) (A - \rho I)^k e_\rho(A) \big|_{E_{\text{off}}}. \quad (\text{GC.21})$$

Here $e_\rho(A)$ means multiplication by the explicitly constructed remainder (GC.17). Equation (GC.19) proves that (GC.21) is a unital algebra homomorphism on the off-line summand. No diagonalizability of A has entered this construction.

In particular, finite geometric expansion in each nilpotent block proves

$$\begin{aligned} m(A_{\text{off}}) &= \int (A_{\text{off}} - sI)^{-1} d\nu_Z(t), \\ m_n(A_{\text{off}}) &= \sum_j \omega_{n,j} (A_{\text{off}} - \lambda_{n,j} I)^{-1}, \\ \mathcal{E}_n(A_{\text{off}}) &= \sum_{\rho \in Z_{\text{off}}} \sum_{k < d_\rho} \mathcal{E}_n^{[k]}(\rho) (A - \rho I)^k e_\rho(A) \big|_{E_{\text{off}}}. \end{aligned} \quad (\text{GC.22})$$

For the first equality the expansion is

$$(A - sI)^{-1} e_\rho(A) = \sum_{k < d_\rho} \frac{(-1)^k (A - \rho I)^k e_\rho(A)}{(\rho - s)^{k+1}}.$$

Each coefficient is integrable since $\delta(\rho) > 0$, and its integral is exactly $m^{[k]}(\rho)$. This proves the integral identity, rather than assigning a formal meaning to it.

For a concrete bound keep any fixed positive Hermitian form G_* on the original coefficient space and restrict it to E_{off} . When that summand is nonzero, put

$$\delta_* = \min_{\rho \in Z_{\text{off}}} \delta(\rho) > 0, \quad C_{\rho,k} = \left\| (A - \rho I)^k e_\rho(A) \right\|_{E_{\text{off}}} \Big|_{G_*}.$$

The triangle inequality and (GC.16) give, for every $0 < \sigma < \delta_*$,

$$\|m(A_{\text{off}}) - m_n(A_{\text{off}})\|_{G_*} \leq \frac{4aE_a}{\pi} \Gamma\left(\frac{2a\sigma}{\pi}\right) (4n)^{-2a\sigma/\pi} \sum_{\rho \in Z_{\text{off}}} \sum_{k < d_\rho} \frac{C_{\rho,k}}{(\delta_* - \sigma)^k}. \quad (\text{GC.23})$$

Thus the full initial resolvent converges in the original finite coefficient space, with every nilpotent contribution controlled. Applying the displayed operators to the retained vector $e_{\text{off}}c$ gives the corresponding vector statement with the additional factor $\|e_{\text{off}}c\|_{G_*}$.

The polynomial kernel (KR.9) continues to act on all of E , including E_{line} and collisions with quadrature nodes. The off-line estimates above concern the initial observation m_n . Its exact map to the terminal observation F_n is the block and tail comparison (KR.14)–(KR.20), with the factors q_n, κ_n, a_n and the tail denominator still present. The estimate (GC.23) may be inserted into those identities; it does not remove their arithmetic factors or the critical-line summand of (GC.20).

11 The exact terminal step and its retained boundary energy

All measures, packet coordinates and maps in this section are the original arithmetic ones. Thus $s = 1/2 + it$, $d\nu_Z = |g(s)/h(s)|^2 dt/(2\pi)$, $E = \mathbb{C}[s]/(h)$, $A = M_s$, and $c = j_h 1$. Every selected zero occurs with its complete multiplicity in h . Retain the monic q_n , their actual norms κ_n , and

$$a_n = \frac{\kappa_n}{\kappa_{n-1}} > 0, \quad q_{n+1}(s) = (s - b_n)q_n(s) + a_n q_{n-1}(s), \quad b_n + \bar{b}_n = 1. \quad (\text{KT.1})$$

No recurrence coefficient is replaced by a limiting value.

11.1 A step written entirely in the original polynomial packet

For $n \geq d = \deg h$, put

$$v_j = q_j(A)c, \quad K = K_{n-1} = \sum_{j=0}^{n-1} \frac{v_j v_j^*}{\kappa_j} > 0, \quad M_n = \kappa_{n-1}^{-1} K_{n-1}^{-1}. \quad (\text{KT.2})$$

Use the positive form $\langle x, y \rangle_n = x^* M_n y$, conjugate-linear in its first argument, and define

$$u_n = v_n, \quad w_n = v_{n-1}, \quad \alpha_n = \langle w_n, w_n \rangle_n, \quad \beta_n = \langle u_n, u_n \rangle_n, \quad \gamma_n = \langle u_n, w_n \rangle_n. \quad (\text{KT.3})$$

The arithmetic relative allowance at source level $m = n - d - 1$ will be denoted by $\epsilon^{(n)}$. The fixed-section extension at $m = -1$ has already been specified; the original corrected sequence has $n \geq d + 1$. The exact kernel boundary identity gives

$$\epsilon^{(n)} = |\operatorname{Re} \gamma_n| + \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}, \quad \operatorname{Re} \gamma_n = \operatorname{Re} \operatorname{tr} A - \frac{d}{2} =: g_Z. \quad (\text{KT.4})$$

Indeed the kernel weight form is $(u_n w_n^* + w_n u_n^*)/\kappa_{n-1}$. Its possible nonzero generalized eigenvalues are obtained by reversing the two rectangular factors, giving $\operatorname{Re} \gamma_n \pm \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}$. Taking the trace after multiplication by K^{-1} gives the second identity. Cauchy–Schwarz proves the nonnegativity of the radicand and justifies the maximum absolute value in (KT.4).

Theorem 11.1 (Universal scalar and metric step). *Write $a = a_n$, and retain the exact next vector*

$$t_n = (A - b_n I)u_n + a_n w_n = v_{n+1}. \quad (\text{KT.5})$$

Then

$$M_{n+1} = \frac{1}{a} \left(M_n - \frac{M_n u_n u_n^* M_n}{a + \beta_n} \right), \quad (\text{KT.6})$$

$$\begin{aligned} \alpha_{n+1} &= \frac{\beta_n}{a + \beta_n}, \\ \gamma_{n+1} &= \frac{\langle t_n, u_n \rangle_n}{a + \beta_n}, \\ \beta_{n+1} &= \frac{1}{a} \left(\langle t_n, t_n \rangle_n - \frac{|\langle u_n, t_n \rangle_n|^2}{a + \beta_n} \right). \end{aligned} \quad (\text{KT.7})$$

Every denominator is strictly positive. These identities hold also when any selected critical centre equals a quadrature node, and when $q_n(A)$ is singular.

The exact determinant changes of the kernel and original theta metric are

$$\frac{\det K_n}{\det K_{n-1}} = 1 + \frac{\beta_n}{a_n}, \quad \frac{\det G_{m+1}}{\det G_m} = \frac{a_n}{a_n + \beta_n} = 1 - \alpha_{n+1}. \quad (\text{KT.8})$$

Proof. The unchanged kernel update is $K_n = K_{n-1} + u_n u_n^* / \kappa_n$. Its inverse is verified by multiplication:

$$K_n^{-1} = K^{-1} - \frac{K^{-1} u_n u_n^* K^{-1}}{\kappa_n + u_n^* K^{-1} u_n}.$$

Insert $\kappa_n = a \kappa_{n-1}$ and the definition of M_n ; this gives (KT.6). At the next step the ordered vectors in (KT.3) are $u_{n+1} = t_n$, $w_{n+1} = u_n$. Pair each ordered pair with (KT.6). The three resulting expressions are precisely (KT.7), including their conjugations. The denominator is positive because $a > 0$, $\beta_n \geq 0$. The determinant of the rank-one update is $1 + u_n^* K^{-1} u_n / \kappa_n = 1 + \beta_n / a$; this follows directly by expanding the determinant multilinearly in its columns. The original metric is $G_m = U_\varepsilon^* K_{n-1}^{-1} U_\varepsilon$, with the same fixed invertible local unit at adjacent levels. Its determinant ratio is therefore the inverse kernel ratio. Equation (KT.7) gives the last equality in (KT.8). All calculations use polynomial residues at A ; no inverse of $q_n(A)$ has been used. $\square$

11.2 The complete boundary-energy increment

Suppose first that $\beta_n > 0$. The exact orthogonal projection for $\langle \cdot, \cdot \rangle_n$ is

$$\begin{aligned} P_n x &= x - u_n \frac{\langle u_n, x \rangle_n}{\beta_n}, \quad p_n = P_n w_n, \quad z_n = P_n A u_n, \\ \Delta_n &= \alpha_n \beta_n - |\gamma_n|^2 = \beta_n \|p_n\|_n^2, \quad \Omega_n = \beta_n \|z_n\|_n^2. \end{aligned} \quad (\text{KT.9})$$

The projection records a specified map inside the original packet; $A u_n$, its removed coefficient, and its remaining vector z_n are all retained in (KT.9).

Theorem 11.2 (Exact transverse energy law). *At every such step,*

$$\Delta_{n+1} = \frac{\beta_n \|z_n + a_n p_n\|_n^2}{a_n(a_n + \beta_n)}. \quad (\text{KT.10})$$

Consequently

$$\Delta_{n+1} - \Delta_n = \frac{\beta_n}{a_n(a_n + \beta_n)} (\|z_n\|_n^2 + 2a_n \operatorname{Re} \langle z_n, p_n \rangle_n - a_n \beta_n \|p_n\|_n^2). \quad (\text{KT.11})$$

The exact energy law gives the following two-sided bounds by the sharp triangle inequalities:

$$\frac{|a_n\sqrt{\Delta_n} - \sqrt{\Omega_n}|}{\sqrt{a_n(a_n + \beta_n)}} \leq \sqrt{\Delta_{n+1}} \leq \frac{a_n\sqrt{\Delta_n} + \sqrt{\Omega_n}}{\sqrt{a_n(a_n + \beta_n)}}. \quad (\text{KT.12})$$

The actual relative allowance is recovered exactly by

$$\boxed{\epsilon^{(n)} = |g_Z| + \sqrt{g_Z^2 + \Delta_n}}. \quad (\text{KT.13})$$

For a reflection-stable packet with its multiplicities, $g_Z = 0$, so the energy law calculates the square of the actual allowance without a trace term.

Proof. Let $T = \langle t_n, t_n \rangle_n$ and $v = \langle u_n, t_n \rangle_n$. Substitution of (KT.7) gives

$$\alpha_{n+1}\beta_{n+1} - |\gamma_{n+1}|^2 = \frac{\beta_n T - |v|^2}{a_n(a_n + \beta_n)} = \frac{\beta_n \|P_n t_n\|_n^2}{a_n(a_n + \beta_n)}.$$

Since $P_n u_n = 0$, the exact recurrence (KT.5) gives $P_n t_n = P_n A u_n + a_n P_n w_n = z_n + a_n p_n$. This proves (KT.10). Expand the squared norm and subtract $\Delta_n = \beta_n \|p_n\|_n^2$ to prove (KT.11). The triangle inequality and its reverse, applied to these two specific vectors, prove (KT.12). Finally (KT.4) gives (KT.13). Its real trace term is unchanged with n ; equivalently the same trace computation can be performed at each adjacent positive kernel. Reflection permutes all roots and multiplicities, so $2 \operatorname{Re} \operatorname{tr} A = d$ in that case. $\square$

If $\beta_n = 0$, positivity of M_n forces $u_n = v_n = 0$. Then $\gamma_n = 0$, the current kernel weight is zero, $K_n = K_{n-1}$, and (KT.5)–(KT.7) give the entire exceptional step:

$$t_n = a_n w_n, \quad \alpha_{n+1} = \gamma_{n+1} = 0, \quad \beta_{n+1} = a_n \alpha_n, \quad \Delta_n = \Delta_{n+1} = 0, \quad \epsilon^{(n)} = \epsilon^{(n+1)} = 0. \quad (\text{KT.14})$$

Moreover $v_n = 0$ is exactly the divisibility $h \mid q_n$. Since every root of q_n is simple and lies on the Mellin line, this particular event forces the selected packet to consist of simple line roots. It does not remove the resulting packet or the next vectors in (KT.14).

11.3 The exact rational terminal step and its polynomial continuation

Where $q_n(A)$ and $q_{n+1}(A)$ are invertible, use the actual terminal maps and Gram from Section 9:

$$T_n = F_n(A), \quad r_n = T_n c, \quad Q_n = q_n(A), \quad K_{n-1} = \kappa_{n-1}^{-1} Q_n S_n Q_n^*.$$

The map Q_n is the exact isometry

$$Q_n : (E, S_n^{-1}) \longrightarrow (E, M_n), \quad Q_n c = u_n, \quad Q_n r_n = w_n. \quad (\text{KT.15})$$

Indeed $Q_n^* M_n Q_n = S_n^{-1}$ follows by inverting the displayed Gram congruence, and the vector identity uses $F_n = q_{n-1}/q_n$. This proves the precise relation between the rational and polynomial forms of (KT.3)–(KT.13).

The terminal recurrence gives

$$T_{n+1} = (A - b_n I + a_n T_n)^{-1}, \quad \boxed{S_{n+1} = T_{n+1} (c c^* + a_n S_n) T_{n+1}^*}. \quad (\text{KT.16})$$

For a direct full-jet proof, the scalar kernel of a terminal observation is

$$H_{F_n}(z, w) = \frac{F_n(z) + \overline{F_n(w)}}{z + \overline{w} - 1}.$$

The quotient denotes its removable polynomial/rational continuation where numerator and denominator both vanish. It is identically the positive resolvent Gram $\sum_j \sigma_{n,j}/[(z - \lambda_{n,j})(\bar{w} - \bar{\lambda}_{n,j})]$. Substitution of $F_{n+1} = (z - b_n + a_n F_n)^{-1}$ gives

$$H_{F_{n+1}}(z, w) = F_{n+1}(z) \overline{F_{n+1}(w)} [1 + a_n H_{F_n}(z, w)]. \quad (\text{KT.17})$$

This identity holds first away from its removable sets and hence through them by continuation. Taking the complete residue on both variables proves (KT.16). It also follows from the original kernel update and $Q_{n+1}^{-1} Q_n = T_{n+1}$. Thus a quadrature collision is handled by the polynomial formulas (KT.2)–(KT.14), which retain all coordinates; (KT.15) states exactly the isometry on the open set where the rational presentation is defined.

In this presentation the determinant still retains the entire polynomial factor:

$$\frac{\det S_{n+1}}{\det S_n} = \frac{a_n^{d-1}(a_n + \beta_n)}{|\det(A - b_n I + a_n T_n)|^2}, \quad \det(A - b_n I + a_n T_n) = \prod_{\rho \in Z} \left(\frac{q_{n+1}(\rho)}{q_n(\rho)} \right)^{m_\rho}. \quad (\text{KT.18})$$

The determinant of $a_n S_n + cc^*$ gives the first formula. The second follows by evaluating the polynomial ratio on each full triangular Taylor block $\rho I + N_\rho$. The determinant has the stated multiplicity even when the block is not semisimple.

11.4 The extra jets required to recover the metric from the tail

There is an exact map from the scalar tail identity to the matrix energy, with a specified number of Taylor coefficients. For (KT.19)–(KT.22), take centres at which q_n is nonzero, so F_n is holomorphic there. Every off-line centre meets this condition. At a critical quadrature intersection the universal polynomial formulas (KT.2)–(KT.14) retain the complete packet. The terminal function satisfies $\overline{F_n(1 - \bar{z})} = -F_n(z)$, by its positive spectral sum on the line. Thus

$$H_{F_n}(z, w) = \frac{F_n(z) - F_n(1 - \bar{w})}{z - (1 - \bar{w})}. \quad (\text{KT.19})$$

For centres ρ, σ , and Taylor orders $0 \leq k < m_\rho, 0 \leq l < m_\sigma$, let $S_n^{\text{jet}}(\rho, k; \sigma, l)$ denote the coefficient of $(z - \rho)^k (\bar{w} - \bar{\sigma})^l$ in this kernel. In particular, at a reflection pair,

$$\boxed{S_n^{\text{jet}}(\rho, k; \sigma, l) = (-1)^l F_n^{[k+l+1]}(\rho) \quad \text{if } \rho = 1 - \bar{\sigma}.} \quad (\text{KT.20})$$

Here $F^{[j]} = F^{(j)}/j!$. To prove the formula, put $z = \rho + x, 1 - \bar{w} = \rho - y$ in (KT.19), and expand

$$\frac{x^r - (-y)^r}{x + y} = \sum_{j=0}^{r-1} x^{r-1-j} (-y)^j.$$

The coefficient of $x^k y^l$ is supplied exactly by $r = k + l + 1, j = l$, proving (KT.20) without dividing by zero.

For $D_{\rho\sigma} = \rho + \bar{\sigma} - 1 \neq 0$, the corresponding complete coefficient is

$$\begin{aligned} S_n^{\text{jet}}(\rho, k; \sigma, l) &= \sum_{i=0}^k F_n^{[i]}(\rho) \frac{(-1)^{k-i+l} \binom{k-i+l}{k-i}}{D_{\rho\sigma}^{k-i+l+1}} \\ &\quad + \sum_{j=0}^l \overline{F_n^{[j]}(\sigma)} \frac{(-1)^{k+l-j} \binom{k+l-j}{k}}{D_{\rho\sigma}^{k+l-j+1}}. \end{aligned} \quad (\text{KT.21})$$

This is the full product of the numerator Taylor series and the geometric expansion of $(D_{\rho\sigma} + x + y)^{-1}$. Both constant terms in the numerator are retained. If $V_Z : E \rightarrow \bigoplus_{\rho} \mathbb{C}^{m_{\rho}}$ is the fixed complete Taylor-coordinate map, then

$$S_n^{\text{jet}} = V_Z S_n V_Z^*, \quad S_n = V_Z^{-1} S_n^{\text{jet}} (V_Z^{-1})^*. \quad (\text{KT.22})$$

This follows by applying V_Z to each resolvent column in the positive Gram representation of S_n .

To specify the domain of the analytic extraction, let $h_{\text{off}} = \prod_{\text{Re } \rho \neq 1/2} (s - \rho)^{m_{\rho}}$. The full original algebra projects onto $E_{\text{off}} = \mathbb{C}[s]/(h_{\text{off}})$; the kernel is the retained critical-line CRT summand, with its original action of A . Above the off-line summand use the doubled algebra $\hat{E}_{\text{off}} = \mathbb{C}[s]/(h_{\text{off}}^2)$. Its exact reduction sequence of $\mathbb{C}[s]$ -modules is

$$0 \longrightarrow E_{\text{off}} \xrightarrow{[p] \mapsto [h_{\text{off}} p]} \hat{E}_{\text{off}} \xrightarrow{[p] \mapsto [p]} E_{\text{off}} \longrightarrow 0. \quad (\text{KT.23})$$

The first map is well-defined and injective because $h_{\text{off}}^2 \mid h_{\text{off}} p$ holds exactly when $h_{\text{off}} \mid p$; its image is precisely the kernel of the reduction. All maps commute with multiplication by s . The reduction is also a unital algebra map. The image of the injection is the square-zero ideal $(h_{\text{off}})/(h_{\text{off}}^2)$: the product of the images of $[p]$ and $[q]$ is $[h_{\text{off}}^2 pq] = 0$. This records its complete inherited multiplication as well as its module isomorphism with E_{off} . The nilpotent lengths in the middle term are exactly $2m_{\rho}$, and reduction retains the lower m_{ρ} coefficients. At reflected actual zeros, the full multiplicities are equal, by the functional equation and conjugation of g . Thus (KT.20) uses at most order $2m_{\rho} - 1$, exactly a coefficient of this doubled algebra. Formula (KT.21) uses only the stated lower coefficients. Equations (KT.20)–(KT.23) prove the precise passage from terminal Taylor data to the full metric.

Write $\mathcal{E}_n = m - m_n$ for the initial observation error, and retain the genuine tail ζ_n from (KR.14). For every off-line point its real part has the sign of $\text{Re } z - 1/2$, so $\zeta_n \neq 0$. The exact identity (KR.16) also gives

$$\mathcal{E}_n(z) = \frac{(-1)^n \kappa_n}{q_n(z)^2} H_n^{\text{obs}}(z), \quad H_n^{\text{obs}}(z) = \frac{1}{\kappa_n} \int \frac{|q_n(s)|^2}{z - s} d\nu_Z.$$

The real part of H_n^{obs} has that same strict sign: its defining measure is positive of total mass one. Hence $\mathcal{E}_n(z) \neq 0$ off the line as well. Solving the actual tail equation, before taking any residues, therefore gives the exact extraction

$$F_n(z) = \frac{1}{a_n} \left(\frac{(-1)^n \kappa_n}{q_n(z)^2 \mathcal{E}_n(z)} - \frac{1}{\zeta_n(z)} \right). \quad (\text{KT.24})$$

It holds in the complete doubled algebra at the off-line centres. Every reciprocal there is explicit: if $f(\rho + x) = \sum_{j=0}^{2m_{\rho}-1} f_j x^j$ modulo $x^{2m_{\rho}}$, with $f_0 \neq 0$, the inverse coefficients are

$$b_0 = f_0^{-1}, \quad b_k = -f_0^{-1} \sum_{j=1}^k f_j b_{k-j} \quad (1 \leq k < 2m_{\rho}). \quad (\text{KT.25})$$

Multiplication of the two finite series proves all coefficients. The already proved bounds (GC.13)–(GC.16) apply to every one of the error coefficients required here. Equations (KT.24)–(KT.25), then (KT.20)–(KT.22), give a complete finite extraction of the terminal metric on E_{off} from the initial error and genuine tail data. In the retained monomial quotient coordinates, set $A_{\text{off}} = M_s$, $\pi_{\text{off}}([p]_h) = [p]_{h_{\text{off}}}$, and $c_{\text{off}} = [1] = \pi_{\text{off}} c$. Explicitly, retain the same ν_Z , q_n , nodes and terminal weights, and put

$$S_n^{\text{off}} = \sum_j \sigma_{n,j} (A_{\text{off}} - \lambda_{n,j} I)^{-1} c_{\text{off}} c_{\text{off}}^* (A_{\text{off}}^* - \bar{\lambda}_{n,j} I)^{-1}. \quad (\text{KT.26})$$

For the original quotient map $\pi_{\text{off}} : E \rightarrow E_{\text{off}}$, the resolvent intertwining identity gives $S_n^{\text{off}} = \pi_{\text{off}} S_n \pi_{\text{off}}^*$ whenever the full rational Gram is defined. The right side of (KT.26) remains defined at

every off-line centre even when the full packet has a critical quadrature collision. No replacement of the measure by $|g/h_{\text{off}}|^2 dt/(2\pi)$ occurs in this projection. The full polynomial quadrature formula, applied to these projected residue columns, proves even at such a collision that

$$\pi_{\text{off}} K_{n-1} \pi_{\text{off}}^* = \kappa_{n-1}^{-1} q_n(A_{\text{off}}) S_n^{\text{off}} q_n(A_{\text{off}})^*. \quad (\text{KT.27})$$

These extraction formulas retain the inverse error, the inverse tail, q_n , κ_n , and a_n , including their cancellations. The initial error bound alone is not used to replace any of those terms. The polynomial metric step (KT.6) continues to include the critical-line summand, where the off-line resolvent domain ends.

11.5 Actual quartet packets close the energy in two volume ratios

Assume here that the selected actual packet is closed under both $\rho \mapsto \bar{\rho}$ and $\rho \mapsto 1 - \bar{\rho}$, with the complete multiplicities. These are exact invariances of the specified divisor; no value of $\text{Re } \rho$ has been imposed.

Theorem 11.3 (Two consecutive arithmetic volume contractions). *Let G_{-1} be the fixed-section metric already identified with $U_\varepsilon^* K_{d-1}^{-1} U_\varepsilon$, and define the actual positive determinant ratios*

$$r_m = \frac{\det G_m}{\det G_{m-1}} \quad (m \geq 0).$$

Then $0 < r_m \leq 1$, and for every $m \geq 0$, with $n = d + m + 1$,

$$\epsilon_m^2 = \frac{\kappa_{d+m+1}}{\kappa_{d+m}} \frac{(1 - r_m)(1 - r_{m+1})}{r_{m+1}}. \quad (\text{KT.28})$$

This is the original relative allowance and the original theta Gram sequence, including their local unit and all multiplicities.

Proof. Conjugation invariance of the divisor makes every coefficient of the monic h real. The entire function g obeys $g(\bar{s}) = \overline{g(s)}$, so the unchanged density $|g(1/2 + it)/h(1/2 + it)|^2$ is even in t . For a monic orthogonal polynomial q , form $q^c(s) = \overline{q(\bar{s})}$. Substitution $t \mapsto -t$ and conjugation of each orthogonality equation show that q^c is monic with the same degree and the same lower-degree orthogonality. Positivity of the polynomial Gram proves uniqueness, so $q^c = q$. Thus every q_n has real coefficients in the original variable s . The coefficient of q_n in sq_n is therefore real as well; (KT.1) proves its exact value $b_n = 1/2$. This is a consequence of the given packet symmetries, with every other recurrence coefficient retained.

The original companion matrix A , vector c , residue columns v_j , and Gram matrices K_{n-1} , M_n are real. Therefore the scalar γ_n in (KT.3) is real. The second specified symmetry gives $g_Z = 0$ in (KT.4), so $\gamma_n = 0$ exactly. Hence $\epsilon_m^2 = \alpha_n \beta_n$. At the preceding and following steps (KT.7)–(KT.8) give

$$\alpha_n = 1 - r_m, \quad \alpha_{n+1} = 1 - r_{m+1}, \quad \beta_n = \frac{a_n \alpha_{n+1}}{1 - \alpha_{n+1}}.$$

All these denominators are positive: $G_m > 0$ and $a_n/(a_n + \beta_n) > 0$. Substitution proves (KT.28), including the exceptional zero-column steps. The inequality $0 < r_m \leq 1$ follows directly from the same determinant formula. At $m = 0$ the preceding metric is precisely the stated G_{-1} , so no earlier undefined source level is used. $\square$

11.6 What the recurrence proves about boundary energy

The unchanged arithmetic source has supplied the explicit energy increment (KT.11). Its three terms are the squared transverse A -image, its real pairing with the retained terminal vector, and the specified negative term $-a_n\beta_n\|p_n\|_n^2$. Thus the source recurrence produces an exactly computable boundary energy, including its sign-determining cross term. No sign for their sum has been assumed. Every term is a polynomial residue, an actual moment Gram inverse, or a retained recurrence coefficient. The rational route to those same quantities is (KT.15)–(KT.25). The original theta representative and its boundary are recovered from the preceding kernel-minimum formula and its explicit division by h . Consequently all updates are maps between the original source representatives, with the same quotient class and supported zero. For each support label the displayed linear maps lift by $(\lambda, x) \mapsto (\lambda, Tx)$; external absence remains τ , and a zero vector in that fibre remains its supported zero throughout.

11.7 An exact positive-moment counterexample to consecutive energy monotonicity

This section concerns the moment-to-packet construction for the explicitly specified positive measure below. It proves that positivity, the vertical orthogonal-polynomial recurrence, reflection stability of the packet, and the terminal rank-two identity together allow an increase of the relative weight energy at consecutive degrees. The measure is specified by its nine atoms; no assertion identifies it with the arithmetic measure $d\nu_Z(t) = |g(\frac{1}{2} + it)/h(\frac{1}{2} + it)|^2 dt/(2\pi)$. The exact correspondence with that construction will be the same polynomial residue map, multiplication operator, kernel sum and Hermitian defect, with the measure retained as part of the data throughout.

11.7.1 The measure and every moment needed below

Keep the original coordinate $s = \frac{1}{2} + it$. Define a probability measure ν on this line by the following table. In a row containing two signs, the displayed weight is assigned to each of the two atoms.

t	weight at each atom	
0	115/288	
$\pm 1/4$	29/120	
$\pm 1/2$	13/240	(TEC.1)
$\pm 3/4$	11/2520	
± 1	1/6720	

All nine weights are strictly positive, and

$$\frac{115}{288} + \frac{29}{60} + \frac{13}{120} + \frac{11}{1260} + \frac{1}{3360} = 1. \quad (\text{TEC.2})$$

Write $\mu_k = \int t^k d\nu$ and $v = 1/16$. The paired atoms give $\mu_k = 0$ for every odd integer $k \geq 1$, including $k = 1, 3, 5, 7, 9$. For positive integers r , the even-moment formula, with all original atom weights retained, is

$$\mu_{2r} = 4^{-2r} \left(\frac{29}{60} 1^{2r} + \frac{13}{120} 2^{2r} + \frac{11}{1260} 3^{2r} + \frac{1}{3360} 4^{2r} \right). \quad (\text{TEC.3})$$

Here the factor 4^{-2r} comes from the explicitly displayed atom locations in (TEC.1). The four contributions and their sums are

r	$\frac{29}{60}1^{2r}$	$\frac{13}{120}2^{2r}$	$\frac{11}{1260}3^{2r}$	$\frac{1}{3360}4^{2r}$	sum
1	29/60	13/30	11/140	1/210	1
2	29/60	26/15	99/140	8/105	3
3	29/60	104/15	891/140	128/105	15
4	29/60	416/15	8019/140	2048/105	105
5	29/60	1664/15	72171/140	32768/105	939

(TEC.4)

Consequently

$$(\mu_0, \mu_2, \mu_4, \mu_6, \mu_8, \mu_{10}) = (1, v, 3v^2, 15v^3, 105v^4, 939v^5). \quad (\text{TEC.5})$$

Equations (TEC.2)–(TEC.5), together with cancellation of each odd moment between the two atoms, verify all moments through degree ten.

11.7.2 The actual monic orthogonal polynomials of this measure

Use the inner product

$$\langle P, Q \rangle_\nu = \int \overline{P(\frac{1}{2} + it)} Q(\frac{1}{2} + it) d\nu(t), \quad x = s - \frac{1}{2}. \quad (\text{TEC.6})$$

The following table retains the monic polynomials in the original s -coordinate and gives their actual squared norms.

k	$q_k(s)$	$\kappa_k = \langle q_k, q_k \rangle_\nu$
0	1	1
1	x	v
2	$x^2 + v$	$2v^2$
3	$x^3 + 3vx$	$6v^3$
4	$x^4 + 6vx^2 + 3v^2$	$24v^4$
5	$x^5 + 10vx^3 + 15v^2x$	$114v^5$

(TEC.7)

Proof. For an explicit verification, define real polynomials

$$\begin{aligned} P_0(t) &= 1, & P_1(t) &= t, & P_2(t) &= t^2 - v, \\ P_3(t) &= t^3 - 3vt, & P_4(t) &= t^4 - 6vt^2 + 3v^2, \\ P_5(t) &= t^5 - 10vt^3 + 15v^2t. \end{aligned} \quad (\text{TEC.8})$$

Direct substitution gives $q_k(\frac{1}{2} + it) = i^k P_k(t)$. Thus

$$\langle q_j, q_k \rangle_\nu = (-i)^j i^k \int P_j(t) P_k(t) d\nu(t). \quad (\text{TEC.9})$$

If j, k have different parity, the integrand on the right is odd and its integral is zero. The remaining six off-diagonal integrals, using (TEC.5), are precisely

$$\begin{aligned} \int P_0 P_2 d\nu &= v(1 - 1) = 0, \\ \int P_0 P_4 d\nu &= v^2(3 - 6 + 3) = 0, \\ \int P_2 P_4 d\nu &= v^3(15 - 21 + 9 - 3) = 0, \\ \int P_1 P_3 d\nu &= v^2(3 - 3) = 0, \\ \int P_1 P_5 d\nu &= v^3(15 - 30 + 15) = 0, \\ \int P_3 P_5 d\nu &= v^4(105 - 195 + 135 - 45) = 0. \end{aligned} \quad (\text{TEC.10})$$

For the diagonal integrals, direct polynomial multiplication gives

$$\begin{aligned}
\int P_0^2 d\nu &= 1, & \int P_1^2 d\nu &= v, \\
\int P_2^2 d\nu &= v^2(3 - 2 + 1) = 2v^2, \\
\int P_3^2 d\nu &= v^3(15 - 18 + 9) = 6v^3, \\
\int P_4^2 d\nu &= v^4(105 - 180 + 126 - 36 + 9) = 24v^4, \\
\int P_5^2 d\nu &= v^5(939 - 2100 + 1950 - 900 + 225) = 114v^5.
\end{aligned} \tag{TEC.11}$$

In (TEC.9), the diagonal phase is $(-i)^{k_i k} = 1$. These equations therefore prove every entry of the Gram matrix asserted in (TEC.7). Each polynomial in that table is monic. A nonzero polynomial of degree at most five has at most five distinct roots, so it cannot vanish at all nine atoms in (TEC.1). Positivity of the weights makes the inner product positive definite on polynomials of degree at most five. If two monic polynomials of the same degree were both orthogonal to every lower degree polynomial, their difference would have lower degree and zero squared norm. Positive definiteness forces that difference to vanish. Thus (TEC.7) consists of the actual monic orthogonal polynomials and norms of the stated measure, including the fifth norm. $\square$

Polynomial multiplication in (TEC.7) also proves

$$sq_k = q_{k+1} + \frac{1}{2}q_k - kvq_{k-1} \quad (0 \leq k \leq 4), \tag{TEC.12}$$

where the last term is omitted for $k = 0$. For $1 \leq k \leq 4$, the displayed coefficient is exactly $a_k = \kappa_k / \kappa_{k-1} = kv = k/16 > 0$. Thus (TEC.12) has the required vertical sign and $b_k + \overline{b_k} = 1$, with $b_k = 1/2$. The fifth norm in (TEC.7) is $\kappa_5 = 114/16^5 = 57/524288$; its value was derived from the actual tenth moment $939/16^5$.

11.7.3 The retained packet and its exact evaluation isomorphism

Set

$$h(s) = (s - \frac{3}{4})(s - \frac{1}{4}) = s^2 - s + \frac{3}{16}, \quad E = \mathbb{C}[s]/(h), \quad j_h : \mathbb{C}[s] \longrightarrow E. \tag{TEC.13}$$

Both packet centres lie in $0 < \operatorname{Re} s < 1$, and the reflection $\rho \mapsto 1 - \overline{\rho}$ exchanges them. They are distinct, and each has its complete multiplicity one. In the retained monomial basis $(1, s)$ of E , the unit and multiplication operator are

$$c = j_h 1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad A = M_s = \begin{pmatrix} 0 & -3/16 \\ 1 & 1 \end{pmatrix}. \tag{TEC.14}$$

Indeed $s^2 = s - 3/16$ in this quotient, which proves both columns of the multiplication matrix.

The complete evaluation map is

$$\begin{aligned}
T : E &\longrightarrow \mathbb{C} \oplus \mathbb{C}, & [p] &\longmapsto (p(3/4), p(1/4)), \\
T &= \begin{pmatrix} 1 & 3/4 \\ 1 & 1/4 \end{pmatrix}, & T^{-1} &= \begin{pmatrix} -1/2 & 3/2 \\ 2 & -2 \end{pmatrix}.
\end{aligned} \tag{TEC.15}$$

Both evaluations vanish on the ideal (h) , so the map is well defined. Evaluations preserve addition, multiplication and the unit. For any pair (u_+, u_-) , the polynomial

$$2u_+(s - \frac{1}{4}) + 2u_-(\frac{3}{4} - s) \tag{TEC.16}$$

evaluates to that pair. It has the coefficients given by T^{-1} , and multiplication of the two matrices in (TEC.15) gives the identity. This proves the typed algebra isomorphism, including its inverse. Evaluation of sp also proves

$$TA = \tilde{A}T, \quad \tilde{A} = \text{diag}(3/4, 1/4), \quad Tc = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (\text{TEC.17})$$

In the original quotient E one has $x^2 = v$. Reducing the explicit polynomials of (TEC.7), with no deletion of their supported residue classes, gives

k	$v_k := j_h q_k$ in the monomial basis $(1, s)$
0	$(1, 0)^T$
1	$(-1/2, 1)^T$
2	$(1/8, 0)^T$
3	$(-1/8, 1/4)^T$
4	$(5/128, 0)^T$
5	$(-13/256, 13/128)^T$

(TEC.18)

For example $q_3 = 4vx = x/4$, $q_4 = 10v^2 = 5/128$, and $q_5 = 26v^2x = (13/128)x$ in E . These calculations establish every entry of (TEC.18) directly from the original ideal relation.

11.7.4 The original kernels, terminal defects and exact energies

For $1 \leq N \leq 4$, define

$$\begin{aligned} K_N &= \sum_{k=0}^N \frac{v_k v_k^*}{\kappa_k}, \\ L_N &= AK_N + K_N A^* - K_N, \\ \epsilon_N &= \left\| K_N^{-1/2} L_N K_N^{-1/2} \right\|. \end{aligned} \quad (\text{TEC.19})$$

All adjoints in (TEC.19) are conjugate transposes in the explicitly retained monomial coordinates. The columns v_0, v_1 are linearly independent. If a nonzero column u had $u^* K_N u = 0$, the positive sum in (TEC.19) would force both $v_0^* u = 0$ and $v_1^* u = 0$, a contradiction. Thus each K_N is positive definite, so its positive inverse square root and the displayed Hermitian operator norm are well defined.

Using (TEC.7) and (TEC.18), the successive kernels are

$$\begin{aligned} K_1 &= \begin{pmatrix} 5 & -8 \\ -8 & 16 \end{pmatrix}, & K_2 &= \begin{pmatrix} 7 & -8 \\ -8 & 16 \end{pmatrix}, \\ K_3 &= \begin{pmatrix} 53/3 & -88/3 \\ -88/3 & 176/3 \end{pmatrix}, & K_4 &= \begin{pmatrix} 131/6 & -88/3 \\ -88/3 & 176/3 \end{pmatrix}. \end{aligned} \quad (\text{TEC.20})$$

In particular the last step is the exact positive rank-one update

$$K_4 - K_3 = \frac{v_4 v_4^*}{\kappa_4} = \frac{25}{6} cc^*. \quad (\text{TEC.21})$$

Multiplication by the matrix in (TEC.14) gives

$$L_3 = \begin{pmatrix} -20/3 & 20/3 \\ 20/3 & 0 \end{pmatrix}, \quad L_4 = \begin{pmatrix} -65/6 & 65/6 \\ 65/6 & 0 \end{pmatrix}. \quad (\text{TEC.22})$$

These defects satisfy the original terminal identity, with its exact coefficients and signs:

$$L_N = \frac{v_{N+1} v_N^* + v_N v_{N+1}^*}{\kappa_N} \quad (N = 3, 4). \quad (\text{TEC.23})$$

For $N = 3$, the numerator calculated from (TEC.18) is

$$\begin{pmatrix} -5/512 & 5/512 \\ 5/512 & 0 \end{pmatrix}, \quad \kappa_3 = 3/2048.$$

For $N = 4$, it is

$$\begin{pmatrix} -65/16384 & 65/16384 \\ 65/16384 & 0 \end{pmatrix}, \quad \kappa_4 = 3/8192.$$

Division gives exactly the two matrices in (TEC.22), proving (TEC.23) without an appeal to a numerical rank calculation.

Theorem 11.4 (Strict increase at two consecutive permitted levels). *For the positive measure (TEC.1) and the complete reflection-stable packet (TEC.13), the energies in (TEC.19) satisfy*

$$\epsilon_3^2 = \frac{25}{99}, \quad \epsilon_4^2 = \frac{4225}{15136}, \quad \epsilon_4^2 - \epsilon_3^2 = \frac{39875}{1498464} > 0. \quad (\text{TEC.24})$$

In the indexing $N = d + m$, with $d = \dim_{\mathbb{C}} E = 2$, these are the consecutive levels $m = 1$ and $m = 2$, both with $m \geq 0$.

Proof. Transport the two forms by the proved isomorphism (TEC.15):

$$\begin{aligned} \tilde{K}_3 &= TK_3T^* = \begin{pmatrix} 20/3 & -2/3 \\ -2/3 & 20/3 \end{pmatrix}, \\ \tilde{K}_4 &= TK_4T^* = \begin{pmatrix} 65/6 & 7/2 \\ 7/2 & 65/6 \end{pmatrix}. \end{aligned} \quad (\text{TEC.25})$$

By (TEC.17), the transported defect is

$$\tilde{L}_N = TL_NT^* = \tilde{A}\tilde{K}_N + \tilde{K}_N\tilde{A}^* - \tilde{K}_N. \quad (\text{TEC.26})$$

For either matrix $\tilde{K} = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$ in (TEC.25), one has

$$\tilde{L} = \text{diag}(a/2, -a/2), \quad \det(\tilde{L} - \lambda\tilde{K}) = \lambda^2(a^2 - b^2) - a^2/4. \quad (\text{TEC.27})$$

To check that this calculation gives the norm defined in the original coordinates, set $H = K^{-1/2}LK^{-1/2}$. Then

$$\det(H - \lambda I) = \det(K)^{-1} \det(L - \lambda K), \quad \det(T(L - \lambda K)T^*) = |\det T|^2 \det(L - \lambda K). \quad (\text{TEC.28})$$

The two determinant factors are nonzero and independent of λ . Consequently the eigenvalues of the Hermitian matrix H equal the roots of (TEC.27), with their multiplicities. The roots are the opposite real numbers $\pm a/(2\sqrt{a^2 - b^2})$. A Hermitian matrix has operator norm equal to the greatest absolute value of its eigenvalues, so

$$\epsilon^2 = \frac{a^2}{4(a^2 - b^2)}. \quad (\text{TEC.29})$$

For $N = 3$, substitution of $a = 20/3, b = -2/3$ gives $\epsilon_3^2 = (400/9)/(4 \cdot 396/9) = 25/99$. For $N = 4$, substitution of $a = 65/6, b = 7/2$ gives $\epsilon_4^2 = (4225/36)/(4 \cdot 3784/36) = 4225/15136$. Their difference, retaining the two denominators, is

$$\frac{4225 \cdot 99 - 25 \cdot 15136}{15136 \cdot 99} = \frac{418275 - 378400}{1498464} = \frac{39875}{1498464} > 0. \quad (\text{TEC.30})$$

Both energies are nonnegative, and their squared values have strictly positive difference, so $\epsilon_4 > \epsilon_3$. The original coordinate characteristic polynomials provide the same calculation:

$$\begin{aligned} \det(L_3 - \lambda K_3) &= 176\lambda^2 - 400/9, \\ \det(L_4 - \lambda K_4) &= (3784/9)\lambda^2 - 4225/36. \end{aligned} \quad (\text{TEC.31})$$

Finally $N = 3 = 2 + 1$ and $N = 4 = 2 + 2$, proving the stated degree range without using the lower degree $N = 1$. $\square$

11.7.5 The exact scope of the obstruction

For this measure the construction is the composite of the polynomial source with the original residue quotient

$$\mathcal{P}_N \hookrightarrow \mathbb{C}[s] \xrightarrow{j_h} E, \quad E \xrightarrow{T} \mathbb{C} \oplus \mathbb{C}, \quad (\text{TEC.32})$$

and multiplication is intertwined exactly by (TEC.17). Its source inner product is (TEC.6); its monic source Gram matrix is $\text{diag}(\kappa_0, \dots, \kappa_N)$; its kernel is the full sum (TEC.19). Thus it has precisely the vertical recurrence, positive kernel update, complete packet quotient and terminal defect maps appearing in the arithmetic moment construction. Equations (TEC.24) and (TEC.30) prove that these data alone do not imply nonincreasing consecutive energy. An argument for that property for ν_Z must therefore establish and use additional identities or estimates of its stated arithmetic density. The present example establishes the indicated obstruction to a universal recurrence-based argument; it asserts no consecutive increase for ν_Z .

The companion exact checker verifies 88 rational and complex-rational identities in both ordinary Python and optimized Python. It checks the atom weights and relevant moments, source Gram matrix, packet isomorphism, residue columns, kernel updates, terminal identities and generalized characteristic polynomials. Those finite checks supplement the complete computations above.

12 The exact metric departure and the cost of complete jets

Retain the original packet $E = \mathbb{C}[s]/h$, its polynomial coordinates and $A = M_s$. Write its eigenvalues with their algebraic multiplicities as $\rho_1, \dots, \rho_d$, and put $\delta_j = \text{Re } \rho_j - 1/2$. No eigenvalue or repeated factor is discarded. For a positive Hermitian matrix G in these coordinates define

$$\begin{aligned} W_G &= A^*G + GA - G, \quad S_G = G^{-1/2}W_GG^{-1/2}, \quad \epsilon_G = \|S_G\|_{\text{op}}, \\ \mathfrak{d}_G(A) &= \text{Tr}(G^{-1}A^*GA) - \sum_{j=1}^d |\rho_j|^2. \end{aligned} \quad (\text{MD1})$$

The positive square root is used only for the explicit isometry $x \mapsto G^{1/2}x$ from the specified metric space (E, G) to the Euclidean coordinate space. The original A, G , their entries and their maps remain those in (MD1).

Theorem 12.1 (Exact departure identity). *The number $\mathfrak{d}_G(A)$ is real and nonnegative, and*

$$\text{Tr } S_G = 2 \sum_j \delta_j, \quad \text{Tr}(S_G^2) = 4 \sum_j \delta_j^2 + 2\mathfrak{d}_G(A). \quad (\text{MD2})$$

*It vanishes precisely when A is normal for the specified metric, that is, when $A(G^{-1}A^*G) = (G^{-1}A^*G)A$. For the original orthogonal representatives R_m , their matrices G_m and rank-two forms W_m , one has exactly*

$$\epsilon_m = \left| \sum_j \delta_j \right| + \sqrt{2 \sum_j \delta_j^2 + \mathfrak{d}_{G_m}(A) - \left(\sum_j \delta_j \right)^2}. \quad (\text{MD3})$$

For a reflection-stable packet this becomes

$$\boxed{\epsilon_m^2 = 2 \sum_{\rho \in Z} m_\rho (\text{Re } \rho - 1/2)^2 + \mathfrak{d}_{G_m}(A).} \quad (\text{MD4})$$

Proof. Let $C = G^{1/2}AG^{-1/2}$. Then $S_G = C + C^* - I$ and $\text{Tr}(G^{-1}A^*GA) = \text{Tr}(C^*C)$. For completeness, choose a unit eigenvector of C , extend it to an orthonormal basis, and repeat on the lower-right compression. Induction gives a unitary matrix U with upper-triangular $T = U^*CU$ whose diagonal lists the same eigenvalues with their algebraic multiplicities. Unitary invariance of the trace gives

$$\mathfrak{d}_G(A) = \sum_{i < j} |T_{ij}|^2 \geq 0. \quad (\text{MD5})$$

If it is zero, T is diagonal and C is normal. Conversely, an upper-triangular normal matrix is diagonal: its first row norm equals its first column norm, eliminating the strict first-row entries, and induction eliminates every remaining such entry. Thus normality of C is equivalent to the stated equality for A , through the specified isometry.

Expanding $(C + C^* - I)^2$ under the trace gives

$$\text{Tr}(S_G^2) = 2 \text{Tr}(C^*C) + 2 \text{Re} \text{Tr}(C^2) - 4 \text{Re} \text{Tr} C + d.$$

The traces of C and C^2 are respectively $\sum_j \rho_j$ and $\sum_j \rho_j^2$, even in the presence of nilpotents. Since $2|\rho|^2 + 2 \text{Re} \rho^2 - 4 \text{Re} \rho + 1 = 4(\text{Re} \rho - 1/2)^2$, this proves (MD2). The trace formula follows directly from $S_G = C + C^* - I$.

For G_m , Section 5 proved $\text{rank } S_{G_m} \leq 2$. List its two possibly nonzero real eigenvalues a, b , appending zero if necessary (also when $d = 1$). Equations (MD2) give

$$a + b = 2 \sum_j \delta_j, \quad (a - b)^2/4 = 2 \sum_j \delta_j^2 + \mathfrak{d}_{G_m}(A) - \left(\sum_j \delta_j\right)^2.$$

Thus the radicand is nonnegative, and $\max(|a|, |b|) = |(a + b)/2| + |a - b|/2$, proving (MD3). Reflection-stability pairs δ with $-\delta$, with the same multiplicities, proving (MD4). $\square$

The actual arithmetic kernel of Section 8 also gives this quantity without changing the local unit. Indeed $G_m = U_\varepsilon^* K_N^{-1} U_\varepsilon$, $N = d + m$, and $U_\varepsilon A = AU_\varepsilon$. Cyclicity of trace gives

$$\mathfrak{d}_{G_m}(A) = \text{Tr}(K_N A^* K_N^{-1} A) - \sum_j |\rho_j|^2. \quad (\text{MD6})$$

To verify the cancellation, substitute $G_m^{-1} = U_\varepsilon^{-1} K_N (U_\varepsilon^*)^{-1}$ in (MD1), commute A with U_ε and A^* with U_ε^* , and cancel the outer similar factors inside the trace. This calculation removes no source representative: their exact recovery remains (TB.7)–(TB.9).

12.1 A quantitative constraint on every retained nilpotent chain

For a selected root ρ of full order r , let

$$I_\rho : \mathbb{C}[z]/z^r \longrightarrow E$$

be the original Chinese-remainder inclusion: the ρ -component is the given Taylor polynomial in $z = s - \rho$, and every other component is zero. Its Gram matrix is $G_\rho = I_\rho^* G I_\rho$, in the basis $1, z, \dots, z^{r-1}$. Put $N_\rho z^k = z^{k+1}$, with $z^r = 0$. The exact intertwining relation is $AI_\rho = I_\rho(\rho I + N_\rho)$.

Theorem 12.2 (Full-jet metric inequalities). *For every positive original metric G , write $\delta = \text{Re} \rho - 1/2$. Then $\epsilon_G \geq 2|\delta|$, and for all $0 \leq k < k + \ell < r$,*

$$\frac{(G_\rho)_{k+\ell, k+\ell}}{(G_\rho)_{k, k}} \leq \left[\frac{r}{2} (\epsilon_G^2 - 4\delta^2) \right]^\ell. \quad (\text{MD7})$$

In particular, for $r \geq 2$,

$$\kappa_2(G_\rho) \geq \left[\frac{r}{2} (\epsilon_G^2 - 4\delta^2) \right]^{-(r-1)}. \quad (\text{MD8})$$

The bracket is strictly positive when $r \geq 2$. Here κ_2 is the ratio of the largest and smallest eigenvalues in the specified Taylor coordinates.

Proof. The definition of ϵ_G is the two-sided form inequality $-\epsilon_G G \leq W_G \leq \epsilon_G G$. On an eigenvector it gives $\epsilon_G \geq 2|\delta|$. Pull this same inequality back by I_ρ . Let $L = G_\rho^{1/2} N_\rho G_\rho^{-1/2}$ and $T = 2\delta I + L + L^*$. The pulled-back inequality gives $\|T\| \leq \epsilon_G$. Nilpotence retains $\text{Tr } L = \text{Tr}(L^2) = 0$, whence

$$\text{Tr } T = 2r\delta, \quad \|L\|_F^2 = \frac{1}{2} \text{Tr}(L + L^*)^2 = \frac{1}{2} \text{Tr } T^2 - 2r\delta^2 \leq \frac{r}{2}(\epsilon_G^2 - 4\delta^2). \quad (\text{MD9})$$

The first equality for the Frobenius norm follows by expanding the square and using the two zero traces. The final inequality uses the r real eigenvalues of T , each bounded in modulus by ϵ_G . For the original metric norm, $\|N_\rho\|_{G_\rho} = \|L\| \leq \|L\|_F$. Apply its ℓ -th power to the specified basis vector z^k . Since $N_\rho^\ell z^k = z^{k+\ell}$, the resulting squared norm inequality is exactly (MD7).

When $r \geq 2$, $N_\rho \neq 0$, so (MD9) makes the bracket strictly positive. Put $k = 0, \ell = r - 1$. The Rayleigh inequalities give $\kappa_2(G_\rho) \geq (G_\rho)_{0,0}/(G_\rho)_{r-1,r-1}$; combine with (MD7) to obtain (MD8). $\square$

All entries in (MD7) for the original sequence are explicitly $I_\rho^* U_\epsilon^* K_N^{-1} U_\epsilon I_\rho$. Thus this theorem supplies constraints on the actual moment calculation, with its original Taylor factors. At a critical root of order $r \geq 2$, a sequence with relative control approaching zero necessarily has the stated diverging Taylor-metric condition number. The surviving jets have not been erased; the inequalities calculate their relative metric scales. This is compatible with the complete graph and weighted-jet maps in Section 15, which retain those same coordinates.

13 Coherent packets, joint tensor layers, and their exact spectral cost

This section integrates the complete arithmetic interpolation construction of the 12 September coherent-interpolation delivery, whose source revision is 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc, and proves its comparison with the preceding kernel and departure calculations. All zeros in a packet retain their complete multiplicities. A reflected packet is required only in statements explicitly using its reflection. The original spaces $V, \mathcal{B}$, map Θ , generator $D = -x\partial_x$, and Hilbert observation $L^2(dx)$ remain fixed.

13.1 The arithmetic unit and the coherent one-variable maps

For $h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$, write

$$d = \deg h, \quad E_h = \mathbb{C}[s]/h, \quad A_h = M_s, \quad v_h = g/h, \quad v_h = j_h v_h, \quad \varepsilon_h = v_h^{-1}, \quad g = 2\xi. \quad (\text{CT.1})$$

Complete multiplicities make v_h invertible. Let $F_h \in \mathcal{B}$ be the original inverse satisfying $h(D)F_h = f_0$ and $\mathcal{M}F_h = g/h$, and put $\mathcal{T}_h P = P(D)F_h$. The integral construction of F_h , its estimates at both ends, and injectivity of the Mellin transform are proved in Section 5. They give

$$\mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h P = v_h[P]_h. \quad (\text{CT.2})$$

Let $C_N : \mathcal{P}_N \rightarrow E_h$ be the full remainder map in the original monomial coordinates, let $M_{h,N}$ be the Gram matrix of $\mathcal{P}_N$ for $d\nu_h = |g(1/2 + it)/h(1/2 + it)|^2 dt/(2\pi)$, and use U_a for multiplication by a on E_h . The two kernel matrices occurring in the constructions have the exact types

$$K_N = C_N M_{h,N}^{-1} C_N^*, \quad J_{h,N} = U_{v_h} C_N, \quad \mathcal{C}_{h,N} = U_{v_h} K_N U_{v_h}^*. \quad (\text{CT.3})$$

Consequently their actual representative and metric coincide:

$$\begin{aligned} M_{h,N}^{-1} J_{h,N}^* \mathcal{C}_{h,N}^{-1} &= M_{h,N}^{-1} C_N^* K_N^{-1} U_{\varepsilon_h}, \\ G_{h,n} &= \mathcal{C}_{h,n+d}^{-1} = U_{\varepsilon_h}^* K_{n+d}^{-1} U_{\varepsilon_h}. \end{aligned} \quad (\text{CT.4})$$

Indeed invert the congruence in (CT.3) and cancel the adjacent inverse factors. The first line is exactly the polynomial numerator in (TB.7)–(TB.9). Its difference from $\text{rem}_h(\varepsilon_h u)$ is divisible by h and has

quotient of degree at most n . Applying (CT.2) gives the same original theta correction. Thus the arithmetic unit remains in every coefficient map and every source representative.

For completeness, the minimum assertion follows directly without an existence assumption. For any positive Gram matrix M and surjective J , set $G = (JM^{-1}J^*)^{-1}$ and $L = M^{-1}J^*G$. Then $JL = I$, $L^*ML = G$, and for $Jz = 0$, $z^*ML = z^*J^*G = 0$. Every feasible vector is $Lu + z$, so

$$(Lu + z)^*M(Lu + z) = u^*Gu + z^*Mz. \quad (\text{CT.5})$$

Positivity makes this the unique minimum. In (CT.3), positivity holds because a nonzero polynomial cannot vanish almost everywhere on the original integration line; surjectivity follows from complete polynomial remainders. The kernel is exactly $h\mathcal{P}_n$ when $N = n + d$. Hence $R_{h,n} = (I - P_{L_n})s_h$, where $s_h = \mathcal{T}_h \text{rem}_h(\varepsilon_h \cdot)$ and $L_n = \Theta \text{span}(\phi_*, \dots, D^n \phi_*)$.

If $h \mid H$ are full packets, put $m = H/h$. The factors h, m are coprime. The original CRT injection $\iota : E_h \rightarrow E_H$ inserts zero jets at the additional centres, and satisfies $A_H \iota = \iota A_h$. Multiplication $P \mapsto mP$ is an isometry of the declared polynomial images, since $(g/H)mP = (g/h)P$; it satisfies

$$\mathcal{T}_H(mP) = \mathcal{T}_h P, \quad J_H \mathcal{T}_H(mP) = \iota J_h \mathcal{T}_h P. \quad (\text{CT.6})$$

At the added centres the second identity follows from the full zeros of g/h ; at the old centres it is the identity of entire functions. Both numerators for $s_H \iota = \mathcal{T}_H(m \text{rem}_h(\varepsilon_h \cdot))$ have degree below $\deg H$ and the same complete jets, proving equality. Since L_n is independent of the packet,

$$R_{H,n} \iota = R_{h,n}, \quad \iota^* G_{H,n} \iota = G_{h,n}, \quad \iota^* W_{H,n} \iota = W_{h,n}. \quad (\text{CT.7})$$

These are congruences, with no interchange of inversion and compression. Their split maps at a retained support λ are $(\lambda, u) \mapsto (\lambda, \iota u)$; all added zero jets are represented zeros at that support.

13.2 The original reflection and the exact diagonal coefficient

The original arithmetic density $|g(1/2 + it)|^2$ is even in t , since the real Taylor coefficients of g give $g(1/2 - it) = \overline{g(1/2 + it)}$. If Q_j is the monic orthogonal polynomial for that density in the variable s , the polynomial $(-1)^j Q_j(1 - s)$ is monic and orthogonal to all lower degrees, by the change $t \mapsto -t$. Uniqueness therefore gives

$$Q_j(1 - s) = (-1)^j Q_j(s), \quad \int t |Q_j(1/2 + it)|^2 |g(1/2 + it)|^2 \frac{dt}{2\pi} = 0. \quad (\text{CT.8})$$

The diagonal recurrence coefficient is thus exactly $\langle Q_j, sQ_j \rangle / \|Q_j\|^2 = 1/2$. For lower degrees the identity $D^* + D = I$ eliminates every coefficient except the adjacent one and gives its sign and actual norm ratio. More explicitly, sQ_j is orthogonal to Q_i for $i \leq j - 2$, because its inner product is the inner product of Q_j with $(1 - s)Q_i$. For $i = j - 1$ the leading term gives coefficient $-\|Q_j\|^2 / \|Q_{j-1}\|^2$. This proves, with the original vectors and norms,

$$Du_j = u_{j+1} + \frac{1}{2}u_j - \frac{\mathfrak{h}_j}{\mathfrak{h}_{j-1}}u_{j-1}, \quad u_j = Q_j(D)f_0, \quad \mathfrak{h}_j = \|u_j\|^2. \quad (\text{CT.9})$$

The final term is omitted at $j = 0$. The even density in (CT.8) is the additional reason the imaginary part of the diagonal coefficient vanishes; the adjoint identity by itself fixes only its real part.

For a packet stable under $\rho \mapsto 1 - \bar{\rho}$, including the same complete orders, the original maps are

$$\mathfrak{j}_h u(s) = \overline{u(1 - \bar{s})} \pmod{h}, \quad \mathfrak{j}_{\mathcal{B}} F(x) = x^{-1} \overline{F(1/\bar{x})}.$$

The second is antiunitary for dx , and $D\mathfrak{j}_{\mathcal{B}} = \mathfrak{j}_{\mathcal{B}}(I - D)$. The identities $h^\dagger = (-1)^d h$ and $g^\dagger = g$ imply $(g/h)^\dagger = (-1)^d g/h$; they give $\mathfrak{j}_{\mathcal{B}} s_h = s_h \mathfrak{j}_h$ by the same complete jets and degree bounds as

above. Equation (CT.9), or invariance of the polynomial space under $s \mapsto 1 - s$, shows $j_{\mathcal{B}}L_n = L_n$. The unique minimum (CT.5) then commutes with these maps. If $j_h u = C_h \bar{u}$,

$$C_h^* G_{h,n} C_h = \overline{G_{h,n}}, \quad A_h C_h = C_h (I - \overline{A_h}), \quad C_h^* W_{h,n} C_h = -\overline{W_{h,n}}. \quad (\text{CT.10})$$

Substitution into a quadratic form proves that an upper bound and its reflected lower bound use this same $G_{h,n}$.

13.3 The joint total-degree source and its relation layer

Fix $k \geq 1$, $M \geq k(d-1)$, and set

$$E^{(k)} = E_h^{\otimes k}, \quad A^{(k)} = \sum_{j=1}^k A_{h,j}, \quad \mathcal{P}_M^{(k)} = \mathbb{C}[s_1, \dots, s_k]_{\leq M}, \quad \mathcal{T}_h^{(k)} P = P(D_1, \dots, D_k) F_h^{\otimes k}.$$

Here $\leq M$ means total degree. The full jet map is reduction by each $h(s_j)$, followed by multiplication by $v_h^{\otimes k}$, and its kernel is

$$\mathcal{I}_{h,k,M} = \mathcal{P}_M^{(k)} \cap (h(s_1), \dots, h(s_k)). \quad (\text{CT.11})$$

Successive monic division in the declared order $s_1, \dots, s_k$ gives $P = \sum_j h(s_j) P_j$ in this kernel, with $\deg P_j \leq M - d$. Division lowers total degree at every replacement, so this bound holds at every division step. In the original complex $[V \rightarrow \mathcal{B}]$, placed in degrees 0, 1, the primitive for the j -th term uses $(-1)^{j-1} \phi_*$ in the j -th factor and F_h in the others, with the polynomial $P_j(D_1, \dots, D_k)$ applied to the whole tensor. The preceding $j-1$ factors have degree one, so the tensor differential supplies the further factor $(-1)^{j-1}$. Its boundary is exactly $+\mathcal{T}_h^{(k)}(h(s_j)P_j)$.

Write $\mathcal{B}_M = \mathcal{T}_h^{(k)} \mathcal{I}_{h,k,M}$ inside $\mathcal{H}_k = L^2(d^k x)$. The arithmetic moment matrix in the retained monomials is

$$(M_{k,M})_{\alpha\beta} = \prod_{j=1}^k \int \overline{s^{\alpha_j}} s^{\beta_j} d\nu_h, \quad G_M = (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1}. \quad (\text{CT.12})$$

Product integration proves the first formula. A polynomial vanishing almost everywhere for this product measure is zero, by successive one-variable uniqueness, proving positivity. The complete remainder basis has total degree at most $k(d-1)$, proving surjectivity. Thus (CT.5) proves

$$R_M = (I - P_{\mathcal{B}_M}) s_h^{\otimes k}, \quad J^{(k)} R_M = I, \quad q^{(k)} R_M = \sigma_h^{\otimes k}. \quad (\text{CT.13})$$

The kernel dimension is $\binom{M+k}{k} - d^k$. Since $\mathcal{T}_h^{(k)}$ is injective by its entire Mellin multiplier, the orthogonal relation layer $\mathcal{E}_M = \mathcal{B}_{M+1} \cap \mathcal{B}_M^\perp$ has exactly

$$\dim \mathcal{E}_M = \binom{M+k}{k-1}. \quad (\text{CT.14})$$

Define all of the following maps on the original $E^{(k)}$:

$$B_M = D^{(k)} R_M - R_M A^{(k)}, \quad Y_M = P_{\mathcal{E}_M} R_M, \quad C_M = (I - P_{\mathcal{B}_M}) B_M. \quad (\text{CT.15})$$

The complete jet of B_M is zero and its numerator has degree at most $M+1$, so $B_M(E^{(k)}) \subseteq \mathcal{B}_{M+1}$. Both Y_M, C_M therefore have target $\mathcal{E}_M$. Integration by parts on these rapidly decreasing functions gives $(D^{(k)})^* + D^{(k)} = kI$. Substitution into $R_M^*(D^{(k)} + (D^{(k)})^*) R_M = kG_M$, followed by orthogonality to $\mathcal{B}_M$, proves

$$\begin{aligned} W_M &= (A^{(k)})^* G_M + G_M A^{(k)} - kG_M = -(Y_M^* C_M + C_M^* Y_M), \\ \Delta_M &= G_M - G_{M+1} = Y_M^* Y_M \succeq 0. \end{aligned} \quad (\text{CT.16})$$

For the second identity use the orthogonal decomposition of $\mathcal{B}_{M+1}$ and $R_{M+1} = R_M - Y_M$. The actual quotient map $[z] \mapsto (I - P_{\mathcal{B}_M})z$ identifies $\mathcal{B}_{M+1}/\mathcal{B}_M$ with $\mathcal{E}_M$, with inverse the inclusion followed by quotient. A transported boundary is zero at its next support and remains represented there; this map sends no supported input to external τ .

For a reflection-stable packet, put $C_{h,k} = C_h^{\otimes k}$. The original product reflection acts on a numerator by $(-1)^{kd} \overline{P(1 - \overline{s_1}, \dots, 1 - \overline{s_k})}$. This retains total degree and preserves the ideal in (CT.11), with its displayed factor. Its action on $\mathcal{H}_k$ is antiunitary. The unique minimum (CT.5) therefore commutes with reflection and gives

$$C_{h,k}^* G_M C_{h,k} = \overline{G_M}, \quad A^{(k)} C_{h,k} = C_{h,k} (kI - \overline{A^{(k)}}), \quad C_{h,k}^* W_M C_{h,k} = -\overline{W_M}. \quad (\text{CT.16a})$$

If $W_M u = \lambda G_M u$, conjugation followed by $(C_{h,k}^*)^{-1}$ in the last identity gives $W_M C_{h,k} \overline{u} = -\lambda G_M C_{h,k} \overline{u}$. All generalized eigenvalues are real because $G_M > 0$. This invertible antilinear map pairs their full multiplicities. The source-layer maps themselves retain the reflection types. If $j_E = j_h^{\otimes k}$ and $j_{\mathcal{E}}$ is the restriction of the original function reflection to $\mathcal{E}_M$, then

$$j_{\mathcal{E}} Y_M = Y_M j_E, \quad j_{\mathcal{E}} C_M = -C_M j_E. \quad (\text{CT.16b})$$

Both boundary projections commute with reflection, proving the first identity. For the second use $D^{(k)} j_{\mathcal{H}} = j_{\mathcal{H}} (kI - D^{(k)})$, $A^{(k)} j_E = j_E (kI - A^{(k)})$, and $j_{\mathcal{H}} R_M = R_M j_E$ in (CT.15). Expanding its two terms gives $j_{\mathcal{H}} B_M = -B_M j_E$; the projection then proves (CT.16b).

13.4 Nested packets on the joint source: the additional projection

Let $h \mid H$, $q = \deg(H/h)$, $N = M + kq$, and $\iota_k = \iota_{hH}^{\otimes k}$. Multiplication of a numerator by $\prod_{j=1}^k (H/h)(s_j)$ gives a degree-preserving-with-shift map $\mathcal{P}_M^{(k)} \rightarrow \mathcal{P}_N^{(k)}$, with

$$\mathcal{T}_H^{(k)} (\prod_j (H/h)(s_j) P) = \mathcal{T}_h^{(k)} P, \quad J_H^{(k)} \mathcal{T}_h^{(k)} P = \iota_k J_h^{(k)} \mathcal{T}_h^{(k)} P. \quad (\text{CT.17})$$

The second equality follows on pure monomials from (CT.6) and then by linearity. In particular it maps the smaller relation space into the larger one in the same $\mathcal{H}_k$. Let

$$\mathcal{F} = \mathcal{B}_{H,k,N} \cap \mathcal{B}_{h,k,M}^{\perp}, \quad Z = P_{\mathcal{F}} R_{h,k,M}.$$

Since $s_H^{\otimes k} \iota_k = s_h^{\otimes k}$, the exact comparison is

$$\begin{aligned} R_{H,k,N} \iota_k &= R_{h,k,M} - Z, \\ \iota_k^* G_{H,k,N} \iota_k &= G_{h,k,M} - Z^* Z, \\ \iota_k^* W_{H,k,N} \iota_k &= W_{h,k,M} - ((A_h^{(k)})^* Z^* Z + Z^* Z A_h^{(k)} - k Z^* Z). \end{aligned} \quad (\text{CT.18})$$

Indeed nested orthogonal projections commute, and their difference is $P_{\mathcal{F}}$; apply them to the common section to prove the first line. Pythagoras proves the second; intertwining of $A^{(k)}$ and direct expansion prove the third. Also

$$\dim \mathcal{F} = \binom{N+k}{k} - (\deg H)^k - \binom{M+k}{k} + d^k. \quad (\text{CT.19})$$

This is the difference of the exact kernel dimensions. For $k = 1$, $M = n + d$, both boundary spaces are the original L_n , so $\mathcal{F} = 0$ and (CT.18) recovers (CT.7). For joint tensors (CT.18) retains every additional allowed boundary and calculates its effect on the control form.

13.5 Positive layer rank and the complete tensor spectral departure

List $\rho_1, \dots, \rho_d$ with all algebraic multiplicities, put $\delta_i = \operatorname{Re} \rho_i - 1/2$, $\sigma_1 = \sum_i \delta_i$, $\sigma_2 = \sum_i \delta_i^2$, and define

$$\begin{aligned} S_M &= G_M^{-1/2} W_M G_M^{-1/2}, \quad \epsilon_M = \|S_M\|_{\text{op}}, \quad r_M = \operatorname{rank} \Delta_M, \\ \mathfrak{d}_M^{(k)} &= \operatorname{Tr}(G_M^{-1} (A^{(k)})^* G_M A^{(k)}) - \sum_{i_1, \dots, i_k} \left| \sum_{j=1}^k \rho_{i_j} \right|^2, \\ \mathcal{D}_k &= \sum_{i_1, \dots, i_k} \left| \sum_{j=1}^k \delta_{i_j} \right|. \end{aligned} \tag{CT.20}$$

All sums are ordered and retain repeated entries. The isometry $x \mapsto G_M^{1/2} x$ in this calculation leaves the original jet coordinates, source functions, metric and layer maps unchanged. For a map into $\mathcal{E}_M$, the Frobenius symbol below denotes its Hilbert–Schmidt norm for the layer’s inherited $L^2(d^k x)$ inner product.

Theorem 13.1 (Exact tensor departure and relative layer cost). *One has $\mathfrak{d}_M^{(k)} \geq 0$, and exactly*

$$\begin{aligned} \operatorname{Tr} S_M &= 2kd^{k-1}\sigma_1, \\ \operatorname{Tr}(S_M^2) &= 4\{kd^{k-1}\sigma_2 + k(k-1)d^{k-2}\sigma_1^2\} + 2\mathfrak{d}_M^{(k)}. \end{aligned} \tag{CT.21}$$

The positive and negative inertias of W_M each obey

$$n_+(W_M), n_-(W_M) \leq \min(\operatorname{rank} Y_M, \operatorname{rank} C_M), \quad \operatorname{rank} Y_M = r_M \leq \binom{M+k}{k-1}. \tag{CT.22}$$

The complete ordered spectral cost satisfies

$$\boxed{\mathcal{D}_k \leq \frac{1}{2}\|S_M\|_1 \leq \|Y_M G_M^{-1/2}\|_F \|C_M G_M^{-1/2}\|_F}, \quad \mathcal{D}_k \leq r_M \epsilon_M. \tag{CT.23}$$

In particular, the actual source matrices satisfy

$$\mathcal{D}_k^2 \leq \operatorname{Tr}(G_M^{-1} \Delta_M) \operatorname{Tr}(G_M^{-1} C_M^* C_M). \tag{CT.24}$$

If the packet is reflection-stable, put $p_M = n_+(W_M) = n_-(W_M)$. Then

$$p_M \leq \min\{r_M, \operatorname{rank} C_M, \lfloor d^k/2 \rfloor\}, \quad \mathcal{D}_k \leq p_M \epsilon_M. \tag{CT.24a}$$

In this case (CT.21) becomes $\operatorname{Tr}(S_M^2) = 4kd^{k-1}\sigma_2 + 2\mathfrak{d}_M^{(k)}$. When $p_M > 0$, it implies

$$\epsilon_M^2 \geq \frac{2kd^{k-1}\sigma_2 + \mathfrak{d}_M^{(k)}}{p_M}. \tag{CT.25}$$

Proof. Triangularize A_h over $\mathbb{C}$. The tensor product of this basis makes $A^{(k)}$ triangular, with diagonal entries $\sum_j \rho_{i_j}$. Thus these are all its eigenvalues with algebraic multiplicity; its nilpotent entries have not been discarded. Now apply the unitary triangularization argument of (MD5) to $T = G_M^{1/2} A^{(k)} G_M^{-1/2}$. The sum of squared absolute values of its strictly upper triangular entries is exactly $\mathfrak{d}_M^{(k)}$, proving nonnegativity. Since $S_M = T + T^* - kI$, expansion of its squared trace gives

$$\operatorname{Tr}(S_M^2) = 4 \sum_{i_1, \dots, i_k} \left(\sum_j \delta_{i_j} \right)^2 + 2\mathfrak{d}_M^{(k)}.$$

The k diagonal terms in this ordered sum contribute $kd^{k-1}\sigma_2$; the $k(k-1)$ ordered cross terms contribute $k(k-1)d^{k-2}\sigma_1^2$. The linear trace is computed in the same way. This proves (CT.21), including $k = 1$.

Write $\hat{Y} = Y_M G_M^{-1/2}$ and $\hat{C} = C_M G_M^{-1/2}$. Then $S_M = -(\hat{Y}^* \hat{C} + \hat{C}^* \hat{Y})$. Its form is zero on $\ker \hat{Y}$. A positive definite subspace of dimension greater than $\text{rank } \hat{Y}$ intersects this kernel nontrivially, a contradiction; the same argument holds for a negative definite subspace and with $\hat{C}$ in place of $\hat{Y}$. Sylvester inertia under the displayed invertible congruence proves the first part of (CT.22). The identity $\Delta_M = Y_M^* Y_M$ gives equality of their kernels, hence their ranks, proving the rest.

In the unitary triangular basis of T , the diagonal of S_M consists of $2 \sum_j \delta_{ij}$. For any Hermitian matrix S , its spectral decomposition gives $|S_{ii}| \leq \sum_a |\lambda_a| |U_{ia}|^2$. Sum over i to obtain $\sum_i |S_{ii}| \leq \sum_a |\lambda_a| = \|S\|_1$. This proves the first inequality in (CT.23). For finite matrices B, C with a common row space, $\|B^* C\|_1 \leq \|B\|_F \|C\|_F$: choose a singular-value decomposition of $B^* C$, use its left and right singular vectors to express its trace norm as $|\text{Tr}(UB^* C)|$ for a unitary U , and apply Cauchy-Schwarz to the entries of BU^* and C . The triangle inequality now proves the second inequality of (CT.23). Finally (CT.22) bounds the number of nonzero eigenvalues of S_M by $2r_M$, so $\|S_M\|_1 \leq 2r_M \epsilon_M$. Squaring the middle bound and writing its two Frobenius norms as traces proves (CT.24).

For a reflected packet $\sigma_1 = 0$, with every multiplicity paired. The exact reflection (CT.16a) pairs the nonzero eigenvalues of S_M , giving equal positive and negative counts p_M . Equation (CT.22) proves the bounds on p_M , and there are exactly $2p_M$ nonzero eigenvalues of S_M , each of square at most ϵ_M^2 . Their trace norm is at most $2p_M \epsilon_M$, proving (CT.24a); their squared trace proves (CT.25). If $p_M = 0$, the pairing implies $W_M = 0$ directly, and (CT.21) states that all the nonnegative terms vanish; no division by p_M is made. $\square$

Every quantity in (CT.24) is attached to the same arithmetic moment matrix and its next original relation layer. In particular, $0 \preceq \Delta_M \prec G_M$, because $G_{M+1} > 0$, so all eigenvalues of $G_M^{-1/2} \Delta_M G_M^{-1/2}$ lie in $[0, 1)$, and its trace is strictly smaller than r_M when $r_M > 0$. Its actual value is retained in (CT.24). Equations (CT.21)–(CT.25) calculate necessary relative layer costs and the full nilpotent departure for the constructed tensor family. They assert no unproved upper bound on these arithmetic quantities.

There is an exact determinant refinement of the same source loss. For $r_M > 0$, put

$$\pi_M = \frac{\det G_{M+1}}{\det G_M}, \quad \tau_M = \text{Tr}(G_M^{-1} \Delta_M), \quad \chi_M = \text{Tr}(G_M^{-1} C_M^* C_M).$$

Then

$$1 - \pi_M \leq \tau_M \leq r_M(1 - \pi_M^{1/r_M}) < r_M, \quad \boxed{\mathcal{D}_k^2 \leq \tau_M \chi_M \leq r_M(1 - \pi_M^{1/r_M}) \chi_M.} \quad (\text{CT.26})$$

To prove every factor, let $t_1, \dots, t_{r_M} \in (0, 1)$ be the positive eigenvalues of $G_M^{-1/2} \Delta_M G_M^{-1/2}$. The other eigenvalues are zero, so $\tau_M = \sum_i t_i$ and $\pi_M = \prod_i (1 - t_i)$. Induction on the number of factors proves $\prod_i (1 - t_i) \geq 1 - \sum_i t_i$. For positive numbers with a fixed sum, their product is at most the product with all entries equal: a maximum exists on the closed nonnegative simplex, lies in its interior when the sum is positive, and any unequal pair can be replaced by its average to increase the product, since $((a+b)/2)^2 - ab = (a-b)^2/4$. Apply this fact to $1 - t_i$ to obtain $\pi_M \leq (1 - \tau_M/r_M)^{r_M}$, proving both bounds. The last two inequalities in (CT.26) now follow from (CT.24). For $r_M = 1$, $\tau_M = 1 - \pi_M$ exactly; for $r_M = 0$, $\Delta_M = 0$, $\pi_M = 1$, and $W_M = 0$, so this case requires no exponent $1/r_M$.

14 Density of the original joint boundaries and determinant dissipation

Retain the complete packet h , its degree $d \geq 1$, the arithmetic unit v_h , and every original map in Section 13. In particular

$$s_j = \frac{1}{2} + it_j, \quad v_h = g/h, \quad g = 2\xi, \quad d\nu_h(t) = |v_h(\frac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad E^{(k)} = E_h^{\otimes k}, \quad \mathcal{H}_k = L^2(d^k x). \quad (\text{JD.1})$$

The one-variable cyclic-density argument in the coherent-interpolation source has a full joint-source extension. We prove it here, including the exact relation between the disappearing Hilbert representatives and their unchanged finite jets.

14.1 Polynomial density with the original weighted measures

Lemma 14.1. *Let μ be a finite positive Borel measure on $\mathbb{R}^k$ such that $\int e^{a \sum_j |t_j|} d\mu < \infty$ for some $a > 0$. The complex polynomials in $t_1, \dots, t_k$ are dense in $L^2(\mu)$.*

Proof. Suppose $f \in L^2(\mu)$ is orthogonal to every polynomial. Define the finite complex measure $d\eta = \bar{f} d\mu$. For $0 < b < a/2$, Cauchy–Schwarz gives

$$\int e^{b \sum_j |t_j|} d|\eta| \leq \|f\|_{L^2(\mu)} \left(\int e^{2b \sum_j |t_j|} d\mu \right)^{1/2} < \infty. \quad (\text{JD.2})$$

Thus $\Phi(z) = \int e^{i \sum_j z_j t_j} d\eta(t)$ is holomorphic in the connected tube $|\text{Im } z_j| < b$. On every smaller closed tube, each differentiated integrand is dominated by the integrable function in (JD.2), multiplied by a constant; the strict exponent margin absorbs each monomial. This justifies every derivative and the convergent Taylor series on small polydiscs. All derivatives at zero vanish by polynomial orthogonality. The Taylor series therefore vanishes near zero, and successive overlapping polydiscs along any path in the tube show $\Phi = 0$ throughout it. In particular the Fourier transform of η vanishes on $\mathbb{R}^k$.

For clarity, uniqueness here follows from an explicit approximate identity. Put

$$\gamma_\epsilon(x) = (4\pi\epsilon)^{-k/2} e^{-|x|^2/(4\epsilon)} = (2\pi)^{-k} \int e^{-\epsilon|u|^2} e^{iu \cdot x} du. \quad (\text{JD.3})$$

The second equality is the product Gaussian integral, obtained in one variable by differentiating its Fourier integral in x , integrating by parts in u , and evaluating its value at zero. The Gaussian integrability and finiteness of $|\eta|$ justify Fubini. Consequently $\eta * \gamma_\epsilon = 0$. For any continuous compactly supported φ , integration and Fubini give $\int (\varphi * \gamma_\epsilon) d\eta = 0$. These convolutions converge uniformly to φ : split the integral at a fixed small radius, use uniform continuity within it, and use the Gaussian tail and boundedness outside it. Hence $\int \varphi d\eta = 0$. Uniqueness of finite Borel measures from their continuous compactly supported integrals gives $\eta = 0$, and therefore $f = 0$ almost everywhere. The orthogonal complement of the polynomial closure is zero, proving the lemma. $\square$

The actual measure ν_h has every exponential moment of order less than $\pi/2$, with its full factor $dt/(2\pi)$, as proved in (GC.2). Multiplying it by any fixed polynomial in t preserves the same open range of exponential moments: choose a strictly larger exponent still below $\pi/2$ and absorb the polynomial into the difference. Product integration therefore verifies the lemma for both measures

$$\nu_h^{\otimes k}, \quad d\mu_{h,k}(t) = |h(s_1)|^2 \prod_{j=1}^k d\nu_h(t_j). \quad (\text{JD.4})$$

No factor h has been discarded. Its finitely many zeros on the first integration line have measure zero. The multiplication map

$$U_h : L^2(\mu_{h,k}) \longrightarrow L^2(\nu_h^{\otimes k}), \quad U_h P = h(s_1)P \quad (\text{JD.5})$$

is a unitary surjection: its norm identity follows directly from (JD.4), and its inverse is division by $h(s_1)$ away from that null set, with an arbitrary value on the null set. Polynomials in $s_j = 1/2 + it_j$ are precisely polynomials in t_j , since these coordinate changes are invertible. Thus (JD.5) and the lemma prove density of $h(s_1)\mathbb{C}[s_1, \dots, s_k]$ in $L^2(\nu_h^{\otimes k})$. This dense space is contained in the original joint ideal $(h(s_1), \dots, h(s_k))$.

14.2 Strong limits of the actual source representatives

The Mellin Plancherel map, with its original sign and measure, is

$$(\mathcal{M}_k F)(t) = \int_{(0,\infty)^k} F(x) \prod_{j=1}^k x_j^{-1/2+it_j} d^k x, \quad \mathcal{M}_k : \mathcal{H}_k \longrightarrow L^2\left(\frac{d^k t}{(2\pi)^k}\right). \quad (\text{JD.6})$$

Indeed the substitutions $x_j = e^{u_j}$ send F to $e^{\sum_j u_j/2} F(e^{u_1}, \dots, e^{u_k})$ isometrically in $L^2(d^k u)$, and (JD.6) becomes its Fourier transform with kernel $e^{it \cdot u}$. The inverse has kernel $e^{-it \cdot u}$ and factor $(2\pi)^{-k}$. Multiplication by $\prod_j v_h(s_j)$ is a unitary surjection from $L^2(\nu_h^{\otimes k})$ to the target in (JD.6). For surjectivity divide by this product almost everywhere: the entire nonzero function v_h has only isolated zeros on the line, and the product zero set is null. Composing with the inverse of (JD.6) gives the unitary extension

$$\mathcal{U}_{h,k} : L^2(\nu_h^{\otimes k}) \longrightarrow \mathcal{H}_k, \quad \mathcal{U}_{h,k} P = \mathcal{T}_h^{(k)} P \quad \text{for every polynomial } P. \quad (\text{JD.7})$$

The polynomial identity follows from the actual Mellin transform of F_h and D_j , so this is an extension of the original source map, not a change of representative.

Theorem 14.2 (Joint boundary density and exact finite-jet graph closure). *For each $k \geq 1$, the original total-degree boundary spaces $\mathcal{B}_M$ satisfy*

$$\overline{\bigcup_{M \geq k(d-1)} \mathcal{B}_M} = \mathcal{H}_k, \quad P_{\mathcal{B}_M} \longrightarrow I \quad \text{strongly.} \quad (\text{JD.8})$$

For the actual coherent minima of (CT.13), in the original fixed finite monomial tensor coordinates,

$$\|R_M\|_{E^{(k)} \rightarrow \mathcal{H}_k} \longrightarrow 0, \quad G_M \downarrow 0, \quad \det G_M \longrightarrow 0. \quad (\text{JD.9})$$

Here $G_M > 0$ at every finite M , and the middle limit is both decreasing in the positive-semidefinite order and convergent in matrix operator norm. Every finite jet is still retained: $J^{(k)} R_M = I$ for every M .

Define J_{pol} on the dense original source $\mathcal{D}_{h,k} = \mathcal{T}_h^{(k)} \mathbb{C}[s_1, \dots, s_k]$ by its complete arithmetic jets, including multiplication by $v_h^{\otimes k}$. Its graph has exactly the closure

$$\overline{\text{Graph}(J_{\text{pol}})} = \mathcal{H}_k \oplus E^{(k)}. \quad (\text{JD.10})$$

As a closed linear relation this has the split exact sequence

$$0 \longrightarrow E^{(k)} \xrightarrow{u \mapsto (0,u)} \overline{\text{Graph}(J_{\text{pol}})} \xrightarrow{(F,u) \mapsto F} \mathcal{H}_k \longrightarrow 0, \quad F \longmapsto (F, 0). \quad (\text{JD.11})$$

Proof. Every polynomial in $h(s_1)\mathbb{C}[s_1, \dots, s_k]$ belongs to one finite total-degree ideal $\mathcal{I}_{h,k,M}$. Its density proved in (JD.5), followed by the unitary map (JD.7), proves the first assertion in (JD.8). Given $F \in \mathcal{H}_k$ and $\epsilon > 0$, choose $b \in \mathcal{B}_N$ with $\|F - b\| < \epsilon$. For every $M \geq N$, the distance from F to $\mathcal{B}_M$ is at most that norm, so $\|(I - P_{\mathcal{B}_M})F\| < \epsilon$. This proves strong convergence.

Apply it to each of the d^k images of the fixed monomial basis under $s_h^{\otimes k}$. The identity $R_M = (I - P_{\mathcal{B}_M})s_h^{\otimes k}$ makes each column tend to zero. For a vector with coordinate norm one, Cauchy-Schwarz bounds its image by the square root of the sum of the squared column norms, proving the first limit in (JD.9). Since $G_M = R_M^* R_M$ and $G_M - G_{M+1} = Y_M^* Y_M$, the norm limit and monotonicity follow. The determinant limit follows by continuity of the determinant in these same finite coordinates. Positivity and $J^{(k)} R_M = I$ are the proved finite interpolation identities; none of these finite algebraic equations is lost in taking the Hilbert norm limit.

The kernel of J_{pol} is exactly the union of the $\mathcal{B}_M$, so its graph closure contains $\mathcal{H}_k \oplus \{0\}$ by (JD.8). For any $u \in E^{(k)}$, the vectors $(R_M u, u)$ belong to its graph and converge to $(0, u)$ by (JD.9). The closure is a linear subspace, and therefore contains every (F, u) ; the reverse inclusion is its declared ambient space. This proves (JD.10), and the maps, exactness and section in (JD.11) follow immediately by evaluating their two coordinates. In particular the closure is a relation with vertical fibre $E^{(k)}$, and it cannot be the graph of a single-valued operator because $d^k > 0$. $\square$

There are corresponding exact limits for the original weight form and both interpolation kernels. Write $U = U_{v_h^{\otimes k}}$, let K_M be the remainder-only polynomial kernel on the joint source, and let $\mathcal{C}_M = G_M^{-1} = U K_M U^*$ be the complete arithmetic interpolation matrix. In the same fixed coordinates,

$$\begin{aligned} \|W_M\| &\leq (2\|A^{(k)}\| + k)\|G_M\| \longrightarrow 0, \\ \lambda_{\min}(\mathcal{C}_M) &= \|G_M\|^{-1} \longrightarrow \infty, \\ \lambda_{\min}(K_M) &\geq \frac{1}{\|U\|^2 \|G_M\|} \longrightarrow \infty. \end{aligned} \tag{JD.9a}$$

The first line follows by the triangle inequality applied to the definition of W_M . Positivity of G_M proves the second. For the third, use $K_M = U^{-1} \mathcal{C}_M U^{-*}$ and $\|U^{-*} z\| \geq \|z\|/\|U\|$. Thus these coercive limits include every retained local jet, and the arithmetic unit remains in the comparison. The relative weight is related to its absolute form by the exact congruence

$$W_M = G_M^{1/2} S_M G_M^{1/2}, \quad \lambda_{\min}(G_M) \epsilon_M \leq \|W_M\| \leq \|G_M\| \epsilon_M. \tag{JD.9b}$$

The upper bound is submultiplicativity; applying it instead to $S_M = G_M^{-1/2} W_M G_M^{-1/2}$ gives the lower bound. Equations (JD.9a)–(JD.9b) retain the metric factors needed to pass from the proved absolute limits to the relative quantity.

The complementary quotient of the same closed relation is also explicit:

$$\frac{\overline{\text{Graph}(J_{\text{pol}})}}{\mathcal{H}_k \oplus \{0\}} \xrightarrow{\sim} E^{(k)}, \quad [(F, u)] \longmapsto u, \quad u \longmapsto [(0, u)]. \tag{JD.11a}$$

The kernel, well-definedness and both inverse identities follow by subtracting the two displayed pairs. With the product Hilbert norm the quotient norm is exactly $\|u\|$, since $\inf_{H \in \mathcal{H}_k} (\|F - H\|^2 + \|u\|^2)^{1/2} = \|u\|$, attained at $H = F$. Thus every completed-boundary class retains its original finite jet through an explicit isometric quotient.

This limit has an exact Split-Zero morphism. For a complex vector space X , use its one-support lift $\hat{X} = \{\tau_X\} \sqcup X^\bullet$, with τ_X the additive identity, vector addition within $X^\bullet$, and the original $G(\mathbb{C})$ -scalar action. In particular $0_X^\bullet \neq \tau_X$. Any linear map $T : X \rightarrow Y$ lifts to $\hat{T}(\tau_X) = \tau_Y$ and $\hat{T}(u^\bullet) = (Tu)^\bullet$; direct substitution checks addition and every scalar, including the supported scalar zero and external scalar zero. Give the supported fibre its norm topology and τ_X its isolated topology. Equation (JD.9) then says exactly

$$\hat{R}_M \longrightarrow \hat{0}, \quad \hat{0}(\tau_E) = \tau_{\mathcal{H}}, \quad \hat{0}(u^\bullet) = 0_{\mathcal{H}}^\bullet \quad (u \in E^{(k)}). \tag{JD.12}$$

Thus the limiting morphism retains the support of every input. Equation (JD.11) identifies all finite jets in the vertical fibre of the closed relation; it supplies the full typed relation between these supported Hilbert limits and the unchanged algebraic jets.

14.3 The accumulated determinant cost of the same boundary sequence

Continue to use the original ordered spectral sum and layer energy

$$\mathcal{D}_k = \sum_{i_1, \dots, i_k} \left| \sum_{j=1}^k (\operatorname{Re} \rho_{i_j} - 1/2) \right|, \quad \chi_M = \operatorname{Tr}(G_M^{-1} C_M^* C_M), \quad \tau_M = \operatorname{Tr}(G_M^{-1} (G_M - G_{M+1})). \quad (\text{JD.13})$$

The list $\rho_1, \dots, \rho_d$ includes every algebraic multiplicity. Define $\pi_M = \det G_{M+1} / \det G_M$. For integers $k(d-1) \leq a < b$, the following exact finite inequality holds whenever $\mathcal{D}_k > 0$:

$$\mathcal{D}_k^2 \sum_{M=a}^{b-1} \frac{1}{\chi_M} \leq \sum_{M=a}^{b-1} \tau_M \leq \log \frac{\det G_a}{\det G_b}. \quad (\text{JD.14})$$

All denominators in this statement are proved positive. Indeed (CT.24) gives $\mathcal{D}_k^2 \leq \tau_M \chi_M$, with both factors nonnegative; strict positivity of the left side forces strict positivity of each factor. Let $t_{M,j} \in [0, 1)$ be the full eigenvalue list of $G_M^{-1/2} (G_M - G_{M+1}) G_M^{-1/2}$, including zeros. The finite original matrices give exactly

$$\tau_M = \sum_j t_{M,j}, \quad \pi_M = \prod_j (1 - t_{M,j}), \quad -\log \pi_M - \tau_M = \sum_j \int_0^{t_{M,j}} \frac{u}{1-u} du \geq 0. \quad (\text{JD.15})$$

The first two identities are congruence and determinant identities; the last follows by integrating the derivative of $-\log(1-t) - t$. Sum (CT.24) divided by χ_M , then sum (JD.15). The product of the determinant ratios telescopes without a change of coordinates, proving (JD.14).

If $\mathcal{D}_k = 0$, every repeated tuple $(i, \dots, i)$ has zero summand, hence $\operatorname{Re} \rho_i = 1/2$ for every retained zero. Conversely these equalities give $\mathcal{D}_k = 0$. The second inequality in (JD.14) and the exact remainder (JD.15) hold also in this case; no quotient $0/0$ is introduced. Equation (JD.9) proves that the right-hand side tends to infinity as $b \rightarrow \infty$ for each fixed a . Equations (JD.13)–(JD.15) retain the actual rate through χ_M , the complete relative Gram losses and their explicit logarithmic remainder. They impose no unproved upper estimate on that rate.

The same sequence supplies two further exact energy identities:

$$G_a = \sum_{M=a}^{\infty} Y_M^* Y_M, \quad \sum_{M=a}^{\infty} \|Y_M\|_{\text{HS}}^2 = \operatorname{Tr} G_a, \quad \sum_{M=a}^{\infty} \tau_M = \infty. \quad (\text{JD.16})$$

The first series converges in matrix operator norm, by telescoping $G_M - G_{M+1} = Y_M^* Y_M$ and the proved limit $G_M \rightarrow 0$. Taking traces of its finite partial sums proves the second identity; all summands are nonnegative. To prove the third, suppose instead that $\sum_M \tau_M$ were finite. Then beyond some index all $\tau_M \leq 1/2$, so every $0 \leq t_{M,j} \leq 1/2$. Integrating $(1-u)^{-1} \leq 2$ over $[0, t_{M,j}]$ gives $-\log(1-t_{M,j}) \leq 2t_{M,j}$. Hence the tail sum of $-\log \pi_M$ would be bounded by $2 \sum_M \tau_M < \infty$. The finite initial terms are finite because $G_M > 0$. Telescoping would therefore bound $\log(\det G_a / \det G_b)$ uniformly in b , contradicting $\det G_b \rightarrow 0$. This proves the divergence, including packets entirely on the critical line and every retained nilpotent chain. The equality $\tau_M = \|Y_M G_M^{-1/2}\|_{\text{HS}}^2$ identifies the relative energy in this divergence with the same layer map appearing in the convergent absolute energy series.

15 Which completion retains the arithmetic jets?

This section extends the Hilbert/jet calculation of the September 12 orthogonal-boundary note. All changes of topology are specified maps on the same arithmetic functions. The theta relation space, the operator $D = -x\partial_x$, and $g = 2\xi$ are retained.

15.1 Logarithmic coordinates and the original measure

Let $\mathcal{B}, V, \Theta$ have the definitions in Section 5. Put $\mathcal{H} = L^2(\mathbb{R}_+, dx)$. The map

$$\mathcal{L}F(y) = e^{y/2}F(e^y), \quad \mathcal{L}^{-1}f(x) = x^{-1/2}f(\log x)$$

is unitary from $\mathcal{H}$ to $L^2(\mathbb{R}, dy)$, since $dx = e^y dy$. Direct differentiation gives

$$\mathcal{L}D\mathcal{L}^{-1} = \frac{1}{2} - \partial_y, \quad \mathcal{M}F(\rho) = \int_{\mathbb{R}} (\mathcal{L}F)(y) e^{(\rho-1/2)y} dy. \quad (\text{JT1})$$

On $\mathcal{H}$, D denotes the closed maximal distributional operator which $\mathcal{L}$ identifies with $1/2 - \partial_y$ on the Sobolev domain $H^1(\mathbb{R})$. For an integer $r \geq 0$, let $\mathcal{H}^{(r)}$ be the domain of the first r powers of D in $\mathcal{H}$, with norm $\|F\|_r^2 = \sum_{k=0}^r \|D^k F\|_{\mathcal{H}}^2$. The Mellin-line transform makes this the norm with density $w_r(t) = \sum_{k=0}^r |1/2 + it|^{2k}$ relative to $dt/(2\pi)$.

Theorem 15.1 (Finite differential graph norms and the complete jet graph). *Let Z be a nonempty finite packet of actual nontrivial zeros with full multiplicities, $E_Z = \mathbb{C}[t]/h_Z$, and $J_Z : \mathcal{B} \rightarrow E_Z$ the complete Mellin-jet map. For every finite r ,*

$$\overline{\text{span}\{D^j f_0 : j \geq 0\}}^{\mathcal{H}^{(r)}} = \mathcal{H}^{(r)}, \quad f_0 = \Theta\phi_*, \quad (\text{JT2})$$

$$\overline{\{(F, J_Z F) : F \in \mathcal{B}\}}^{\mathcal{H}^{(r)} \oplus E_Z} = \mathcal{H}^{(r)} \oplus E_Z. \quad (\text{JT3})$$

Thus J_Z is not closable in this norm. The completed theta-boundary graph is $\mathcal{H}^{(r)} \oplus 0$; its quotient in (JT3) is E_Z through $[(F, u)] \mapsto u$.

Proof. For each $0 < c < \pi/4$, the actual source gives $|g(1/2 + it)| \leq C_c e^{-c|t|}$. Indeed $e^{z/2}f_0(e^z)$ is holomorphic in $|\text{Im } z| < \pi/4$: its Gaussian series and derivatives converge uniformly on compact subsets since $\text{Re}(e^{2z}) > 0$. On smaller horizontal strips it decays faster than every exponential at the positive end. Poisson gives $Jf_0 = f_0$, so analytic continuation gives $e^{z/2}f_0(e^z) = e^{-z/2}f_0(e^{-z})$; this controls the negative end. Shift the Fourier contour to height c or $-c$. The vertical integrals vanish by that decay, proving the bound.

The measure $d\mu_r = w_r(t)|g(1/2 + it)|^2 dt/(2\pi)$ has a positive exponential moment. If $u \in L^2(\mu_r)$ is orthogonal to every polynomial, $d\nu = \bar{u} d\mu_r$ is a finite complex measure with a smaller positive exponential moment, by Cauchy-Schwarz. Its Fourier transform is analytic in a strip, and every derivative at zero vanishes. Analytic continuation makes it zero there. Fourier uniqueness gives $\nu = 0$, hence $u = 0$. For this uniqueness step, convolve with Gaussian approximate identities: each convolution is in L^2 , since $\|\nu * \gamma_\epsilon\|_2 \leq \|\nu\|_{\text{TV}} \|\gamma_\epsilon\|_2$; its transform is zero, so Plancherel makes it zero, and weak convergence then gives $\nu = 0$. Polynomials are therefore dense in $L^2(\mu_r)$. Multiplication by $g(1/2 + it)$ is an isometry onto $L^2(w_r dt/(2\pi))$: divide almost everywhere, since this nonzero analytic function vanishes on a discrete, measure-zero set. The Mellin transform of $D^j f_0$ is $(1/2 + it)^j g(1/2 + it)$, proving (JT2).

Every theta polynomial has zero J_Z -jet. Thus the closure in (JT3) contains $(h, 0)$ for every h . The actual packet section $R_{\text{ref}} : E_Z \rightarrow \mathcal{B}$ satisfies $J_Z R_{\text{ref}} = 1$. Approximate $R_{\text{ref}} u$ in $\mathcal{H}^{(r)}$ by theta polynomials $p_n(D)f_0$. Then $(R_{\text{ref}} u - p_n(D)f_0, u) \rightarrow (0, u)$. This proves (JT3). For $u \neq 0$ it prevents closability. The boundary and quotient statements follow from the coordinate projections. $\square$

Corollary 15.2 (Differential on the completed graph). *The continuous map*

$$(D, A) : \mathcal{H}^{(r+1)} \oplus E_Z \longrightarrow \mathcal{H}^{(r)} \oplus E_Z, \quad (F, u) \mapsto (DF, Au) \quad (\text{JT4})$$

extends the original $(F, J_Z F) \mapsto (DF, J_Z DF)$. *On the quotient by the boundary closure it is exactly* $A = M_t$.

Proof. The norm definitions bound $D : \mathcal{H}^{(r+1)} \rightarrow \mathcal{H}^{(r)}$; the jet identity $J_Z D = A J_Z$ follows by retaining the complete Taylor polynomial of $s\mathcal{M}F(s)$. Apply (JT3). $\square$

Its split observation records $(e_{\mathcal{H}}, u^\bullet)$ when the Hilbert component tends to its supported zero and the jet survives. The projections compare this point to $e_{\mathcal{H}}$ and to $u^\bullet$; they do not identify it with (τ, τ) .

15.2 Exponential weights and exact jet norms

For $a \geq 0$, retain the original functions and introduce

$$\mathcal{H}_a = L^2(\mathbb{R}_+, e^{2a|\log x|} dx), \quad \|F\|_a^2 = \int_{\mathbb{R}} |\mathcal{L}F(y)|^2 e^{2a|y|} dy. \quad (\text{JT5})$$

The identity inclusions $\mathcal{B} \hookrightarrow \mathcal{H}_a \hookrightarrow \mathcal{H}_0$ are continuous. Source seminorms with powers exceeding $a + 1/2$ control both ends. Smooth functions compactly supported in log coordinates prove density.

Theorem 15.3 (Exact threshold, including the boundary). *Put $\ell_{\rho,k}(F) = (\mathcal{M}F)^{(k)}(\rho)/k!$ and $b = \text{Re } \rho - 1/2$, with $k \geq 0$. The functional on $\mathcal{B}$ extends continuously to $\mathcal{H}_a$ if and only if $|b| < a$. Its exact squared norm is*

$$\|\ell_{\rho,k}\|^2 = \frac{(2k)!}{(k!)^2} \left(\frac{1}{[2(a-b)]^{2k+1}} + \frac{1}{[2(a+b)]^{2k+1}} \right). \quad (\text{JT6})$$

Proof. Inner products are conjugate-linear in their first argument. The Riesz vector in log coordinates must be

$$r_{\rho,k}(y) = \frac{y^k}{k!} e^{(\bar{\rho}-1/2)y} e^{-2a|y|}.$$

Its norm squared is $(k!)^{-2} \int y^{2k} e^{2by-2a|y|} dy$. The two half-lines converge exactly when $|b| < a$; their gamma integrals give (JT6). If $|b| \geq a$, testing any putative Riesz representative against compactly supported smooth functions forces it to be this same non-square-integrable density, a contradiction. The equality case is included: one end has no exponential decay. $\square$

Theorem 15.4 (Complete confluent Cauchy Gram). *For distinct centres ρ_i with $|\text{Re } \rho_i - 1/2| < a$ and orders $0 \leq k < m_i$, let $J_a : \mathcal{H}_a \rightarrow \mathbb{C}^d$ be their Taylor-jet map. In the original Taylor coordinates its Gram $C_a = J_a^* J_a$ is*

$$(C_a)_{(i,k),(j,l)} = \frac{(k+l)!}{k!l!} \left(\frac{1}{(2a-z_{ij})^{k+l+1}} + \frac{(-1)^{k+l}}{(2a+z_{ij})^{k+l+1}} \right), \quad z_{ij} = \rho_i + \bar{\rho}_j - 1. \quad (\text{JT7})$$

It is positive definite. The exact right inverse $S_a = J_a^ C_a^{-1}$ satisfies*

$$J_a S_a = 1, \quad \|S_a u\|_a^2 = u^* C_a^{-1} u, \quad \|F\|_a^2 = \|(1 - S_a J_a)F\|_a^2 + (J_a F)^* C_a^{-1} (J_a F). \quad (\text{JT8})$$

Proof. The entry is $\langle r_{\rho_i,k}, r_{\rho_j,l} \rangle_a$; integrating $y^{k+l} e^{z_{ij}y-2a|y|} / (k!l!)$ on both half-lines proves (JT7) with its stated sign. The vectors are independent. After removing their common nonzero factor, a dependence is an exponential polynomial $\sum_i P_i(y) e^{\bar{\rho}_i y} = 0$. For each i , apply $\prod_{j \neq i} (\partial_y - \bar{\rho}_j)^{m_j}$. On the i -th polynomial factor every nonzero constant plus derivative is invertible on the finite-degree polynomial space; hence $P_i = 0$. Thus $C_a > 0$. Now $S_a J_a$ is the orthogonal projection onto $\text{range } J_a^* = (\ker J_a)^\perp$. Multiplying the specified matrices proves all of (JT8). $\square$

For an actual packet the unchanged theta subspace lies in $\ker J_a$. Consequently there is an onto continuous quotient map

$$\mathcal{H}_a/\overline{\Theta V}^{\mathcal{H}_a} \longrightarrow \mathbb{C}^d, \quad [F] \mapsto J_a F, \quad \ker = (\ker J_a)/\overline{\Theta V}^{\mathcal{H}_a}. \quad (\text{JT9})$$

Its surjectivity follows from (JT8). Define the retained coordinate map

$$T : E_Z \longrightarrow \mathbb{C}^d, \quad T[p] = (p^{(k)}(\rho_i)/k!)_{i,k}.$$

Then $J_a|_{\mathcal{B}} = TJ_Z$, and the exact representative difference is $R_{\text{ref}}T^{-1} - S_a : \mathbb{C}^d \rightarrow \ker J_a$. Here R_{ref} is the original source map; S_a is a Hilbert map, with no claim it lies in $\mathcal{B}$. In E_Z coordinates the observation Gram is $C_E = T^{-1}C_a(T^{-1})^*$ and the right inverse is $S_a T$.

15.3 The energy term carried by the weighted topology

Theorem 15.5 (Weighted boundary identity on the actual representatives). *On $\mathcal{B}$, with $(\Sigma H)(x) = \text{sgn}(\log x)H(x)$,*

$$\langle DF, H \rangle_a + \langle F, DH \rangle_a - \langle F, H \rangle_a = 2a \langle F, \Sigma H \rangle_a. \quad (\text{JT10})$$

*For an actual packet representative $R : E_Z \rightarrow \mathcal{B}$ with $DR - RA = B = \Theta k$, put $G_a = R^*R$, $T_a = R^*\Sigma R$. Adjoints use exactly $\mathcal{H}_a$. Then*

$$A^*G_a + G_aA - G_a = 2aT_a - (B^*R + R^*B), \quad -G_a \leq T_a \leq G_a. \quad (\text{JT11})$$

For $Au = \rho u \neq 0$,

$$2 \operatorname{Re} \rho - 1 = 2a \frac{u^*T_a u}{u^*G_a u} - 2 \operatorname{Re} \frac{\langle Ru, Bu \rangle_a}{\|Ru\|_a^2}. \quad (\text{JT12})$$

Proof. In log coordinates integrate $-(\bar{f}h)'e^{2a|y|}$ on the two half-lines. Source decay kills the terms at infinity; those at zero cancel because the weight is continuous there. The weak derivative of the weight is $2a \operatorname{sgn}(y)e^{2a|y|}$. This proves (JT10) with its positive sign. Substitute $DR = RA + B$ and move both boundary terms to obtain (JT11). Multiplication by sign is a self-adjoint contraction, proving the form bounds. Evaluation on u proves (JT12). $\square$

At $a = 0$, (JT11) is the source boundary-control formula. At $a > 0$, the topology making a jet continuous has its explicitly retained energy term $2aT_a$. Off-line jets cannot remain continuous as $a \downarrow 0$. For line-centred jets, all $a > 0$ work but (JT6) diverges as $a \downarrow 0$. These are exact comparisons on the same Mellin map and differential, without a weight theorem assumed.

15.4 The tessarine coordinate in the actual weight form

Retain $c_{-+} = 4(\beta - 1/2)$, $c_{--} = 4i\gamma$ for $\rho = \beta + i\gamma$. For $p > 1$ and $|\psi| = 1$,

$$z = \psi \exp\left(\frac{\log p}{4}(c_{-+} + c_{--})\right), \quad \frac{u^*(A^*G + GA - G)u}{u^*G u} = \frac{c_{-+}}{2}, \quad |b| < a \iff |c_{-+}| < 4a. \quad (\text{JT13})$$

The first equality is exactly $z = \psi p^{\rho-1/2}$; the second is evaluation on $Au = \rho u$. Thus the original sector, local holonomy radius, arithmetic weight form and jet-continuity strip have specified comparison maps. The scalar evaluations do not replace the original matrix or its support-labelled source.

16 The current arithmetic route and its next calculation

The source chain now supplies an actual $\mathbb{C}[t]$ -module cohomology Q , a finite invariant packet inside that cohomology, a specified nonzero extension class, an original theta boundary for its representative defect, and an exact arithmetic formula for the relative weight form. The local continuation makes that last formula an explicit confluent interpolation problem. These constructions are logically prior to any purity conclusion. The subsequent continuation now supplies certified entries of the original arithmetic moment matrix, the exact finite-compression and infinite-tail resolvent comparison, and a convergence rate for the initial resolvent including its complete off-line jets.

16.1 A single chain of exact comparison maps

For each actual finite packet, retaining every full order, the maps proved in this reader are

$$E = \mathbb{C}[t]/h \xrightarrow{R_{\text{ref}}} \mathcal{B} \xrightarrow{q} Q, \quad J_Z R_{\text{ref}} = 1, \quad D R_{\text{ref}} - R_{\text{ref}} A = \Theta \phi_* \ell.$$

Every admitted polynomial correction keeps these same quotient and jet maps. At its distinguished orthogonal level let $B_{N,u}(t) = (p_{N,u}(t) - R_Z(u)(t))/h(t)$, a polynomial of degree at most m . The complete calculation is

$$\begin{aligned} d\nu_Z(t) &= \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{dt}{2\pi}, \quad N = d + m, \quad C_N P = j_h P, \quad K_N = C_N (H_N^\nu)^{-1} C_N^*, \\ p_{N,u} &= (H_N^\nu)^{-1} C_N^* K_N^{-1} U_\varepsilon u, \quad R_m u = R_{\text{ref}} u + \Theta(B_{N,u}(D) \phi_*), \\ G_m &= U_\varepsilon^* K_N^{-1} U_\varepsilon, \quad W_m = U_\varepsilon^* K_N^{-1} (A K_N + K_N A^* - K_N) K_N^{-1} U_\varepsilon. \end{aligned} \quad (\text{R1})$$

The middle line evaluates the specified polynomial at the original D ; every source vector lies in V .

Write $v_n = j_h q_n$ and $\kappa_n = \|q_n\|_{\nu_Z}^2$. The original arithmetic measure now determines the exact quantity

$$\epsilon_m = \frac{|\operatorname{Re}(v_N^* K_N^{-1} v_{N+1})| + \sqrt{(v_{N+1}^* K_N^{-1} v_{N+1})(v_N^* K_N^{-1} v_N) - (\operatorname{Im}(v_N^* K_N^{-1} v_{N+1}))^2}}{\kappa_N}. \quad (\text{R2})$$

The proofs in Section 8 establish (R1)–(R2) before passing to cohomology. For a reflection-stable packet the first absolute-value term is zero by the exact trace identity. The same sequence obeys

$$\epsilon_m \geq \sum_{\rho \in Z} m_\rho \left| \operatorname{Re} \rho - \frac{1}{2} \right|. \quad (\text{R3})$$

Thus the rank-two boundary detects the combined displacement of the entire packet, not only its largest individual displacement. The estimate concerns the distinguished R_m sequence. The larger family of all polynomial source corrections has the separately proved metric image and sharp infimum in Section 7; those results do not identify its arbitrary member with R_m .

The full two-row source observation in Section 8 is injective after the original A -iterates are retained. It therefore offers a finite arithmetic calculation preserving every nilpotent jet: compute the source moments, the adjacent monic polynomials and their complete residue vectors, rather than discard a component whose scalar trace vanishes.

16.2 Topology, duality and the two sides of a weight estimate

The actual source moments now have the absolutely convergent incomplete-gamma expression and explicit double-tail bound of Section 6. Its certified six-by-six Gram matrix concerns the original $|2\xi|^2 dt/(2\pi)$ measure. The exact identity (AM.19) supplies the original monic recurrence norms through degree five, with interval enclosures.

For the $|g/h|^2$ kernel, Section 9 proves a universal polynomial quadrature formula, including centres that coincide with a quadrature node. On the precise nonintersection domain its alternate expression is

$$K_{n-1} = \kappa_{n-1}^{-1} q_n(A) S_n q_n(A)^*, \quad AS_n + S_n A^* - S_n = cr_n^* + r_n c^*, \quad r_n = F_n(A)c.$$

The exact Schur comparison (KR.15)–(KR.20) then joins the initial and terminal observations to the actual infinite-dimensional tail. Section 10 proves the explicit initial-resolvent rate (GC.8) and its complete finite-algebra bound (GC.23). The projection to that off-line domain retains the critical summand as its stated kernel. All factors in the Schur identity remain in converting the initial estimate to the terminal form that determines ϵ_m .

In particular, Section 12 identifies exactly the two contributions in a reflection-stable packet:

$$\epsilon_m^2 = 2 \sum_{\rho \in Z} m_\rho (\operatorname{Re} \rho - 1/2)^2 + \mathfrak{d}_{G_m}(A).$$

Its full-jet inequalities also quantify the required metric ratios along each retained nilpotent chain. These identities do not infer a relative estimate from absolute matrix shrinkage.

The theta-polynomial span is dense in every finite differential graph norm. The completed jet graph, however, is $\mathcal{H}^{(r)} \oplus E$, with quotient E , not a quotient of $\mathcal{H}^{(r)}$ in which the arithmetic component has vanished. This supplies the exact comparison between the shrinking Hilbert representative and its unchanged cohomological class.

The exponential observation norm supplies a second exact comparison. A Mellin jet at ρ is bounded precisely in the strip

$$|\operatorname{Re} \rho - \tfrac{1}{2}| < a,$$

with the factorial norm in Theorem 15.3. The original differential then has the energy term $2a \operatorname{sgn}(\log x)$ in Theorem 15.5. Consequently stronger topology and weight control are related by that explicit term; choosing a positive observation norm does not remove it.

For any packet Z , reflection gives the conjugate-linear ring isomorphism

$$\dagger_Z : E_Z \longrightarrow E_{Z^\dagger}, \quad f^\dagger(s) = \overline{f(1 - \bar{s})}, \quad Z^\dagger = \{1 - \bar{\rho} : \rho \in Z\},$$

with all multiplicities retained. In fact $h_Z^\dagger = (-1)^d h_{Z^\dagger}$; this proves well-definedness, and applying the same operation twice is the identity. Restrict the following internal residue pairing to reflection-stable Z , so that this map is an involution on E_Z . The arithmetic residue form is the source form

$$\mathcal{R}_Z(f, u) = \sum_{\rho \in Z} \operatorname{Res}_\rho \frac{f^\dagger u}{g} ds.$$

Its Jacobian contraction is

$$\mathcal{R}_Z(f, g'u) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} u(\rho). \quad (\text{R4})$$

Indeed locally $g = z^{m_\rho} w(z)$ gives $g'/g = m_\rho/z + w'/w$, proving (R4) with multiplicities. The trace form pairs reflected centres and can retain nontrivial null directions from nilpotents. In the retained basis $1, t, \dots, t^{d-1}$, define

$$S_{ij} = \sum_{\rho \in Z} \operatorname{Res}_\rho \frac{(s^i)^\dagger s^j}{g(s)} ds, \quad 0 \leq i, j < d, \quad J_g = M_{jhg'}.$$

Changing representatives by h makes the local quotient by g holomorphic, proving that this matrix represents the intrinsic form. By (R4) the trace pairing has matrix SJ_g . Its exact comparison to the positive source metric is therefore $G_m^{-1}SJ_g$, since $f^*G_m(G_m^{-1}SJ_g)u = f^*SJ_gu$. Its positivity is an arithmetic question about this operator, not a consequence of the positive Gram construction.

The finite-field two-sided weight theorem remains a source for methods, with its compact-support comparison, duality and tensor hypotheses. This reader has not constructed a realization meeting those geometric hypotheses for the entire arithmetic quotient. It has instead calculated the current characteristic-zero objects and their original maps, including the local-cohomology spectral extraction and meromorphic complement documented in the complete finite-comparison appendix.

16.3 Provenance and mathematical status

The coherent interpolation now has the full coefficient identity $J_{h,N} = U_{v_h} C_N$, with $v_h = j_h(g/h)$ and $\varepsilon_h = v_h^{-1}$. Equations (CT.3)–(CT.7) prove equality with (R1), including the original representative, and compatibility with the complete CRT injections between finite packets. The actual reflection preserves the unique minimum and proves the same-metric signed pairing of (CT.10) and (CT.16a).

On the original tensor complex, the complete calculation is

$$\begin{aligned} E^{(k)} &= E_h^{\otimes k}, & A^{(k)} &= \sum_{j=1}^k A_{h,j}, & \mathcal{I}_{h,k,M} &= \mathcal{P}_M^{(k)} \cap (h(s_1), \dots, h(s_k)), \\ R_M &= (I - P_{\mathcal{B}_M}) s_h^{\otimes k}, & G_M &= (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1}, \\ W_M &= -(Y_M^* C_M + C_M^* Y_M), & G_M - G_{M+1} &= Y_M^* Y_M, & Y_M, C_M &: E^{(k)} \longrightarrow \mathcal{E}_M, \\ \dim \mathcal{E}_M &= \binom{M+k}{k-1}. \end{aligned} \tag{R5}$$

Section 13 proves the original ordered division, both cochain signs, every layer map, and the extra orthogonal projection required when tensor packets are enlarged. Its formula (CT.18) calculates the resulting change in both G and W .

For a reflection-stable packet, let p_M be either signed inertia of W_M , and let $\mathfrak{d}_M^{(k)}$ be its complete tensor departure from (CT.20). The same source layers give

$$\begin{aligned} \mathcal{D}_k &\leq p_M \epsilon_M, & p_M \epsilon_M^2 &\geq 2kd^{k-1} \sum_{\rho \in Z} m_\rho (\operatorname{Re} \rho - 1/2)^2 + \mathfrak{d}_M^{(k)}, \\ p_M &\leq \min\{\operatorname{rank} Y_M, \operatorname{rank} C_M, \lfloor d^k/2 \rfloor\}, & \mathcal{D}_k^2 &\leq \operatorname{Tr}(G_M^{-1}(G_M - G_{M+1})) \operatorname{Tr}(G_M^{-1} C_M^* C_M). \end{aligned} \tag{R6}$$

The zero-inertia case is proved directly in (CT.25). The full ordered sum $\mathcal{D}_k$ and all multiplicities are retained in (CT.20). The joint density and exact graph closure in Section 14 show $G_M \downarrow 0$ for this actual total-degree sequence and identify every retained jet in the vertical fibre of (JD.11). The same section proves the coercive inverse-interpolation limit with its arithmetic unit, the supported-zero limit morphism, and the accumulated determinant cost (JD.14)–(JD.15). These formulas connect the relative layer energy to the exact rate of determinant loss through the same original matrices.

The September 11 support, Rees, comparison and audit packages are retained in the companion source directory. The current analytic baseline is the September 12 sequence: finite operator comparison (PR 8), chain descent (PR 9), global retraction (PR 10), homotopy control (PR 11), formal tau recovery (PR 12) orthogonal boundary control (PR 13), and coherent interpolation with joint tensor boundaries (PR 14). Part II includes the complete selected source proof notes with their exact revisions. Historical statements that a map remains unconstructed belong to their stated source dates; a later explicit construction in this reader or the source chain supersedes that particular statement.

The new local proofs are the full support-index equivalence, the corrected carrier and sector comparisons, the arithmetic confluent-kernel minimum and complete two-row source observation, the classification of source-metric images and aggregate control, and the graph/weighted-jet comparison. The next tranche adds the certified original theta-moment series and its full truncation bound, the confluent quadrature and finite/terminal/tail resolvent identities, the explicit initial convergence rate, and the metric-departure and nilpotent-chain identities. The recovered June 24 historical

packet calculation is reproduced with a full-jet comparison in Section 2. Its exact A -invariant saturation is every off-axis local factor; the quotient, involutions, source-metric transport and split-zero endomorphism descent are proved there. The original transcript excerpt and its response locators are retained in the source package. This is an explicitly recorded historical recovery, with its new arithmetic saturation proof distinguished from the original permutation calculation. The subsequent historical reconstruction in Section 3 proves the inverse map from a complete local Hurwitz zero-cluster germ to every moment of its original compact probability measure, followed by the explicit Bernstein recovery of that measure. The monic cluster polynomial retains arbitrary zero multiplicity. The same section proves the complete Laurent period maps and their collision kernels. For two distinct primes, the joint Laurent map is surjective onto the full finite spectral algebra; Theorem 3.1 gives every primary kernel and the inverse original spectral coordinate, including its truncated logarithm on each nilpotent factor. The factorwise Split-Zero lifts and their exact comparison with the synchronized global lift are proved in (S4)–(S5). Their elementary algebraic and analytic steps are written here. No claim of global mathematical priority follows from finding or proving an identity in this research programme.

The finite validation records distinguish exact symbolic identities, rational calibration models, numerical quadratures and intentionally rejected negative controls. They do not count test methods as theorems or imply analytic certification from an L^2 numerical experiment. The supplied formalization records are also kept separate: successful exact-head CI metadata does not turn the unformalized analytic theta arguments into Lean proofs. No Lean run was performed for this continuation.

The current research calculation is (R2) on the actual $|g/h|^2$ -moments, with its determinant update and two-row recurrence, together with the topology constants of Theorem 15.3. No uniform arithmetic estimate forcing (R2) to vanish has been established in this reader. RH, GRH and a global purity theorem are not conclusions of it. The exact maps and source constraints above specify the live route on which tandem work can now continue. The next estimate must propagate (GC.8) through (KR.16), retaining q_n^2 , κ_n , a_n , the terminal ratio F_n and the actual tail denominator. The arithmetic moment formulas give an executable source for those finite data. The exact departure identity and the full-jet metric bounds specify what a resulting relative estimate would control in the original packet; no component is removed by changing observation.

The next step of that propagation is now calculated in Section 11. For $u_n = q_n(A)c$, $w_n = q_{n-1}(A)c$, and $M_n = \kappa_{n-1}^{-1}K_{n-1}^{-1}$, let $\beta_n = u_n^* M_n u_n$, and let P_n be the exact M_n -orthogonal projection with kernel $\mathbb{C}u_n$ when $\beta_n > 0$. Put $p_n = P_n w_n$, $z_n = P_n A u_n$. The universal step is

$$\begin{aligned} M_{n+1} &= a_n^{-1} \left(M_n - \frac{M_n u_n u_n^* M_n}{a_n + \beta_n} \right), & \frac{\det G_{m+1}}{\det G_m} &= \frac{a_n}{a_n + \beta_n}, \\ \Delta_{n+1} &= \frac{\beta_n \|z_n + a_n p_n\|_n^2}{a_n(a_n + \beta_n)}, & \epsilon^{(n)} &= |gz| + \sqrt{g_Z^2 + \Delta_n}, \quad gz = \operatorname{Re} \operatorname{tr} A - d/2. \end{aligned} \quad (\text{R7})$$

Here $m = n - d - 1$. The zero-vector exceptional step is given completely in (KT.14), so critical centres and every polynomial coincidence remain in (R7)'s source calculation. The alternate terminal Gram follows by the exact isometry (KT.15) and recurrence (KT.16). The full scalar-to-matrix extraction uses the doubled off-line algebra and the explicit sequence (KT.23); at a reflected pair its matrix coefficient is exactly $(-1)^l F_n^{[k+l+1]}(\rho)$. Finally (KT.24)–(KT.25) recover these coefficients from the original initial error and tail, with both reciprocal series and all polynomial/norm factors displayed. The remaining energy estimate is therefore attached to the three explicit terms of (KT.11), including their actual cross pairing, on the same arithmetic moment sequence.

For a packet closed under both conjugation and reflection, the original real-coefficient calculation in (KT.28) closes this step in terms of consecutive determinant ratios. With the fixed section metric G_{-1} retained, put $r_m = \det G_m / \det G_{m-1}$ for $m \geq 0$. Then the actual arithmetic allowance obeys exactly

$$\boxed{\epsilon_m^2 = a_{d+m+1} \frac{(1 - r_m)(1 - r_{m+1})}{r_{m+1}}.} \quad (\text{R8})$$

The coefficient reality and full reflection trace prove $\gamma_n = 0$ in the original coordinates; (KT.7)–(KT.8) then prove (R8), including zero-step cases. This places the relative quartet control directly on the same monic norm ratios and consecutive Gram determinants as the certified moment problem.

This also gives a direct arithmetic bound for the original tesserine violation channel. Retain the original quartet element

$$Q(\rho) = \rho 1 + (1 - \rho)\sigma + \bar{\rho}\kappa + (1 - \bar{\rho})\sigma\kappa, \quad \rho = \beta + i\gamma,$$

and its character $c_{-+}(\rho) = 2(2\beta - 1)$, as calculated in the carrier section. If $\gamma \neq 0$, $\beta \neq 1/2$, and Z is this entire four-point zero orbit with common full order μ , then $d = 4\mu$ and for every $m \geq 0$,

$$\boxed{\mu^2 |c_{-+}(\rho)|^2 \leq \epsilon_m^2 = a_{4\mu+m+1} \frac{(1 - r_m)(1 - r_{m+1})}{r_{m+1}}.} \quad (\text{R9})$$

To prove every coefficient, conjugation and the functional equation preserve the full order, and the four real displacements are $\beta - 1/2, \beta - 1/2, -(\beta - 1/2), -(\beta - 1/2)$. The proved aggregate bound (R3) is therefore $\epsilon_m \geq 4\mu|\beta - 1/2| = \mu|2(2\beta - 1)|$. Squaring and using (R8) gives (R9). The sector morphism on the original supported quartet is precisely

$$G(\chi_{-+}) : G(\mathbb{C}[K_4]) \longrightarrow G(\mathbb{C}), \quad Q(\rho)^\bullet \longmapsto c_{-+}(\rho)^\bullet, \quad \tau \longmapsto \tau.$$

Here $\chi_{-+} = \text{ev}_{-1,+1}$ is the original group character extended linearly; it preserves multiplication because the character does, and hence its displayed G -lift is a semiring morphism. At $\beta = 1/2$ this morphism takes the supported value e ; the same original quartet has phase channel $c_{--} = 4i\gamma$. The orbit then has two distinct points when $\gamma \neq 0$, so its degree is 2μ and (R8) applies with that exact degree; the left-hand side of the sector bound is zero. Thus the arithmetic volume comparison and the supported critical value use the same typed character map, with the phase coordinate retained.

Part II

Complete source proof notes and chronology

A PR 8: Finite operator comparison

Source attribution. Web contribution, pull request #8, [Zeta research repository](#).

Exact revision. 25689c5376166df37801229bc275ff5a896007d9

Source file. workbenches/tau-base-cohomology/FINITE_OPERATOR_COMPARISON.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 57502c7fe6a7e1de239c94c3479abbb6a70b4718b551ad1bb1e26e2d3e4f4c22

A.1 Exact finite operator quotients of the tau-base theta cohomology

Owner-directed continuation, 12 September 2026. Mathematical derivation and research draft; no new analytic theorem below is claimed to have been checked by Lean.

A.1.1 0. Inputs, provenance, and the new result

This note continues TAU_BASE_MODEL.md, rather than moving the programme back to a finite-field variety. Its arithmetic coefficient object is the original split semiring

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e = 0_R^\bullet.$$

The geometric base remains the one-point absolute pointed base described there. The quotient

$$p_{\mathbb{Z}} : G(\mathbb{Z}) \longrightarrow \mathbb{Z}$$

and its square with $G(\mathbb{C}) \rightarrow \mathbb{C}$ remain part of the marked object. No ideal norm, Euler factor, scaling parameter, Gaussian, or completion factor is changed here.

The supplied derived formalization (PR #6, 5f6ee6575550d8a20c6fc3990359cb4f67855f80) provides the actual reconstruction, internal quotient, homology-kernel equivalence, and compatible kernel actions. Its finite-jet module supplies the balanced tensor-quotient comparison with the split-reflection square. Those results are reused; this note does not duplicate their Lean implementation.

The new analytic step computes the previously only surjective finite Mellin-jet observation as an exact operator quotient. For every polynomial h whose roots lie in the open critical strip, it constructs

$$Q/h(D)Q \xrightarrow{\sim} E_h/g_h E_h, \quad Q[h(D)] \xrightarrow{\sim} \ker(g_h : E_h \rightarrow E_h),$$

with $E_h = \mathbb{C}[t]/(h)$ and $g = 2\xi$. It then constructs finite spectral projectors on the original Q , not only quotients of Q , and identifies the torsion subsheaf of the balanced analytic realization with $\mathcal{O}/(2\xi)$. Their kernels, original supported zeros, nilpotent actions, and trace-producing residue pairing are all retained.

The general existence of spectral realizations of zeta zeros is prior literature (not a novelty claim here): see R. Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277. The contribution of this note is the explicit comparison in the specified workbench, together with its original SplitZero quotients and its complete finite kernel calculation.

A.1.2 1. The exact arithmetic complex and its operator

Retain

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0\}.$$

The source maps, with their original factors, are

$$D = -x\partial_x, \quad \Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}.$$

The Fourier transform is $\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx$. Poisson and the two moment conditions put ΘV in $\mathcal{B}$. Also $D\Theta = \Theta D$, and D preserves V .

Set

$$A = \mathbb{C}[t], \quad t \cdot F = DF, \quad t \cdot \phi = D\phi,$$

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}] \text{ in degrees } 0, 1, \quad Q = \mathcal{B}/\Theta V.$$

The existing injectivity proof of Θ gives $H^0(C_+) = 0$ and $H^1(C_+) = Q$.

This is the one-leg subcomplex of the actual tau-base chart complex

$$C_{+-} = [V \oplus V \xrightarrow{\Theta\phi - \Theta\widehat{\psi}} \mathcal{B}].$$

Its inclusion is $\phi \mapsto (\phi, 0)$ in degree zero and the identity in degree one. The quotient complex is the second copy of V in degree zero. The explicit section is $\psi \mapsto (\psi, \psi)$. The operator on that second copy is $1 - D$, because $D\widehat{\psi} = (1 - D)\psi$. Thus the additional joint-support degree-zero cohomology is retained in this split exact sequence; it is not included in Q and then silently discarded.

The actual source multiplier is

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

$$H_\phi(s) = c(s)M_+\phi(s), \quad c(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)}, \quad M_+\phi(s) = \int_0^\infty \phi(x) x^s \frac{dx}{x}.$$

On the open strip $\mathcal{S} = \{0 < \Re s < 1\}$, $c(s)$ is holomorphic and nowhere zero. The source identity is

$$\mathcal{M}\Theta\phi = gH_\phi.$$

The specific function

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}$$

has $H_{\phi_*} = 1$. Consequently

$$H_{P(D)\phi_*}(s) = P(s), \quad \mathcal{M}\Theta(P(D)\phi_*) = g(s)P(s).$$

These are identities with the stated Gaussian and $g = 2\xi$; no rescaling is performed.

A.1.3 2. A continuous inverse to the actual Euler differential

For any $\rho \in \mathbb{C}$, define on the subspace $\{H \in \mathcal{B} : \mathcal{M}H(\rho) = 0\}$

$$S_\rho H(x) = x^{-\rho} \int_x^\infty H(t) t^{\rho-1} dt.$$

All powers use the real logarithm on $x, t > 0$. Direct differentiation gives

$$(D - \rho)S_\rho H = H.$$

Endpoint control. Write $x = e^y$, $h(y) = H(e^y)$, and $\sigma = \Re \rho$. At $y \geq 0$, the displayed tail and a bound $|h(t)| \leq C_N e^{-Nt}$ give

$$|S_\rho H(e^y)| \leq \frac{C_N}{N - \sigma} e^{-Ny} \quad (N > \sigma).$$

At $y \leq 0$, the zero moment changes the same expression into the negative lower tail. A bound $|h(t)| \leq C'_N e^{Nt}$ gives

$$|S_\rho H(e^y)| \leq \frac{C'_N}{N + \sigma} e^{Ny} \quad (N + \sigma > 0).$$

Differentiating uses $(S_\rho H)'_y = -\rho S_\rho H - H(e^y)$, so every derivative has the same family of estimates. This proves membership in $\mathcal{B}$ and continuity in its stated seminorms. The homogeneous solution $x^{-\rho}$ fails the required rapid decrease unless its coefficient is zero. Thus $D - \rho$ is injective, and

$$(D - \rho)\mathcal{B} = \ker(F \mapsto \mathcal{M}F(\rho)).$$

Mellin intertwining gives

$$\mathcal{M}(S_\rho H)(s) = \frac{\mathcal{M}H(s)}{s - \rho},$$

with the removable value filled at $s = \rho$.

There is also a source inverse. For $\rho \in \mathcal{S}$ and $\phi \in V$ with $H_\phi(\rho) = 0$, use the same tail for $x > 0$ and extend evenly. The nonvanishing of $c(\rho)$ gives $M_+\phi(\rho) = 0$. Near zero the formula is

$$S_\rho \phi(x) = - \int_0^1 u^{\rho-1} \phi(xu) du.$$

Differentiation under this integral proves smoothness at zero in every order. The function is even and its value at zero is zero. The upper-tail formula proves Schwartz decay at infinity. Finally

$$(1 - \rho) \int_{\mathbb{R}} S_\rho \phi(x) dx = \int_{\mathbb{R}} \phi(x) dx = 0,$$

so the second moment condition is preserved. This proves the source range identity

$$(D - \rho)V = \ker(\phi \mapsto H_\phi(\rho)) \quad (\rho \in \mathcal{S}).$$

At the level of Taylor coefficients $\phi(x) = \sum_{j \geq 1} a_{2j} x^{2j}$, the local expression is $-\sum_{j \geq 1} a_{2j} x^{2j} / (\rho + 2j)$, interpreted to each finite Taylor order with its smooth remainder. All denominators and the minus sign are retained. Endpoints 0, 1 are not included in this source range theorem.

A.1.4 3. Every finite operator test, without choosing zeros first

Choose any finite set of distinct centres $\rho_1, \dots, \rho_r \in \mathcal{S}$ and positive integers m_i . Define the particular polynomial

$$h(t) = \prod_{i=1}^r (t - \rho_i)^{m_i}, \quad d = \deg h, \quad E_h = A/(h).$$

This defines the test; it does not presume that any centre is a zeta zero. The Chinese-remainder map is

$$\theta_h : E_h \xrightarrow{\sim} \prod_i \mathbb{C}[z_i]/(z_i^{m_i}), \quad [P(t)] \mapsto [P(\rho_i + z_i)].$$

For a holomorphic germ family f at the centres, $j_h f$ denotes its Taylor jets transported back by θ_h^{-1} . In particular $g_h = j_h g$ retains the original local units of g .

Iterate Section 2 through the factors of h . Uniqueness makes the inverse independent of their chosen order. This gives the continuous inverse

$$S_h : \ker(j_h \mathcal{M}) \longrightarrow \mathcal{B}, \quad h(D)S_h = \text{id}.$$

The exact range identities are

$$h(D)\mathcal{B} = \ker(j_h \mathcal{M}), \quad h(D)V = \ker(j_h H).$$

Surjectivity on the source follows from $H_{P(D)\phi_*} = P$. Surjectivity of the target jet map follows directly from finite Mellin interpolation: on an interval in $y = \log x$, the functions $e^{\rho_i y} y^j / j!$ are linearly independent. For a nonnegative smooth bump η positive on a subinterval, the Gram matrix

$$G_{(i,j),(k,l)} = \int \eta(y) e^{(\rho_i + \bar{\rho}_k)y} \frac{y^{j+l}}{j!l!} dy$$

is positive definite. Multiplying its inverse into the prescribed jet vector produces a compactly supported smooth preimage. This is only an interpolation device; it does not define or alter the arithmetic Weil pairing.

Consequently the vertical arrows in

$$\begin{array}{ccc} V/h(D)V & \xrightarrow{\bar{\theta}} & \mathcal{B}/h(D)\mathcal{B} \\ \alpha_h \downarrow \wr & & \wr \downarrow \beta_h \\ E_h & \xrightarrow{\times g_h} & E_h \end{array}$$

are actual A -module isomorphisms:

$$\alpha_h[\phi] = j_h H_\phi, \quad \beta_h[F] = j_h \mathcal{M}F.$$

An inverse to α_h sends the unique polynomial representative P of degree less than d to $[P(D)\phi_*]$. An inverse to β_h is any of the above explicit interpolation lifts, taken modulo $h(D)\mathcal{B}$; the class is independent of that choice by the range theorem.

A.1.5 4. The kernel and quotient of the global arithmetic operator

Both V and $\mathcal{B}$ are torsion-free A -modules: every factor $D - \rho$ is injective, since its homogeneous solution cannot be Schwartz at infinity. Over the PID A , they are flat. The bounded complex C_+ is therefore a valid flat model for derived tensoring. The displayed isomorphism computes

$$C_+ \otimes_A^{\mathbf{L}} E_h \simeq [E_h \xrightarrow{\times g_h} E_h]$$

in degrees zero and one. It is the base-change of the actual theta complex, not a newly declared spectral complex with an assigned differential.

The original exact sequence $0 \rightarrow V \xrightarrow{\Theta} \mathcal{B} \rightarrow Q \rightarrow 0$ also gives

$$0 \rightarrow Q[h(D)] \xrightarrow{\kappa_h} E_h \xrightarrow{\times g_h} E_h \rightarrow Q/h(D)Q \rightarrow 0.$$

Here $Q[h(D)] = \ker(h(D) : Q \rightarrow Q)$. The maps are explicit:

- Given $[F] \in Q[h(D)]$, uniquely write $h(D)F = \Theta\phi$. Then $\kappa_h[F] = j_h H_\phi$.
- The quotient observation sends $[F]$ to $j_h \mathcal{M}F$ modulo $g_h E_h$.
- For $P \in \ker(g_h : E_h \rightarrow E_h)$, choose its polynomial representative of degree less than d and set

$$\partial_h(P) = [S_h \Theta(P(D)\phi_*)] \in Q[h(D)].$$

The last expression is defined because its theta input has zero jets modulo h . Replacing P by $P + hR$ changes the result by $\Theta(R(D)\phi_*)$. Thus it is independent of representatives and is the inverse of κ_h onto its kernel. The two injectivity and surjectivity assertions also follow directly by lifting representatives in the preceding exact sequence; no dimension-only argument is needed.

All maps commute with D , with $U_a F(x) = F(x/a)$, and with compactly supported multiplicative convolution. On E_h , the scaling action is multiplication by $j_h(a^s)$, including its entire finite nilpotent Taylor polynomial.

At one centre, retain $g(\rho + z) = u(z)z^r$, where $u(0) \neq 0$ (and $r = 0$ is allowed). For test order m , the two cohomology modules have dimension $\min(m, r)$. When $r = 0$, multiplication by g_h is invertible. The derived amplitude complex is then acyclic, while its reconstructed supported homology still has each labelled internal zero.

The degree-zero term is $Q[h(D)]$; the degree-one term is $Q/h(D)Q$. Their finite alternating traces cancel. The arithmetic packet trace below is the degree-one action, or equivalently its explicitly constructed direct summand in the original Q . The canonical truncation triangle retains the degree-zero term rather than relabelling it as zero.

A.1.6 5. Exact finite spectral projectors on the original cohomology

Now let Z be any finite set of actual nontrivial zeros and retain their full multiplicities r_ρ . Put

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{r_\rho}, \quad E_Z = A/(h_Z), \quad v_Z(s) = g(s)/h_Z(s).$$

The quotient v_Z is entire and its germs are nonzero at every point of Z . Hence $v_{Z,h} = j_{h_Z} v_Z$ is an invertible element of E_Z . This inverse retains the actual Taylor coefficients of g/h_Z ; it is not set to one.

Here $g_{h_Z} = 0$, and the exact sequence becomes

$$Q[h_Z(D)] \xrightarrow[\kappa_Z]{\sim} E_Z, \quad Q/h_Z(D)Q \xrightarrow[\beta_Z]{\sim} E_Z.$$

There is a further map that must be calculated, not assumed: the inclusion of the kernel into Q , followed by the quotient. For $P \in E_Z$,

$$\mathcal{M}(S_{h_Z}\Theta(P(D)\phi_*)) = \frac{gP}{h_Z} = v_Z P.$$

Therefore this kernel-to-quotient comparison is precisely

$$\boxed{\beta_Z q_Z \partial_Z = M_{v_{Z,h}} : E_Z \xrightarrow{\sim} E_Z.}$$

Define

$$\sigma_Z = \partial_Z \circ M_{v_{Z,h}}^{-1} : E_Z \longrightarrow Q,$$

$$\boxed{\Pi_Z = \sigma_Z J_Z : Q \longrightarrow Q, \quad J_Z[F] = j_{h_Z} \mathcal{M}F.}$$

Then

$$\boxed{J_Z \sigma_Z = \text{id}, \quad \Pi_Z^2 = \Pi_Z, \quad \text{im } \Pi_Z = Q[h_Z(D)], \quad \ker \Pi_Z = h_Z(D)Q.}$$

This gives the actual equivariant direct sum

$$\boxed{Q = Q[h_Z(D)] \oplus h_Z(D)Q.}$$

A representative formula is completely determined. For $F \in \mathcal{B}$, let $P_{Z,F}$ be the unique polynomial of degree less than $\deg h_Z$ representing $v_{Z,h}^{-1} j_{h_Z} \mathcal{M}F$. Then

$$\boxed{\Pi_Z[F] = [S_{h_Z}\Theta(P_{Z,F}(D)\phi_*)].}$$

For a simple actual zero ρ , this specializes to

$$\Pi_{\{\rho\}}[F] = \frac{\mathcal{M}F(\rho)}{g'(\rho)} \left[x^{-\rho} \int_x^\infty \Theta\phi_*(t) t^{\rho-1} dt \right].$$

The displayed denominator is the actual $g'(\rho) = 2\xi'(\rho)$; it is not omitted. The bracketed class is an actual nonzero eigenclass in Q , proved by the exact sequence, even though its representative is only an eigenvector modulo the theta relation.

For finite packets the projectors satisfy

$$\Pi_Z \Pi_W = \Pi_{Z \cap W}, \quad \Pi_{Z \cup W} = \Pi_Z + \Pi_W \quad (Z \cap W = \emptyset).$$

These follow from the direct sums, coprimality of disjoint packet polynomials, and commutation with D . Inclusions of finite packets therefore give compatible invariant subspaces and quotient observations. No convergence of an infinite sum of projectors is asserted.

A.1.7 6. Carry every projector through the original split quotient

For any vector space M the single-active-fibre lift is $M^\tau = \{\tau\} \sqcup \{m^\bullet : m \in M\}$. For a submodule N use exactly

$$q_N^{\text{sp}} : M^\tau \rightarrow (M/N)^\tau, \quad m^\bullet \mapsto [m]^\bullet, \quad \tau \mapsto \tau.$$

This is the original coequalizer of the relation inclusion and its internal-zero companion. Its universal property is the one implemented in PR #6, including arbitrary noncancellative target semi-modules.

Consequently the finite observation has the split-linear equivalence

$$\boxed{(Q/h(D)Q)^\tau \xrightarrow{\sim} G(E_h/g_h E_h), \quad [F]^\bullet \mapsto (j_h \mathcal{M} F \bmod g_h E_h)^\bullet.}$$

The domain is being used as a split semimodule, not as an unconstructed ring. A class in $h(D)Q$ maps to the supported zero, and external absence alone maps to external absence.

Lift the projector as

$$\tilde{\Pi}_Z(\lambda, [F]) = (\lambda, \Pi_Z[F]).$$

This formula applies to the actual nonempty support faces of the tau-base theta model; their degree-one transports are the stated identities of Q . At bottom it is the original zero map of the bottom module. The full reconstruction interface permits a nonzero bottom fibre when a subsequent diagram requires one; it is not silently specialized away on an opposite-index dual diagram.

For disjoint packets,

$$\boxed{\tilde{\Pi}_Z \tilde{\Pi}_W = \varepsilon, \quad \varepsilon(\lambda, x) = (\lambda, 0).}$$

This is the supported lift of the zero linear operator, not the constant external-zero map. More uniformly, take $\Pi_\emptyset = 0_Q$ and lift it to ε . The scalar representation

$$G(\text{End}_{\mathbb{C}} Q) \longrightarrow \text{End}_{G(\mathbb{C})}(Q^\tau)$$

sends $f^\bullet$ to its split lift and sends τ to the constant- τ map. It is a unital semiring morphism and sends the supported scalar zero to ε .

Retain smaller admitted source labels For a subspace $W \subset V$, first use its specified enlargement $W_D = \mathbb{C}[D]W$ and keep the map

$$\mathcal{B}/\Theta W \longrightarrow Q_{W_D} := \mathcal{B}/\Theta W_D, \quad \ker = \Theta W_D/\Theta W.$$

For a D -stable W , define the ideal

$$J_{W,h} = \{j_h H_\phi : \phi \in W\} \subseteq E_h.$$

It is an ideal because $H_{P(D)\phi} = PH_\phi$ and every element of E_h has a polynomial representative. The same target range calculation gives

$$\boxed{Q_W/h(D)Q_W \cong E_h/(g_h J_{W,h}).}$$

Enlarging to $W + \mathbb{C}[D]\phi_*$ makes $J_{W,h} = E_h$. The exact finite comparison kernel is

$$\boxed{g_h E_h / (g_h J_{W,h}).}$$

Thus the finite equality for the full arithmetic relation space has a precise smaller-label comparison. It is not asserted for every original W without this kernel.

A.1.8 7. The Hausdorff observation and the remaining global kernel

The projectors just constructed have continuous finite-rank representatives on $\mathcal{B}$: J_Z is continuous, E_Z is finite dimensional, and S_{h_Z} is continuous on its zero-jet domain. They annihilate ΘV , hence also its closure. They therefore descend to

$$Q^H = \mathcal{B} / \overline{\Theta V}.$$

The finite observation still has kernel exactly $h_Z(D)Q^H$. Indeed

$$\ker J_Z = h_Z(D)\mathcal{B} + \Theta V,$$

and continuity gives $\overline{\Theta V} \subset \ker J_Z$. Hence adding the closure to the right side leaves it unchanged. The same finite section yields a continuous direct sum

$$Q^H = \sigma_Z(E_Z) \oplus h_Z(D)Q^H.$$

The algebraic packet $Q[h_Z(D)]$ injects into this Hausdorff quotient: its composite with J_Z is already an isomorphism onto E_Z . No conclusion about vanishing of the entire algebraic-to-Hausdorff kernel follows from this finite statement.

Let

$$\mathcal{R}_\infty = \bigcap_{Z \text{ finite}} \ker \Pi_Z = \bigcap_{Z \text{ finite}} h_Z(D)Q.$$

The map $Q \rightarrow \prod_\rho \mathcal{O}_\rho / (g)$ has exactly this kernel. Every nonzero polynomial whose roots are in $\mathcal{S}$ acts bijectively on $\mathcal{R}_\infty$. To verify this, its kernel on Q lies in the finite packet at its zero centres; its cokernel is supported there as calculated in Section 4. On the complementary summand it is bijective, and its inverse preserves all other commuting finite projectors. This proves injectivity and surjectivity on the intersection.

The continuation retains $\mathcal{R}_\infty$ and its scaling action. Finite packets do not prove it zero. A topological spectral-synthesis theorem or a global cohomological trace argument must address that specific invariant module rather than infer its disappearance from finite-jet isomorphisms.

7.1 The global balanced kernel has a computed local-cohomology complement Work now on the analytic strip $\mathcal{S}$, with its sheaf $\mathcal{O}$ and sheaf $\mathcal{K}$ of meromorphic functions. Form the sheafified balanced realization

$$\mathcal{M}_Q = \mathcal{O} \otimes_A Q, \quad \beta : \mathcal{M}_Q \longrightarrow \mathcal{O}/(g), \quad b \otimes [F] \longmapsto [bMF].$$

Let $\text{tors}_\mathcal{O}(\mathcal{M}_Q)$ denote its torsion subsheaf: a germ is in it when some nonzero holomorphic germ annihilates it. This is torsion over the specified analytic coefficient ring; its connection to the original split support is the reconstruction of the inclusion and quotient maps below.

Theorem. The actual Mellin map restricts to an isomorphism

$$\boxed{\beta|_{\text{tors}} : \text{tors}_\mathcal{O}(\mathcal{M}_Q) \xrightarrow{\sim} \mathcal{O}/(g).}$$

Moreover, $\mathcal{N} = \ker \beta$ has a canonical $\mathcal{K}$ -module structure, and the displayed map canonically splits the sequence:

$$\boxed{\mathcal{M}_Q \cong \mathcal{O}/(g) \oplus \mathcal{N}, \quad \mathcal{K} \otimes_{\mathcal{O}} \mathcal{N} \cong \mathcal{N}.}$$

Proof at every stalk. Fix $\rho \in \mathcal{S}$ and set $x = s - \rho$. If $g(\rho) \neq 0$, Section 4 with $h = t - \rho$ proves that $D - \rho$ is bijective on Q . Consequently x is invertible on $\mathcal{M}_{Q,\rho}$, and this module is a module over the meromorphic field $\mathcal{O}_\rho[1/x]$. Its torsion is zero, as is $\mathcal{O}_\rho/(g)$.

If $r = \text{ord}_\rho g > 0$, the finite projector for this one full packet gives

$$Q = Q[(D - \rho)^r] \oplus (D - \rho)^r Q.$$

On the second summand, $(D - \rho)^r$ is bijective: it is injective by the direct-sum intersection, and

$$(D - \rho)^r Q = (D - \rho)^{2r} Q$$

follows by applying $(D - \rho)^r$ to the direct sum. Hence $D - \rho$ itself is bijective there. Tensoring the direct sum with $\mathcal{O}_\rho$ gives a first summand $\mathcal{O}_\rho/(x^r)$, on which β is the isomorphism calculated in Section 5, and a second summand on which x is invertible. The map β kills the second summand because it is in $x^r \mathcal{M}_{Q,\rho}$, while x^r annihilates its target. Every nonzero germ of $\mathcal{O}_\rho$ is a unit times a power of x , so that second summand is a module over $\mathcal{K}_\rho$ and is torsion-free.

This proves the stated torsion isomorphism at every stalk. It glues without choosing neighbourhood projectors: both the torsion subsheaf and the restricted map β are intrinsic. The inverse of that isomorphism, followed by the torsion inclusion, is the global section of β . Local inverses to nonzero holomorphic scalars on $\mathcal{N}$ are unique, so they glue to its $\mathcal{K}$ -module structure. This proves the theorem.

7.2 The extraction is an actual local-cohomology map At the same stalk the principal-ideal local-cohomology complex is the explicit localization complex

$$R\Gamma_{(x)}(\mathcal{M}_{Q,\rho}) = [\mathcal{M}_{Q,\rho} \longrightarrow \mathcal{M}_{Q,\rho}[1/x]], \quad \text{degrees } 0, 1.$$

The decomposition just proved computes it:

$$\boxed{R\Gamma_{(x)}(\mathcal{M}_{Q,\rho}) \xrightarrow{\sim} (\mathcal{O}_\rho/(g))[0].}$$

The finite torsion summand maps to zero upon localization; the meromorphic summand maps isomorphically to itself. The map to the displayed target is β in degree zero and zero in degree one. Thus the supported arithmetic divisor is the actual cohomological image of the localization kernel, not a prescribed deletion of a list of spectral points.

Before analytic extension there is the corresponding operator-localization map

$$Q \longrightarrow A[h_Z^{-1}] \otimes_A Q,$$

with kernel $Q[h_Z(D)]$ and cokernel zero, because $h_Z(D)$ is invertible on its complementary summand. Its supported lift sends that kernel to the internal supported zero, not to external absence.

At every original label λ , reconstruct the localization complex using

$$d(\lambda, m) = (\lambda, m/1), \quad \varepsilon(\lambda, m) = (\lambda, 0).$$

Its internal degree-zero cycles are exactly the reconstructed torsion module. Its internal first cohomology is the support skeleton, since the ordinary first cohomology vanishes. The map $p_{\mathbb{Z}}$ remains

in the structural coefficient square throughout; x is the analytic image of the operator $t - \rho$, not a replacement for the scalar e or τ .

The theorem does not assert $\mathcal{N} = 0$. It computes what local cohomology does to it: localization is an isomorphism on that module, so it contributes no cohomology to the stated local-support complex. Its possible contribution to a different global trace functor has not been declared absent.

A.1.9 8. The residue duality is now inside an actual derived finite comparison

Assume the centres and their orders in h are stable under $\iota(s) = 1 - \bar{s}$. For germs define

$$f^\dagger(s) = \overline{f(1 - \bar{s})}.$$

The chosen monic polynomial satisfies $h^\dagger = (-1)^d h$. Retain that factor. Define the perfect finite algebra pairing

$$R_h(f, u) = \sum_i \text{Res}_{s=\rho_i} \frac{f^\dagger(s)u(s)}{h(s)} ds.$$

It satisfies $\overline{R_h(u, f)} = (-1)^{d+1} R_h(f, u)$. No phase is inserted to change this sign. Since $g^\dagger = g$, multiplication by g_h is self-adjoint for this pairing. Therefore it induces a perfect pairing

$$\overline{\text{coker } g_h} \otimes \ker g_h \longrightarrow \mathbb{C}.$$

Well-definedness is the identity $R_h(ga, u) = R_h(a, gu)$; nondegeneracy follows by taking the annihilator of the image in the finite perfect algebra pairing. Through Section 4 this pairs the two actual cohomology groups of $C_+ \otimes_A^{\mathbf{L}} E_h$.

For an exact packet h_Z , set $v_{Z,h} = j_h(g/h_Z)$ as above and

$$j_Z : A_Z \rightarrow \ker g_h = E_Z, \quad [f] \mapsto [j_h(h_Z/g)f].$$

Here $A_Z = \prod_{\rho \in Z} \mathcal{O}_\rho / (g) \cong E_Z$ by the specified local quotient maps; h_Z/g is holomorphic at every centre. The local units are not replaced by one. The exact contraction is

$$R_h(f, j_Z k) = \sum_{\rho \in Z} \text{Res}_\rho \frac{f^\dagger k}{g} ds =: R_Z(f, k).$$

Thus the previously constructed arithmetic residue form is the duality pairing of the actual finite operator comparison after the displayed unit morphism.

The Jacobian map is still $J_g[k] = [g'k]$. Its contraction is

$$R_h(f, j_Z J_g k) = R_Z(f, J_g k) = \sum_{\rho \in Z} r_\rho \overline{f(1 - \bar{\rho})} k(\rho) = \text{Tr}(M_{f^\dagger k} | A_Z).$$

In particular the factor 2 in $g = 2\xi$ is retained in both the denominator and the Jacobian. The trace contraction is invariant under their combined explicitly specified transformation, not by deleting that factor in only one place.

Every map lifts to the original support fibres. Dual support arrows are precomposition maps on the opposite index category. A vanishing pairing value is the supported scalar $e_{\mathbb{C}}$; an absent argument gives τ .

A.1.10 9. Cohomological placement and arithmetic traces

On the tau-base chart use the one-leg sheaf $\mathcal{T}_+$ and its A -action. The bounded projective resolution from TAU_BASE_MODEL.md gives the canonical map

$$\mathbf{A}_\tau(\mathcal{T}_+ \otimes_A^{\mathbf{L}} E_h) \xrightarrow{\sim} \mathbf{A}_\tau(\mathcal{T}_+) \otimes_A^{\mathbf{L}} E_h \xrightarrow{\sim} [E_h \xrightarrow{g_h} E_h].$$

These isomorphisms use finite evaluation sums and the displayed free/flat resolutions. They are not compact-support base-change assertions for a different arithmetic scheme.

The packet projector is now an actual finite-rank endomorphism of the original $H_\tau^1 = Q$ (and of each of its retained support fibres). For an admissible smooth compactly supported multiplicative test k ,

$$H_k F(x) = \int_0^\infty k(a) F(x/a) \frac{da}{a}$$

commutes with the projector. Its ordinary finite-rank trace is

$$\boxed{\mathrm{Tr}(H_k \Pi_Z | Q) = \sum_{\rho \in Z} r_\rho \mathcal{M}k(\rho).}$$

This is a finite-rank trace on the original cohomology, using its explicit finite invariant image. It is stronger than declaring that a surjective finite observation has a trace. Infinite trace summation still requires the specified topology and admissible test class.

On r -fold products over the tau-point, the already calculated external tensor map gives $H^r = Q^{\otimes r}$. The operator $\Pi_Z^{\otimes r}$ is a finite projector onto $Q[h_Z]^{\otimes r}$. The full nilpotent scaling action there is

$$a^{\rho_1 + \dots + \rho_r} \exp \left((\log a) \sum_{j=1}^r N_{\rho_j, j} \right).$$

For a repeated exponent the exact modulus comparison remains

$$\log \frac{|a^{r\rho}|^2}{a^r} = r(2\Re\rho - 1) \log a.$$

This supplies actual finite invariant objects on which a Deligne-style tensor estimate would have to act. It does not prove that estimate. The source's opposite weight bounds, its global dualizing theory, and the admissibility of its geometric hypotheses have not been inferred from these finite isomorphisms.

A.1.11 10. Status and publication boundary

Established in this note, with written proofs: the differential range identities on the stated spaces; the exact finite operator quotient and torsion kernel; the derived finite comparison; the analytic torsion-subsheaf and local-cohomology identification with $\mathcal{O}/(2\xi)$; finite spectral direct summands with explicit projectors; the original split lifts and smaller-label kernels; the residue/Jacobian contraction on that comparison; and compatible finite-rank arithmetic traces and tensor actions.

Not claimed: a global injectivity theorem for the full balanced analytic realization; vanishing of $\mathcal{R}_\infty$; identification of the entire four-point chart category with an arithmetic curve; a full compact-support/ordinary-cohomology realization satisfying the hypotheses of Weil II; positivity of the unshifted Weil form; RH or GRH. The finite tests are not a Lean or analytic proof certificate.

The prior base note and this continuation are published as additive research material beside PR #6. The original scalar core, formal modules, frozen reader and all existing branches remain unchanged. New derivations are AI-assisted work under the owner's programme; the structural source and its Lean implementation retain their own attribution. No copyrighted source corpus or private conversation is included.

References and source interfaces

- Owner's SplitZero programme: the original scalar, support fibre and quotient constructions; PR #6 at 5f6ee6575550d8a20c6fc3990359cb4f67855f80, especially SplitZeroHomology.lean, SplitZeroComplex.lean, SplitZeroInternalQuotient.lean, and SplitZeroFiniteJets.lean.
- TAU_BASE_MODEL.md: the actual characteristic-zero chart, cohomology, right-adjoint and tensor constructions used here. It remains an authored research model, not an accepted geometric RH realization.
- Deligne, P., *La conjecture de Weil. II* (1980), sections 3.3.4-3.3.6 and 6.2.1-6.2.7. Selected supplied TeX passages were reread for the intended comparison/duality architecture; no source proof is republished or claimed fully audited.
- Meyer, R., *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277. Prior spectral-realization context; no theorem from it is silently used to identify the test spaces here.
- The Stacks Project, Derived tensor product, Tag 06XY. The general bounded-flat-complex construction used in Section 4.

B PR 9: Chain descent

Source attribution. Web contribution, pull request #9, [Zeta research repository](#).

Exact revision. 8f99b94d306c3b1aaa817d52c57fa6fd298d511c

Source file. workbenches/tau-chain-descent/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 1ab370cf2db9fce70020bebbab76d3dbd3bd37d39e28c83114eeb0eccfa2f0b3

B.1 Chain-level arithmetic descent over the tau base

Owner-directed research continuation, 12 September 2026. Draft proofs for review; no new Lean certification or RH claim.

B.1.1 0. Fixed inputs and publication scope

This is an additive continuation of Zeta PR #8 at 25689c5376166df37801229bc275ff5a896007d9, particularly workbenches/tau-base-cohomology/FINITE_OPERATOR_COMPARISON.md (Git blob 68f59d098d60708f8e899bb960abbd1135ebb3b4). Its sections 1–5 supply the stated analytic range theorem and the finite packet maps. The new result here is their chain-level descent, its actual extension class, and the connection of that class to residue duality and the arithmetic trace.

The source programme retains

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

$$p_R(\tau) = 0_R, \quad p_R(r^\bullet) = r.$$

The absolute pointed base and the coefficient square remain

$$\mathfrak{b}_\tau = \operatorname{Spec} \mathbf{F}_{1,\tau}, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

The target of the first vertical map remains the infinite ring of integers. Every new arrow below is a coefficient or cochain construction over this marked base. No base field, ideal norm, Euler factor, or completion constant is changed.

The supplied derived note and PR #6 provide `Recovery.recoverEquiv`, `Relations.coequalizer_universal`, the internal-homology coequalizer, `ChainMap.homologyKernelEquiv`, and `homology_int_ertwines`. Their algebraic conclusions are used at their stated types. No existing formal module is modified. Analytic inputs from PR #8 remain written research arguments; they are not promoted to Lean theorems.

B.1.2 1. The actual arithmetic source and packet

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for every } b \in \mathbb{Z}, k \geq 0\}.$$

Retain

$$D = -x\partial_x, \quad \Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x},$$

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad A = \mathbb{C}[t], \quad t \cdot F = DF.$$

The one-leg tau-chart complex is

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad |V| = 0, \quad |\mathcal{B}| = 1, \quad Q = H^1(C_+).$$

The injectivity of theta gives $H^0(C_+) = 0$. The two-leg complex is

$$C_{+-} = [V \oplus V \longrightarrow \mathcal{B}], \quad (\phi, \psi) \mapsto \Theta\phi - \Theta\widehat{\psi}.$$

Its source operator is $(D, 1 - D)$; the inclusion $C_+ \rightarrow C_{+-}$ is $\phi \mapsto (\phi, 0)$ and the identity in degree one. It leaves the extra degree-zero copy of V in C_{+-} present.

Use the unchanged arithmetic multiplier

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

The input function is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2},$$

and the established identities are

$$\mathcal{M}\Theta\phi = gH_\phi, \quad H_{\phi_*} = 1, \quad H_{P(D)\phi_*}(s) = P(s).$$

Fix a nonempty finite packet Z of actual nontrivial zeros, with full orders m_ρ . Set

$$h(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E = A/(h), \quad v(s) = g(s)/h(s).$$

The specific monic polynomial defines this finite test; it does not rescale g . Write j_h for the complete Taylor-jet/Chinese-remainder map at these centres, and retain the invertible element

$$v_h = j_h v \in E^\times, \quad \varepsilon_Z = v_h^{-1}.$$

For $u \in E$, let $R_Z(u) \in \mathbb{C}[t]_{<d}$ be the unique polynomial representing $\varepsilon_Z u$.

The input range theorem supplies the inverse $S_h^{\mathcal{B}}$ of $h(D)$ on the zero-jet range, and the analogous source inverse S_h^V . For one factor,

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy.$$

The stated source inverse uses $0 < \Re \rho < 1$ and the zero moment. Iteration supplies S_h . Define the actual maps

$$J : \mathcal{B} \rightarrow E, \quad JF = j_h \mathcal{M}F,$$

$$s : E \rightarrow \mathcal{B}, \quad s(u) = S_h^{\mathcal{B}} \Theta(R_Z(u)(D)\phi_*).$$

The inverse is defined because $gR_Z(u)$ has all required zero jets. Directly,

$$\mathcal{M}s(u) = vR_Z(u), \quad Js = 1_E, \quad J\Theta = 0, \quad h(D)s(u) = \Theta(R_Z(u)(D)\phi_*). \quad (1)$$

These equations fix the representative map, including all derivatives of g/h . On Q , the existing projector is $\Pi_Z = \bar{s}\bar{J}$, with image $Q[h(D)]$ and kernel $h(D)Q$.

B.1.3 2. The exact equivariant extension and its class

Define an actual subspace of the original test-function space:

$$\mathcal{B}_Z = \{F \in \mathcal{B} : [F] \in Q[h(D)]\}.$$

Then

$$\mathcal{B}_Z = \Theta V \oplus s(E) \quad \text{as complex vector spaces,}$$

with maps

$$(\phi, u) \mapsto \Theta\phi + s(u), \quad F \mapsto (\Theta^{-1}(F - sJF), JF). \quad (2)$$

For the inverse, $F - sJF$ has zero class in Q by the packet projector theorem; injectivity of theta makes its source unique. The intersection is zero by $Js = 1$ and $J\Theta = 0$.

The original operator D preserves $\mathcal{B}_Z$, and there is a short exact sequence of A -modules

$$0 \longrightarrow V \xrightarrow{\Theta} \mathcal{B}_Z \xrightarrow{J} E \longrightarrow 0. \quad (3)$$

The arrows in (3), not just the three modules, specify its extension class. Compute it using the resolution

$$0 \longrightarrow A \xrightarrow{\times h} A \longrightarrow E \longrightarrow 0.$$

The resulting isomorphism $\text{Ext}_A^1(E, V) \cong V/h(D)V$ sends an extension to $h(D)b$ in the source, where b is any lift of $1 \in E$. Fixing this explicit rule fixes its sign. Under the existing map $[\phi] \mapsto j_h H_\phi$, the class of (3) is

$$[\mathcal{B}_Z] = \varepsilon_Z = j_h(h/g) \in E^\times. \quad (4)$$

Proof: use the lift $s(1)$ and (1). Its source boundary is $R_Z(1)(D)\phi_*$, whose H -jet is ε_Z .

In particular, (3) has no A -linear section. This also follows immediately from the injectivity of $h(D)$ on $\mathcal{B}$: an A -linear map $E \rightarrow \mathcal{B}$ has image killed by $h(D)$ and is therefore zero. The map s in (2) is the explicit complex-linear section, and the next section computes exactly how it fails to intertwine the arithmetic action. This non-splitting occurs for every nonempty packet, including critical-line packets; it is not an RH-failure criterion.

The general use of extension classes is standard homological algebra (Stacks, Tags 010I and 06XP). Formula (4) calculates the specific extension arising from this theta complex.

B.1.4 3. The scaling defect is an original theta boundary

For $P \in A$ and $u \in E$, define

$$q_P(u) = \frac{P(t)R_Z(u) - R_Z(Pu)}{h(t)} \in A, \quad k_P(u) = q_P(u)(D)\phi_* \in V. \quad (5)$$

The numerator is divisible by h , since both of its residues are $P\varepsilon_Z u$. Using (1) and the injectivity of $h(D)$ on $\mathcal{B}$ gives

$$\boxed{P(D)s - sM_P = \Theta k_P.} \quad (6)$$

No quotient has yet been applied in (6). The right side is an actual theta function. The composition law is

$$\boxed{k_{PQ} = P(D)k_Q + k_P M_Q.} \quad (7)$$

It follows either by expanding the numerator of (5), or by expanding the commutator in (6) and using injectivity of theta.

For $P = t$, the numerator of (5) has degree at most d . Hence

$$\ell_Z(u) = [t^{d-1}]R_Z(u), \quad q_t(u) = \ell_Z(u), \quad \boxed{Ds(u) - s(tu) = \ell_Z(u)\Theta\phi_*} \quad (8)$$

Thus, in the exact coordinates (2), the original operator is

$$\boxed{D_{\mathcal{B}_Z} = \begin{pmatrix} D_V & \phi_*\ell_Z \\ 0 & M_t \end{pmatrix}.} \quad (9)$$

Its off-diagonal map is rank one and its coefficient is determined by the retained local arithmetic unit. It is not chosen to improve a spectral estimate.

For one simple zero,

$$h = t - \rho, \quad \varepsilon_Z = 1/g'(\rho),$$

$$s(1) = \frac{1}{g'(\rho)} x^{-\rho} \int_x^\infty \Theta\phi_*(y) y^{\rho-1} dy,$$

$$\boxed{(D - \rho)s(1) = \frac{1}{g'(\rho)} \Theta\phi_*} \quad (10)$$

The coefficient is $1/(2\xi'(\rho))$, with the original factor two retained.

The full dilation group For $a > 0$, retain $U_a F(x) = F(x/a)$ and $A_a = M_{j_h(a^s)}$ on E . Define

$$k_a(u) = S_h^V (U_a(R_Z(u)(D)\phi_*) - R_Z(A_a u)(D)\phi_*). \quad (11)$$

The source of S_h^V has zero H -jets because $a^s \varepsilon_Z u$ and $\varepsilon_Z A_a u$ have equal jets. All powers use the real logarithm of a . The exact identities are

$$U_a s - s A_a = \Theta k_a, \quad k_{ab} = U_a k_b + k_a A_b, \quad k_1 = 0. \quad (12)$$

The same construction applies to a smooth compactly supported multiplicative test f : replace U_a by $H_f = \int f(a) U_a da/a$ and A_a by $M_{j_h \mathcal{M} f}$. The source integral belongs to V ; its H -jets give the same cancellation. These are actual linear maps, not a choice of a formal matrix on E alone.

B.1.5 4. A derived equivariant inclusion with its full homotopy

Put $P_Z = sJ : \mathcal{B} \rightarrow \mathcal{B}$. It is a finite-rank idempotent, and $P_Z\Theta = 0$. Consequently

$$\mathfrak{p}_Z = (0, P_Z) : C_+ \rightarrow C_+ \quad (13)$$

is a cochain idempotent. Its image is the two-term complex $[0 \rightarrow s(E)]$, complex-linearly isomorphic to $E[-1]$.

For the operator $P(D)$ on C_+ , define a degree-minus-one map with sole component

$$\mathbf{H}_P^1 = k_P J : \mathcal{B} \rightarrow V.$$

Equation (6) gives

$$\boxed{P(D)\mathfrak{p}_Z - \mathfrak{p}_Z P(D) = d\mathbf{H}_P + \mathbf{H}_P d.} \quad (14)$$

In degree one this is $\Theta k_P J$; in degree zero it is zero because $J\Theta = 0$. For dilation use $\mathbf{H}_a^1 = k_a J$. Equation (12) is the coherence law of those homotopies.

There is also a strictly A -linear derived presentation. Set

$$C_Z = [V \xrightarrow{\Theta} \mathcal{B}_Z].$$

In the coordinates (2) its differential is $\phi \mapsto (\phi, 0)$ and its action is (9). The maps

$$\iota : C_Z \hookrightarrow C_+, \quad \pi : C_Z \rightarrow E[-1], \quad \pi^0 = 0, \quad \pi^1 = J \quad (15)$$

are strictly A -linear cochain maps, and π is a quasi-isomorphism. Thus the actual equivariant inclusion is the roof

$$\boxed{E[-1] \xleftarrow[\simeq]{\pi} C_Z \xrightarrow{\iota} C_+.} \quad (16)$$

The complex-linear deformation retraction also has an exact contracting homotopy: on $\mathcal{B}_Z$, send $\Theta\phi + s(u)$ to $\phi \in V$. Its equation is $1 - s\pi = dH + Hd$ (with s interpreted as the degree-one cochain section). This homotopy is not claimed A -linear; the A -linear replacement is the roof (16).

The strictly A -linear map $j : C_+ \rightarrow E[-1]$ is J in degree one and zero in degree zero. It satisfies $j\iota = \pi$. This gives a split retract in $D(A)$ without falsely asserting an A -linear section into $\mathcal{B}$.

On the two-leg complex use $\mathfrak{p}_Z^0 = 0$ on $V \oplus V$, $\mathfrak{p}_Z^1 = P_Z$, and $\mathbf{H}_a^1(F) = (k_a JF, 0)$. The source dilation is $(U_a, aU_{1/a})$, so (14) holds with its original two-leg differential. The inclusion $C_+ \rightarrow C_{+-}$ commutes with these cochain projectors. The extra degree-zero cohomology of C_{+-} is killed by this projector; it remains in the complementary complex.

B.1.6 5. The mixed-support homotopies require the actual synchronization map

Let $L = \mathcal{P}(\{+, -\})$ be the original chart-support lattice. Its three active complexes are

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad C_- = [V \xrightarrow{-\Theta\mathcal{F}} \mathcal{B}], \quad C_{+-} = [V \oplus V \xrightarrow{\Theta(\phi - \hat{\psi})} \mathcal{B}].$$

Here $\mathcal{F}$ is the specified Fourier transform, with $\mathcal{F}^2 = 1$ on even functions. The source inclusions into the joint fibre are $v \mapsto (v, 0)$ and $v \mapsto (0, v)$; degree-one transports are identities. The bottom complex is zero in this particular model. The new formalization allows a nonzero bottom fibre in the general reconstruction; no assertion about all diagrams is obtained by specializing this model.

The reconstructed operations are

$$(\lambda, x) + (\mu, y) = (\lambda \vee \mu, \rho_{\lambda, \lambda \vee \mu} x + \rho_{\mu, \lambda \vee \mu} y),$$

$$e(\lambda, x) = (\lambda, 0), \quad \tau(\lambda, x) = (\perp, 0). \quad (17)$$

The cochain projector $\tilde{p}_Z$ is zero-in-its-fibre in degree zero and P_Z in degree one. It commutes with the support transports. So does the actual action: U_a on the plus source, $aU_{1/a}$ on the minus source, their direct sum on the joint source, and U_a at every active overlap.

The objectwise homotopies are, however, different maps. With $b = k_a JF$, they are

$$H_{a,+}^1(F) = b, \quad H_{a,-}^1(F) = -\hat{b}.$$

Their images in the joint source are $(b, 0)$ and $(0, -\hat{b})$. The difference is the explicit cycle

$$(b, \hat{b}) \in \ker d_{+-} = H^0(C_{+-}). \quad (18)$$

For nonzero k_a , there is no label-preserving family of homotopies compatible with both source inclusions: the joint component would have to equal both $(b, 0)$ and $(0, -\hat{b})$. This says exactly which naturality square fails. It does not discard the cohomology or the arithmetic action.

The replacement is the programme's actual synchronization. Define the support map

$$r_L(\perp) = \perp, \quad r_L(\lambda) = \{+, -\} \quad (\lambda \neq \perp).$$

In degree zero use literal multiplication by $E = (1, e)$ on the original two-coordinate support module:

$$\mathfrak{r}^0(a, b) = (a + eb, b + ea).$$

In degree one define $\mathfrak{r}^1(\lambda, F) = (r_L \lambda, F)$ for active labels and preserve global zero. An absent source coordinate becomes its supported vector zero, so the source differential has unchanged amplitude. Hence

$$d\mathfrak{r}^0 = \mathfrak{r}^1 d, \quad \mathfrak{r}^2 = \mathfrak{r}. \quad (19)$$

This uses the operation verified in the synchronization attachment, now on the actual complex. The fixed active fibre is exactly C_{+-} . On degree-one homology, the map is

$$(\lambda, [F]) \mapsto (\{+, -\}, [F]).$$

It has no nonzero amplitude kernel, because all degree-one transports are the identity of Q . The original mask is retained through

$$x \mapsto (\text{supp}(x), \mathfrak{r}x).$$

This map is injective in each degree of this chart complex: in degree zero the coordinate inclusions are injective, and in degree one the transport is the identity. This injectivity is a proved property of these maps, not an assumption for general diagrams with noninjective transports.

There are now two actual split-linear homotopies with the same support map r_L . Define

$$\tilde{H}_a^{\text{left},1}(\lambda, F) = (\{+, -\}, (k_a JF, 0)),$$

$$\tilde{H}_a^{\text{right},1}(\lambda, F) = (\{+, -\}, (0, -\mathcal{F}k_a JF)),$$

and send global zero to global zero. Both satisfy

$$\tau(\tilde{U}_a \tilde{p}_Z + (-1)^{\bullet} \tilde{p}_Z \tilde{U}_a) = \tilde{d} \tilde{H}_a + \tilde{H}_a \tilde{d}. \quad (20)$$

In degree one this is (6); in degree zero it follows from $Jd = 0$. The two choices differ by the retained cycle (18). No averaging or removal of that cycle is performed. This is the precise role of synchronization here: it permits a common supported comparison fibre for the homotopy, while the original mask and the difference of homotopies remain recorded.

In the simple-zero case (10), the defect in any source support is the nonzero chain amplitude $\Theta(\phi_*/g'(\rho))$. Its internal homology class is the zero of that support, not external absence.

For every chain comparison $f : C \rightarrow C'$, the supplied formalized kernel equivalence remains

$$\{z \in Z(C) : f(z) \in B(C')\} / B(C) \xrightarrow{\sim} \ker H(f).$$

Applied to π in (15), it computes the zero kernel in degree one from its actual exact sequence. Applied to support synchronization, it yields the degree-one assertion just proved and the separately retained degree-zero data. It is not applied to a different global comparison without carrying that comparison's maps.

For any smaller D -stable admitted source $W \subset V$, the map $Q_W \rightarrow Q$ has kernel $\Theta V / \Theta W$. A homotopy value may not already lie in W . The explicit replacement is enlargement by the actual source values, with this quotient map and its kernel retained. The present chart uses full V , where every displayed homotopy is defined.

B.1.7 6. The same extension class produces the arithmetic duality factor

Assume Z is stable under $\rho \mapsto 1 - \bar{\rho}$. Then

$$f^\dagger(s) = \overline{f(1 - \bar{s})}, \quad h^\dagger = (-1)^d h.$$

The perfect finite pairing is

$$R_h(f, u) = \sum_{\rho \in Z} \text{Res}_{s=\rho} \frac{f^\dagger(s) u(s)}{h(s)} ds,$$

with exact symmetry $\overline{R_h(u, f)} = (-1)^{d+1} R_h(f, u)$.

Identify E with the actual divisor algebra $A_Z = \prod_{\rho \in Z} \mathcal{O}_\rho / (g)$ by polynomial evaluation and the complete Chinese-remainder jets. The nonzero local units in g/h make this an isomorphism. The operator represented by the extension class is

$$M_{\varepsilon_Z} : E \rightarrow E.$$

Substituting (4) into the residue formula gives

$$R_h(f, M_{\varepsilon_Z} u) = \sum_{\rho \in Z} \text{Res}_\rho \frac{f^\dagger u}{g} ds = R_Z(f, u). \quad (21)$$

This is independent of representatives: changing ε_Z by h times a germ adds a holomorphic differential. Thus the actual extension class is precisely the unit morphism converting the polynomial-resolution residue form into the arithmetic residue form.

Let $J_g = M_{j_h g'}$. Then

$$\boxed{R_h(f, M_{\varepsilon_Z} J_g u) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} u(\rho) = \text{Tr}(M_{f^\dagger u} \mid E).} \quad (22)$$

Locally $g = z^m w(z)$ gives $g'/g = m/z + w'/w$. This proves the formula and retains every multiplicity. The product $\varepsilon_Z J_g$ carries both factors associated to $g = 2\xi$.

The pairing-valued support map sends a zero value to $e_{\mathbb{C}}$ and an absent argument to τ . Its original input support and relation fibre remain available upstream. Formula (22) is a calculated trace, not a positivity assertion.

Reflection is strict for this specified section Define conjugate-linear involutions

$$j_B F(x) = x^{-1} \overline{F(1/x)}, \quad j_V \phi = \widehat{\phi}, \quad j_E u = u^\dagger.$$

They satisfy $\Theta j_V = j_B \Theta$ and $J j_B = j_E J$. The actual source function is Fourier-fixed: writing $\phi_* = D(D-1)e^{-\pi x^2}$, Fourier sends the two factors to $(1-D)(-D) = D(D-1)$ and fixes the Gaussian. It is real, so $j_V \phi_* = \phi_*$.

Since $\varepsilon_Z^\dagger = (-1)^d \varepsilon_Z$ and reflection preserves polynomial degree,

$$R_Z(j_E u) = (-1)^d (R_Z(u))^\dagger.$$

Uniqueness of the inverse of $h(D)$ on its range gives

$$j_B S_h^{\mathcal{B}} = (-1)^d S_h^{\mathcal{B}} j_B.$$

Both factors $(-1)^d$ are retained in the substitution, and their product is one. Therefore

$$\boxed{j_B s = s j_E, \quad j_B P_Z = P_Z j_B.} \quad (23)$$

This is an actual equality for this section; it would not follow for an arbitrary choice of representatives. Its compatibility with the nonzero dilation homotopy is

$$\boxed{j_V k_a = a k_{1/a} j_E.} \quad (24)$$

Indeed, apply j_B to (12), use $j_B U_a = a U_{1/a} j_B$ and (23), then use injectivity of theta. The split lift is semilinear through $r^\bullet \mapsto \bar{r}^\bullet$, which fixes both e and τ .

B.1.8 7. A finite-rank Lefschetz trace on the actual chain complex

For $f \in C_c^\infty(\mathbb{R}_{>0})$, let U_f be its convolution action on C_+ . The cochain endomorphism

$$p_Z U_f = (0, s J H_f)$$

has finite-rank components. Its cochain supertrace is therefore defined without an infinite-dimensional regularization:

$$\boxed{\text{Str}(p_Z U_f \mid C_+) = -\text{Tr}(s J H_f \mid \mathcal{B}) = -\sum_{\rho \in Z} m_\rho \mathcal{M}f(\rho).} \quad (25)$$

The finite-rank identity $\text{Tr}(AB) = \text{Tr}(BA)$ reduces the middle expression to $\text{Tr}(J H_f s \mid E)$; this is multiplication by $j_h \mathcal{M}f$. Equation (25) equals the supertrace of the induced operator on homology.

On C_{+-} its degree-zero contribution is still zero, since the same cochain projector has zero degree-zero component.

The boundary representatives in (6), (14), and (20) are retained even though their contribution to homology is zero. The finite-rank trace does not delete their complex or assign its zero to external absence.

For the n -fold tensor complex over the tau-base product, use the original coefficient tensor over $\mathbb{C}$ and its differential

$$d(x_1 \otimes \cdots \otimes x_n) = \sum_j (-1)^{\sum_{i < j} |x_i|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_n.$$

The projector $\mathfrak{p}_Z^{\otimes n}$ has image $E^{\otimes n}[-n]$. For independently specified tests $f_1, \dots, f_n$,

$$\text{Str} \left(\bigotimes_j (\mathfrak{p}_Z \cup_{f_j}) \right) = (-1)^n \prod_j \sum_{\rho \in Z} m_\rho \mathcal{M} f_j(\rho). \quad (26)$$

Let $A = U_a \mathfrak{p}_Z$ and $B = \mathfrak{p}_Z U_a$ as cochain maps. A homotopy for $A^{\otimes n} - B^{\otimes n}$ is

$$\sum_{j=1}^n B^{\otimes(j-1)} \otimes H_a \otimes A^{\otimes(n-j)},$$

with the displayed Koszul sign on homogeneous inputs when the degree-minus-one map crosses the preceding factors. The telescoping identity and (14) prove it. The full support mask maps to the product of its input masks; vanishing products of represented amplitudes stay supported.

B.1.9 8. What now enters the Weil II programme

The structural base, the infinite quotient, and the actual theta complex are unchanged. The new objects are the extension (3), its unit class (4), its synchronized homotopy (20), its strict equivariant derived roof (16), and the chain trace (25). The arithmetic residue unit in (21) is now supplied by that very extension rather than appended without its source.

Deligne's inspected comparison-image argument (Weil II 3.3.4–3.3.6) requires two weight bounds on the same cohomological image. The inspected derived formulation (6.2.1–6.2.7) retains the dualizing operation, its degree and Tate shifts, and its exchange of direct-image functors. This continuation supplies coherent arithmetic maps and finite trace operations for the tau model; it does not transfer the finite-field hypotheses or assert those two bounds.

The retained error is explicit: it is the original boundary Θ_{k_a} on representatives, with the nonzero Ext class (4). It is removed at homology by the original supported quotient, not by making its input absent. A geometric comparison can now be tested for compatibility with this class and the residue map (21). A complex-linear section that ignores (6) fails that test.

No infinite sum of packet projectors is asserted to converge. The existing meromorphic/kernel summand from PR #8 remains in its own exact sequence, and a different global trace functor must retain or analyze its contribution. The source's finite-jet isomorphisms and the new finite-rank chain trace do not by themselves set that summand to zero.

B.1.10 9. Verification and references

The new finite checker tests polynomial remainder identities, the extension class, rank-one scaling defects, cochain homotopies, dilation cocycles, residue contraction, tensor signs, trace identities, and

original support operations. These finite algebraic models are not numerical tests of zeta zeros and not certificates for the analytic range theorem. No Lean execution is claimed for this continuation.

Read source interfaces:

- PR #8 at the pinned revision above, `FINITE_OPERATOR_COMPARISON.md`, sections 1–10; `TAU_BASE_MODEL.md` is its already-published model.
- Owner’s `SplitZero_Derived_Mathematics_2026-09-12(1).md`, sections 1–7. In particular its allowance of a nonzero bottom fibre and the exact homology-kernel/action theorems are retained.
- `formal/splitzero/SplitZeroHomology.lean`, lines 175–end, read at the same revision; blob `7b9e5c49a494dd0c751133e0234b3e01048f7b67`.
- Supplied Deligne TeX excerpts: French segment 2, original lines 882–950; segment 3, original lines 675–756. The used equations are the image bounds and duality operations, not every assertion in the historical transcription. No source corpus is republished with this contribution.
- The Stacks Project, Tags 010I and 06XP, for extension classes and their projective-resolution interpretation.
- R. Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, `arXiv:math/0412277`, for prior spectral-realization context. No global priority claim is made here.

New calculations are AI-assisted work under the owner’s programme. Existing mathematical files, formalizations, licensing, and source attribution remain untouched.

C PR 10: Global retraction

Source attribution. Web contribution, pull request #10, [Zeta research repository](#).

Exact revision. d26c283c2a58da57bb5a5a4d4bc8669019f6ebb7

Source file. workbenches/tau-global-retraction/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 496f3ec752e3e67ddafaae7423feb300b1faa789ca6907872f8d539bdf6e1859

C.1 A global retraction of the original theta complex over the tau base

Owner-directed continuation of Zeta PR #9, 12 September 2026. New derivations are submitted as research proofs for review, not as Lean-certified analytic theorems.

C.1.1 0. Fixed source, result, and provenance

The base is PR #9 at 8f99b94d306c3b1aaa817d52c57fa6fd298d511c; its chain-descent note has Git blob 86db838e79d10a8fe7d4376303706c4ea64570a1. Its source and the owner's derived SplitZero attachment remain the inputs. This continuation constructs an inverse on the entire original theta image, not only on a chosen finite spectral packet.

Retain

$$\begin{aligned}
 G(R) &= \{\tau\} \sqcup \{r^\bullet : r \in R\}, & e_R &= 0_R^\bullet, \\
 \mathfrak{b}_\tau = \operatorname{Spec} \mathbf{F}_{1,\tau}, & \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} & (1)
 \end{aligned}$$

The first vertical target is still infinite. The geometric base, the original additive laws, and the arithmetic quotient are unchanged. The auxiliary cutoffs below are parameters of an explicit continuous representative map; they do not change an Euler factor, the theta relation space, or the completed zeta function.

Result. For every pair of the cutoffs specified in section 3 there is an explicitly constructed continuous complex-linear map

$$\Lambda_{\alpha,\beta} : \mathcal{B} \longrightarrow V, \quad \boxed{\Lambda_{\alpha,\beta}\Theta = \operatorname{id}_V}. \quad (2)$$

Consequently the actual image ΘV is closed and complemented in the stated strong-Schwartz topology. The original quotient $Q = \mathcal{B}/\Theta V$ is Hausdorff. The note constructs its global section, the full supported cochain homotopy, the exact change of the finite sections in #9, its continuous dual sequence, and the completed tensor powers. The proof does not assume RH, simplicity of any zero, or vanishing of the separate balanced-Mellin kernel.

Spectral realizations on nuclear Frechet spaces are prior work, notably R. Meyer, arXiv:math/0412277. That paper's abstract was consulted for scope, but no theorem from an unread body of that paper is imported. The proof below is given for the workbench's precise spaces. No global novelty claim is made.

C.1.2 1. Spaces and all transforms

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : p_{b,k}(F) := \sup_{x>0} x^b |D^k F(x)| < \infty \text{ for every } b \in \mathbb{Z}, k \geq 0\}, \quad D = -x\partial_x. \quad (3)$$

These seminorms give $\mathcal{B}$ its Frechet topology; under $x = e^y$ they are the weighted uniform derivative seminorms on smooth functions of y . Completeness follows by uniform convergence on compact sets of every derivative, followed by the weighted bounds. V has its closed-subspace Schwartz topology.

The original maps are

$$\begin{aligned} \Theta\phi(x) &= \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx) = 2 \sum_{n \geq 1} \phi(nx), \\ \mathcal{F}\phi(\xi) &= \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad \mathcal{F}^{-1}u(x) = \int_{\mathbb{R}} u(\xi) e^{2\pi i x \xi} d\xi, \\ JF(x) &= x^{-1}F(1/x), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \end{aligned} \quad (4)$$

Poisson summation, with both source moments zero, gives

$$\Theta\mathcal{F} = J\Theta, \quad J^2 = 1, \quad DJ = J(1 - D). \quad (5)$$

The theta sum and each derivative converge absolutely on closed subsets of $x > 0$. At infinity, Schwartz estimates give all $p_{b,k}$ bounds. At zero use (5) and the same estimates on $\hat{\phi}$. This proves continuity $\Theta : V \rightarrow \mathcal{B}$. Also J is continuous on $\mathcal{B}$: $p_{b,k}(JF)$ is bounded by the binomial sum of $p_{1-b,j}(F)$, $0 \leq j \leq k$.

The original complex is $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, over $A = \mathbb{C}[t]$ with t acting as D . The two-leg tau chart retains $C_{+-} = [V^2 \rightarrow \mathcal{B}]$ with differential $\Theta(\phi - \hat{\psi})$ and source action $(D, 1 - D)$.

C.1.3 2. Arithmetic inversion on the two exterior regions

Let μ be the ordinary Mobius function, characterized by

$$\sum_{d|n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

For $F \in \mathcal{B}$ define, for $x \neq 0$,

$$\boxed{\mathcal{U}F(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) F(n|x|)}. \quad (6)$$

The factor $1/2$ is the inverse of the full nonzero-integer theta sum in (4). The map is complex-linear into smooth even functions on $\mathbb{R} \setminus \{0\}$. It is not asserted to be smooth at zero for arbitrary F .

Here are the required estimates. For $j \geq 0$, let $P_j(T) = (-1)^j T(T+1) \cdots (T+j-1)$, with $P_0 = 1$. Then $x^j \partial_x^j = P_j(D)$. If $C_{N,j}(F)$ is the sum of the $p_{N,k}(F)$ weighted by the absolute coefficients of P_j , then, for $x > 0$ and integer $N > 1$,

$$|\partial_x^j \mathcal{U}F(x)| \leq \frac{1}{2} \zeta(N) C_{N,j}(F) x^{-N-j}. \quad (7)$$

Indeed each summand is bounded by $C_{N,j}(F) n^{-N} x^{-N-j}/2$. This proves convergence of every derivative uniformly for $|x| \geq \delta > 0$, and gives rapid decay there. It also proves continuity of the cut-off exterior maps used below.

For every $\phi \in V$,

$$\boxed{\mathcal{U}\Theta\phi(x) = \phi(x), \quad \mathcal{U}J\Theta\phi(x) = \widehat{\phi}(x) \quad (x \neq 0).} \quad (8)$$

For the first equality the double series is absolutely convergent for each $x > 0$: its absolute value is bounded by $\sum_{k \geq 1} d(k) |\phi(kx)|$, which is finite by Schwartz decay. Rearrange by $k = mn$ and use the displayed divisor identity. Evenness handles negative x . The second equality follows from (5) and the first equality applied to $\widehat{\phi} \in V$.

In the split lift, every $\mu(n)$ is the supported integer $\mu(n)^\bullet$, including $\mu(n) = 0$. A convergent sum within a specified active fibre begins at that fibre's zero and is taken in that fibre's vector-space topology. Formula (8) thus lifts to a support-preserving equality. A computed cancellation is not external absence.

C.1.4 3. The reconstruction operator, with the strict contraction proved

Choose and retain real, even $\alpha, \beta \in C_c^\infty(\mathbb{R})$ satisfying $0 \leq \alpha, \beta \leq 1$ and equal to 1 on neighborhoods of zero. Their supports and all values are part of the auxiliary data. Work on $\mathcal{H} = L^2(\mathbb{R}, dx)$ with the Fourier convention in (4).

Define bounded operators

$$A_\alpha = M_\alpha, \quad B_\beta = \mathcal{F}^{-1} M_\beta \mathcal{F}, \quad T_{\alpha,\beta} = A_\alpha B_\beta. \quad (9)$$

Lemma. $T_{\alpha,\beta}$ is compact and $c_{\alpha,\beta} := \|T_{\alpha,\beta}\| < 1$.

Proof. Its kernel is $\alpha(x)\check{\beta}(x-y)$, whose squared integral is $\|\alpha\|_2^2 \|\beta\|_2^2$. Hence it is Hilbert-Schmidt. Both factors are contractions. If the norm were 1, compactness would supply a unit vector f attaining it. Equality in

$$\|A_\alpha B_\beta f\| \leq \|B_\beta f\| \leq \|f\|$$

forces $\widehat{f}$ to be supported where $\beta = 1$; there $B_\beta f = f$. The first equality forces f to be supported where $\alpha = 1$. Both supports are compact. The inverse Fourier transform of compactly supported $\widehat{f} \in L^2$ is entire, by differentiation under its finite-interval integral. Its continuous restriction agrees with the compactly supported f , so it vanishes on an open real interval and hence identically. This contradicts $\|f\| = 1$. Thus $c_{\alpha,\beta} < 1$.

Consequently the exact Neumann inverse exists:

$$\boxed{(1 - T_{\alpha,\beta})^{-1} = \sum_{k \geq 0} T_{\alpha,\beta}^k, \quad \|(1 - T_{\alpha,\beta})^{-1}\| \leq \frac{1}{1 - c_{\alpha,\beta}}.} \quad (10)$$

No uniform bound as the cutoffs vary is claimed. In particular the gap $1 - c_{\alpha,\beta}$ is retained, not replaced by a positive universal constant.

The operator $T_{\alpha,\beta}$ maps L^2 continuously to the Schwartz space. For each derivative, compact Fourier support and Cauchy-Schwarz bound $\partial^j B_\beta f$ uniformly by a constant times $\|f\|_2$; multiplication by the smooth compactly supported α gives every Schwartz seminorm.

Define the two exterior data and their combination:

$$a_F = (1 - \alpha)\mathcal{U}F, \quad b_F = (1 - \beta)\mathcal{U}JF,$$

$$Y_{\alpha,\beta}F = a_F + \alpha\mathcal{F}^{-1}b_F. \quad (11)$$

The exterior terms are extended by zero near the origin, where their cutoff is identically zero. Estimates (7) prove that both are even Schwartz functions and depend continuously on F . Therefore $Y : \mathcal{B} \rightarrow \mathcal{S}(\mathbb{R})^{\text{even}}$ is continuous.

Put

$$\Phi_{\alpha,\beta}F = (1 - T_{\alpha,\beta})^{-1}Y_{\alpha,\beta}F. \quad (12)$$

This initially defines an L^2 vector, but the identity $\Phi F = YF + T\Phi F$ proves it is Schwartz. If q is any Schwartz seminorm and $q(Tu) \leq C_q\|u\|_2$, then

$$q(\Phi F) \leq q(YF) + \frac{C_q}{1 - c_{\alpha,\beta}}\|YF\|_2. \quad (13)$$

This is an explicit continuity estimate. Both sides are even because the cutoff operators preserve parity.

For $F = \Theta\phi$, equations (8) give

$$Y\Theta\phi = (1 - A_\alpha)\phi + A_\alpha(1 - B_\beta)\phi = (1 - T_{\alpha,\beta})\phi,$$

hence

$$\boxed{\Phi_{\alpha,\beta}\Theta\phi = \phi.} \quad (14)$$

C.1.5 4. Keep both original moment conditions

For arbitrary F , (12) need not satisfy the two source moments. The correction is the following specified projection, not an alteration of V .

Retain

$$\gamma_0(x) = e^{-\pi x^2}, \quad \gamma_2(x) = 2\pi x^2 e^{-\pi x^2}.$$

Their moment columns $(\phi(0), \int \phi)$ are $(1, 1)$ and $(0, 1)$. Define

$$\mathbf{m}(\phi) = (\phi(0), \int_{\mathbb{R}} \phi), \quad j_0(a, b) = a\gamma_0 + (b - a)\gamma_2,$$

$$P_V\phi = \phi - j_0\mathbf{m}(\phi) = \phi - \phi(0)\gamma_0 - \left(\int \phi - \phi(0)\right)\gamma_2. \quad (15)$$

Then $\mathbf{m}j_0 = 1$, $P_V^2 = P_V$, $\text{im } P_V = V$, and $P_V|_V = 1$. All maps are continuous.

The promised map is

$$\boxed{\Lambda_{\alpha,\beta} = P_V(1 - T_{\alpha,\beta})^{-1}Y_{\alpha,\beta} : \mathcal{B} \rightarrow V.} \quad (16)$$

Equation (14) and $P_V|_V = 1$ prove (2). The discarded moment vector has the recorded map $m\Phi : \mathcal{B} \rightarrow \mathbb{C}^2$ and the explicit correction $j_0 m\Phi$. It vanishes on the actual theta image. No endpoint constant was absorbed into the zeta function.

C.1.6 5. The full quotient is closed and has a continuous section

Write $\Lambda = \Lambda_{\alpha,\beta}$, and define

$$P_\Theta = \Theta\Lambda, \quad K = 1_{\mathcal{B}} - \Theta\Lambda. \quad (17)$$

Direct substitution gives

$$P_\Theta^2 = P_\Theta, \quad K^2 = K, \quad \ker K = \Theta V, \quad \text{im } K = \ker \Lambda. \quad (18)$$

Since K is continuous, its kernel is closed. Thus $Q = \mathcal{B}/\Theta V$ with its actual quotient topology is Frechet and Hausdorff. If $q : \mathcal{B} \rightarrow Q$ is the original quotient map, the continuous section is

$$\boxed{s : Q \rightarrow \mathcal{B}, \quad s[F] = KF, \quad qs = 1, \quad \Lambda s = 0.} \quad (19)$$

The explicit topological direct-sum isomorphism is

$$\boxed{V \oplus Q \xrightarrow{\sim} \mathcal{B}, \quad (\phi, u) \mapsto \Theta\phi + su, \quad F \mapsto (\Lambda F, qF).} \quad (20)$$

For a convergent sequence $\Theta\phi_n \rightarrow F$ in $\mathcal{B}$, one can check the closedness without any abstract theorem: $\phi_n = \Lambda\Theta\phi_n \rightarrow \Lambda F$ in V , so $F = \Theta\Lambda F$ by continuity of theta.

This computes the earlier Hausdorff arrow:

$$Q^{\text{alg}} = \mathcal{B}/\Theta V \longrightarrow Q^{\text{H}} = \mathcal{B}/\overline{\Theta V}, \quad [F] \mapsto [F], \quad (21)$$

as an isomorphism with zero amplitude kernel. It does not say the map from Q into its product of spectral jets is injective. The latter map and the balanced analytic map still have their own named kernels:

$$Q \rightarrow \prod_{\rho} \mathcal{O}_{\rho}/(g), \quad \mathcal{O} \otimes_{\mathbb{C}[t]} Q \xrightarrow{\beta} \mathcal{O}/(g).$$

No claim that $\mathcal{R}_{\infty}$ or $\ker \beta$ vanishes is inferred from (21).

Smaller admitted relations For any specified subspace $W \subset V$, retain

$$Q_W = \mathcal{B}/\Theta W \longrightarrow Q, \quad \ker = \Theta V/\Theta W \cong V/W. \quad (22)$$

The isomorphism sends $[\phi] \mapsto [\Theta\phi]$. It is algebraic for arbitrary W ; when W is closed, (20) makes it a topological isomorphism onto the kernel. The pullback of the global section has the representative relation kernel in (22); the full-source result is not silently asserted with W in place of V .

C.1.7 6. Perform the original internal quotient and synchronization

On the single-leg map, or a fixed-index diagram of that same map, the split lifts are

$$\tilde{\Theta}(\lambda, \phi) = (\lambda, \Theta\phi), \quad \tilde{\Lambda}(\lambda, F) = (\lambda, \Lambda F), \quad \tilde{K}(\lambda, F) = (\lambda, KF). \quad (23)$$

They preserve external bottom and commute with scalar reflection. In particular

$$\tilde{K}\tilde{\Theta}(\lambda, \phi) = (\lambda, 0_{\mathcal{B}}) = z_{V, \mathcal{B}}(\lambda, \phi).$$

Here $z_{V, \mathcal{B}}$ is the label-preserving zero map from the source fibre to the target fibre. The mixed chart, whose minus restriction is $-\Theta\mathcal{F}$, is treated separately by the explicit maps (25)-(28), without changing that restriction. The actual chart has zero bottom fibre; a general nonzero bottom fibre is acted on by its stated linear map, rather than identified with the single global zero.

The supported image projection is idempotent, and the quotient map is exactly the coequalizer of the theta relation inclusion and its supported-zero companion. The implementation's universal property applies against all split-linear targets. A theta boundary is sent to $e_\lambda = (\lambda, 0)$; it is not sent to external absence.

For the single-leg complex there is an actual continuous contraction

$$C_+ \xrightleftharpoons[s]{q} Q[-1], \quad H^1 = \Lambda,$$

$$\boxed{qs = 1, \quad 1_{C_+} - sq = dH + Hd.} \quad (24)$$

In degree zero this is $\Lambda\Theta = 1_V$; in degree one it is $\Theta\Lambda = 1 - K$. In split notation the subtraction means addition of the supported scalar $(-1)^\bullet$ times the second map. It yields the correct labelled zero wherever the amplitude vanishes.

For the two-leg chart set $H_0 = V[0] \oplus Q[-1]$ with zero differential, and define

$$p_L^0(\phi, \psi) = \psi, \quad p_L^1 = q, \quad i^0(\psi) = (\widehat{\psi}, \psi), \quad i^1 = s,$$

$$H_L^1(F) = (\Lambda F, 0). \quad (25)$$

Then $p_L i = 1$, and $1 - ip_L = dH_L + H_L d$. The degree-zero identity is $(\phi, \psi) - (\widehat{\psi}, \psi) = (\phi - \widehat{\psi}, 0)$. The other leg gives

$$p_R^0(\phi, \psi) = \widehat{\phi}, \quad p_R^1 = q, \quad H_R^1(F) = (0, -\widehat{\Lambda F}). \quad (26)$$

It obeys $1 - ip_R = dH_R + H_R d$. Their homotopies differ by

$$\boxed{H_L^1 F - H_R^1 F = (\Lambda F, \widehat{\Lambda F}),} \quad (27)$$

the actual joint degree-zero cycle, which is retained.

On the complete original support diagram, use literal multiplication by $E_{\text{sync}} = (1, e)$ in degree zero and the source's active-label-to-joint map in degree one:

$$\mathfrak{r}^0(a, b) = (a + eb, b + ea), \quad \mathfrak{r}^1(\lambda, F) = (\{+, -\}, F) \quad (\lambda \neq \perp).$$

The absent input remains absent. The original differential obeys $d\mathfrak{r} = \mathfrak{r}d$ and $\mathfrak{r}^2 = \mathfrak{r}$. Consequently (25) lifts to the fully typed equation

$$\boxed{\mathfrak{r} - \tilde{i}\tilde{p}_L\mathfrak{r} = \tilde{d}\tilde{H}_L + \tilde{H}_L\tilde{d},} \quad (28)$$

where $\tilde{H}_L(\lambda, F) = (\{+, -\}, (\Lambda F, 0))$ for each active label. The right-leg equation is analogous with (26). The original mask is recorded with the injection $x \mapsto (\text{supp } x, \mathfrak{r}x)$ already proved for this chart. Neither synchronization nor the contraction identifies the three supported origins with τ .

C.1.8 7. Global scaling defects, and their agreement with the finite extensions

Let $U_a F(x) = F(x/a)$, $a > 0$, and denote its induced continuous action on Q by $\overline{U}_a$. The section (19) gives a global continuous boundary map

$$\boxed{k_a = \Lambda U_a s : Q \rightarrow V, \quad U_a s - s\overline{U}_a = \Theta k_a.} \quad (29)$$

The proof applies $K + \Theta\Lambda = 1$ to $U_a s$, then uses $qU_a s = \overline{U}_a$. Therefore

$$k_{ab} = U_a k_b + k_a \overline{U}_b, \quad k_1 = 0. \quad (30)$$

The same calculation works for D and compactly supported multiplicative convolution, with $k_D = \Lambda D s$. In the explicit coordinates (20),

$$\boxed{U_a|_{\mathcal{B}} = \begin{pmatrix} U_a|_V & k_a \\ 0 & \overline{U}_a \end{pmatrix}, \quad D|_{\mathcal{B}} = \begin{pmatrix} D|_V & k_D \\ 0 & D_Q \end{pmatrix}.} \quad (31)$$

Thus the topological splitting is not substituted for an equivariant splitting. Equation (29) is the precise comparison between them.

For two cutoff choices, let s_0, s_1 be their sections and set $b_{10} = \Lambda_0 s_1 : Q \rightarrow V$. Then

$$s_1 - s_0 = \Theta b_{10}, \quad k_a^{(1)} = k_a^{(0)} + U_a b_{10} - b_{10} \overline{U}_a. \quad (32)$$

All dependence on the chosen representative map is carried by this displayed coboundary. No canonical cutoff or average over choices is declared.

Restrict back to PR #9 without changing its extension class Use its existing maps $s_Z : E_Z \rightarrow \mathcal{B}$, $\sigma_Z = q s_Z : E_Z \rightarrow Q$, and $J_Z : \mathcal{B} \rightarrow E_Z$. Put $b_Z = \Lambda s_Z$. The new continuous representative of the same finite arithmetic class is

$$s_Z^{\text{new}} = s \sigma_Z = K s_Z = s_Z - \Theta b_Z. \quad (33)$$

If $P(D)s_Z - s_Z M_P = \Theta k_{P,Z}$, direct substitution gives

$$k_{P,Z}^{\text{new}} = k_{P,Z} - P(D)b_Z + b_Z M_P. \quad (34)$$

For $P = h_Z$, this changes the source boundary by $-h_Z(D)b_Z$. It is zero in $V/h_Z(D)V$, so the extension class remains exactly

$$\varepsilon_Z = j_{h_Z}(h_Z/g), \quad g = 2\xi. \quad (35)$$

The polynomial residue form still becomes the arithmetic one by M_{ε_Z} ; its Jacobian contraction remains $M_{\varepsilon_Z} M_{j_h g'}$. No unit, derivative, sign, or factor of 2 is changed.

The two finite cochain projectors differ by the explicit homotopy $-b_Z J_Z$ in degree minus one. Their finite-rank trace difference is zero because $J_Z \Theta = 0$. Hence their cochain Lefschetz trace is the unchanged

$$-\sum_{\rho \in Z} m_\rho \mathcal{M}f(\rho). \quad (36)$$

C.1.9 8. Continuous duality and completed tensor powers of the full object

Use continuous duals, denoted E'_b with the strong topology. Transpose the actual topological splitting (20). It gives

$$\boxed{0 \rightarrow Q'_b \xrightarrow{q'} \mathcal{B}'_b \xrightarrow{\Theta'} V'_b \rightarrow 0, \quad \Theta' \Lambda' = 1.} \quad (37)$$

The continuous cochain-dual complex is $[\mathcal{B}'_b \xrightarrow{\Theta'} V'_b]$ in degrees $-1, 0$. Its cohomology is Q'_b in degree -1 and zero in degree 0 . The transpose contraction proves this directly; no general exactness theorem for arbitrary locally convex quotients is assumed.

The actual finite residue map into that continuous dual is

$$\overline{E_Z} \longrightarrow Q'_b, \quad \bar{f} \longmapsto (u \mapsto R_Z(f, j_Z u)), \quad (38)$$

where $j_Z q = J_Z$. Continuity follows from the finite Mellin jets and finite dimensionality of E_Z . It is injective by their surjectivity and perfect residue duality. Multiplication by g' then gives the arithmetic finite trace pairing, with its previously calculated radical retained.

Let $\widehat{\otimes}_\pi$ denote completed projective tensor product. The comparison from the previous algebraic tensor is its canonical completion map. All chain operators in (24) are continuous. Thus their finite tensor products and homotopies extend to the completed products. With $P = sq$ on C_+ , the homotopy

$$H_n = \sum_{j=1}^n P^{\widehat{\otimes}(j-1)} \widehat{\otimes} H \widehat{\otimes} 1^{\widehat{\otimes}(n-j)} \quad (39)$$

includes $(-1)^{\sum_{i < j} |x_i|}$ on homogeneous inputs. It satisfies

$$dH_n + H_n d = 1 - P^{\widehat{\otimes} n}.$$

Consequently, for the completed external coefficient complex on the same tau-base chart product,

$$\boxed{H^k(C_+^{\widehat{\otimes} n}) = \begin{cases} Q^{\widehat{\otimes} n}, & k = n, \\ 0, & k \neq n. \end{cases}} \quad (40)$$

This follows from a split continuous contraction, not from a blanket claim that completed tensor product preserves every exact sequence.

For the joint chart use (25). Its full cohomology in degree k is the direct sum, indexed by subsets $J \subseteq \{1, \dots, n\}$ with $|J| = k$, of the ordered completed tensors having Q in the slots of J and V in the other slots. In those V slots the scaling action is $aU_{1/a}$, exactly as in the original minus-coordinate representative. The degree-zero contribution is not discarded. Split reconstruction retains each product mask and each internal zero.

C.1.10 9. The pairing calculation is now a specified global boundary problem

The section $s : Q \rightarrow \mathcal{B} \subset L^2(dx)$ supplies a positive definite auxiliary form

$$\mathcal{H}_s(u, v) = \langle su, sv \rangle_{L^2(dx)}.$$

This is not assigned to the arithmetic Weil form. The explicit weight-covariance defect is calculated using (29) and $\langle U_a f, U_a g \rangle = a \langle f, g \rangle$:

$$\boxed{\mathcal{H}_s(\overline{U}_a u, \overline{U}_a v) - a \mathcal{H}_s(u, v) = -\langle U_a s u, \Theta k_a v \rangle - \langle \Theta k_a u, U_a s v \rangle + \langle \Theta k_a u, \Theta k_a v \rangle.} \quad (41)$$

The original source boundary and both of its cross-pairings remain. Being a boundary in Q does not assert these pairings are zero in the representative Hilbert space. In a split lift each computed zero is its supported value e .

For a reduced eigenline of the actual dilation representation with exponent ρ , the left side at $u = v$ is $(a^{2\rho} - a) \mathcal{H}_s(u, u)$. Formula (41) locates that defect in the full global boundary cocycle, with the original operator retained.

A concrete comparison explaining which topology must be kept In fact ΘV is dense in $L^2(\mathbb{R}_{>0}, dx)$, although it is closed in $\mathcal{B}$. Here is a proof retaining the map between them. Set $b_* = \Theta \phi_*$ and

$$W : L^2(dx) \xrightarrow{\sim} L^2(dy), \quad W F(y) = e^{y/2} F(e^y).$$

The Fourier transform in y , with kernel $e^{-2\pi i \nu y}$, gives

$$\mathcal{F}_y W b_*(\nu) = g(1/2 - 2\pi i \nu).$$

This is nonzero almost everywhere, because g is a nonzero entire function. The span of all translates of $W b_*$ is dense in $L^2(dy)$: a vector orthogonal to all translates has Fourier transform whose product with $\overline{\mathcal{F}_y W b_*}$ is an L^1 function with zero Fourier transform, hence vanishes almost everywhere. The nonvanishing just stated then makes that vector zero. Since $W U_a b_* = a^{1/2} (W b_*)(y - \log a)$ and $U_a b_* \in \Theta V$, density follows.

The exact quotient comparison is therefore

$$\boxed{Q = \mathcal{B}/\Theta V \longrightarrow L^2(dx)/\overline{\Theta V}^{L^2} = 0.} \quad (42)$$

Its kernel is all of Q . Its split lift sends every supported input to the supported zero in $G(0) = \mathbb{B}$, while keeping external absence separate. The constructive replacement is the Frechet quotient and the explicit global section (19), not a claim that the ordinary Hilbert quotient secretly retains those classes.

C.1.11 10. Scope of the progress toward the Deligne comparison

The selected supplied Weil II passages 3.3.4-3.3.6 put two bounds on the same compact-to-ordinary image; 6.2 retains the dualizing operation and its shifts. This continuation does not apply those bounds over a new base without their geometric proof. It makes the existing characteristic-zero arithmetic complex globally split in its precise topological category, gives its continuous dual and completed products, and computes the operator defects on those same objects.

The formerly unresolved algebraic-to-Hausdorff kernel is now zero by (16)-(21). The finite extension class and its residue/Jacobian trace are carried by (33)-(36). The full scaling boundary is (29), its dependence on the auxiliary cutoffs is (32), and its metric defect is (41). These are concrete maps available for a global comparison.

Still retained: the common kernel of all finite spectral observations, the meromorphic summand of the balanced analytic realization, and the requirement to identify the correct arithmetic polarization/weight control on the global geometric image. The density map (42) explains why discarding the stronger topology would remove the entire arithmetic cohomology, not why the supported programme should be abandoned.

The reconstruction estimate contains $1 - c_{\alpha,\beta}$; no asymptotic uniform estimate for it is proved. No global positivity, RH, GRH, or infinite trace-class theorem is claimed. No theorem from the earlier finite-range work is upgraded to a Lean certificate. The new global retraction proof itself uses only the original theta/Poisson identities, absolute Mobius inversion, the displayed Fourier compactness argument, and the two retained moment maps.

C.1.12 11. Verification and references

The written analytic proofs are the evidence for the new analytic statements. The accompanying unittest script tests exact finite identities: divisor cancellation with supported zeros, cutoff-resolvent identities in a finite unitary model, moment corrections, contractions, synchronization masks, operator cocycles, changes of sections, preservation of extension classes, finite trace changes, and completed-tensor algebra before completion. It does not numerically certify a Fourier operator norm, an analytic convergence statement, or a Riemann zero.

Source interfaces read:

- Owner's SplitZero_Derived_Mathematics_2026-09-12(1).md, sections 1-7; the source permits nonzero bottom fibres in its general interface.
- PR #9, workbenches/tau-chain-descent/RESEARCH_NOTE.md, at the commit above; local bytes match its Git blob.
- PR #8's finite comparison as retained and invoked in #9; the sections using s_Z and J_Z are explicitly dependencies of (33)-(38), not inputs to the proof of (2).
- Supplied Weil II TeX: the actual 3.3.1-3.3.6 excerpt, including its finite-type base and duality assumptions. It was read as TeX, not obtained by OCR. No full source-proof audit is claimed.
- Ralf Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277: prior-context abstract only in this continuation. A source-archive retrieval failed; no unread detailed theorem is cited as a premise.

The contribution is AI-assisted under the owner's mathematical direction. Original mathematical files, formal sources, attribution and licenses are unchanged. Only authored exposition, test code and validation records are proposed for public addition; no private transcript or source-literature corpus is published.

D PR 11: Homotopy control

Source attribution. Web contribution, pull request #11, [Zeta research repository](#).

Exact revision. d37eade4f9e6a82c1a88aeff710a3b851a34eb49

Source file. workbenches/tau-homotopy-control/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 9924b358c0365a8cbdcddb3a7496f610c65d64617cd1860679b42e371d2fe88d

D.1 Homotopy choices and source-preserving arithmetic control

Owner-directed SplitZero continuation, 12 September 2026. Base: PR #10, d26c283c2a58da57bb5a5a4d4bc8669019f6ebb7. This integration note records the new derivations; the conversation delivery also contains a longer equation-by-equation exposition. No original scalar definitions or formal modules are changed.

D.1.1 1. Reviewed theorem and analytic specialization

The supplied other-session proof classifies all homotopies for

$$C = [V \oplus V \xrightarrow{d} B], \quad d(v, w) = \Theta(v - \mathcal{F}w), \quad Q = B/\Theta V,$$

where Θ is injective and $\mathcal{F}$ is an R -linear automorphism. For a fixed $\kappa: B \rightarrow V$ with $\kappa \Theta = 0$, every solution of BOTH equations

$$dH = \Theta \kappa, \quad Hd = 0$$

has the unique form

$$H_\alpha(b) = (\kappa b + \alpha q b, \mathcal{F}^{-1} \alpha q b), \quad \alpha : Q \rightarrow_R V.$$

Proof: writing $H=(u,v)$, injectivity gives $u-\mathcal{F}v=\kappa b$. Since $d(x,0)=\Theta x$, $Hd=0$ gives $H \Theta = 0$. Hence $\mathcal{F}v$ factors uniquely through q , and supplies α . Conversely substitution proves both equations. The reference solution is $H_0=(\kappa, 0)$; the family is an affine space under $\text{Hom}_R(Q, V)$, relative to that reference.

Let $\gamma(v)=(v, \mathcal{F}^{-1} v)$. Then

$$H_\alpha - H_\beta = \gamma(\alpha - \beta)q.$$

The difference is a map $H_1(C) \rightarrow H_0(C)$. There are no degree-minus-two endomorphisms of this two-term coefficient complex. Thus it is a genuine degree-minus-one Hom-complex class, not an erased difference.

The supplied theorem is mathematically correct. Its Lean draft explicitly remains uncompiled, and its earlier failed chart build is not relabelled successful. No Lean or Lake executable is installed in the review environment.

For the actual analytic spaces instantiate $R=\mathbb{C}$. With the quotient topology on Q , the continuous version follows: $\alpha q = \mathcal{F} \text{pr}_2 H$ is continuous, so the quotient property makes α continuous. Conversely the displayed formula sends continuous α to continuous H .

For the polynomial action $A_0 = \mathbb{C}[t]$, Fourier has the exact type

$$\mathcal{F} : V^j \xrightarrow{\sim} V, \quad t \cdot_{V^j} v = (1 - D)v, \quad j(t) = 1 - t,$$

because $F D = (1 - D)F$. Thus d is A_0 -linear from $V + V^\wedge j$ to B . A complex-linear homotopy is not automatically A_0 -linear. The reflected module above is the actual comparison, not a change to the Fourier definition.

D.1.2 2. Original split support on every homotopy

Retain the base b_τ and the square $G(Z) \rightarrow G(C)$ over $Z > C$. The quotient $G(Z) \rightarrow Z$ remains infinite. Both lifted coefficient maps preserve τ and the supported scalar e separately.

Use the original support lattice $L = \{\text{bottom}, +, -, J\}$, $J = \{+, -\}$. Degree-zero fibres are $0, V, V, V + V$ with the coordinate inclusions. Degree-one active fibres are all B , with identity transports. Their differentials are Θ , $-\Theta$, F and d . Retain zero modules in the other degrees; their reconstructed zero map is $z(\lambda, x) = (\lambda, 0)$.

Literal multiplication by $E_{\text{sync}} = (1, e)$ induces

$$r^0(a, b) = (a + eb, b + ea), \quad r_L(\perp) = \perp, \quad r_L(\lambda) = J(\lambda \neq \perp).$$

Every homotopy lifts by

$$\tilde{H}_\alpha(\lambda, b) = (J, (\kappa b + \alpha q b, \mathcal{F}^{-1} \alpha q b)), \quad \lambda \neq \perp; \quad \tilde{H}_\alpha(\tau) = \tau.$$

This is split-linear: two active inputs add at their joined label, both outputs add at J , and their amplitudes obey the two displayed linear formulas. A supported scalar zero gives $(J, (0, 0))$; external τ gives the global zero. Therefore

$$\tilde{d} \tilde{H}_\alpha + \tilde{H}_\alpha \tilde{d} = r \tilde{\Delta}.$$

The difference is $(J, \gamma(\alpha - \beta)qb)$, retaining its degree-zero class. For this chart, the pair $(\text{support}(x), r(x))$ is injective. No such injectivity is assumed for arbitrary diagrams with noninjective transports. For a smaller relation space W , keep $Q \rightarrow W$ with kernel $\Theta V / \Theta W$; q in the formulas means the displayed quotient by the full admitted V .

D.1.3 3. Coherence for all dilations

Let $S_a = U_a$ on V , $T_a = U_a$ on B , $U_a F(x) = F(x/a)$, and T_{bar}_a the quotient action. The actual cochain action is

$$\mathsf{T}_a^0 = (S_a, \mathcal{F}^{-1} S_a \mathcal{F}) = (U_a, a U_{1/a}), \quad \mathsf{T}_a^1 = T_a.$$

It satisfies $T_a^0 \gamma = \gamma S_a$ and $q T_a = T_{\text{bar}}_a q$. With the PR #10 section s , take $P = (0, sq)$, and write

$$[\mathsf{T}_a, P]^1 = \Theta \kappa_a, \quad \kappa_a = k_a q, \quad T_a s - s \overline{T}_a = \Theta k_a.$$

The commutator product identity gives $\kappa_{ab} = S_a \kappa_b + \kappa_a T_b$. The family of homotopies is composition-coherent,

$$H_{ab} = \mathsf{T}_a H_b + H_a \mathsf{T}_b,$$

exactly when

$$\alpha_{ab} = S_a \alpha_b + \alpha_a \bar{T}_b.$$

Proof: the reference family $(\kappa_a, 0)$ is coherent. Subtract it and use injectivity of γ and surjectivity of q . No action or boundary is omitted.

For a smoothly parameterized continuous-linear family with $L = (d/dt)|_0 \alpha(e^t)$, the exact formula is

$$\alpha_{e^t} = \int_0^t S_{e^{t-r}} L \bar{T}_{e^r} dr.$$

This is a vectorwise integral in the complete Frechet source, with the original oriented integral for negative t . Differentiating the composition equation gives the formula; splitting the integral proves its converse. In particular $\alpha_a = S_a b - b \bar{T}_a$ has infinitesimal parameter $D_V b - b D_Q$.

D.1.4 4. A homotopy parameter leaves the control form fixed

Fix an actual finite arithmetic packet with full multiplicities,

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad E_Z = \mathbb{C}[t]/(h_Z), \quad A = M_t.$$

Use the existing $R: E_Z \rightarrow B$ and $J_Z: B \rightarrow E_Z$ with

$$qR = \sigma_Z, \quad J_Z R = 1, \quad J_Z \Theta = 0, \quad J_Z D = A J_Z, \quad DR - RA = \Theta k.$$

The cochain projector $P_R = (0, R|_Z)$ has commutator $(0, \Theta k|_Z)$, so the reviewed theorem applies with $\kappa = k|_Z$. The quotient $\text{jet } J_{\text{bar}}_Z$ satisfies $J_{\text{bar}}_Z \sigma_Z = 1$, hence any $\alpha_E: E_Z \rightarrow V$ extends to $\alpha_E J_{\text{bar}}_Z$ on Q . Restriction to R gives

$$H_{\alpha_E} R = (k + \alpha_E, \mathcal{F}^{-1} \alpha_E), \quad dH_{\alpha_E} R = \Theta k.$$

Take $H = L^2(R_+, dx)$ with the original inner product, conjugate-linear in the first variable. Define

$$G_R = R^* R, \quad B_R = \Theta k, \quad W_R = -(R^* B_R + B_R^* R).$$

The composite of actual maps

$$\text{Hom}_{\mathbb{C}}(E_Z, V) \longrightarrow \text{Hom}(E_Z, V^2) \xrightarrow{d \circ (-)} \text{Hom}(E_Z, H) \xrightarrow{B \mapsto -(R^* B + B^* R)} \text{Herm}(E_Z)$$

is constant at W_R . Its last map is real-linear. The retained H^0 ambiguity does not change this boundary contraction.

Integration by parts on the actual source functions gives

$$A^* G_R + G_R A - G_R = W_R.$$

The boundary term $[x \text{ conjugate}(F)H]_0^\infty$ vanishes by the stated decay. No claim that a θ boundary is orthogonal to a representative is used.

D.1.5 5. The actual representative-change family

Every representative of the same inclusion σ_Z is uniquely

$$R_b = R + \Theta b, \quad b : E_Z \rightarrow_{\mathbb{C}} V.$$

Indeed $q(R'-R)=0$ gives values in ΘV , and the PR #10 retraction gives the unique $b=\Lambda(R'-R)$. Finite source dimension gives continuity. Global continuous sections have the identical formula with $b:Q \rightarrow V$ continuous.

The complete changes are

$$k_b = k + D_V b - bA,$$

$$\delta G_b = R^* \Theta b + (\Theta b)^* R + (\Theta b)^* \Theta b,$$

$$\boxed{W_{R_b} - W_R = A^* \delta G_b + \delta G_b A - \delta G_b.}$$

Their extension-coordinate map is $U_b(v,u)=(v+bu,u)$, and

$$U_b^{-1} \begin{pmatrix} D_V & k \\ 0 & A \end{pmatrix} U_b = \begin{pmatrix} D_V & k + D_V b - bA \\ 0 & A \end{pmatrix}.$$

Lift that automorphism on the common active fibre $(V+E_Z)^{\tau}$. For the original masks, first keep the pair $(\text{support}(x), r(x))$; apply the shear on the joint fibre, and recover the original amplitudes by U_b^{-1} and the original mask by its recorded component. No inverse to the unrecorded support collapse is asserted.

The projectors obey $P_{(R_b)} - P_R = \text{partial } L_b$ with $L_b^{-1}(F) = (b|_Z F, 0)$. Their finite cochain traces agree because

$$\text{Tr}(\Theta b J_Z H_f) = \text{Tr}(J_Z H_f \Theta b) = 0.$$

The cochain supertrace sign remains -1 in degree one. The arithmetic extension class changes by $h_Z(D)b(1)$, so its class in $V/h_Z(D)V$ remains $\epsilon_Z = j_h(h_Z/(2\pi))$. The residue/Jacobian trace form is consequently unchanged.

D.1.6 6. Finite actual source ansatz and one master Gram matrix

Retain $\phi = (4\pi^2 x^4 - 6\pi x^2) \exp(-\pi x^2)$, $\phi_j = D^j \phi$, and

$$\Phi_m : \mathbb{C}^{m+1} \rightarrow V, \quad (c_j) \mapsto \sum_{j=0}^m c_j \phi_j.$$

Their actual Mellin identities are $M \Theta \phi_j(s) = 2\pi i(s)^j$. They remain in the original two-moment source and are linearly independent. The source section $R_0 = s_Z$ has the exact rank-one defect $DR_0 - R_0 A = \Theta \phi^* \ell_Z$, with the unchanged arithmetic remainder functional ℓ_Z .

Define maps, including the target enlargement,

$$I_m e_j = e_j, \quad S_m e_j = e_{j+1}, \quad I_m, S_m : \mathbb{C}^{m+1} \rightarrow \mathbb{C}^{m+2}, \quad D\Phi_m = \Phi_{m+1} S_m.$$

For B in $\text{Mat}_{((m+1) \times d)}(\mathbb{C})$, $d = \dim E_Z$, put

$$R_B = R_0 + \Theta \Phi_m B, \quad K_B = e_0 \ell_Z + S_m B - I_m B A.$$

Then

$$DR_B - R_B A = \Theta \Phi_{m+1} K_B, \quad q R_B = \sigma_Z, \quad J_Z R_B = 1.$$

Take the actual column map and its Gram matrix

$$\mathcal{T}_m : E_Z \oplus \mathbb{C}^{m+2} \rightarrow L^2(\mathbb{R}_+, dx), \quad (u, c) \mapsto R_0 u + \Theta \Phi_{m+1} c, \quad M_m = \mathcal{T}_m^* \mathcal{T}_m.$$

M_m is positive definite. Applying J_Z to a vanishing combination gives $u=0$; Mellin then gives $c=0$. Its entries are the convergent integrals of the displayed arithmetic functions, with the original measure.

Define

$$X_B = \begin{pmatrix} I_d \\ I_m B \end{pmatrix}, \quad Y_B = \begin{pmatrix} 0 \\ K_B \end{pmatrix}.$$

The complete control calculation is

$$\boxed{G_B = X_B^* M_m X_B, \quad W_B = -(X_B^* M_m Y_B + Y_B^* M_m X_B) = A^* G_B + G_B A - G_B.}$$

This restricts representative choice to an actual finite family of theta boundaries. It is not an arbitrary positive matrix ansatz, and it is not asserted to exhaust all continuous source corrections. All packet coordinates, multiplicities, local units and finite source-degree changes are retained.

D.1.7 7. Error transport, duality, and the remaining estimate

With actual enclosures $||M - \hat{M}|| \leq \eta_M$, $||X - \hat{X}|| \leq \eta_X$, $||Y - \hat{Y}|| \leq \eta_Y$, put $x = ||\hat{X}||$, $y = ||\hat{Y}||$, $m_0 = ||\hat{M}||$. Product estimates give

$$\eta_G = \eta_M (x + \eta_X)^2 + m_0 (2x\eta_X + \eta_X^2),$$

$$\eta_W = 2[\eta_M (x + \eta_X)(y + \eta_Y) + m_0 (x\eta_Y + y\eta_X + \eta_X \eta_Y)].$$

These include coefficient errors, not just quadrature errors. Exact certificates for $\hat{G} - \eta_G I > 0$ and

$$\epsilon \hat{G} \pm \hat{W} - (\epsilon \eta_G + \eta_W) I \succeq 0$$

imply the actual two-sided control. This is an error-transfer theorem; arithmetic enclosures have not been computed here.

The unchanged residue matrix S satisfies $S^* = -S$ and $A^* S + S A = S$. Its exact dual transport is

$$G_B^D = S^* G_B^{-1} S, \quad W_B^D = -S^* G_B^{-1} W_B G_B^{-1} S.$$

Thus an upper certificate gives a lower certificate on the same arithmetic packet. The Weil form stays $S J_g$, with metric comparison $G_B^{-1} S J_g$; no positivity is assigned by choosing G_B .

For tensor degree n , retain $A_n = \sum_j 1$ tensor A tensor 1 and $G_{(B,n)} = G_B^{\wedge}(\text{tensor } n)$. The product control is $\sum_j G_B$ tensor W_B tensor G_B . Changing a homotopy by $\gamma \alpha q$ changes

the tensor homotopy by its Koszul-signed insertions; its differential is zero, hence it does not change that control form.

The existing orbit construction on this actual tensor image gives

$$G_{B,n,T} = \int_0^T e^{-nt} e^{tA_n^*} G_{B,n} e^{tA_n} dt,$$

$$A_n^* G_{B,n,T} + G_{B,n,T} A_n - n G_{B,n,T} = e^{-nT} e^{TA_n^*} G_{B,n} e^{TA_n} - G_{B,n}.$$

Every occurrence of $G_{B,n}$ now has the explicitly constrained source in section 6.

A nonzero boundary need not have nonzero control. The exact calibration $D_{\text{amb}} = \text{diag}(1/2+i, 1/2+2i)$, $\Theta(v) = (v, 0)$, $A = 1/2+2i$, $R(u) = (u, u)$ has boundary $\Theta(-i)$, Gram 2, and control zero. It is an algebraic calibration, not a zeta packet. The real-linear map $B \mapsto (R^*B + B^*R)$ explains which nonzero boundaries can have zero control.

Conversely every eigenvector $Av = \rho v$ obeys

$$\frac{v^* W_B v}{v^* G_B v} = 2\Re \rho - 1.$$

This ensures that changing representatives has not changed a spectral coordinate. The analytic target is now to bound the actual matrices from section 6, or their tensor-orbit endpoints, with all packet and tensor dependence retained. No uniform purity estimate, RH, GRH, or global spectral-kernel vanishing is proved here.

D.1.8 8. Validation and attribution

The attachment supplies the homotopy-classification theorem and the uncompiled draft. This note verifies its proof and adds the continuous/twisted interface, supported reconstruction, coherent-family condition, and the connection to the actual control matrices. The original owner-directed SplitZero construction remains the source.

The independent checker has sixteen exact unittest methods, including finite parameter families. They passed normally and under python -O with identical successful JSON records. Intentional negative controls failed in both modes. The previous control checker also passed its sixteen methods. These finite checks cover algebraic models and polynomial-times-exponential integral calibrations, not evaluated zeta packets or analytic/Lean certification.

General references: The Stacks Project, Tags 0A8H and 0115 for the Hom-complex differential and homotopy, and 010I and 06XP for extension classes. The formulas above are proved on the specified programme objects; those references are not imported as a number-field weight theorem. Only authored notes, test code and validation records are published; no private transcript, source-literature corpus, or font files are included.

E PR 13: Orthogonal boundary control

Source attribution. Web contribution, pull request #13, [Zeta research repository](#).

Exact revision. e7ba4c29d4503bf20d06cb70c93c3347f199f1b4

Source file. workbenches/tau-orthogonal-boundary-control/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 9d364d9cc2f6840af8eb39555e4b89622acd7714fbcab7ca135e8ec0ce25ab0b

E.1 Orthogonal theta boundaries and rank-two arithmetic control

Owner-directed SplitZero continuation, 12 September 2026. Base: PR #11, d37eade4f9e6a82c1a88aeff710a3b851a34eb49. This is an additive research contribution with written proofs for review, not a new Lean certificate for the analytic results. The conversation delivery contains a longer numbered LaTeX/HTML version, the full reading record, and exploratory arithmetic moments.

E.1.1 1. Reviewed input and exact types

The supplied other-session theorem is correct: for injective $\Theta: V \rightarrow B$, an R -linear automorphism F , $Q = B/\Theta V$, and $\kappa: B \rightarrow V$ with $\kappa \Theta = 0$, all homotopies satisfying BOTH $dH = \Theta \kappa$ and $Hd = 0$, for $d(v, w) = \Theta(v - Fw)$, are

$$H_\alpha(b) = (\kappa b + \alpha q_b, F^{-1} \alpha q_b), \quad \alpha: Q \rightarrow R V.$$

Writing $H = (u, v)$ gives $u - Fv = \kappa$. Since $d(x, 0) = \Theta x$, $Hd = 0$ gives $H \Theta = 0$; hence Fv descends uniquely to α . Substitution proves the converse. The difference is $\gamma(\alpha - \beta)q$ with $\gamma(v) = (v, F^{-1}v)$. The solution set is affine relative to H_0 , not generally a vector space with the zero homotopy as origin.

The attachment's uncompiled-draft status has now been superseded by the other session's successful PR #12 at 5d2772ea0a16d177ac3e01ff70394b90ba95a232. The actual SplitZeroTauHomotopy.lean, completed job 103535438942, and its full log were read. Both cochain equations and the inverse equivalence passed the specified kernel checks. Retraction and quotient-topology interfaces also passed with their explicit hypotheses. This is the other session's formalization; neither the actual analytic theta construction nor the new Gram/control results below are certified by it. No local Lean/Lake execution is claimed. See COORDINATION_UPDATE.md for exact scope. PR #11 already supplies the continuous specialization, the Fourier twist over $C[t]$, support-changing lifts, coherent dilations and representative-change formulas. Those are retained inputs, not new results here.

Keep the original tau base and the square $G(Z) \rightarrow G(C)$ over $Z \rightarrow C$, with $p: G(Z) \rightarrow Z$ still infinite, $e = \text{ofR}(0)$, and tau the external scalar zero. No Boolean localization replaces the base quotient. All quotient relations below use the original internal coequalizer and preserve the support label.

Retain V, B and the actual operators of PRs #8-#11:

```
V={even Schwartz phi on R: phi(0)=0 and integral(phi)=0};
B={smooth F on R+: sup x^b |D^k F(x)| < infinity for all integer b and k>=0};
D=-x d/dx; Theta phi(x)=2 sum_(n>=1) phi(nx);
M F(s)=integral_0^infinity F(x)x^s dx/x;
phi_*= (4 pi^2 x^4 - 6 pi x^2) exp(-pi x^2);
phi_j=D^j phi_*; f_j=Theta phi_j=D^j f_0;
M f_j(s)=s^j g(s), g=2 xi.
```

The Hilbert observation is $H=L^2(\mathbb{R}_+, dx)$, conjugate-linear in the first argument. Polynomial packet coordinates are retained, not changed to orthonormal coordinates.

For an actual finite packet with all multiplicities, let $E=C[t]/h_Z$, $A=M_t$. The previous analytic constructions supply

$R_\theta: E \rightarrow B$, $J_Z: B \rightarrow E$, $qR_\theta = \sigma_Z$, $J_{ZR_\theta} = 1$, $J_Z \theta = 0$,
 $DR_\theta - R_\theta A = f_\theta \in \ell_Z$.

The exact ℓ_Z comes from $\epsilon_Z = j_h(h/(2\xi))$; at a simple zero its value on 1 is $1/(2\xi'(\rho))$. These analytic constructions remain the cited research-proof dependencies, not theorems certified by the finite tests here.

E.1.2 2. An actual minimum over the admitted theta relations

For $m \geq 0$ define

$W_m = \text{span}(\phi_0, \dots, \phi_m)$;
 $\phi_m: C^{(m+1)} \rightarrow V$, $c \mapsto \sum c_j \phi_j$;
 $F_m = \theta \phi_m: C^{(m+1)} \rightarrow H$;
 $H_m = F_m^* F_m$; $C_m = F_m^* R_\theta$; $G_\theta = R_\theta^* R_\theta$.

The ϕ_j are independent: Mellin sends a relation to $g(s)P(s)=0$. Thus H_m is positive definite. Here C_m is a cross-Gram matrix $E \rightarrow C^{(m+1)}$, not the complex of that name in the endpoint workbench.

Set

$B_m = -H_m^{-1} C_m$;
 $\pi_m = F_m H_m^{-1} F_m^*$;
 $R_m = R_\theta + \theta \phi_m B_m = (1 - \pi_m) R_\theta$.

The correction is a finite original theta boundary, so R_m lands in B , $qR_m = \sigma_Z$, and $J_{ZR_m} = 1$ at every m . Direct multiplication proves

$F_m^* R_m = 0$;
 $G_m = R_m^* R_m = G_\theta - C_m^* H_m^{-1} C_m \succ 0$.

Positive definiteness follows from $J_{ZR_m} = 1$. For any $B: E \rightarrow C^{(m+1)}$,

$(R_\theta + F_m B)^* (R_\theta + F_m B)$
 $= G_\theta + (B - B_m)^* H_m (B - B_m)$.

This is the minimum of the Hilbert Gram within the stated source family, not yet a minimum of its relative weight error. Also $(1 - \pi_m)(R_\theta + F_m B') = R_m$: changing the initial representative by a relation already in W_m leaves the result unchanged.

E.1.3 3. The complete control form has rank at most two

Let $I_m, S_m: C^{(m+1)} \rightarrow C^{(m+2)}$ be $I_m e_j = e_j$ and $S_m e_j = e_{(j+1)}$. Define

$K_m = e_\theta \in \ell_Z + S_m B_m - I_m B_m A$.

Then

$DR_m - R_m A = F_{(m+1)} K_m = \theta (\phi_{(m+1)} K_m)$.

No boundary component is discarded in this identity. Put $b_m = e_m^* B_m: E \rightarrow C$, and $z_m = R_m^* f_{(m+1)}: C \rightarrow E$. Integration by parts on the stated source functions gives

$$\langle DF, H \rangle + \langle F, DH \rangle = \langle F, H \rangle,$$

with vanishing boundary term $[x \text{ conjugate}(F)H]_0^\infty$. Hence

$$\begin{aligned} W_m &= A^* G_m + G_m A - G_m \\ &= -(R_m^* F_{(m+1)} K_m + K_m^* F_{(m+1)}^* R_m) \\ &= -z_m b_m - b_m^* z_m^*. \end{aligned}$$

The last equality uses $F_m^* R_m = 0$ and the exact last row of K_m , which is b_m . Thus $\text{rank}(W_m) \leq 2$ for every packet dimension and every finite m . The pairings with the first $m+1$ theta directions vanish; those supported vectors themselves are retained.

Define the actual scalar quantities

$$\begin{aligned} u_m &= z_m^* G_m^{-1} z_m \geq 0; \\ v_m &= b_m^* G_m^{-1} b_m \geq 0; \\ c_m &= b_m^* G_m^{-1} z_m. \end{aligned}$$

The nonzero spectrum of $G_m^{-1} W_m$ is the nonzero spectrum of

$$-\begin{bmatrix} c_m & v_m \\ u_m & \text{conjugate}(c_m) \end{bmatrix}.$$

This follows by factoring through the two-column map $[z_m, b_m^*]$ and using the equality of nonzero spectra of the two compositions. Its trace is $-2 \text{Re}(c_m)$, determinant $|c_m|^2 - u_m v_m$. Therefore the least symmetric control allowance is

$$\begin{aligned} \epsilon_m &= |\text{Re}(c_m)| + \sqrt{u_m v_m - (\text{Im}(c_m))^2}, \\ -\epsilon_m G_m &\leq W_m \leq \epsilon_m G_m. \end{aligned}$$

Cauchy-Schwarz makes the radicand nonnegative. The formula also covers rank one. A reflection-stable full packet has

$$\text{trace}(G_m^{-1} W_m) = 2 \text{Re}(\text{trace } A) - \dim E = 0,$$

so $\text{Re}(c_m) = 0$. This is an exact trace identity, not a choice of coordinates. For every eigenvector $Av = \rho v$,

$$(v^* W_m v) / (v^* G_m v) = 2 \text{Re}(\rho) - 1.$$

Thus the relative control still detects precisely the original spectral weight.

E.1.4 4. The next relation is a rank-one update with its own retained coordinate

Set

$$\begin{aligned} a_{(m+1)} &= (1 - \Pi_m) f_{(m+1)} = \Theta \psi_{(m+1)}, \\ \psi_{(m+1)} &= \phi_{(m+1)} - \Pi_m H_m^{-1} F_m^* f_{(m+1)}, \\ \eta_{(m+1)} &= \|a_{(m+1)}\|^2 > 0. \end{aligned}$$

Then $R_m^* a_{(m+1)} = z_m$, and orthogonal decomposition gives

$$\begin{aligned} \Pi_{(m+1)} &= \Pi_m + a_{(m+1)} \eta_{(m+1)}^{-1} a_{(m+1)}^*; \\ R_{(m+1)} &= R_m - a_{(m+1)} \eta_{(m+1)}^{-1} z_m^*; \\ G_{(m+1)} &= G_m - \eta_{(m+1)}^{-1} z_m z_m^*. \end{aligned}$$

The control update is obtained by applying the same real-linear operator $X \mapsto A^*X + XA - X$ to the last identity.

The original quotient transport is

$$Q_W = B/\Theta W_m \rightarrow Q_{W(m+1)} = B/\Theta W_{(m+1)},$$

$$\text{kernel} = \Theta W_{(m+1)} / \Theta W_m \simeq W_{(m+1)} / W_m.$$

That kernel is one-dimensional. The square $E \rightarrow Q_W$ by R_m and $E \rightarrow Q_{W(m+1)}$ by $R_{(m+1)}$ commutes. The new relation becomes the supported zero only through the actual next-level quotient; it is not another scalar adjunction at the bottom.

There is a form on the entire quotient:

$$H_{\text{cal}_m}([F], [G]) = \langle (1 - \Pi_m)F, (1 - \Pi_m)G \rangle.$$

It is positive definite since the kernel of $1 - \Pi_m$ in B is exactly ΘW_m . Let $\lambda_{(m+1)}[F] = \langle a_{(m+1)}, F \rangle$. This is well defined on Q_W and

$$H_{\text{cal}_m}(x, y) = H_{\text{cal}_{(m+1)}}(qx, qy) + \eta_{(m+1)}^{-1} \text{conjugate}(\lambda(x)) \lambda(y).$$

The map $(q, \lambda): Q_W \rightarrow Q_{W(m+1)}$ plus C is a linear isomorphism. Its inverse is

$$([F], c) \mapsto [F - a \eta^{-1} \langle a, F \rangle + a \eta^{-1} c]_W.$$

Changing F by a relation at level $m+1$ changes this expression only by a relation at level m . This supplies the exact metric comparison between levels instead of treating the quotient transports as isometries.

Every displayed linear map has its original fixed-support lift. On the full source chain, reconstructed addition uses the join $\max(m, n)$ and the above transports. The scalar e sends $(m, [F])$ to $(m, [0])$; τ sends it to external absence. The two-leg primitives are $(k_m u, 0)$ and $(0, \text{Fourier}^{-1} k_m u)$, where $k_m = \Phi_{(m+1)} K_m$. Synchronization by the original $E_{\text{sync}} = (1, e)$ supplies their common joint support. Their difference remains the H_0 cycle $(k_m u, \text{Fourier}^{-1} k_m u)$. The other session's $\text{Hom}(Q, H_0)$ freedom leaves their differential and hence W_m unchanged.

E.1.5 5. Actual arithmetic moments and their truncation allowance

The matrices H_m come from the original arithmetic functions. Since $JF(x) = x^{-1} F(1/x)$ is unitary on $L_2(dx)$, $Jf_0 = f_0$ and $JD = (1 - D)J$,

$$H_{ij} = \int_{-\infty}^{\infty} [\text{conjugate}(f_i) f_j + \text{conjugate}(f_i^{\text{ref}}) f_j^{\text{ref}}] dx,$$

$$f_j^{\text{ref}} = (1 - D)^j f_0.$$

This retains both multiplicative ends and permits absolutely convergent Gaussian sums on $x \geq 1$. Write

$$\phi_j(x) = P_j(x) \exp(-\pi x^2),$$

$$P_0 = 4 \pi^2 x^4 - 6 \pi x^2,$$

$$P_{(j+1)} = 2 \pi x^2 P_j - x P_j',$$

$$P_j = \sum_r c_{(j, r)} x^{(2r)},$$

$$c_{(j+1, r)} = 2 \pi c_{(j, r-1)} - 2r c_{(j, r)}.$$

For $f_j^{\text{N}} = 2 \sum_{(n=1)}^N P_j(nx) \exp(-\pi n^2 x^2)$, put

$$C_j = \sum_r |c(j,r)| \sqrt{\Gamma(2r+1/2) / (2 \pi^{(2r+1/2)})}.$$

Then

$$\begin{aligned} & \|f_j - f_j^N\|_{L^2([1, \infty))} \\ & \leq 2 C_j (N+1)^{-1/2} \exp(-\pi(N+1)^2/2) \\ & \quad / [1 - \exp(-\pi(2N+3)/2)]. \end{aligned}$$

Proof: on $x \geq 1$, $2 \pi n^2 x^2 \geq \pi n^2 + \pi n^2 x^2$. Integrating each monomial's squared bound gives $n^{-1/2} \exp(-\pi n^2/2)$ times the displayed gamma constant. Minkowski and a geometric bound on consecutive n terms give the formula. The reflected derivative retains the absolute binomial sum of these allowances.

The finite integral entries use exactly

$$\begin{aligned} & \int_1^\infty x^{(2k)} \exp(-c x^2) dx \\ & = \Gamma(k+1/2, c) / (2 c^{(k+1/2)}), \quad c = \pi(n^2 + n'^2). \end{aligned}$$

For truncation tail norms δ_i, δ_j , each inner-product error is bounded by $\delta_i \|f_j^N\| + \|f_i^N\| \delta_j + \delta_i \delta_j$, plus its reflected contribution. Numerical evaluation errors for the finite expression still require their own enclosure. The optional conversation computation is explicitly exploratory, not interval-certified.

The exact consistency check is $H_{(i+1,j)} + H_{(i,j+1)} = H_{ij}$. Cross-Gram columns $F_m \cdot R_0$ remain their actual source integrals and are not replaced by arbitrary matrices.

E.1.6 6. The same source proves a strong topology comparison

For this actual f_0 ,

$$\text{closure span}\{D^j f_0 : j \geq 0\} \text{ in } L^2(\mathbb{R}_+, dx) = H.$$

Proof. The explicit unitary Mellin-line map sends F to $M F(1/2+it)$ in $L^2(\mathbb{R}, dt/(2\pi))$. It sends D to $1/2+it$ and f_j to $(1/2+it)^j g(1/2+it)$. The log-space function $h(z) = \exp(z/2) f_0(\exp z)$ is holomorphic in $|\text{Im } z| < \pi/4$ by its Gaussian series. For $|\text{Im } z| \leq a < \pi/4$ it decays superexponentially as $\text{Re } z$ tends to positive infinity. Poisson gives $h(z) = h(-z)$, so the same holds at the other end. Fourier contour displacement therefore gives $|g(1/2+it)| \leq C_a \exp(-a|t|)$.

The measure $d\mu = |g(1/2+it)|^2 dt/(2\pi)$ has a positive exponential moment. If an $L^2(\mu)$ function is orthogonal to all polynomials, multiplying μ by its conjugate gives a finite complex measure with a smaller positive exponential moment. Its Fourier transform is analytic in a strip and all derivatives at zero vanish. Analyticity and uniqueness of the Fourier transform imply that measure is zero. Thus polynomials are dense in $L^2(\mu)$. Multiplication by g is unitary from this weighted space onto $L^2(dt/(2\pi))$, because the nonzero analytic function g on the line is nonzero almost everywhere. This proves the assertion without RH.

As a result, for a fixed finite packet,

$$\begin{aligned} & R_m \rightarrow 0 \text{ in } \text{Hom}(E, H), \quad G_m \rightarrow 0 \text{ and } W_m \rightarrow 0, \\ & \text{while } q_{R_m} = \sigma_Z \text{ and } J_{ZR_m} = 1 \text{ for every } m. \end{aligned}$$

This specifies why unscaled smallness of W_m cannot be used alone as weight control: its denominator G_m is also shrinking. The exact relative formula in section 3 must be retained.

The corrective map is the Hilbert/jet graph:

$$\begin{aligned} & \text{closure}\{(F, J_{ZF}) : F \in B\} = H \text{ plus } E; \\ & \text{closure}\{(\Theta \phi, 0) : \phi \in V\} = H \text{ plus } 0. \end{aligned}$$

Indeed finite theta polynomials give every $(h,0)$ in the closure, and the sequence $(R_m u, u)$ gives $(0, u)$. For a specified positive metric on E , completion in $\|F\|_2^2 + \|J_Z F\|_E^2$ is this graph closure, and the quotient by the boundary closure is E by $[(h, u)] \rightarrow u$. The added metric is retained supplied data, not an implied positivity theorem for the Weil pairing.

Its split observation is

$$((R_m u)^\bullet, u^\bullet) \rightarrow (e_H, u^\bullet)$$

in the product of the stated support-fibre topologies. The arithmetic component survives above a supported Hilbert zero. External absence is (τ, τ) .

E.1.7 7. Duality, trace and scope

The original residue map has matrix S with $S^* = -S$ and $A^*S + SA = S$. Its pullback of the dual metric is

$$\begin{aligned} G_m^* D &= S^* G_m^{-1} S, \\ W_m^* D &= -S^* G_m^{-1} W_m G_m^{-1} S. \end{aligned}$$

Thus the upper-to-lower comparison is retained on the same arithmetic packet. The Weil pairing stays $S J_g$, with explicit metric comparison $G_m^{-1} S J_g$. Its positivity is not imposed by choosing a positive Gram matrix.

All representative changes above are original theta boundaries. Their extension class $j_h(h/(2\xi))$ is unchanged; the induced finite cochain projectors differ by the homotopy $(b)_Z, 0$. The trace difference is zero because $J_Z \Theta = 0$. Degree-one and tensor Koszul signs remain those of the source complex. On a raw tensor metric, the inserted W_m terms give the copied allowance $n \epsilon_m$; tensoring alone does not improve it.

The new results are the actual orthogonal source choice, rank-two boundary formula, exact relative certificate, rank-one source-kernel recurrence, arithmetic moment tail bound, and the full Hilbert/jet topology comparison. No uniform arithmetic purity estimate, global spectral-kernel vanishing, RH or GRH is asserted.

The thirteen finite exact tests check the encoded algebra and declared calibration models, not infinite analytic arguments or actual zeta zeros. The complete cochain-homotopy theorem remains the other session's result, now supported by the independently inspected successful PR #12 run. Its certificate does not extend to the new analytic control results. Existing Lean modules and earlier validation receipts are untouched.

References: live PR #11 at the pinned head; successful PR #12 and source/job references in COORDINATION_UPDATE.md; the supplied `Tau_Deligne_Control` and `Tau_Global_Comparison` source notes; the supplied other-session homotopy proof; the previously retained Weil II 3.3.4-3.3.6 and 6.2 duality excerpts. Standard orthogonal-polynomial context is NIST DLMF 18.2; the density argument needed here is proved above for the actual arithmetic measure.

F Derived Split-Zero mathematics

Source attribution. Web contribution retained on the pre-integration main revision.

Exact revision. 14c69d604044b848d64318751a7bee19edfd9aba

Source file. formal/splitzero/DERIVED_MATHEMATICS.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 6a290e193161eb09f49b6faba578193fdaf28b5ae3a97340d1c41a40aa99e695

F.1 SplitZero: reconstruction, internal homology, and balanced finite jets

Owner-directed formalization continuation, 12 September 2026.

F.1.1 Scope and provenance

This contribution extends formal/splitzero on the owner's Zeta workbench. Its base is the synchronization PR #5, commit 236f0a8370477991e2d0cf334ad1ebfe4f3f8b17, itself based on the structural PR #4. The original scalar core and all five preceding extension sources remain unchanged. There is no Sneed-side submission or integration in this continuation.

The mathematical source is the owner's SplitZero programme, including the supplied continuation's internal homology and balanced-Mellin sections. The implementation below distinguishes source-derived structure, ordinary reusable algebra, and the new explicit comparison example. It makes no global priority claim and does not treat the number of Lean declarations as a number of mathematical discoveries. No private conversation or source-literature corpus is republished.

The exact build certificate for a revision is the completed SplitZero\derived\constructions workflow on that commit, including direct source checks and its generated AuditDerived.lean. The PR records the accepted execution revision and run. A failed earlier development run is not a certificate for later source.

F.1.2 1. Reverse reconstruction, with actual scalar action

Let R be a commutative ring, L a join-semilattice with bottom, and let D consist of R -modules V_i with linear transition maps

$$\begin{aligned} \text{rho_ij} &: V_i \rightarrow V_j \quad (i \leq j), \\ \text{rho_ii} &= \text{id}, \quad \text{rho_jk} \circ \text{rho_ij} = \text{rho_ik}. \end{aligned}$$

No finiteness, field, injectivity, or nontriviality assumption is imposed. Construct

$$\text{Total}(D) = \text{disjoint union of } V_i \text{ over } i \text{ in } L.$$

The global zero is (bottom,0), and

$$\begin{aligned} (i,x) + (j,y) &= (i \text{ join } j, \text{rho_i,i-join-j}(x) + \text{rho_j,i-join-j}(y)). \\ \text{tau} \cdot (i,x) &= (\text{bottom},0). \\ \text{ofR}(r) \cdot (i,x) &= (i, r \cdot x). \end{aligned}$$

`LinearDiagram.totalAddCommMonoid` and `LinearDiagram.totalModule` construct the actual instances and prove every law. For associativity, transport both sides to $i \text{ join } j \text{ join } k$. Linearity and transition composition reduce both to the same three-term sum. The scalar laws follow in the supported case from the R -module laws and in the absent cases from the chosen global zero. The supported-zero scalar sends (i,x) to $(i,0)$.

The bottom fibre is allowed to be nonzero. Specializing it to the zero module recovers the source's carrier consisting of one absent point and the non-bottom fibres. This is an explicit generalization, not an assumption that the bottom fibre in every later dual diagram vanishes. Opposite-index duality itself is not constructed here.

Fixed-index natural families of linear maps produce actual $G(R)$ -linear maps between reconstructed totals. Identity and composition are proved. More substantially, `Recovery.recoverEquiv` reconstructs the intrinsic support fibres of **any** existing $G(R)$ -semimodule M and gives a full $G(R)$ -linear equivalence back to M :

$$(1,x) \mapsto x, \quad m \mapsto (e\ m, m).$$

Its addition proof uses the original absorption identity $m + e\ m = m$. It is not merely a bijection of carriers. A bundled equivalence of the two full categories, including arbitrary changes of support index and both natural inverse comparisons, is a separate library interface not claimed by this implementation.

F.1.3 2. The internal quotient is characterized by a universal property

Let B_i be transition-stable submodules of V_i . The quotient diagram has fibres V_i/B_i and the genuinely induced quotient transition maps. The projection

$$q : \text{Total}(D) \rightarrow \text{Total}(D/B), \quad q(i,x) = (i,[x])$$

is a surjective $G(R)$ -linear map. The code proves the exact equality criterion

$$q(i,x)=q(i,y) \text{ iff } x-y \text{ is in } B_i.$$

Different labels cannot become equal under q . In particular a relation maps to its own fibre zero.

The main point is stronger than that observation. Let N be **any** $G(R)$ -semimodule, not assumed additively cancellative, and let $f:\text{Total}(D)\rightarrow N$ be split-linear. If

$$f(i,b)=f(i,0) \text{ for every } b \text{ in } B_i,$$

then there exists a unique split-linear $g:\text{Total}(D/B)\rightarrow N$ with $g\ q=f$. The proof does not subtract elements in N . If $x-y$ is in B_i , use the identity $y+(x-y)=x$ inside V_i and then apply additivity of f . This proves constancy on q -fibres. Surjectivity constructs the descended function; linearity and uniqueness follow by lifting representatives.

`Relations.coequalizer_universal` states this against the actual inclusion of the relation diagram and its supported-zero companion. Thus the universal property is proved against arbitrary semimodule targets, not only against a selected class of fixed-support maps. The implementation gives the complete unbundled universal property; it does not merely define the desired quotient and label it a coequalizer.

F.1.4 3. Actual homology and the exact transported-class kernel

At a degree i of a cochain complex, use the window

$C^{(i-1)} \xrightarrow{d_{\text{Prev}}} C^i \xrightarrow{d_{\text{Next}}} C^{(i+1)}, \quad d_{\text{Next}} d_{\text{Prev}} = 0.$

Window.boundaryToCycles constructs the incoming map into $\ker(d_{\text{Next}})$. Boundaries are its actual linear range, and H is the module quotient of cycles by that range. Window.ofCochain imports these data at every integer degree of a Mathlib CochainComplex; ChainMap.ofCochain imports an actual cochain map without replacing its differentials.

For a chain map $f:C \rightarrow D$, the code constructs

$f_Z : Z(C) \rightarrow Z(D),$
 $H(f) : Z(C)/B(C) \rightarrow Z(D)/B(D).$

It proves identity, composition, and

$H(f)([z]) = 0 \iff \text{there exists } y \text{ with } d_{\text{Prev}_D}(y) = f_i(z).$

Define the module of killed representatives and its original relation submodule:

$\text{Krep} = \{z \text{ in } Z(C) : f_Z(z) \text{ is in } B(D)\},$
 $\text{Bold} = B(C), \text{ viewed as a submodule of Krep.}$

There is a constructed linear equivalence

$\text{Krep}/\text{Bold} \simeq \ker H(f).$

Forward map: the class of z goes to the kernel element $[z]$. It is well-defined because Bold consists exactly of original source boundaries. It is injective because equality of source classes is exactly a difference in Bold. It is surjective because every kernel class has a cycle representative whose image is a boundary. This is ChainMap.homologyKernelEquiv, obtained from a general quotient-kernel equivalence proved in the same file.

The calculation retains actions as well. Given chain endomorphisms a of C and b of D satisfying $f \circ a = b \circ f$, homology_intertwines proves $H(f) \circ H(a) = H(b) \circ H(f)$. kernelAction constructs the restriction of $H(a)$ to $\ker H(f)$. Thus passing to the comparison kernel does not discard a compatible operator or require asserting that the kernel is zero.

F.1.5 4. Internal homology on the complete support diagram

For coherent support-indexed chain windows, the code constructs the diagrams of preceding terms, middle terms, following terms, cycles, boundaries, and homology. Let R_L denote the reverse reconstruction.

The differential identity is typed as

$R_L(d_{\text{Next}}) \circ R_L(d_{\text{Prev}}) = z_L : R_L(C^{(i-1)}) \rightarrow R_L(C^{(i+1)}),$
 $z_L(1, x) = (1, 0).$

The source and target degrees are explicit. A theorem shows that z_L differs from the constant global-zero map whenever L has a non-bottom element. Cycles satisfy the actual equalizer equation $d(x) = e.d(x)$.

The internal homology projection

$q : R_L(Z(C)) \rightarrow R_L(H(C))$

is the coequalizer of the **original incoming differential into cycles** and its supported-zero companion. `homology_coequalizes` and `homology_coequalizer` establish both parts of that universal property against every split-linear target. This is not an assumed homology compatibility of reconstruction.

For every transition $i \leq j$,

$\text{rho}^{\wedge} H_{ij}([z]) = 0$ iff $\text{rho}_{ij}(z)$ is a boundary at j .

If z is not a boundary at i , its retained class $(i, [z])$ is not $(i, 0)$, regardless of whether that transition kills it. The quotient-kernel equivalence of section 3 computes all such classes as a module, not just a yes/no predicate.

F.1.6 5. A concrete comparison that detects more than the support labels

Fix any nontrivial commutative ring k . In the displayed degree let

$C: k \xrightarrow{-\theta} k \xrightarrow{-\theta} k,$
 $D: k \xrightarrow{-a} (a, \theta) \xrightarrow{-} k^2 \xrightarrow{-\theta} k.$

Use zero maps in the outside degrees. The two middle chain maps are

$f(a) = (a, \theta),$ $g(a) = (\theta, a).$

The code proves that the source class $[1]$ is nonzero. Its f -image is the **nonzero vector** $(1, 0)$, but that vector is a boundary in D , so $H(f)([1]) = 0$. In fact $H(f) = 0$ on every source class. The g -image $(0, 1)$ cannot be a boundary, because every boundary has second coordinate zero. Hence $H(g)([1])$ is nonzero and $H(f)$ is not $H(g)$.

The ordinary calculation gives $H(C) = k$ and $H(D) = k$ (the latter through second-coordinate projection). These identifications explain the example; the Lean acceptance tests directly prove the nonzero source class, the nonzero chain image, the zero induced map, the surviving class, and the distinction of induced maps rather than relying on a dimension computation.

Consequently the source and target complexes, individual fibre homology, and terminal homology can be identical while the transition kernel differs. This example is parametric over all nontrivial commutative rings, not a finite numerical experiment. Together with the general assembly theorem it supplies an explicit test of the information retained by support-indexed homology. It is an algebraic test case, not a claimed model of the global theta comparison.

F.1.7 6. Balanced finite jets, including the original split quotient square

Let k and B be commutative rings and fix a $k[t]$ -algebra structure on B . Let $u: k[t] \rightarrow B$ denote that map and

$I_{-}(\text{rho}, m) = ((t - \text{rho})^{\wedge} m), \quad m \geq 0.$

There is an explicit B -algebra equivalence

$B \otimes_{k[t]} (k[t] / I_{-}(\text{rho}, m)) \simeq B / ((u(t) - u(\text{rho}))^{\wedge} m).$

It maps $b \otimes [p]$ to $[u(p)b]$, with inverse $[b]$ to $b \otimes 1$. No flatness assumption is used for this quotient base-change theorem. This is a specialization of the existing Mathlib theorem `Algebra.tensorProduct.quotIdealMapEquivTensorQuot`, not a claim to have independently developed general tensor-product quotient theory.

splitBaseChange lifts this equivalence to the original G-construction on both sides and proves the commutative square with reflect. It preserves τ and e separately and is a genuine semiring equivalence, not just an amplitude comparison. Thus every element of the finite-jet quotient has an inverse image before forgetting split support.

For any $k[t]$ -module V and v in V , the exact balancing calculation is

$$(1-u(t)) \text{ tensor } (t.v) - u(t) \text{ tensor } (v-t.v) = 0.$$

Expanding gives $1 \text{ tensor } (t.v) - u(t) \text{ tensor } v$, which vanishes by the tensor balancing relation. This is the algebraic equation of the source's Mellin repair. The choice $u(t)=s$ and $t.v=Dv$ is an application once those analytic module structures and intertwining maps have been constructed. It does not assert that the entire global balanced kernel vanishes, or that analytic scalar extension is flat without proof.

F.1.8 7. Where Deligne enters the intended RH route

None of sections 1-6 imports a Deligne purity theorem. They provide the algebraic maps with which a proposed realization can now be specified and its kernel retained.

For this programme, the subsequent dependency diagram should distinguish the following mathematical obligations:

1. Construct the actual operator-balanced theta complex at the relevant support labels, and its action-compatible analytic comparison β to the spectral sheaf $O/(2\pi i)$. Identify its kernel or cone using the same maps; a finite-jet isomorphism is not a global injectivity argument.
2. Construct the geometric compact-support and ordinary complexes, their actual comparison c , their correspondence action, and a compatible realization identifying the required spectral sheaf with the image of $H^1(c)$. Any additional kernel/cone contributions to the trace must be kept or proved to vanish.
3. Establish two-sided weight control on that **same image**. In Deligne's geometric setting the compact-support upper estimate and the duality-derived ordinary-cohomology lower estimate meet on the image. The dualizing operation, degree and Tate twists, tensor-power maps, and hypotheses of the estimates are part of this step, not consequences stipulated in the definition of support.
4. Transport the spectral normalization without changing the arithmetic function. For a dilation parameter $a>1$, an eigenvalue a^ρ has modulus $a^{(\text{Re } \rho)}$. A proved weight-one modulus $a^{(1/2)}$, for the representation identified with every relevant spectral zero, then forces $\text{Re } \rho=1/2$. This last calculation does not supply the missing weight estimate; it states the precise final implication of this route.

These are dependencies of the specified route, not a claim that every possible proof of RH must follow it. The newly implemented kernel action is useful in steps 1-2 precisely because it keeps the part killed by a comparison available to the operator and trace calculation.

The supplied Connes-Consani article has an actual quotient-sheaf realization with a specified closure and scaling action. Its Theorem 1.2 realizes **critical zeros**. That theorem alone cannot be substituted for an identification representing all nontrivial zeros in the desired purity argument. This is a concrete source/target distinction for a future comparison map, not a dismissal of the construction.

F.1.9 8. Reading record and references

The six supplied S20 chunks were checked against their individual declared hashes and concatenated in order. The resulting archive has 216,580,466 bytes, 312 members, and SHA-256

e2005bf31e1fcf362765f73c315c855e0f504f24f34d3a0ab188049da522b7c8.

The current top-level readers/english_standalone.html and readers/source_language.html were used for the selected readings, not the nested zero-accepted salvage corpus. The inspected windows include the tensor-square improvement in 3.2.13, the upper-bound statement in 3.3.1, and the duality/image argument in 3.3.4-3.3.6. This is not an independent audit of every page or of the entire historical proof. The Connes-Consani introduction and Theorem 1.2 were read in the supplied volume, printed pages 83-85 (physical PDF pages 94-96). No copyrighted source corpus is included in this contribution.

Deligne, P. (1980). La conjecture de Weil. II. *Publications Mathematiques de l'IHES*, 52, 137-252. <https://doi.org/10.1007/BF02684780>

Connes, A., & Consani, C. (2023). Hochschild homology, trace map and zeta-cycles. In A. Connes, C. Consani, B. I. Dundas, M. Khalkhali, & H. Moscovici (Eds.), *Cyclic cohomology at 40: Achievements and future prospects* (Proceedings of Symposia in Pure Mathematics, Vol. 105, pp. 83-102). American Mathematical Society.

Mathlib contributors. (2026). *Mathlib4*, revision fabf563a7c95a166b8d7b6efca11c8b4dc9d911f. In particular, LinearAlgebra/Quotient/{Defs,Basic}.lean, RingTheory/TensorProduct/Quotient.lean, Algebra/Homology/HomologicalComplex.lean, and Algebra/Category/ModuleCat/Basic.lean.

F.1.10 9. Reproduction

From formal/splitzero on this branch, with elan and Python installed:

```
python3 prepare.py
python3 check_derived.py --prepare
python3 -m unittest -v test_derived_checker.py
lake update
# Verify the resolved Mathlib revision before fetching its cache.
test "$(git -C .lake/packages/mathlib rev-parse HEAD)" = fabf563a7c95a166b8d7b6efca11c8b4dc9d911f
lake exe cache get
lake -KmaxJobs=1 build
for name in SplitZeroReconstruction SplitZeroRecovery SplitZeroHomology SplitZeroInternalQuotient
↪ SplitZeroComplex SplitZeroHomologyExample SplitZeroFiniteJets; do
  lake env lean --trust=0 -DwarningAsError=true "$name.lean"
done
lake env lean --trust=0 -DwarningAsError=true AuditDerived.lean > AuditDerived.log
python3 check_derived.py AuditDerived.log
```

The manifest enumerates the selected declarations, including the principal structures, universal properties, equivalences, and example proofs. Missing, duplicate, unexpected, or nonstandard axiom reports fail. Source checks reject unfinished proofs and native-reduction escapes. The Python tests validate this checking machinery; the mathematical certificate is the successful Lean execution. Build-job totals include dependencies and are not theorem counts.

G PR 14: Coherent arithmetic interpolation and joint tensor boundary control

Source attribution. Complete TeX witness delivered with [web pull request #14](#); the associated Git revision is recorded in its delivery receipt.

Exact revision. 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc

Source file. Tau_Coherent_Interpolation_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 01a4fa76072ca4f3d8266c9fdbf61c816e53d09c673f8b8776b981b736c09e26

Original archive. Tau_Coherent_Interpolation_Control_2026-09-12.zip

Archive SHA-256. a753df8341d05b2fccb6487b1dabc0044153a6c1abcf35092504aadb95c51d26

G.1 Coherent arithmetic interpolation and joint tensor boundary control

Owner-directed continuation, 12 September 2026. Written mathematical arguments for review; the new analytic and matrix results in this note have not been Lean-checked.

G.1.1 0. Accepted inputs and the change in the continuation

The new formalization receipt reports—and the repository metadata and workflow step summaries confirm—a successful check at 5d2772ea0a16d177ac3e01ff70394b90ba95a232 in PR #12. That certificate covers the specified retraction’s algebraic/topological consequences, the complete coefficient homotopy family, and the stated finite-chart dual maps. It does not certify the analytic construction of the theta retraction, support-changing synchronization homotopies, or the matrix estimates developed here. Its branch is left untouched.

The second attachment proves, for an unrestricted finite correction matrix B , the exact formula

$$G_B = G_{\perp,m} + (B + H_m^{-1}C_m)^* H_m (B + H_m^{-1}C_m). \quad (0.1)$$

Once $m+1 \geq \dim E_Z$, every positive metric ray is realized by an actual theta-boundary correction, with the arithmetic jets unchanged. Its further conclusion

$$\inf_B \epsilon_A(G_B) = \max_{\rho \in Z} |2\Re \rho - 1| \quad (0.2)$$

therefore has to be retained. This note does not count unrestricted metric optimization as a new arithmetic estimate. It instead uses the unique least-norm representative for a fixed, increasing space of the original arithmetic boundaries. It proves compatibility of these representatives for all finite packets, an exact full-jet moment-matrix formula, and a joint total-degree tensor construction.

All conclusions are derived on the existing characteristic-zero theta complex over the marked tau-base. None is asserted to be an application of finite-field purity to a space merely renamed tau.

G.1.2 1. Fixed scalar, cochain, and arithmetic maps

Retain the original scalar construction

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

and the entire square

$$\boxed{\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}} \quad (1.1)$$

The target of $p_{\mathbb{Z}}$ is infinite. This quotient remains part of the marked geometry over $\mathfrak{b}_\tau$. Coefficient maps below fix external absence and carry a represented scalar zero to the represented scalar zero of the target.

Use the unchanged spaces

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^k F(x)| < \infty \forall b \in \mathbb{Z}, k \geq 0\}, \quad D = -x\partial_x,$$

$$\Theta\phi(x) = \sum_{m \neq 0} \phi(mx), \quad C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V.$$

The complex is in degrees 0, 1. Write $q : \mathcal{B} \rightarrow Q$. Its action is by the original D , not a shifted operator. Let

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \quad \phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad f_0 = \Theta\phi_*,$$

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad \mathcal{M}f_0 = g. \quad (1.2)$$

The established range identities are used with these exact constants. They give $\mathcal{M}\Theta\phi = gH_\phi$, with H_ϕ entire. In particular every finite jet map used below annihilates ΘV .

For $n \geq 0$, retain

$$V_n = \text{span}\{\phi_*, D\phi_*, \dots, D^n\phi_*\}, \quad L_n = \Theta V_n,$$

$$Q_n = \mathcal{B}/L_n \longrightarrow Q_{n+1}, \quad \ker = L_{n+1}/L_n. \quad (1.3)$$

Their split reconstructions have labels V_n . The transport of a class in L_{n+1}/L_n is the supported zero at V_{n+1} ; it is not external absence. The derivative has type $Q_n \rightarrow Q_{n+1}$ because $DL_n \subseteq L_{n+1}$.

G.1.3 2. A polynomial presentation of the actual finite representatives

Take a nonempty finite set Z of actual zeros of g , with their complete orders m_ρ , and put

$$h(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E_h = \mathbb{C}[t]/(h), \quad v_h(s) = g(s)/h(s). \quad (2.1)$$

The polynomial variable t acts on the source by D and is evaluated at s by Mellin transformation. Full orders make v_h nonzero at all selected centres. Let j_h be the complete Hermite-jet/Chinese-remainder map into E_h and retain

$$v_h = j_h v_h \in E_h^\times, \quad \varepsilon_h = v_h^{-1}. \quad (2.2)$$

The unit ε_h is not the scalar e . Its split scalar is $\varepsilon_h^\bullet$.

There is an actual $F_h \in \mathcal{B}$ with $h(D)F_h = f_0$ and $\mathcal{M}F_h = v_h$. It can be obtained successively from

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy, \quad (D - \rho)S_\rho F = F, \quad (2.3)$$

whenever the required Mellin value vanishes. At zero, replace the integral by its negative from 0 to x , using that vanishing; at infinity use the displayed integral. The defining estimates of $\mathcal{B}$ prove rapid decay of every scaling derivative at both ends. Repeated zero jets permit iteration. Uniqueness follows from injectivity of Mellin and $h\mathcal{M}F = 0 \Rightarrow F = 0$.

Define the linear map

$$\boxed{\mathcal{T}_h : \mathbb{C}[t] \longrightarrow \mathcal{B}, \quad P \longmapsto P(D)F_h.} \quad (2.4)$$

It is injective and satisfies

$$\mathcal{M}\mathcal{T}_h(P) = \frac{g}{h}P, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h\mathcal{T}_h(P) = v_h[P]_h. \quad (2.5)$$

Here $J_h F = j_h \mathcal{M}F$. The first equation retains the whole completed function, not its values only on the critical line.

For $N = n + d$, let $\mathcal{P}_N = \mathbb{C}[t]_{\leq N}$, and define

$$\mathcal{J}_{h,N} : \mathcal{P}_N \twoheadrightarrow E_h, \quad P \longmapsto v_h[P]_h.$$

Its exact sequence is

$$\boxed{0 \longrightarrow \mathcal{P}_n \xrightarrow{\times h} \mathcal{P}_{n+d} \xrightarrow{\mathcal{J}_{h,n+d}} E_h \longrightarrow 0.} \quad (2.6)$$

Applying $\mathcal{T}_h$ sends the left map to the original theta relations L_n . Thus its split internal quotient has exactly the relation $P - P' \in h\mathcal{P}_n$, with the support label retained.

The original source section is

$$s_h(u) = \mathcal{T}_h(\text{rem}_h(\varepsilon_h u)), \quad J_h s_h = 1, \quad q s_h = \sigma_h. \quad (2.7)$$

No arbitrary metric or section has entered (2.6).

G.1.4 3. Exact arithmetic moment matrices and the full-jet interpolation formula

The Hilbert observation is $\mathcal{H} = L^2((0, \infty), dx)$. The precise transform map is

$$\mathcal{H} \xrightarrow{\mathcal{L}} L^2(\mathbb{R}, dy) \xrightarrow{\mathcal{F}_+} L^2(\mathbb{R}, dt/(2\pi)),$$

$$\mathcal{L}F(y) = e^{y/2}F(e^y), \quad \mathcal{F}_+\psi(t) = \int_{\mathbb{R}} \psi(y)e^{ity}dy, \quad (\mathcal{F}_+\mathcal{L}F)(t) = \mathcal{M}F(\tfrac{1}{2} + it). \quad (3.1)$$

Both arrows are unitary with these measures. The inverse Fourier factor is $1/(2\pi)$; no mass of a measure below is set equal to one. The same map carries D to multiplication by $s_t = 1/2 + it$.

Define the finite positive measure

$$d\nu_h(t) = \left| \frac{g(\frac{1}{2} + it)}{h(\frac{1}{2} + it)} \right|^2 \frac{dt}{2\pi}. \quad (3.2)$$

Any apparent singularity at a selected critical-line zero is removed by the entire function v_h . All moments are finite. In the fixed basis $1, t, \dots, t^N$ of $\mathcal{P}_N$, put

$$(M_{h,N})_{ij} = \int_{\mathbb{R}} \overline{s_t^i} s_t^j d\nu_h(t), \quad 0 \leq i, j \leq N. \quad (3.3)$$

In the integrand t is the real integration variable; the polynomial basis is evaluated at s_t . This prevents the polynomial variable from being confused with the spectral-height coordinate.

The matrix $M_{h,N}$ is positive definite. A polynomial with zero norm vanishes almost everywhere on a line on which v_h is nonzero almost everywhere, hence is zero. Write $J_{h,N}$ for the matrix of $\mathcal{J}_{h,N}$ in the fixed quotient basis $1, t, \dots, t^{d-1}$. It includes multiplication by the actual unit v_h .

Set

$$\mathcal{C}_{h,N} = J_{h,N} M_{h,N}^{-1} J_{h,N}^*, \quad C_{h,N} = M_{h,N}^{-1} J_{h,N}^* \mathcal{C}_{h,N}^{-1}. \quad (3.4)$$

These are finite matrices on declared coordinate spaces. $J_{h,N} C_{h,N} = 1$. The actual test-function map is

$$R_{h,n} = \mathcal{T}_h C_{h,n+d} : E_h \longrightarrow \mathcal{B}. \quad (3.5)$$

Theorem. This is precisely the preceding orthogonal representative $(1 - P_n)s_h$, where P_n is the $L^2(dx)$ orthogonal projection onto L_n . In particular,

$$qR_{h,n} = \sigma_h, \quad J_h R_{h,n} = 1, \quad G_{h,n} := R_{h,n}^* R_{h,n} = \mathcal{C}_{h,n+d}^{-1}. \quad (3.6)$$

Proof. Every polynomial with jet u is $C_{h,N}u + z$ with $J_{h,N}z = 0$. The cross term vanishes because

$$z^* M_{h,N} C_{h,N} u = z^* J_{h,N}^* \mathcal{C}_{h,N}^{-1} u = 0.$$

Its norm is therefore $u^* \mathcal{C}_{h,N}^{-1} u + z^* M_{h,N} z$. This proves the unique minimizer. By (2.6), its difference from the degree-below- d lift (2.7) is exactly a member of L_n . It must be $(1 - P_n)s_h$. This also proves (3.6).

Thus the subtraction-of-nearby-Grams formula of the previous note has the exact alternative (3.6): invert the full-jet interpolation matrix. This is a matrix-valued Christoffel construction with the arithmetic local unit retained. The finite least-norm identity itself is standard linear algebra; its application here identifies the actual original representative, source relation, and completed multiplier.

For one simple zero ρ , the same formula is

$$G_{\rho,n} = \frac{1}{|g'(\rho)|^2 \mathcal{K}_{\nu_{\rho,n+1}}(\rho, \rho)}, \quad \mathcal{K}_{\nu,N}(z, w) = (1, z, \dots, z^N) M_N^{-1} (1, w, \dots, w^N)^*. \quad (3.7)$$

The complex conjugate in the second evaluation is part of the displayed kernel. Higher orders use (3.4), not just point evaluations. The factor $g' = 2\xi'$ is retained.

G.1.5 4. All finite packets fit the same source filtration

Let $h \mid H$ be polynomials for nested packets with full orders and set $m = H/h$. Since the newly added centres are disjoint, h and m are coprime. There is a $\mathbb{C}[t]$ -linear CRT injection

$$\iota_{hH} : E_h \hookrightarrow E_H,$$

which inserts zero jets at the new centres. It is multiplication by the appropriate CRT idempotent after lifting a residue; it sends the unit of E_h to that idempotent, so it is not being asserted to be a unital ring map.

The polynomial map $P \mapsto mP$ has the exact properties

$$\mathcal{T}_H(mP) = \mathcal{T}_h(P), \quad J_H \mathcal{T}_h(P) = \iota_{hH} \mathcal{J}_h(P), \quad (4.1)$$

and is isometric from ν_h to ν_H because $|v_H mP| = |v_h P|$. The CRT equations and degree bounds prove

$$s_H \iota_{hH} = s_h.$$

The original L_n is independent of the packet, so

$$\boxed{R_{H,n} \iota_{hH} = R_{h,n}, \quad \iota_{hH}^* G_{H,n} \iota_{hH} = G_{h,n}, \quad \iota_{hH}^* W_{H,n} \iota_{hH} = W_{h,n}.} \quad (4.2)$$

The action also intertwines: $A_H \iota_{hH} = \iota_{hH} A_h$, where $A_h = M_t$. Thus a certificate on the larger packet restricts by this actual congruence to the smaller packet. The inverse interpolation matrices satisfy the corresponding inverse-compression formula; inverse and compression are not commuted.

The finite sections consequently glue on the algebraic direct limit of the actual packet injections. This statement does not assert convergence of an infinite sum of packet projectors or identify the whole Q with that direct limit.

At every support label, lift these maps as $(\lambda, u) \mapsto (\lambda, \iota_{hH} u)$ and $(\lambda, u) \mapsto (\lambda, R_{h,n} u)$. Added zero jets remain represented zeros. The label map in (1.3) is still the map that admits one more source relation.

G.1.6 5. A global density result, with its precise cohomological consequence

The single derivative family $\{D^n f_0 : n \geq 0\}$ is already dense in $\mathcal{H}$.

Proof. Put $d\nu(t) = |g(1/2 + it)|^2 dt / (2\pi)$. It has an exponential moment of every order $c < \pi/2$. To verify this, the alternating-series formula and partial summation give

$$|\zeta(\tfrac{1}{2} + it)| \leq \frac{2|\tfrac{1}{2} + it|}{\sqrt{2} - 1}.$$

The gamma vertical-line asymptotic gives, for sufficiently large $|t|$,

$$|g(\tfrac{1}{2} + it)|^2 \leq C(1 + |t|)^{11/2} e^{-\pi|t|/2}.$$

The constant C is a fixed bound, not a numerical enclosure claimed here. The gamma fact used is NIST DLMF 5.11.9. The same exponential-moment property holds for ν_h , since v_h is entire at the finitely many possible denominator zeros and has polynomially modified tails.

If $f \in L^2(\nu)$ is orthogonal to every polynomial, the finite measure $\bar{f} d\nu$ has a holomorphic Fourier transform in a nonempty horizontal strip, by Cauchy–Schwarz and the exponential moment. All

derivatives at the origin vanish. Analytic uniqueness makes its transform zero; Fourier uniqueness for finite measures makes the measure zero. Thus polynomials are dense in $L^2(\nu)$.

Multiplication by the actual function $g(1/2 + it)$ is a unitary map $L^2(\nu) \rightarrow L^2(dt/(2\pi))$: its inverse divides by g away from its measure-zero set of real-line zeros. The inverse is a map of weighted L^2 spaces, not holomorphic division through an arithmetic zero. Combining it with (3.1) sends polynomials to $P(D)f_0$. This proves the assertion.

It follows for each fixed packet that

$$\boxed{G_{h,n} \longrightarrow 0 \quad \text{in every matrix entry and in operator norm.}} \quad (5.1)$$

Indeed $P_n \rightarrow 1$ strongly on $\mathcal{H}$, and the finite number of columns of s_h is fixed. Since each $G_{h,n}$ is positive definite, the least eigenvalue of its inverse tends to infinity. Every nonzero arithmetic row functional $\ell : E_h \rightarrow \mathbb{C}$ therefore has

$$\boxed{\|\ell\|_{G_{h,n}^\vee}^2 = \ell \mathcal{C}_{h,n+d} \ell^* \longrightarrow \infty.} \quad (5.2)$$

These facts do not kill a cohomology class. The actual maps are

$$\mathcal{B}/L_n \longrightarrow \mathcal{H}/L_n, \quad Q = \mathcal{B}/\Theta V \longrightarrow \mathcal{H}/\overline{\Theta V}^{\mathcal{H}} = 0. \quad (5.3)$$

The first map is injective because L_n is finite-dimensional and consists of the same smooth functions in both spaces. The second map forgets the stronger topology and has the whole Q as kernel. Its split lift lands at the represented zero in $G(0)$, not at τ . $G(0)$ retains both elements. An active class may therefore tend to zero under the L^2 observation while remaining fixed by $qR_{h,n} = \sigma_h$ and $J_h R_{h,n} = 1$.

The constructive consequence is to estimate relative boundary flux against the complete interpolation cost (3.4), not to use the absolute smallness of $G_{h,n}$ as a weight bound.

G.1.7 6. Tail representation and same-metric reflection control

Retain the unscaled orthogonal vectors from the previous calculation:

$$u_0 = f_0, \quad u_n = (1 - P_{n-1})D^n f_0, \quad \mathfrak{h}_n = \|u_n\|^2, \quad (6.1)$$

$$Du_n = u_{n+1} + \frac{1}{2}u_n - \frac{\mathfrak{h}_n}{\mathfrak{h}_{n-1}}u_{n-1} \quad (n \geq 1).$$

The diagonal $1/2$ and the negative last coefficient follow from $D^* = 1 - D$ on these functions. Neither the u_n nor their norms are rescaled. Define $r_n = \mathfrak{h}_n^{-1}u_n^* s_h$. The new density result makes the entire tail formula exact:

$$\boxed{G_{h,n} = \sum_{j>n} \mathfrak{h}_j r_j^* r_j.} \quad (6.2)$$

This series converges in finite matrix norm. Its source map is $u \mapsto (r_j u)_{j>n}$ into the weighted sequence space with norm $\sum_{j>n} \mathfrak{h}_j |c_j|^2$. The map retains every coefficient; it is an isometry for $G_{h,n}$.

The previous boundary calculation, or telescoping (6.1), gives

$$\boxed{W_{h,n} = A_h^* G_{h,n} + G_{h,n} A_h - G_{h,n} = \mathfrak{h}_{n+1}(r_{n+1}^* r_n + r_n^* r_{n+1}).} \quad (6.3)$$

It pairs two explicit maps into L_{n+1}/L_n . At the original support V_n the derivative defect is $-[u_{n+1}]r_n$; at V_{n+1} it becomes the supported zero. This is also the input to the now-checked coefficient homotopy family, with $\kappa_n = K_n J_h$ and $\kappa_n \Theta = 0$.

For a reflection-stable packet define the actual antilinear involution

$$j_h u(s) = \overline{u(1 - \bar{s})} \pmod{h}.$$

On functions use $j_B F(x) = x^{-1} \overline{F(1/x)}$. It is antiunitary on $\mathcal{H}$ and satisfies $Dj_B = j_B(1 - D)$. The specific section (2.7) satisfies $j_B s_h = s_h j_h$: since $h^\dagger = (-1)^d h$, both the unit and the numerator have the corresponding $(-1)^d$ factor. Also $j_B L_n = L_n$, so uniqueness of orthogonal projection gives $j_B R_{h,n} = R_{h,n} j_h$.

Write $j_h u = C_h \bar{u}$ in the retained coordinates. Then

$$\boxed{C_h^* G_{h,n} C_h = \overline{G_{h,n}}, \quad A_h C_h = C_h (I - \overline{A_h}), \quad C_h^* W_{h,n} C_h = -\overline{W_{h,n}}.} \quad (6.4)$$

Consequently the generalized spectrum of $(W_{h,n}, G_{h,n})$ is symmetric under $\lambda \mapsto -\lambda$ **for this same canonical metric**, with full multiplicities retained. An upper inequality $W_{h,n} \preceq \epsilon G_{h,n}$ therefore implies the lower inequality $W_{h,n} \succeq -\epsilon G_{h,n}$ by applying j_h to the input vector.

The residue duality is still the separate explicit morphism with matrix S_h , and its pulled-back metric is $S_h^* G_{h,n}^{-1} S_h$. The relation to the current metric is the linear map $G_{h,n}^{-1} S_h$ into its antilinear dual coordinates. The arithmetic Weil form remains $S_h J_g$; no positivity is supplied merely by (6.4).

For completeness, retain the full rank-two formula

$$a_n = r_n G_{h,n}^{-1} r_n^*, \quad b_n = r_{n+1} G_{h,n}^{-1} r_{n+1}^*, \quad c_n = r_n G_{h,n}^{-1} r_{n+1}^*.$$

The least control excess is

$$\epsilon_{h,n} = \mathfrak{h}_{n+1} \left(|\Re c_n| + \sqrt{a_n b_n - (\Im c_n)^2} \right). \quad (6.5)$$

For reflection-stable Z , the additional proved identity is $\Re c_n = 0$: take the trace of $G_{h,n}^{-1} W_{h,n}$, which is $2\Re \operatorname{Tr} A_h - d = 0$. Thus (6.5) also equals $\mathfrak{h}_{n+1} \sqrt{a_n b_n - (\Im c_n)^2}$. Both equations are retained. Equation (3.6) supplies every occurrence of $G_{h,n}^{-1}$ without a difference of nearly equal Gram matrices.

G.1.8 7. Joint total-degree control on the actual product complex

The tensor degree will be denoted k , independently of the relation degree n and the packet multiplicities. Put

$$E_h^{(k)} = E_h^{\otimes k}, \quad A_h^{(k)} = \sum_{j=1}^k I^{\otimes(j-1)} \otimes A_h \otimes I^{\otimes(k-j)}.$$

The existing product complex over $\mathfrak{b}_\tau$ has differential

$$d(x_1 \otimes \cdots \otimes x_k) = \sum_j (-1)^{\sum_{i < j} |x_i|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_k.$$

Define $\mathcal{P}_M^{(k)} = \mathbb{C}[t_1, \dots, t_k]_{\text{total degree} \leq M}$ for $M \geq k(d-1)$ and

$$\mathcal{T}_h^{(k)} P = P(D_1, \dots, D_k) (F_h^{\otimes k}). \quad (7.1)$$

Its Mellin transform is $P(s_1, \dots, s_k) \prod_j g(s_j)/h(s_j)$. Its full jet map $J_{k,M}$ is reduction modulo each $h(t_j)$ followed by multiplication by $v_h^{\otimes k}$. The target is $E_h^{(k)}$ and this map is surjective. The tensor remainder basis has total degree at most $k(d-1)$.

The exact relation space is

$$\mathcal{I}_{k,M} = \mathcal{P}_M^{(k)} \cap (h(t_1), \dots, h(t_k)). \quad (7.2)$$

Polynomial division in the fixed order $t_1, \dots, t_k$ represents each member as $\sum_j h(t_j)P_j$, with $\deg P_j \leq M-d$. It terminates because every h is monic and replacing a leading power lowers total degree. Under (7.1), each term is an actual top cochain boundary: in the j -th source factor use ϕ_* in place of F_h and give its primitive the factor $(-1)^{j-1}$. The differential contributes the second $(-1)^{j-1}$. The resulting boundary is $+h(t_j)P_j$ with no removed sign.

Thus these joint corrections are maps in the original product complex, not freely chosen positive tensor metrics. Set $\mathcal{B}_{k,M} = \mathcal{T}_h^{(k)} \mathcal{I}_{k,M}$.

The moment matrix on the monomials of total degree at most M is

$$(M_{k,M})_{\alpha\beta} = \prod_{j=1}^k \int \overline{s_t^{\alpha_j}} s_t^{\beta_j} d\nu_h(t). \quad (7.3)$$

Its positive definiteness follows because a polynomial cannot vanish almost everywhere on the full real product unless it is zero. It is a principal submatrix of the stated product moment matrix, not a declaration that the total-degree space is a tensor product of smaller univariate spaces.

The actual joint least-norm representative and its metric are

$$R_{k,M} = \mathcal{T}_h^{(k)} M_{k,M}^{-1} J_{k,M}^* (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1},$$

$$G_{k,M} = (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1}. \quad (7.4)$$

The proof is the same explicit constrained-minimum calculation as (3.6), now with the relation ideal (7.2). In particular its arithmetic class is exactly $\sigma_h^{\otimes k}$ and every jet is fixed.

For a rectangular degree budget N one has the actual inclusion

$$\mathcal{P}_N^{\otimes k} \hookrightarrow \mathcal{P}_{kN}^{(k)}.$$

It carries the old product representative into the feasible set of the new problem. Therefore

$$G_{k,kN} \preceq G_{h,N-d}^{\otimes k} \quad (N \geq d). \quad (7.5)$$

This is an inequality from an inclusion of admissible original boundaries. It is not asserted to imply the same ordering of control forms, whose dependence on A is retained separately.

G.1.9 8. The product error is one new total-degree layer

Let $\mathcal{H}_k = L^2(\mathbb{R}_{>0}^k, d^k x)$, and let $P_{\mathcal{B}_{k,M}}$ be its orthogonal projection onto the finite-dimensional actual boundary space. With the fixed low-degree section $s_h^{\otimes k}$,

$$R_{k,M} = (1 - P_{\mathcal{B}_{k,M}}) s_h^{\otimes k}.$$

Put

$$\mathcal{E}_{k,M} = (1 - P_{\mathcal{B}_{k,M}})\mathcal{B}_{k,M+1}.$$

Because the jet map is already onto at both degrees,

$$\dim \mathcal{E}_{k,M} = \binom{M+k}{k-1}. \quad (8.1)$$

This is $\dim \mathcal{P}_{M+1}^{(k)} - \dim \mathcal{P}_M^{(k)}$; the d^k jet constraints cancel from the dimension difference.

Let $D^{(k)} = \sum_j D_j$. Its induced action on jets is $A_h^{(k)}$. The actual derivative defect lies in $\mathcal{B}_{k,M+1}$:

$$B_{k,M} = D^{(k)} R_{k,M} - R_{k,M} A_h^{(k)}.$$

Define the two explicitly typed maps into the same new relation layer

$$Y_{k,M} = P_{\mathcal{E}_{k,M}} R_{k,M} : E_h^{(k)} \rightarrow \mathcal{E}_{k,M},$$

$$C_{k,M} = (1 - P_{\mathcal{B}_{k,M}}) B_{k,M} : E_h^{(k)} \rightarrow \mathcal{E}_{k,M}.$$

Integration by parts gives $(D^{(k)})^* + D^{(k)} = kI$ on these functions. Orthogonality then proves

$$W_{k,M} := (A_h^{(k)})^* G_{k,M} + G_{k,M} A_h^{(k)} - k G_{k,M} = -(Y_{k,M}^* C_{k,M} + C_{k,M}^* Y_{k,M}), \quad (8.2)$$

$$G_{k,M} - G_{k,M+1} = Y_{k,M}^* Y_{k,M}, \quad \text{rank } W_{k,M} \leq 2 \binom{M+k}{k-1}. \quad (8.3)$$

For (8.3), use the orthogonal decomposition $\mathcal{B}_{k,M+1} = \mathcal{B}_{k,M} \oplus \mathcal{E}_{k,M}$. Its projection sends $R_{k,M}$ to $Y_{k,M}$ and the new representative is $R_{k,M} - Y_{k,M}$.

The source quotient map is the same degree transport as before:

$$\mathcal{B}_{k,M+1}/\mathcal{B}_{k,M} \xrightarrow{\sim} \mathcal{E}_{k,M}, \quad [z] \mapsto (1 - P_{\mathcal{B}_{k,M}})z.$$

A derivative class in this kernel becomes the supported zero only at the next total-degree label. The primitive retains which tensor factor supplied the theta boundary and its cochain sign. Synchronization by $(1, e)$ in each required two-leg factor supplies the common joint support, while the original mask remains recorded.

Thus the product estimate need not simply copy k identical one-factor errors: (8.2) gives a new jointly optimized boundary-layer expression with completely specified moment data and maps. Its rank bound alone does not bound its generalized eigenvalues.

G.1.10 9. Same-metric reflection and residue duality on the product

For a reflection-stable packet, $j_h^{\otimes k}$ is the antilinear involution on $E_h^{(k)}$, defined on pure tensors by applying j_h to every factor. The corresponding function reflection is $F(x_1, \dots, x_k) \mapsto \prod_j x_j^{-1} \overline{F(x_1^{-1}, \dots, x_k^{-1})}$.

It preserves both the total-degree numerator space and its relation ideal. Since $(g/h)^\dagger = (-1)^d (g/h)$, its action on a numerator is $(-1)^{kd} P^\dagger$; this factor is retained. Uniqueness of the minimum in (7.4) proves reflection compatibility of $R_{k,M}$. Therefore the analogue of (6.4) gives symmetry of the generalized spectrum of $(W_{k,M}, G_{k,M})$ on the same metric.

Residue duality remains the explicitly specified map with matrix $S_h^{\otimes k}$ and target line $\mathcal{L}_k$, on which dilation acts by a^k . It satisfies

$$(A_h^{(k)})^* S_h^{\otimes k} + S_h^{\otimes k} A_h^{(k)} = k S_h^{\otimes k}.$$

Any fixed degree-orientation scalar in the cochain evaluation is retained separately; the ordered product of the scalar residue pairings here has matrix exactly $S_h^{\otimes k}$.

Its dual metric and control are

$$G_{k,M}^D = (S_h^{\otimes k})^* G_{k,M}^{-1} S_h^{\otimes k},$$

$$W_{k,M}^D = -(S_h^{\otimes k})^* G_{k,M}^{-1} W_{k,M} G_{k,M}^{-1} S_h^{\otimes k}. \quad (9.1)$$

The Jacobian contraction is still the original multiplication trace, with every multiplicity, and is unchanged by the source boundaries used to construct (7.4). Neither (9.1) nor reflection alone establishes an arithmetic upper bound.

G.1.11 10. Quantitative certificate and the Deligne comparison

For one or several variables write M for the positive moment matrix, J for its fixed full-jet map, and $\mathcal{C} = JM^{-1}J^*$. These actual matrices now replace an unrestricted metric search.

A useful deterministic enclosure is as follows. Suppose a Hermitian approximation $\widehat{M}$ has a verified bound $\|M - \widehat{M}\| \leq \eta$ and $\widehat{M} \succeq \mu I$ for a retained $\mu > \eta$. Then

$$\|M^{-1} - \widehat{M}^{-1}\| \leq \frac{\eta}{\mu(\mu - \eta)}, \quad \|\mathcal{C} - J\widehat{M}^{-1}J^*\| \leq \|J\|^2 \frac{\eta}{\mu(\mu - \eta)}. \quad (10.1)$$

This is the resolvent identity for inverses. The positive lower bound on $\mathcal{C}$ must also be retained before a second inversion. With $\widehat{G}$ and $\widehat{W}$ and verified errors η_G, η_W , the sufficient matrix certificate is

$$\widehat{G} - \eta_G I \succ 0, \quad \epsilon \widehat{G} \pm \widehat{W} - (\epsilon \eta_G + \eta_W) I \succeq 0. \quad (10.2)$$

No numerical interval enclosure for the actual infinite arithmetic moments is claimed merely from a floating-point calculation. The previous exterior-theta sums give another exact representation of their one-variable inputs; all tails and roundoff must be supplied to (10.1).

Deligne's retained Corollaries 3.3.4–3.3.6 place upper and dual lower bounds on the same compact-to-ordinary image; the tensor step in 3.2.13 improves an estimate only after additional geometric control. This note supplies a canonical coherent family, its exact source relation ideal on products, and the same-metric reflection/dual operations with which such an upper estimate can now be tested. It does not import finite-field hypotheses into the tau chart.

For an actual eigenvector $A_h v = \rho v$, every constructed tensor metric still has the exact diagnostic

$$\frac{(v^{\otimes k})^* W_{k,M} v^{\otimes k}}{(v^{\otimes k})^* G_{k,M} v^{\otimes k}} = k(2\Re \rho - 1). \quad (10.3)$$

The quantity to bound is therefore the generalized boundary form (8.2) relative to (7.4). A dimension/rank estimate, a decrease of a Gram matrix, or unrestricted metric realizability is not substituted for that relative estimate. The new constructive route is the exact finite full-jet moment problem together with its next-layer flux, not another homotopy choice.

G.1.12 11. Reading, provenance, and validation

Newly read: both supplied formalization/control notes in full; the preceding local `Tau_Homotopy_Boundary_Control/NOTE.tex`, especially sections 4–10; the retained Deligne English TeX excerpt 3.3.4–3.3.6 and French 6.2.4–6.2.7. This is not a new audit of every page of Weil II. No OCR or PDF extraction was used. Source hashes are in the package reading record.

External context checked: NIST DLMF 5.11.9 for the gamma bound; Connes–Consani–Moscovici, arXiv:2403.01247, abstract, for prior moment/Jacobi constructions from squared absolute local factors; Romik, arXiv:1902.06330, abstract, for prior orthogonal-polynomial expansions of Ξ ; Lasserre–Pauwels, arXiv:1606.03858, abstract, for inverse-moment/Christoffel context. Only the gamma asymptotic is used as an external estimate; the other papers are priority/context references, not sources of an unread theorem. The arguments above prove the specific maps used here.

The accompanying exact checker uses declared finite discrete calibration measures and rational complex polynomial data. It checks the general finite matrix identities, joint relation ranks, signs, full repeated-root reductions, reflection, naturality, and comparison metrics. Its model polynomial is never labelled an arithmetic zeta counterexample. Analytic density, asymptotic gamma bounds, and the original theta range identities are written arguments, not certified by those finite tests. No new Lean execution is claimed. No RH or GRH conclusion is claimed.